

Centered priors

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \mathbf{R}, [\sigma, \tau])$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$\mathbf{R} \sim \text{LKJCorr}(4)$$

Non-centered priors

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha = \bar{\alpha} + v_{-,1}$$

$$\beta = \bar{\beta} + v_{-,2}$$

$$\mathbf{v} = (\text{diag}(\sigma) \mathbf{L} \mathbf{Z})^\top$$

$$\mathbf{Z}_{j,k} \sim \text{Normal}(0, 1)$$

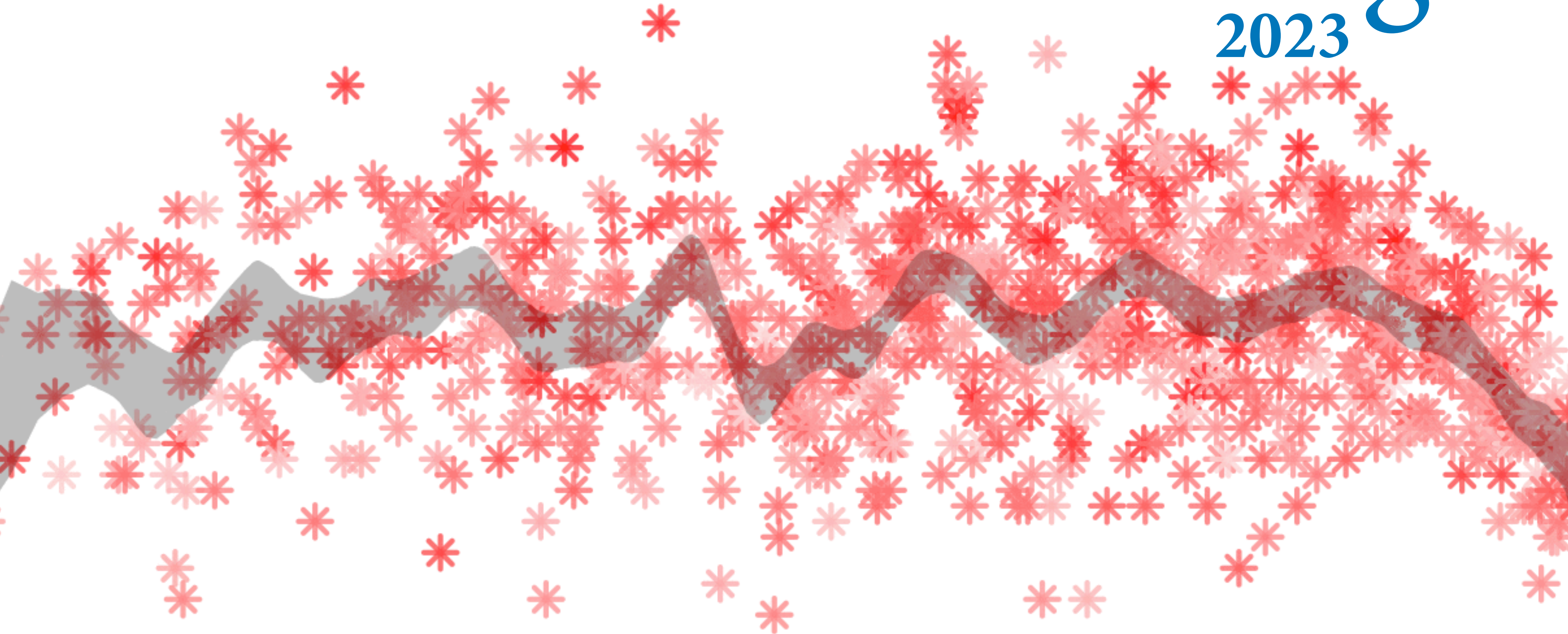
$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\mathbf{R} \sim \text{LKJCorr}(4)$$

Statistical Rethinking

2023



12. Multilevel Models

Repeat observations

12 stories (S)

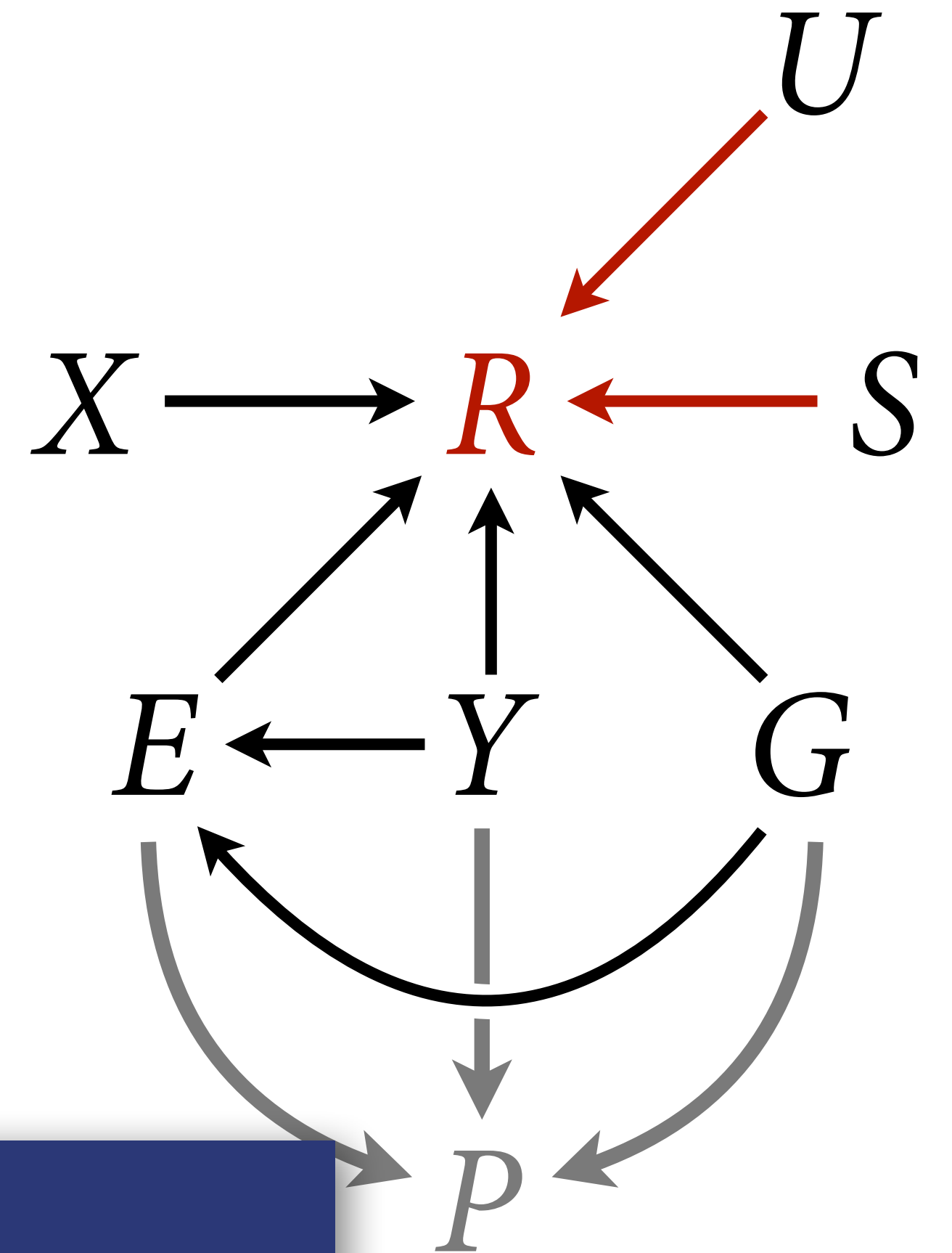
```
> table(d$story)
```

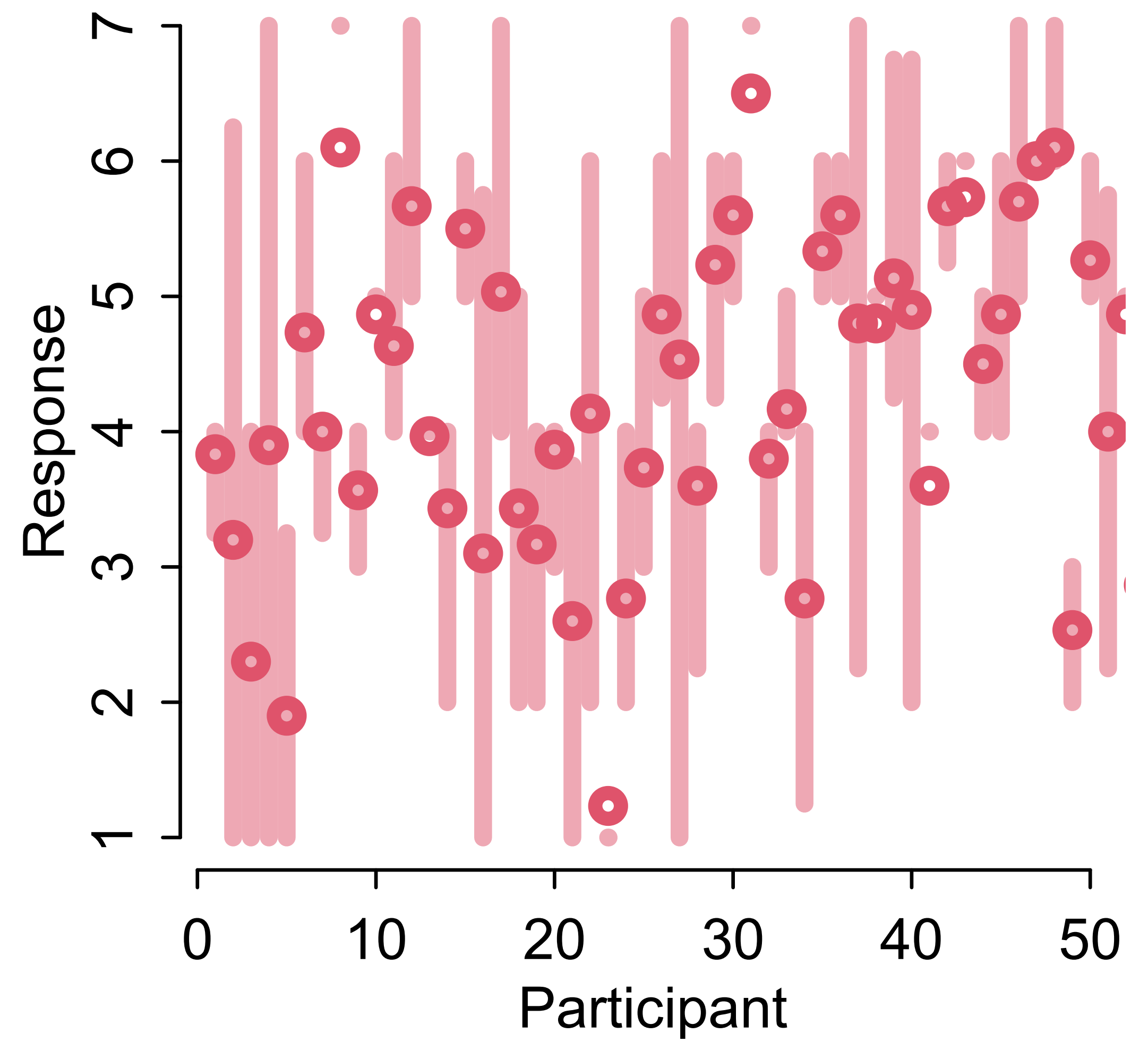
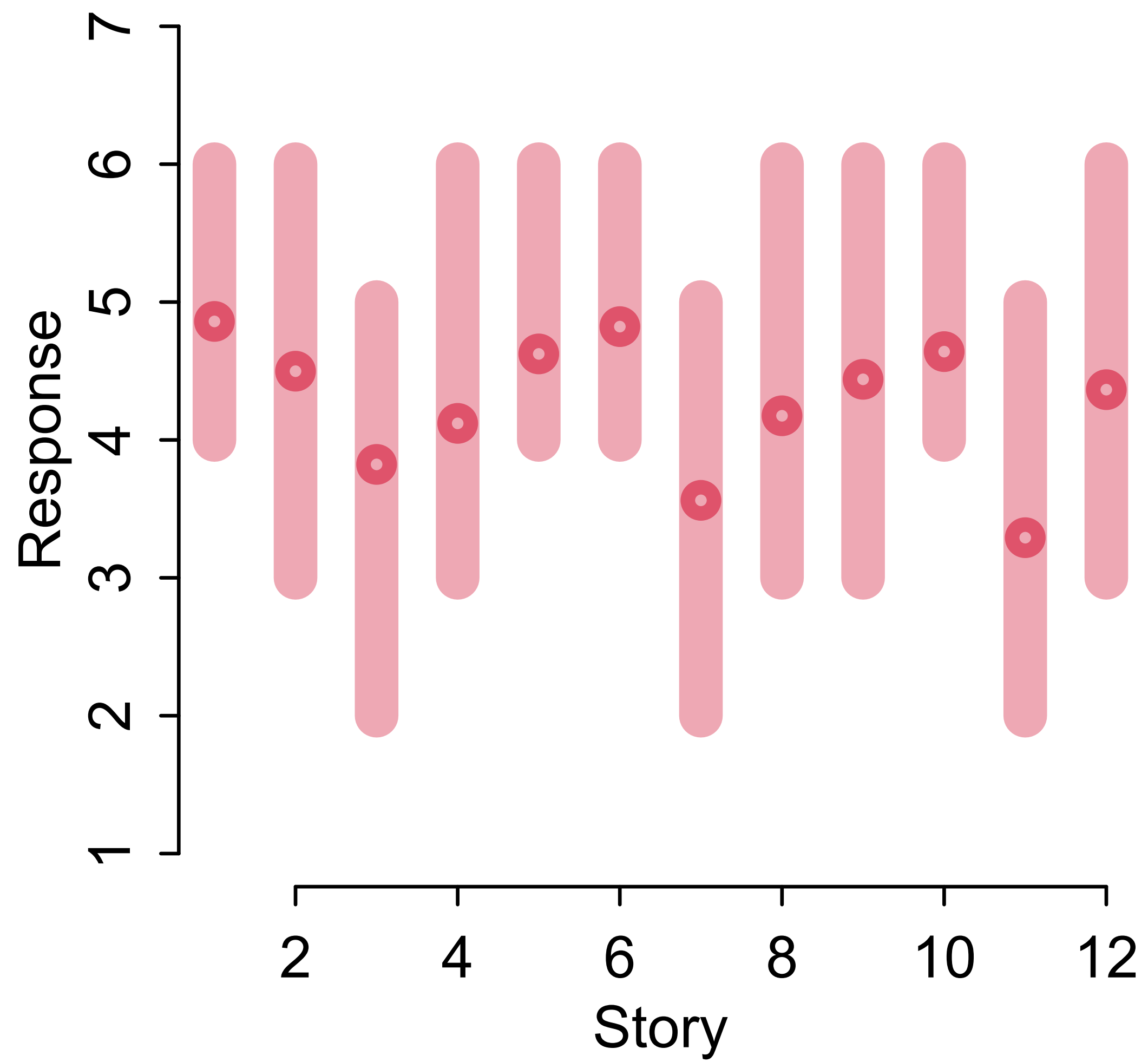
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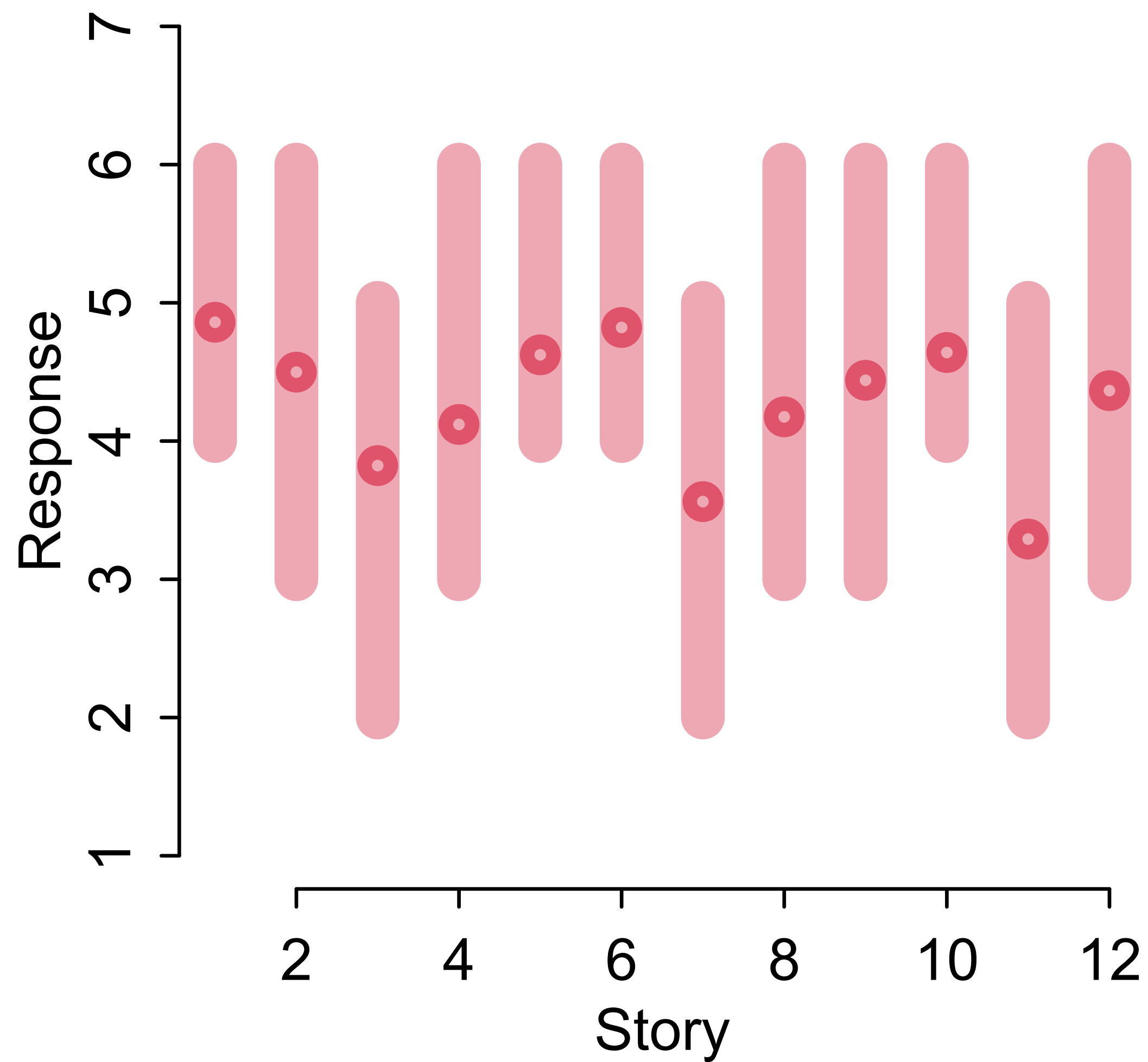
331 individuals (U)

```
> table(d$id)
```

96;434	96;445	96;451	96;456	96;458	96;466	96;467	96;474	96;480	96;481	96;497
30	30	30	30	30	30	30	30	30	30	30
96;498	96;502	96;505	96;511	96;512	96;518	96;519	96;531	96;533	96;538	96;547
30	30	30	30	30	30	30	30	30	30	30
96;550	96;553	96;555	96;558	96;560	96;562	96;566	96;570	96;581	96;586	96;591
30	30	30	30	30	30	30	30	30	30	30







$$R_i \sim \text{OrderedLogit}(\phi_i, \alpha)$$

$$\phi_i = \beta_{S[i]}$$

*This model has
anterograde amnesia*

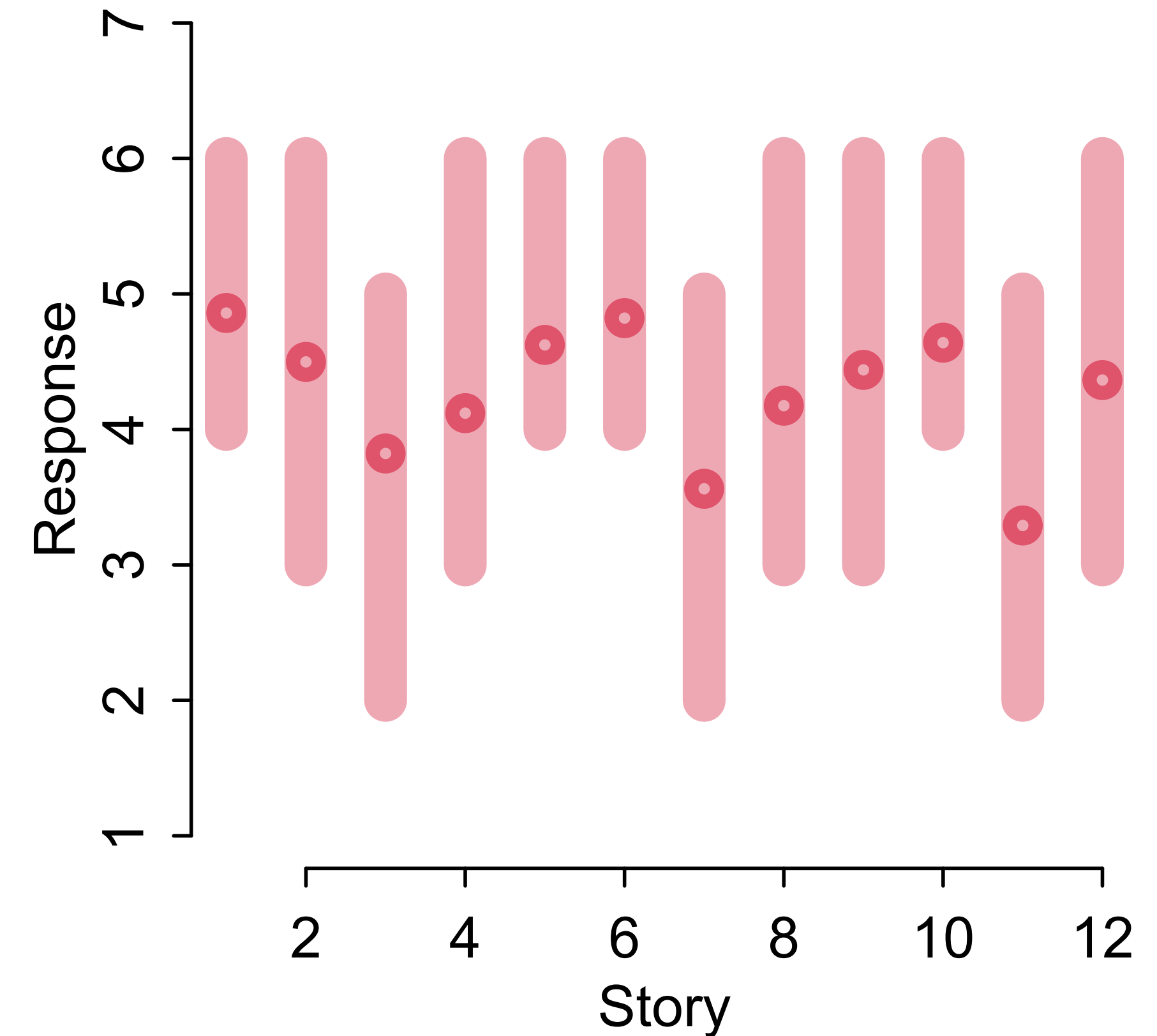
Models With Memory

Multilevel models are models within models

(1) Model observed groups/individuals

(2) Model of population of groups/individuals

The population model creates a kind of memory

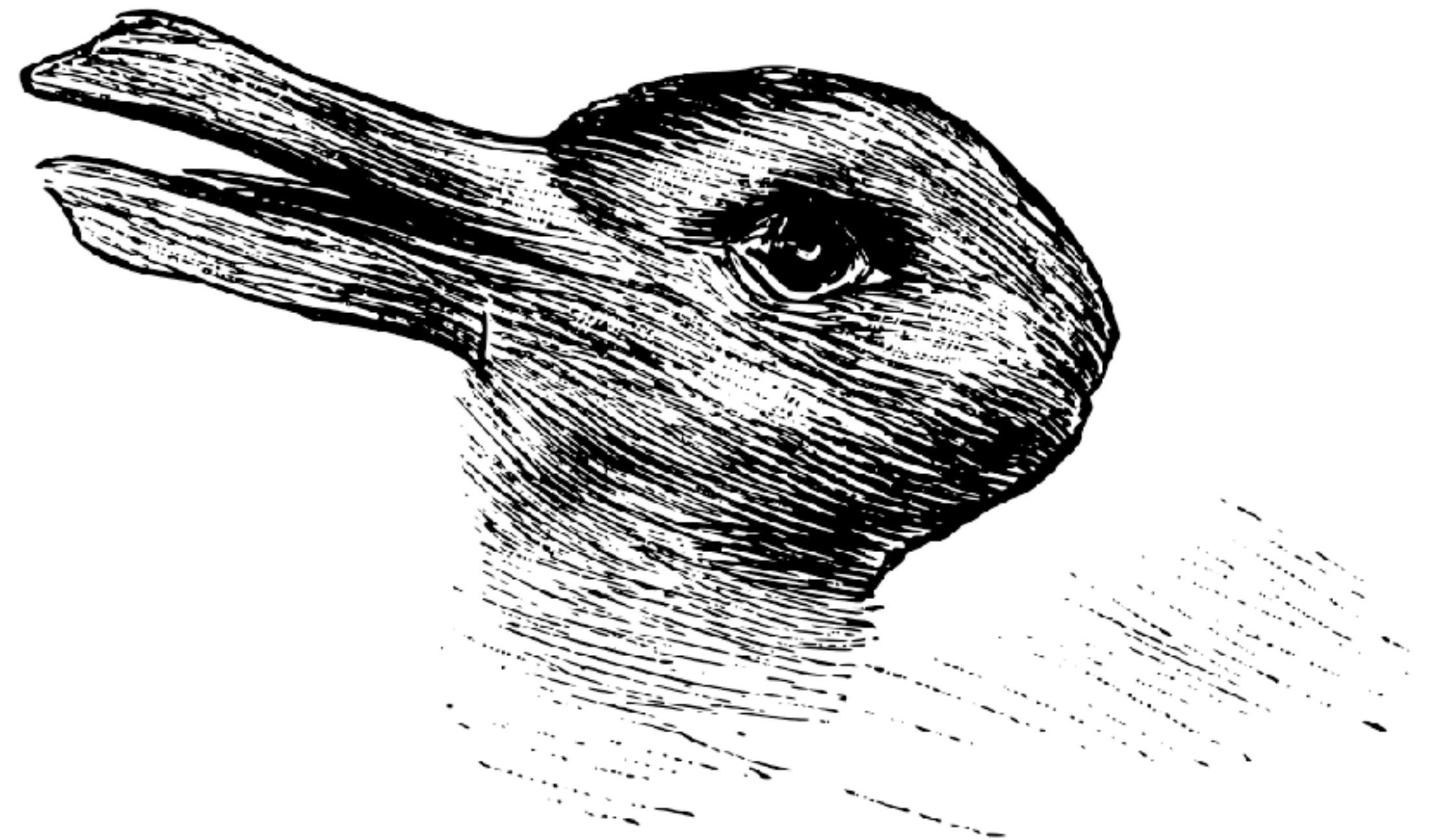


Two Perspectives

(1) Models with memory learn faster, better

(2) Models with memory resist overfitting

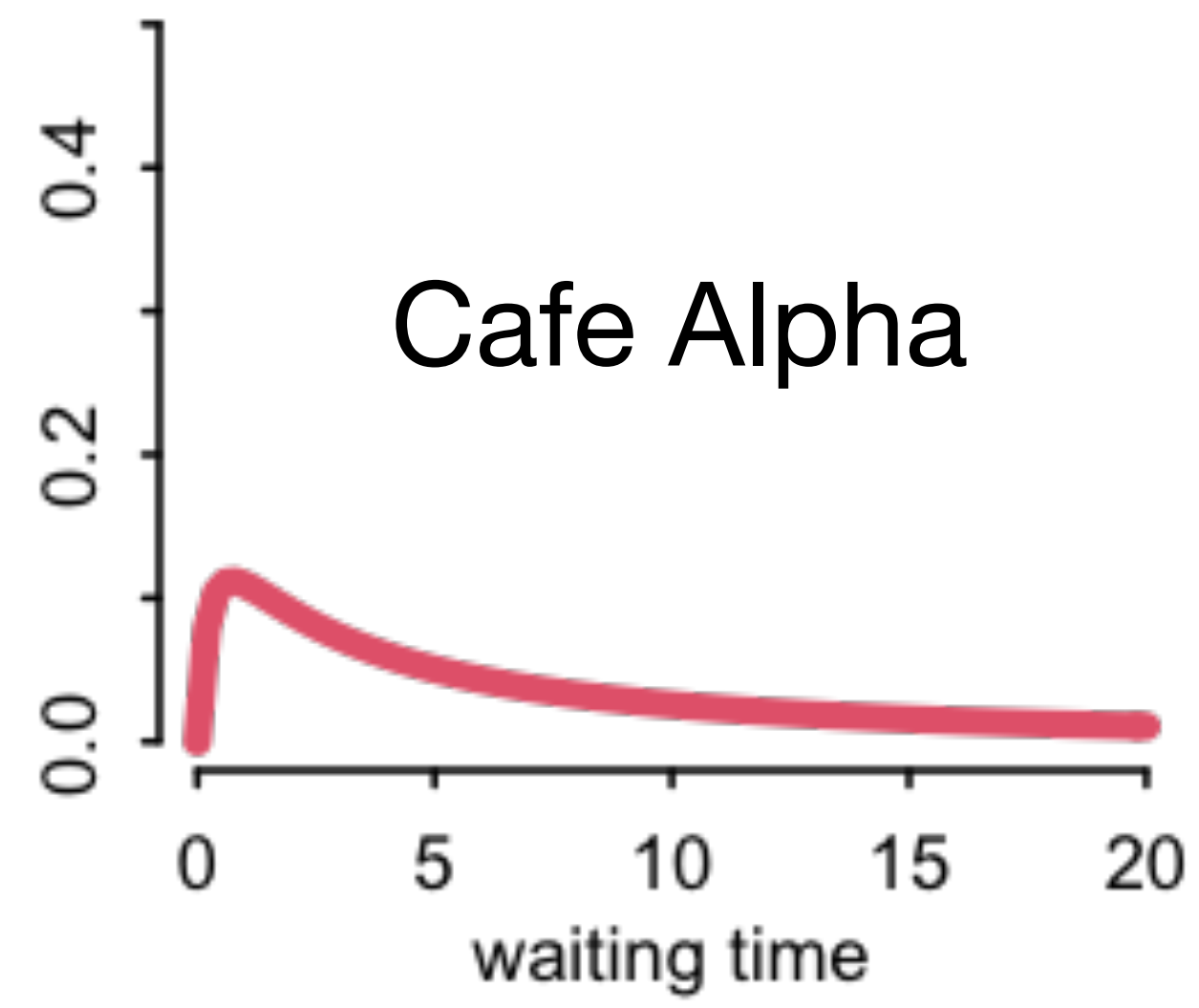
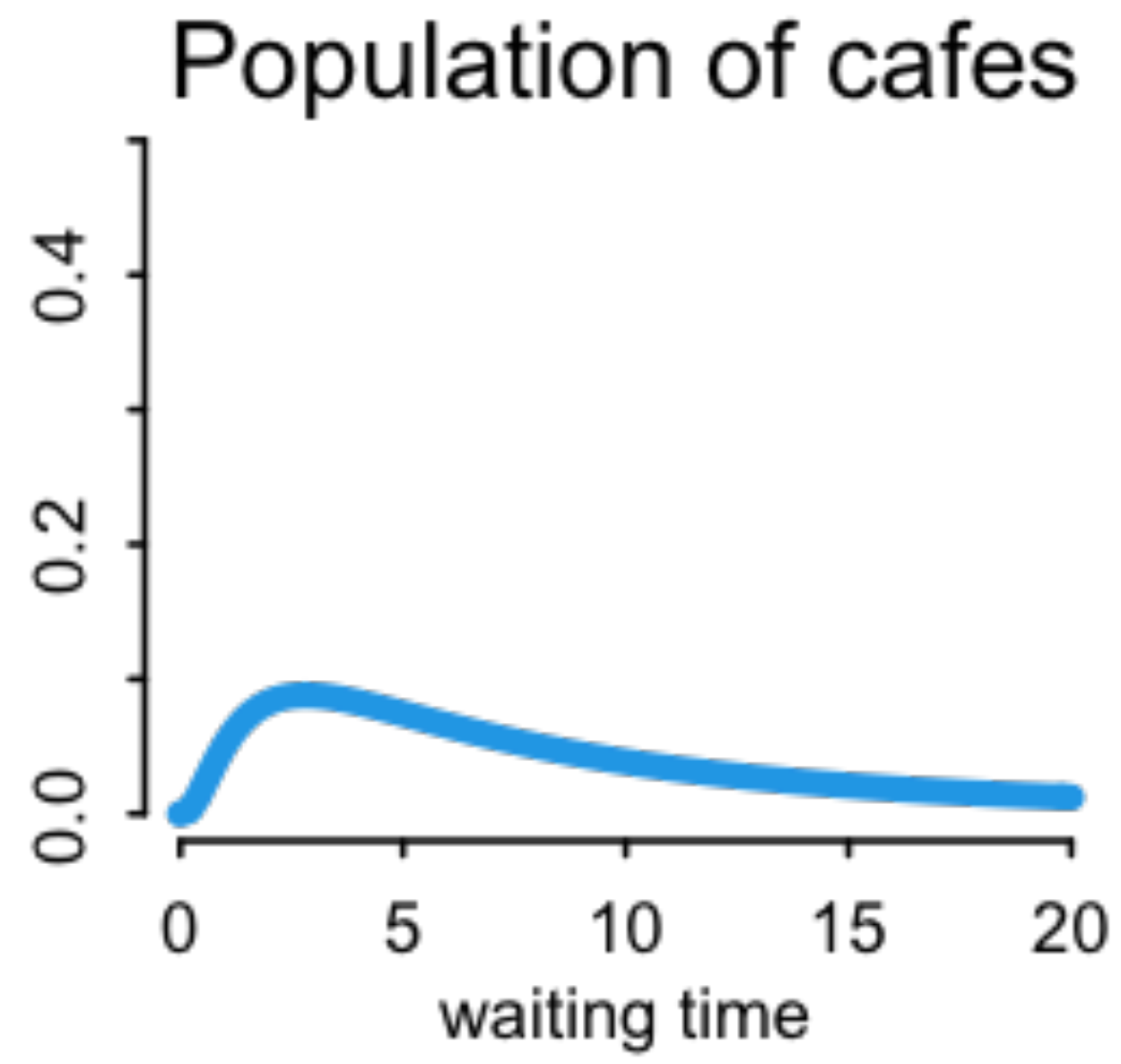
Welche Thiere gleichen ein-
ander am meisten?

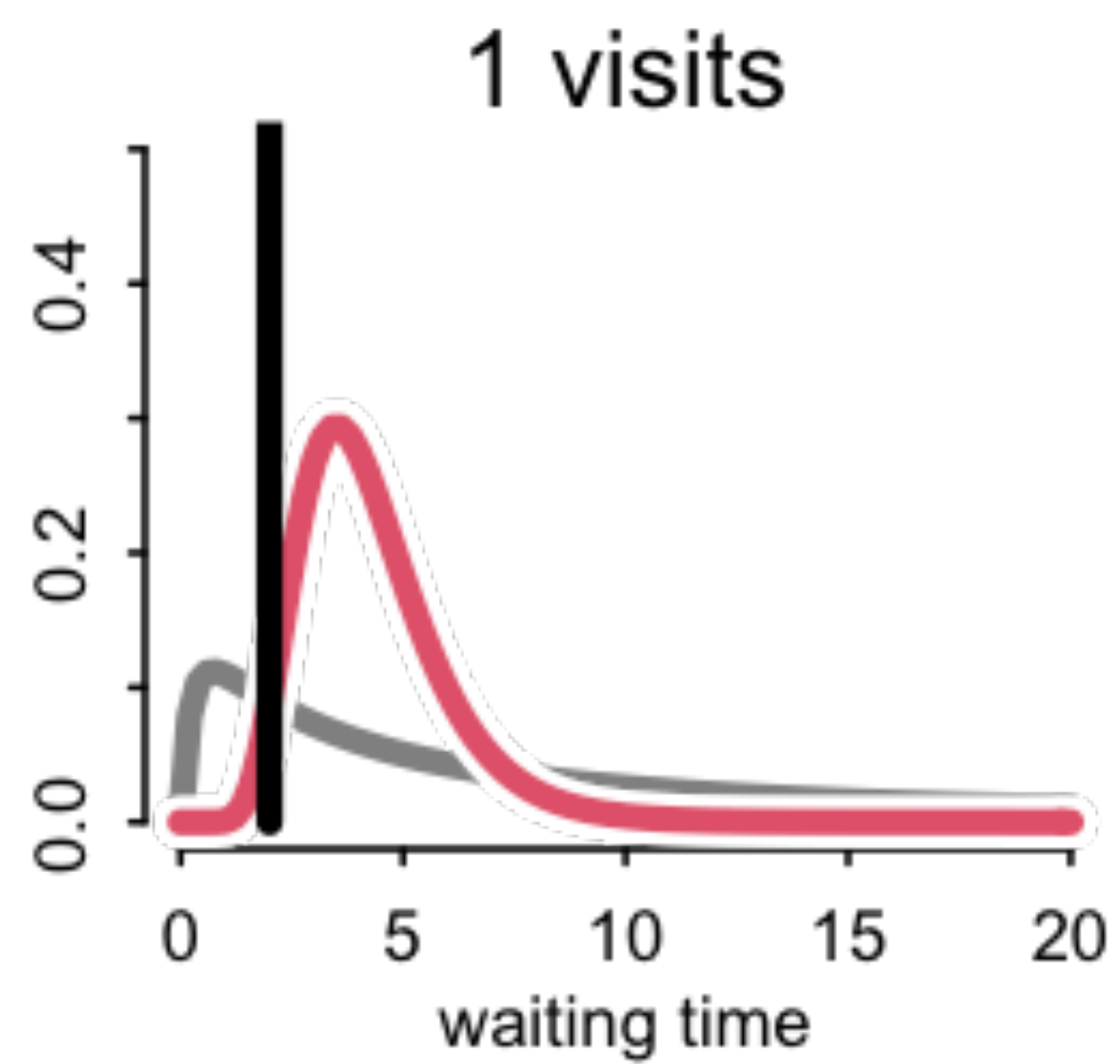
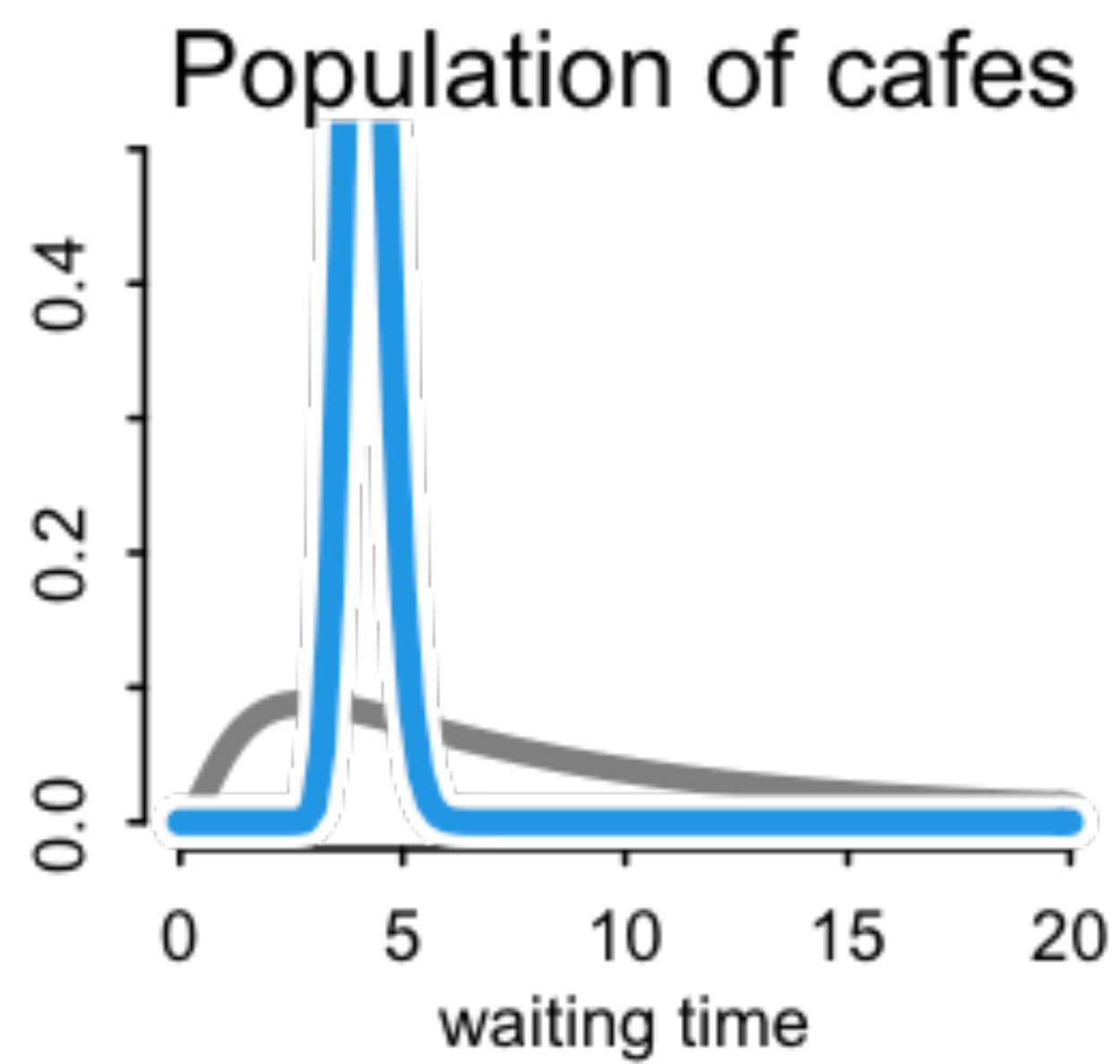


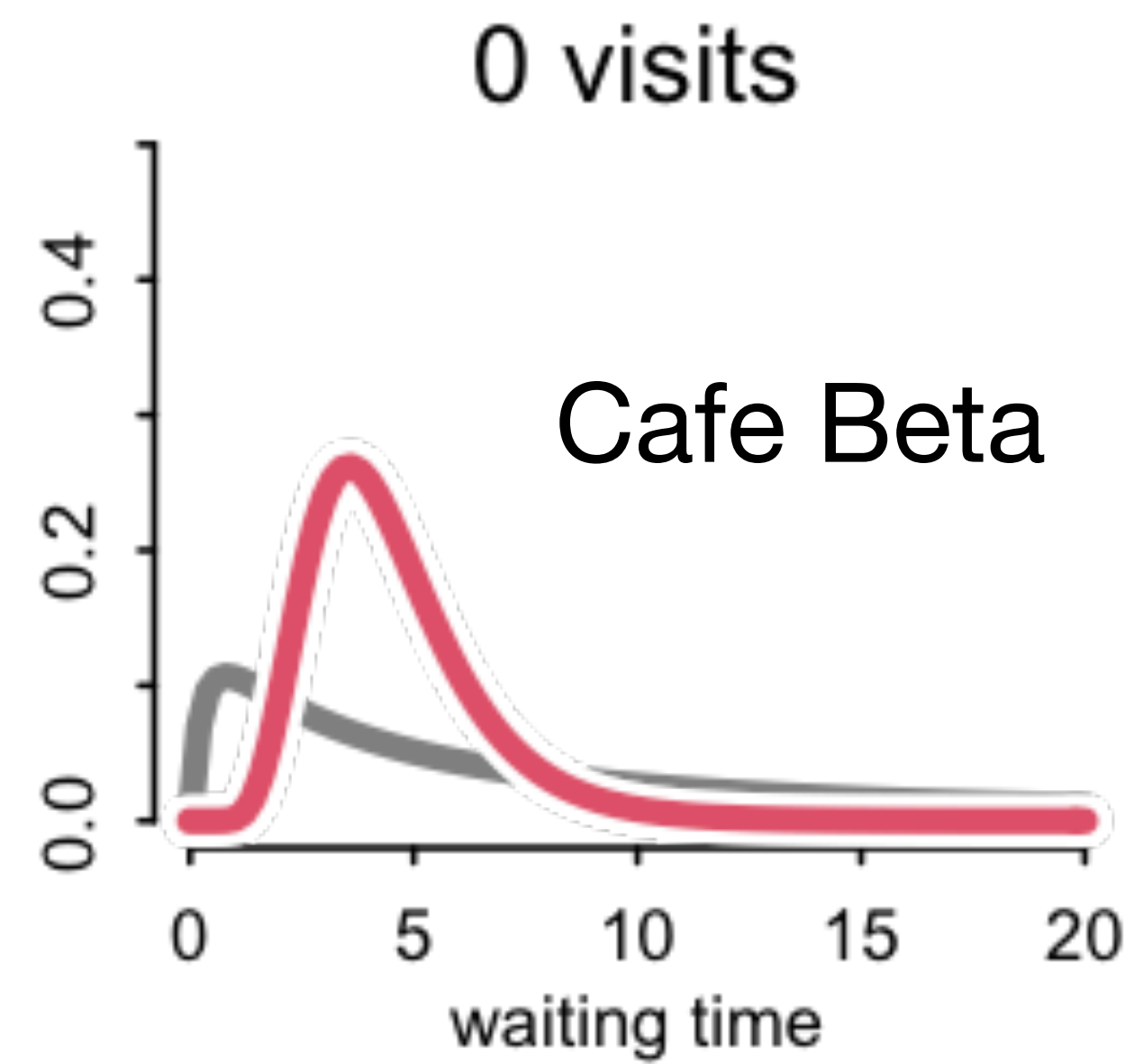
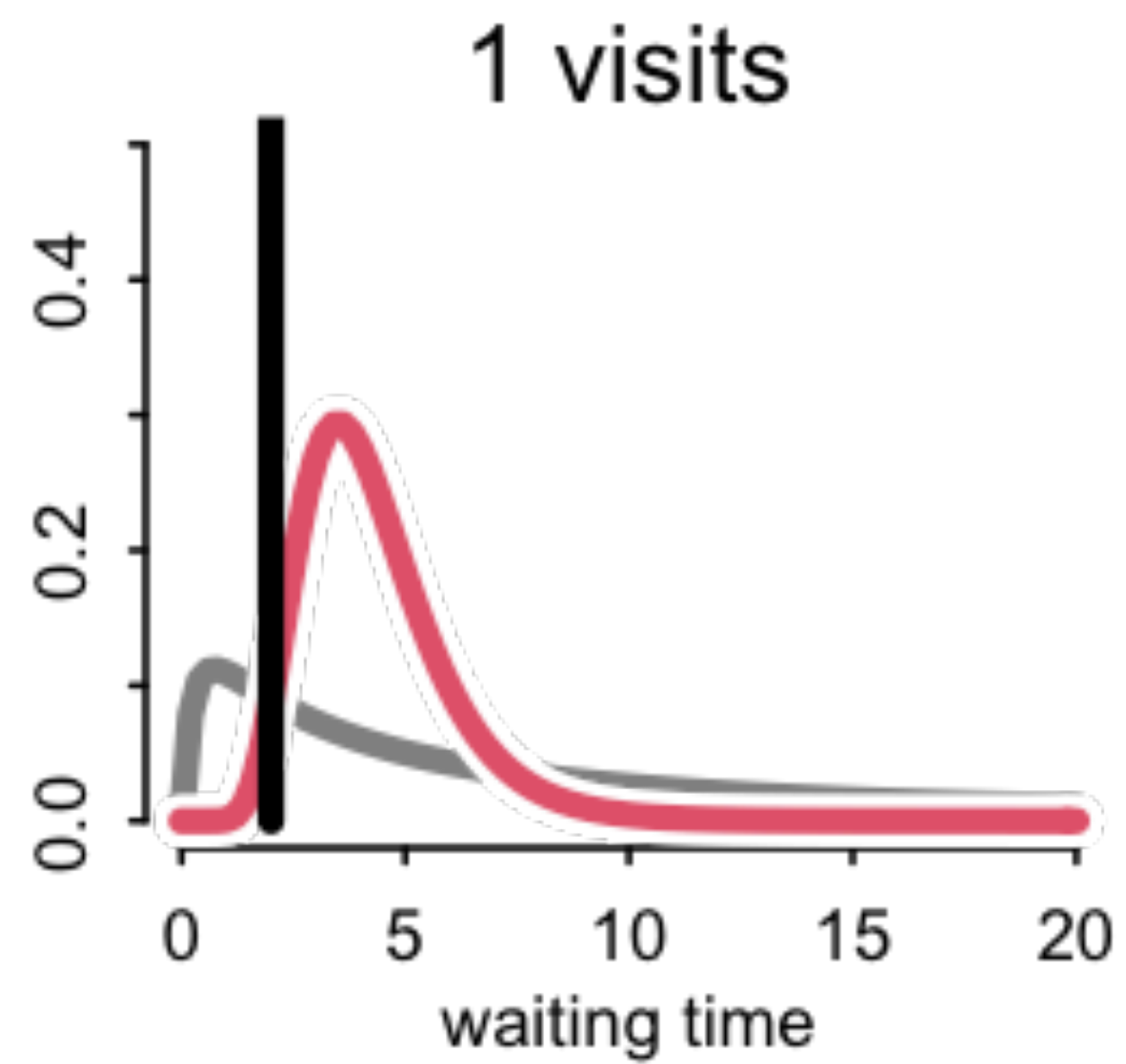
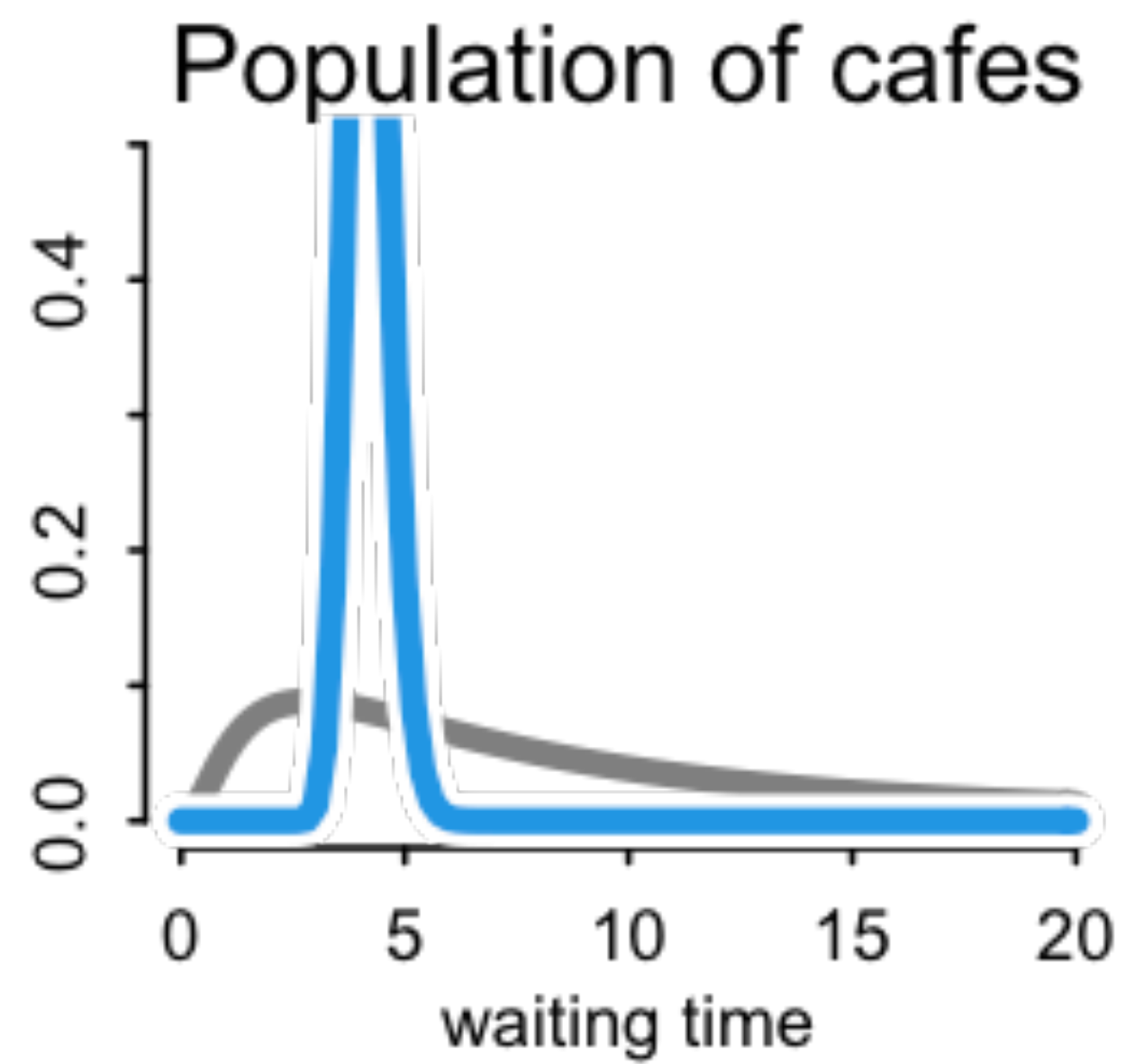
Kaninchen und Ente.

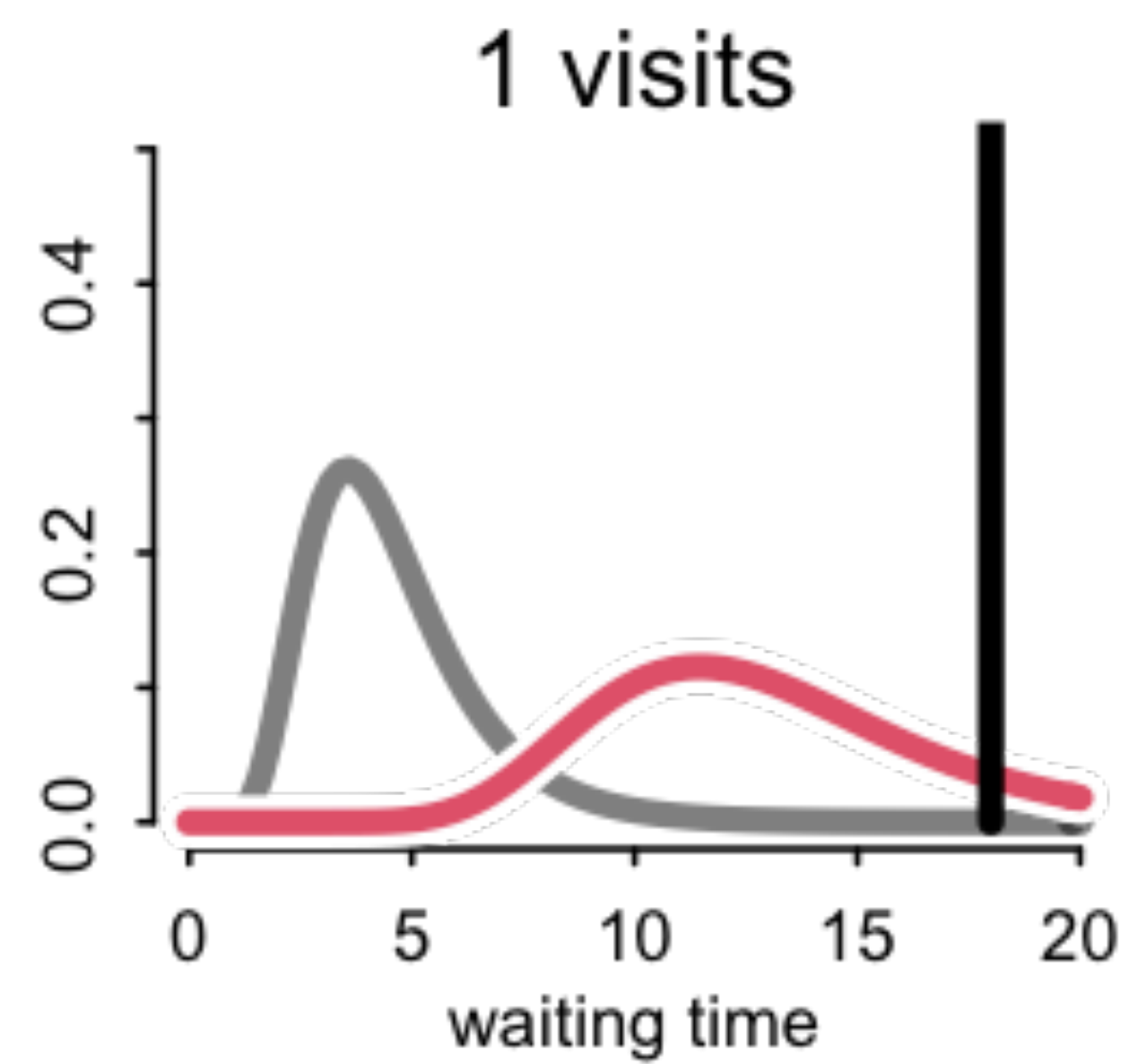
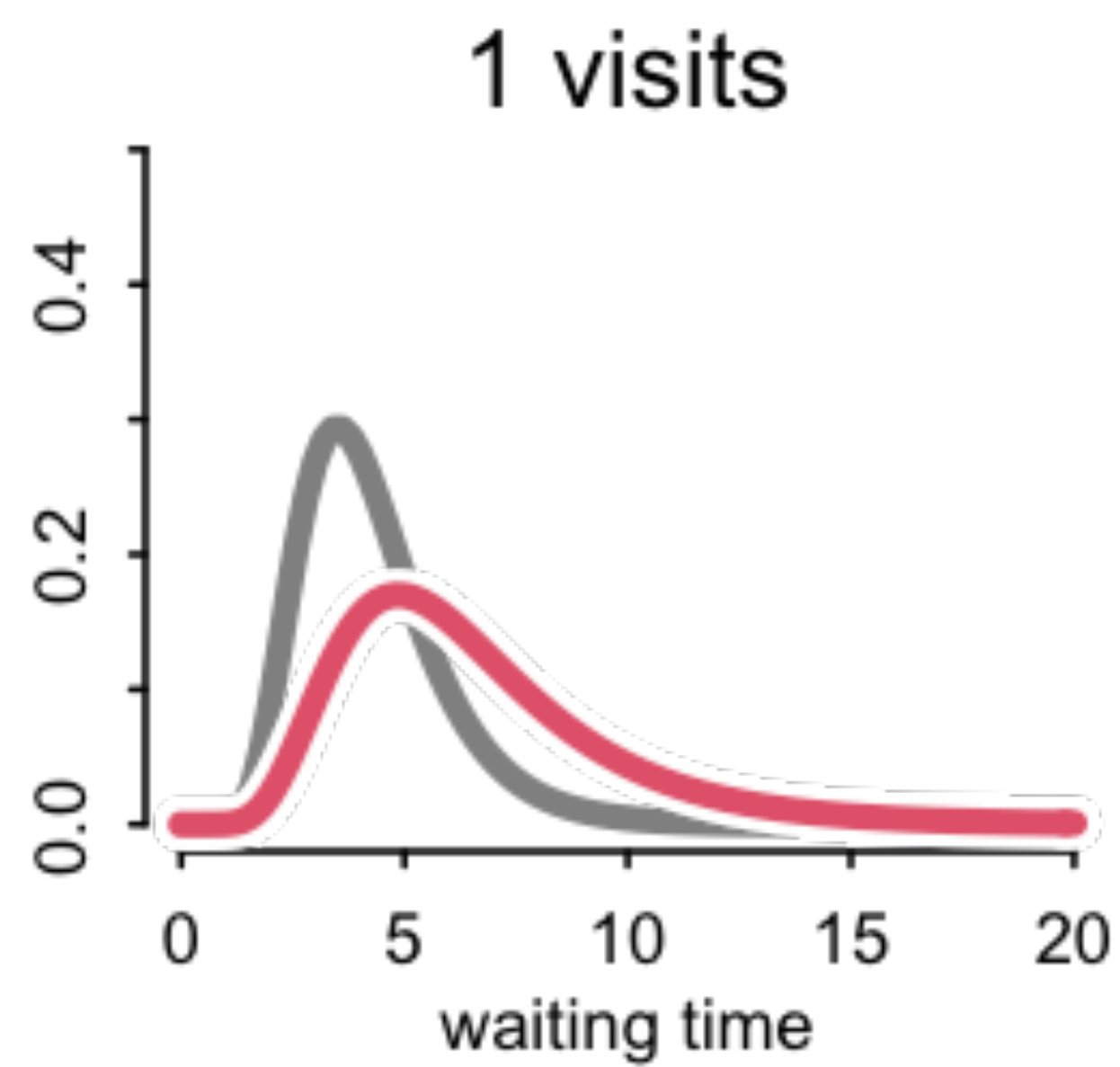
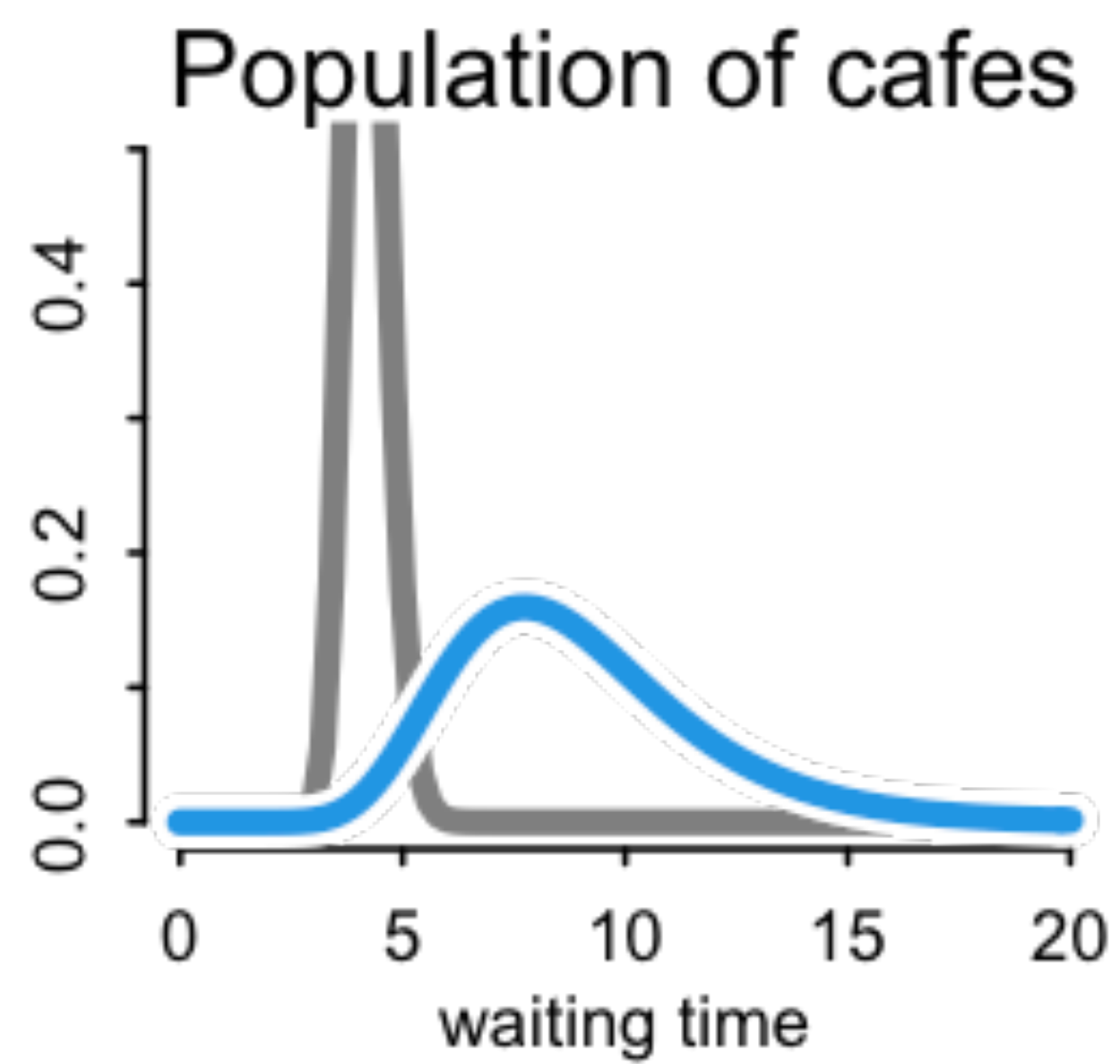


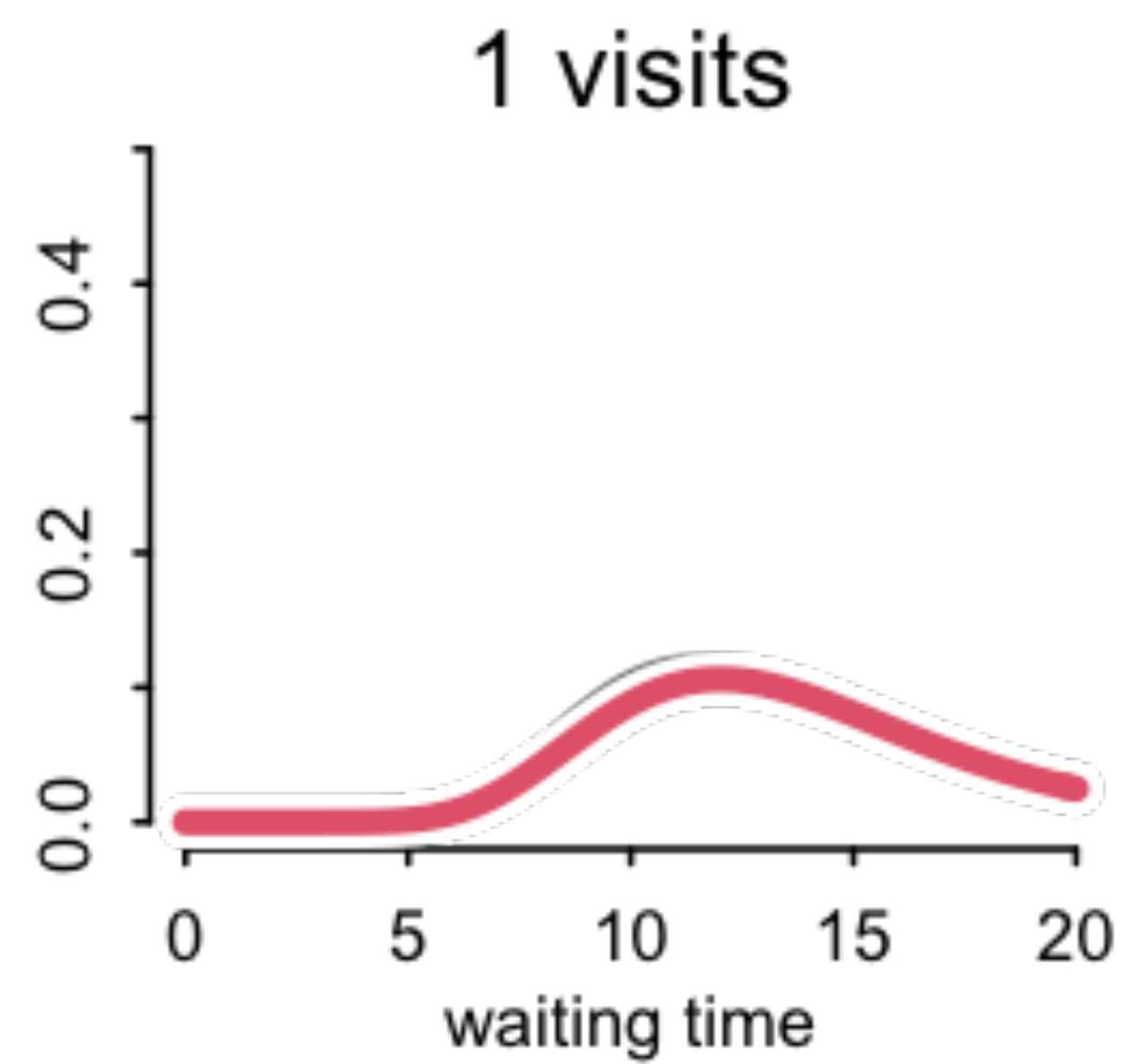
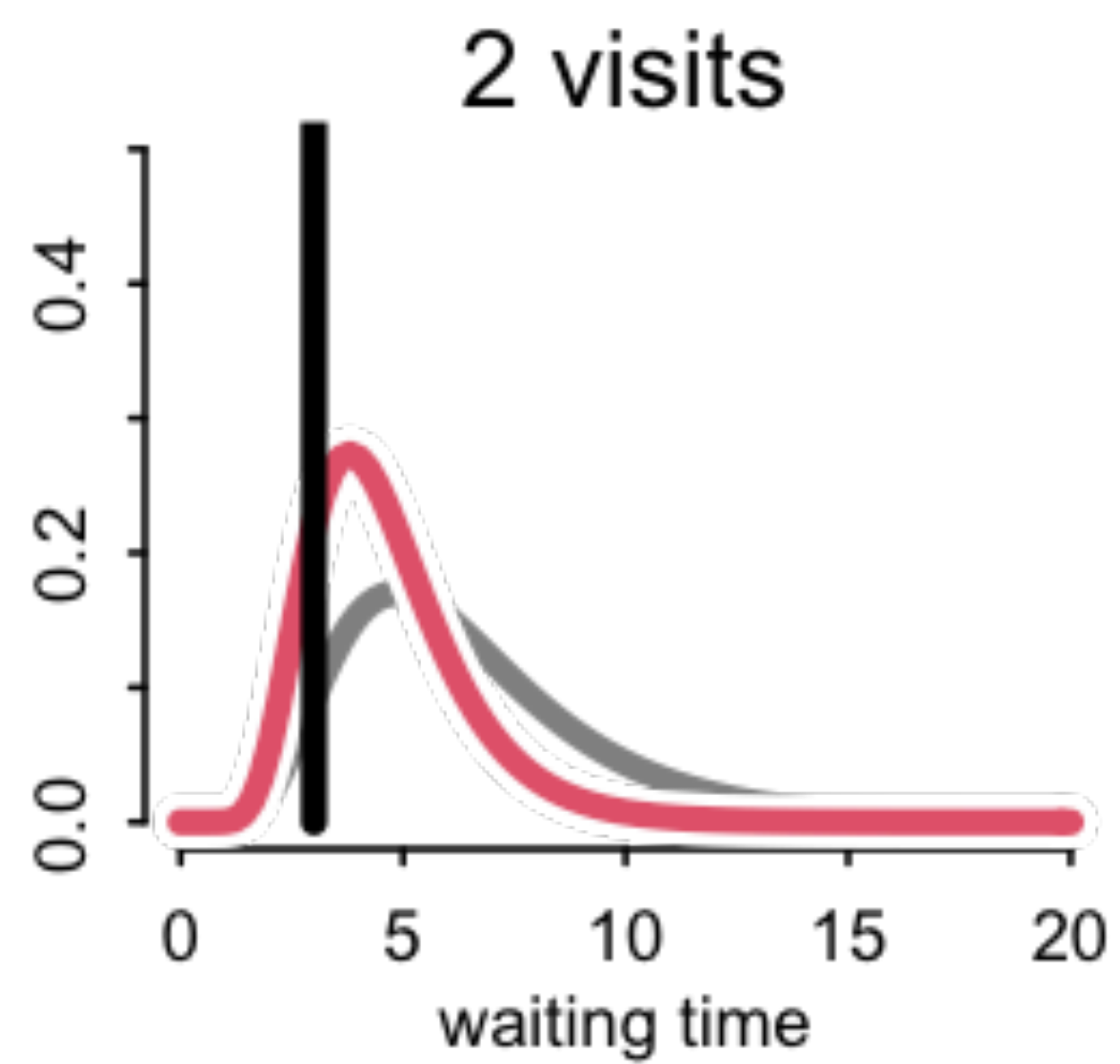
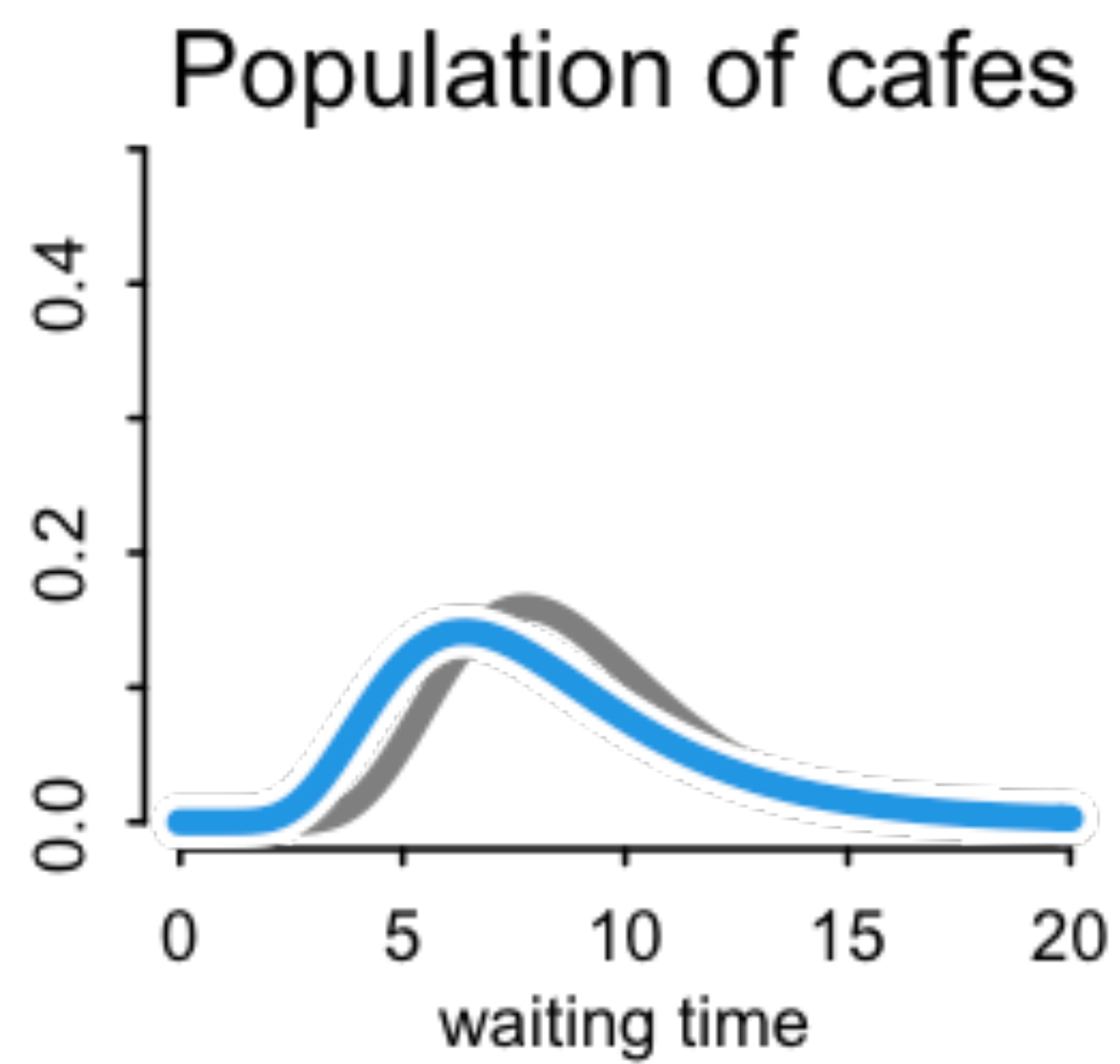
Suppose we want to figure out how long the wait for coffee is at different locations of the same chain

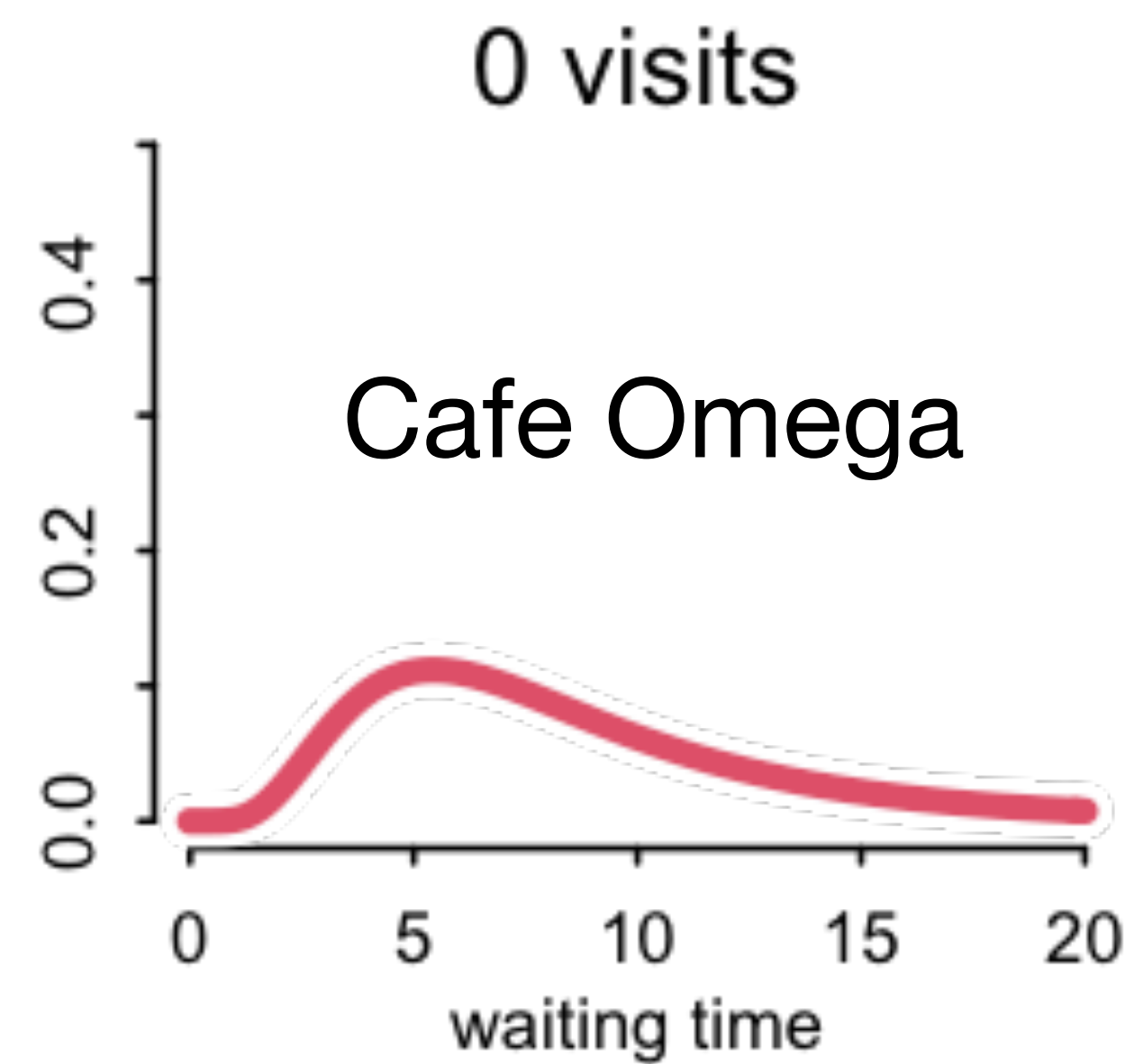
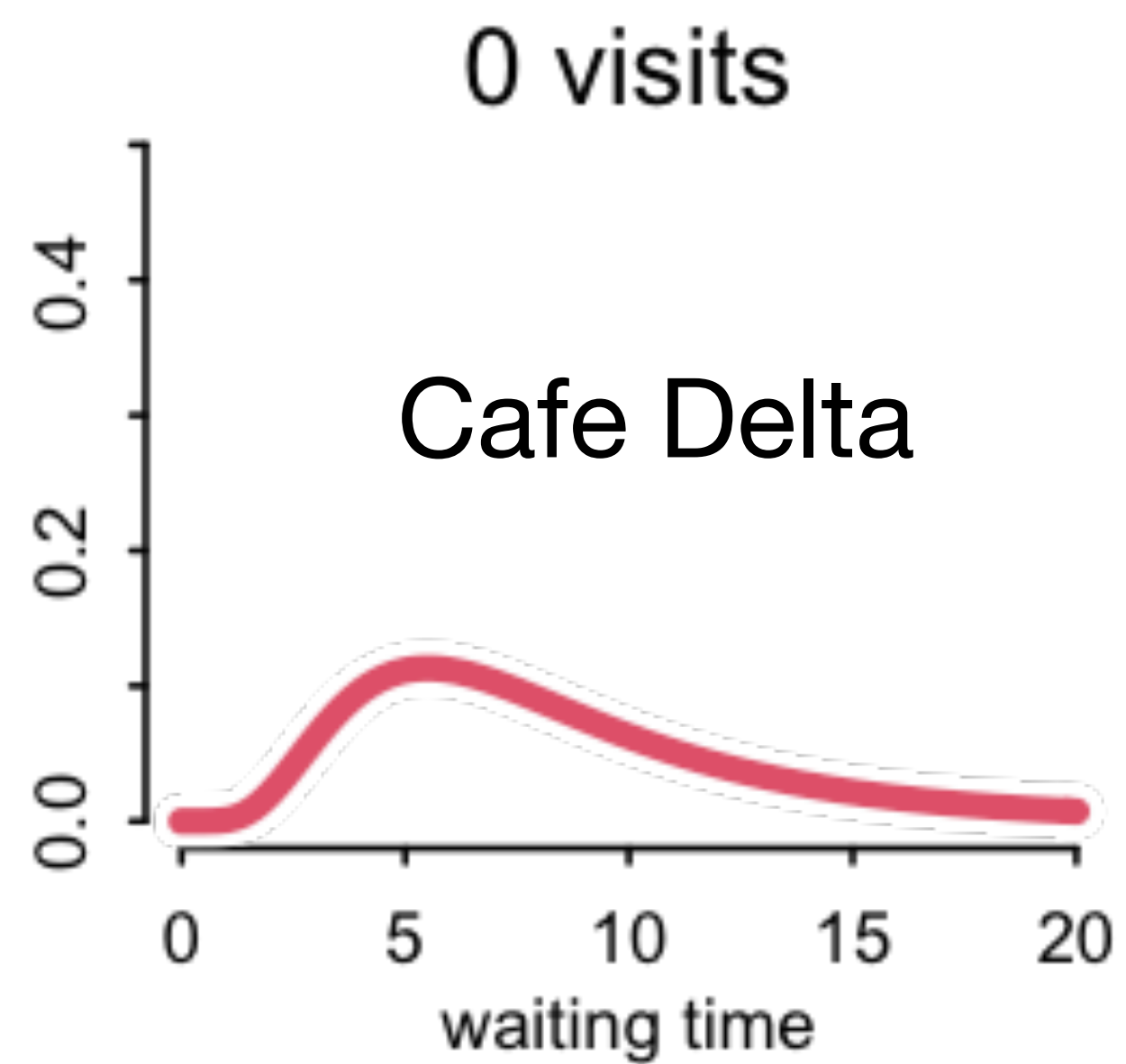
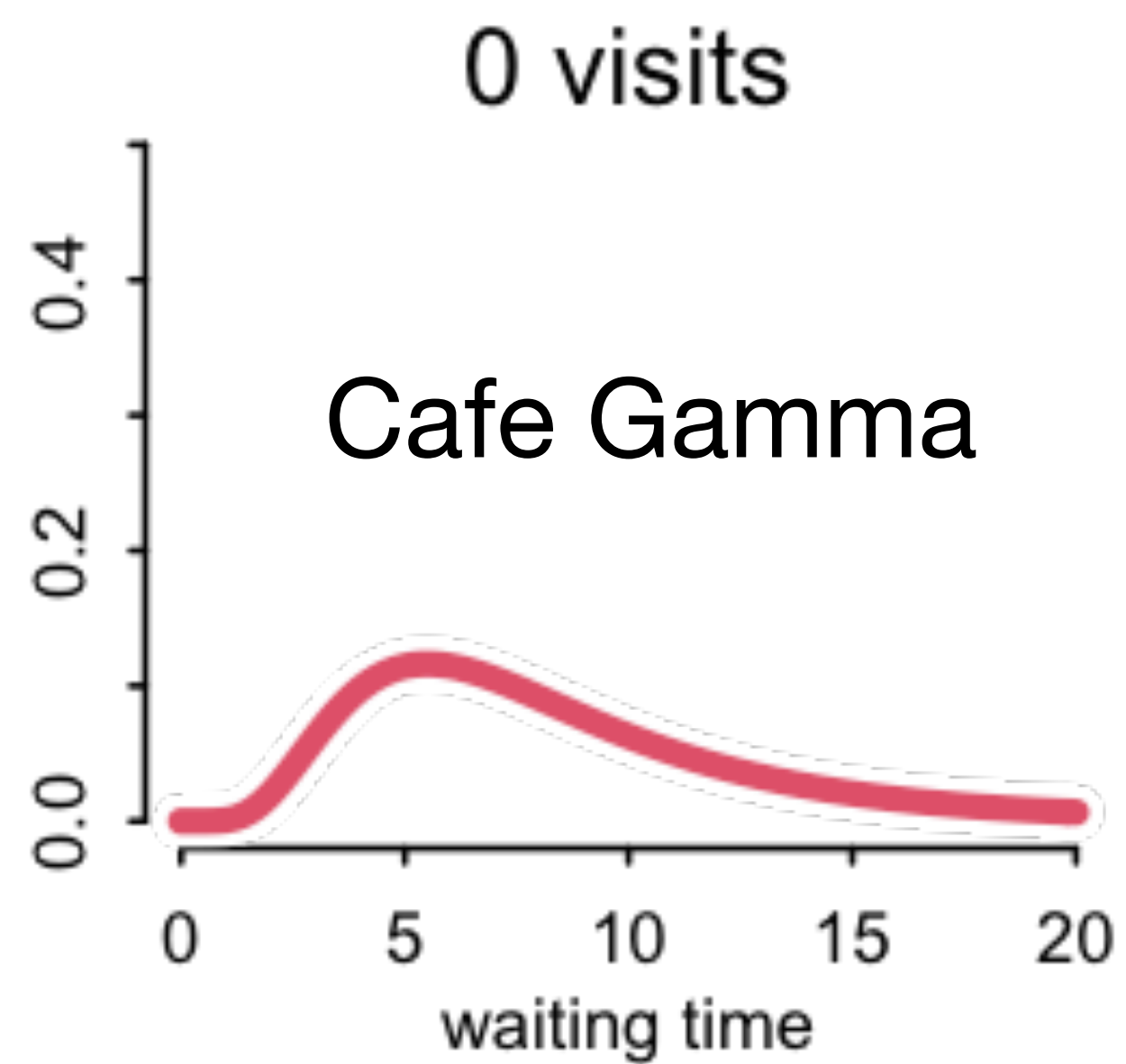
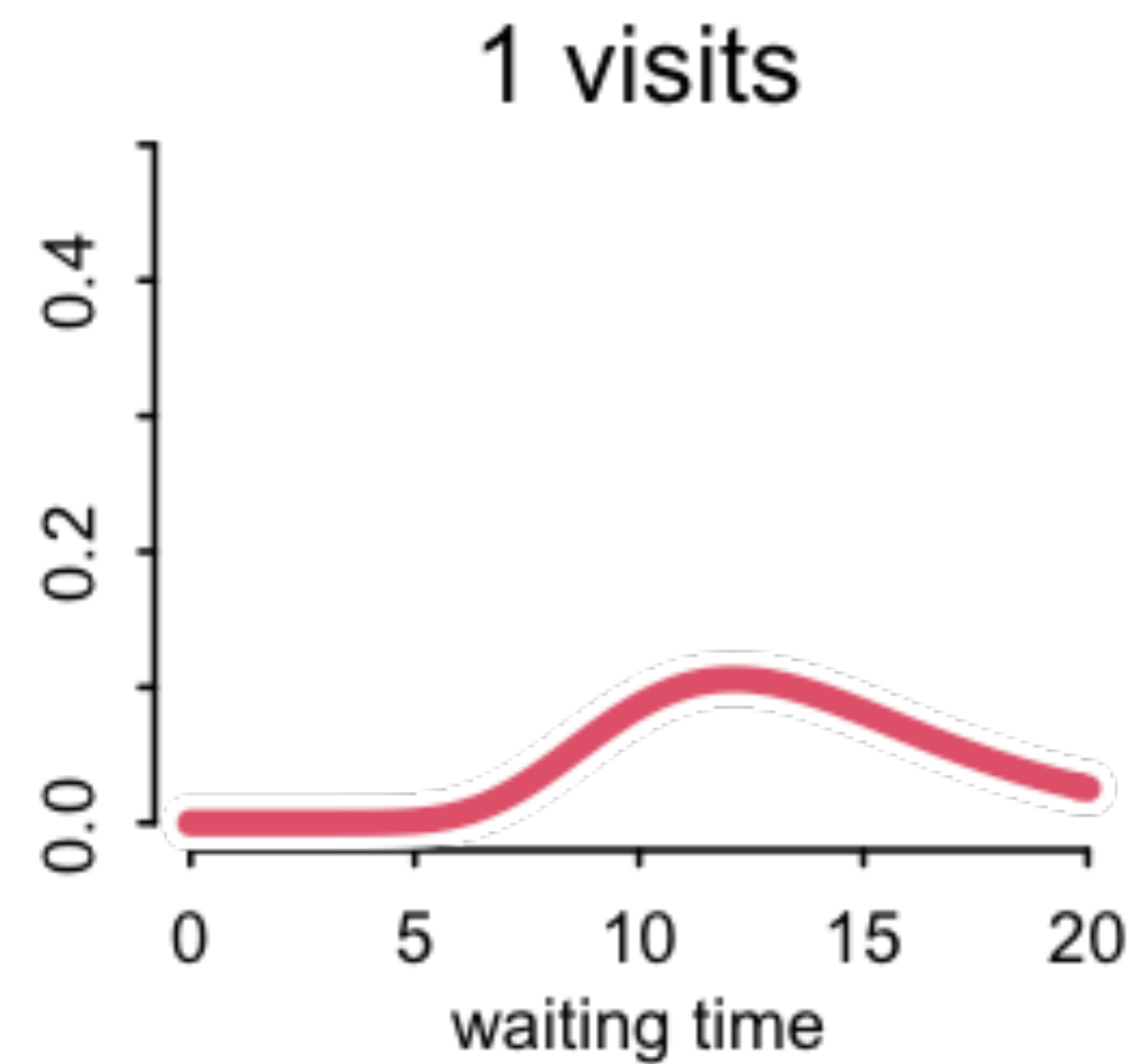
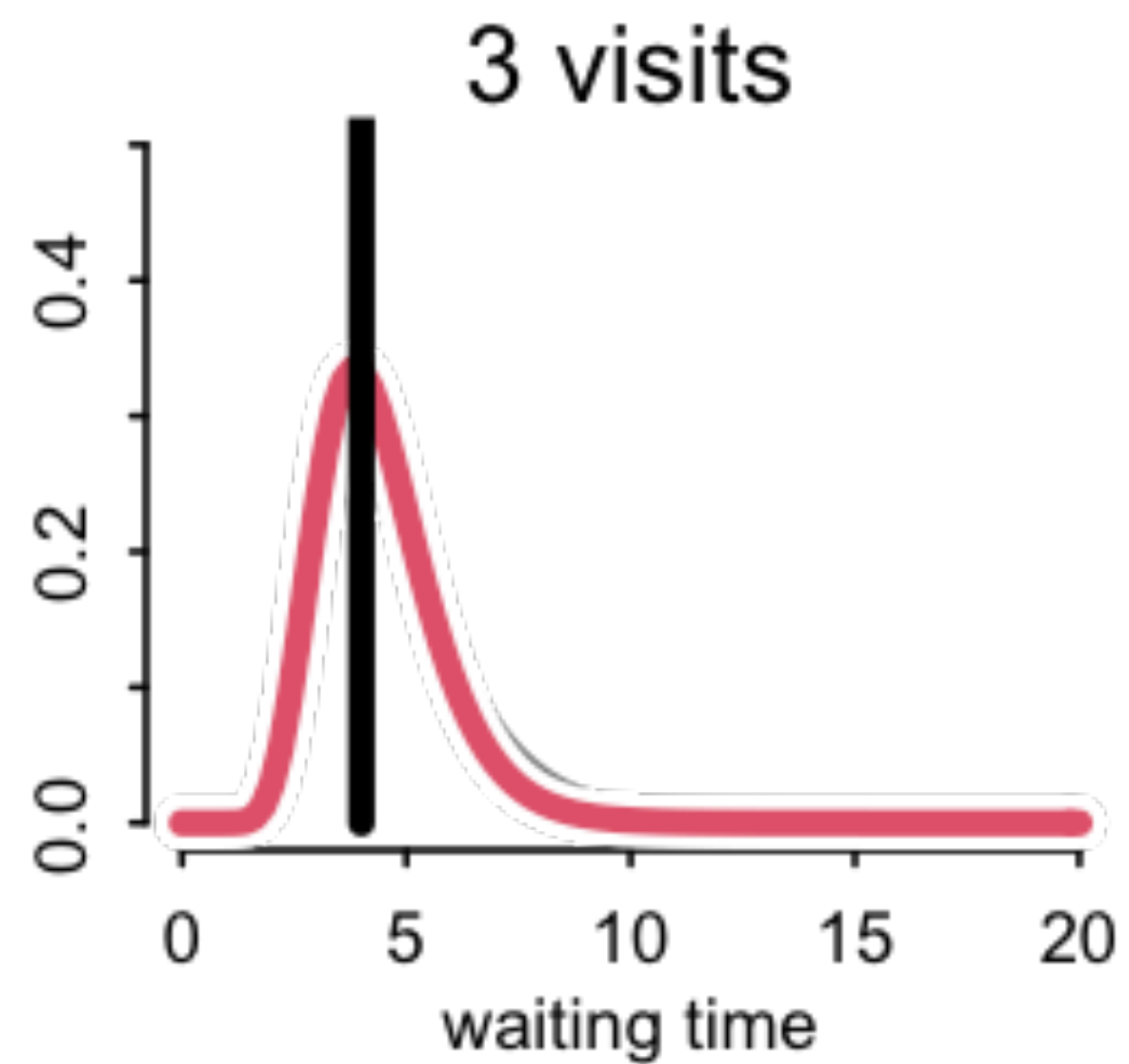
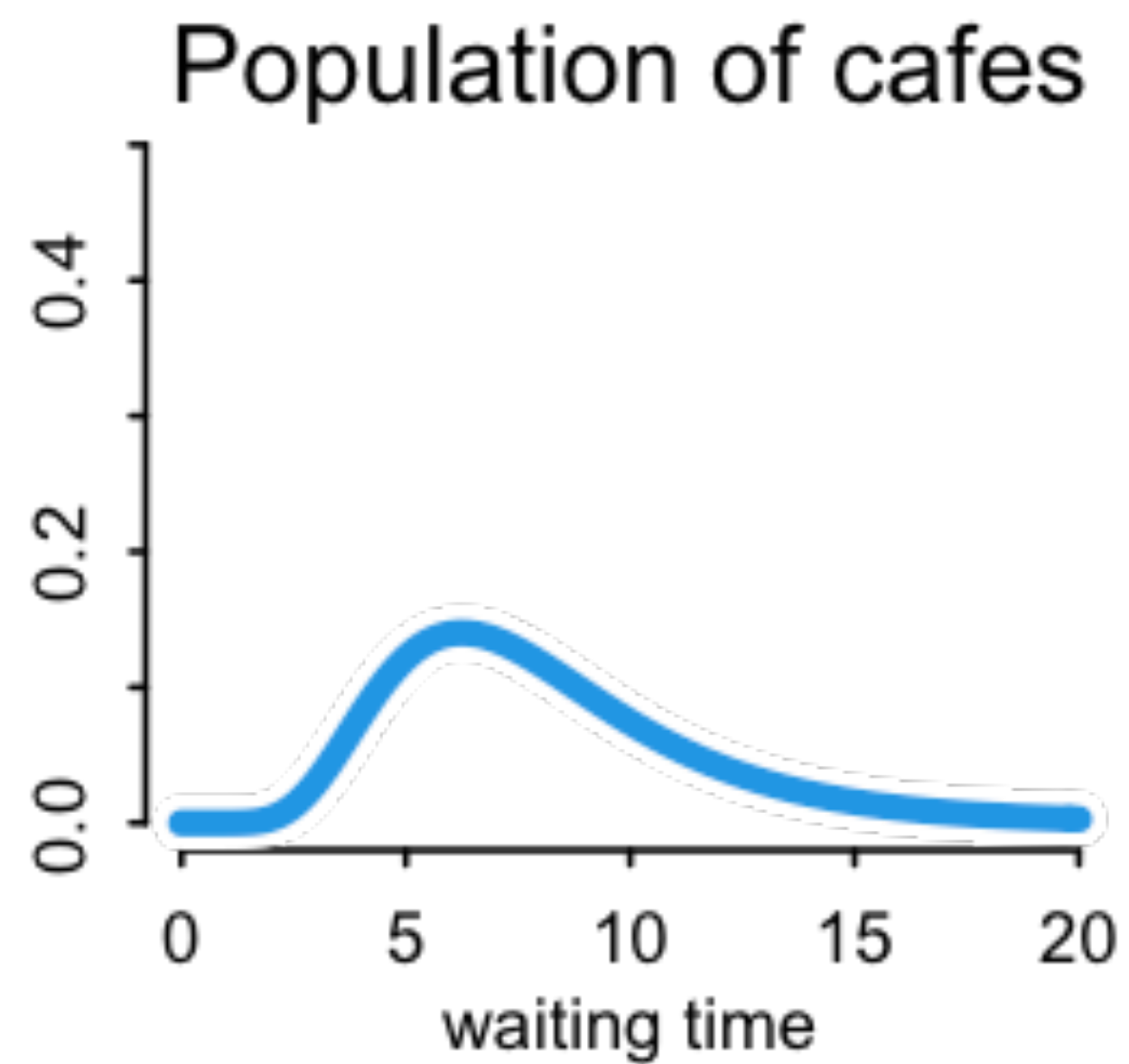




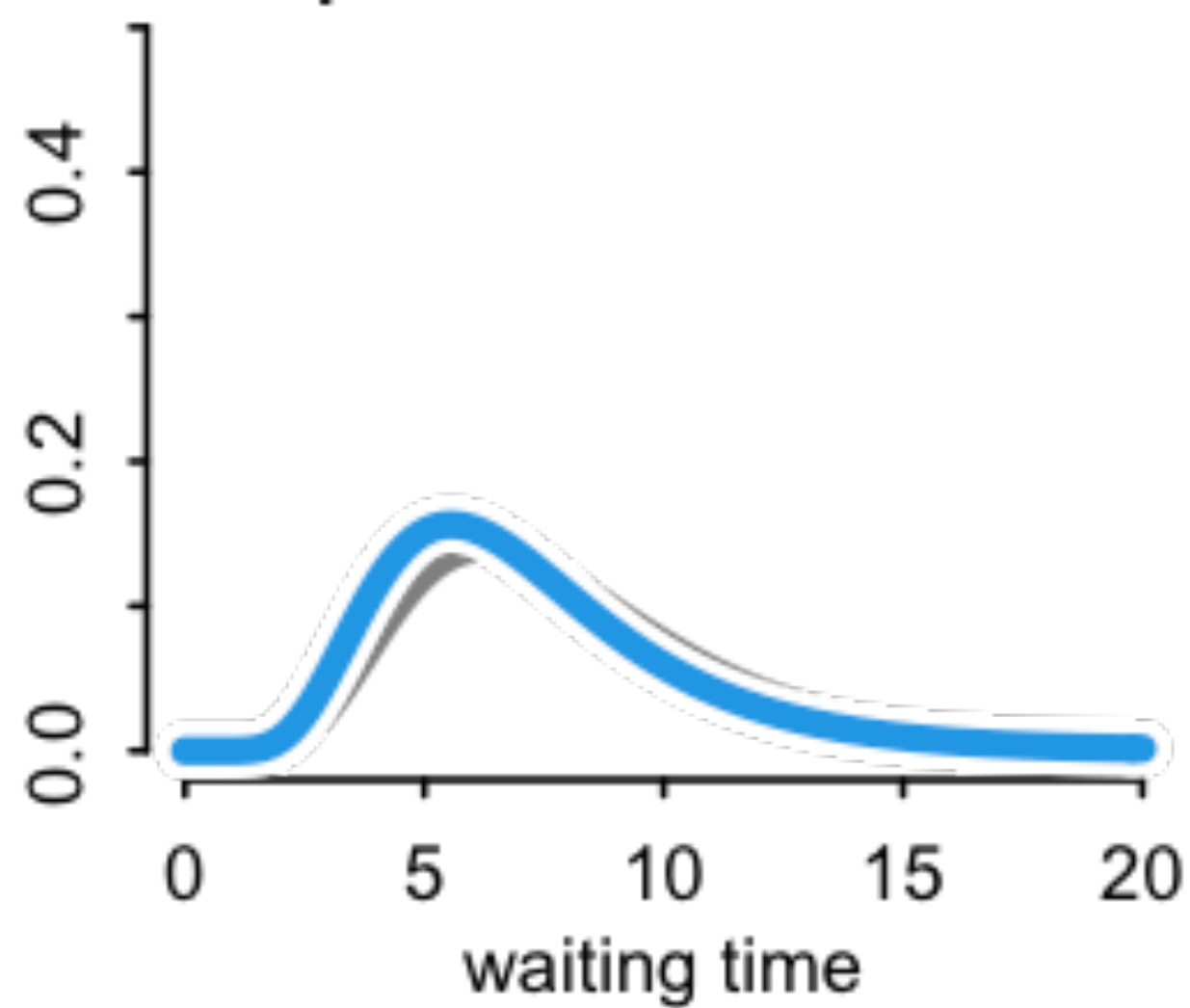




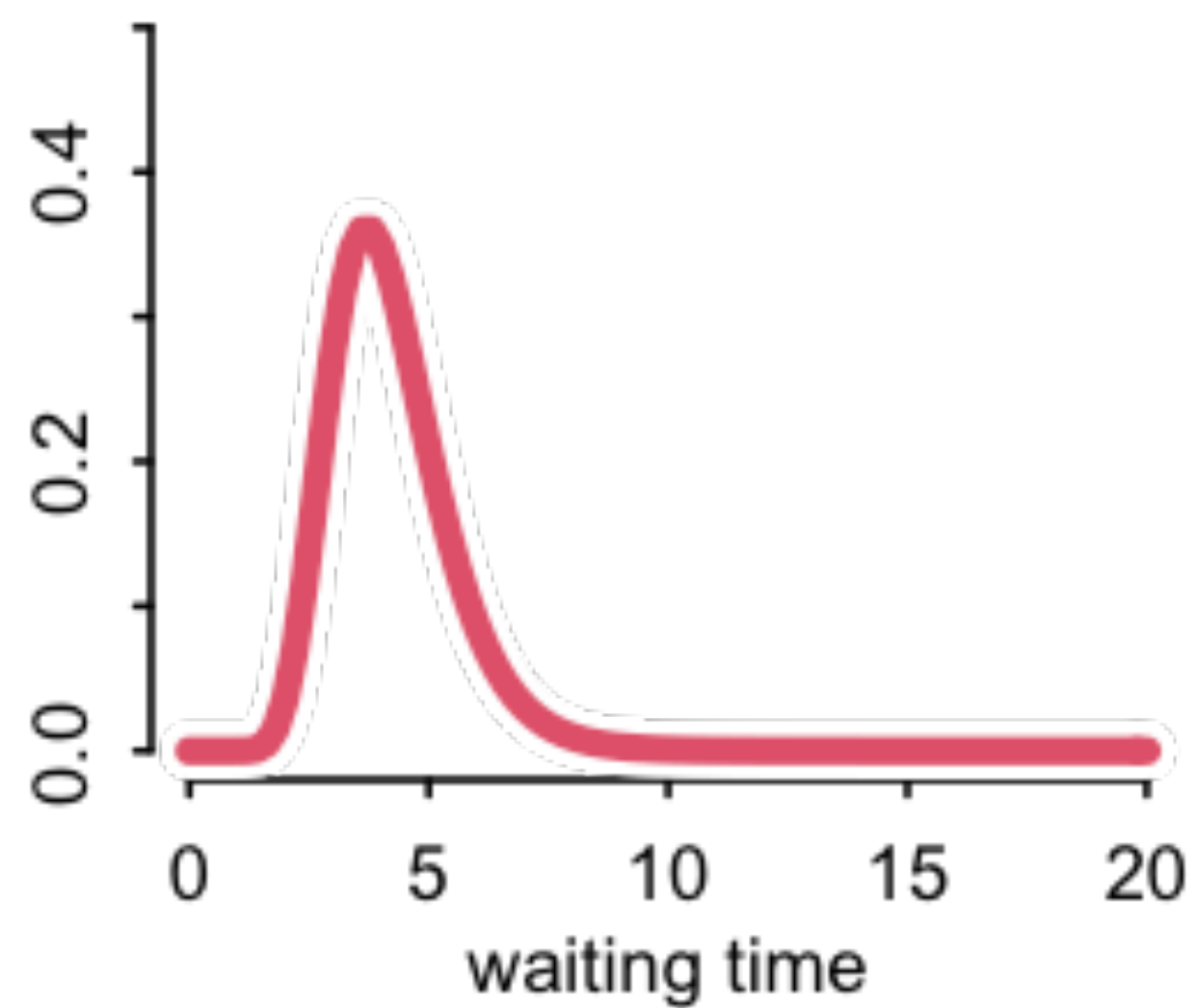




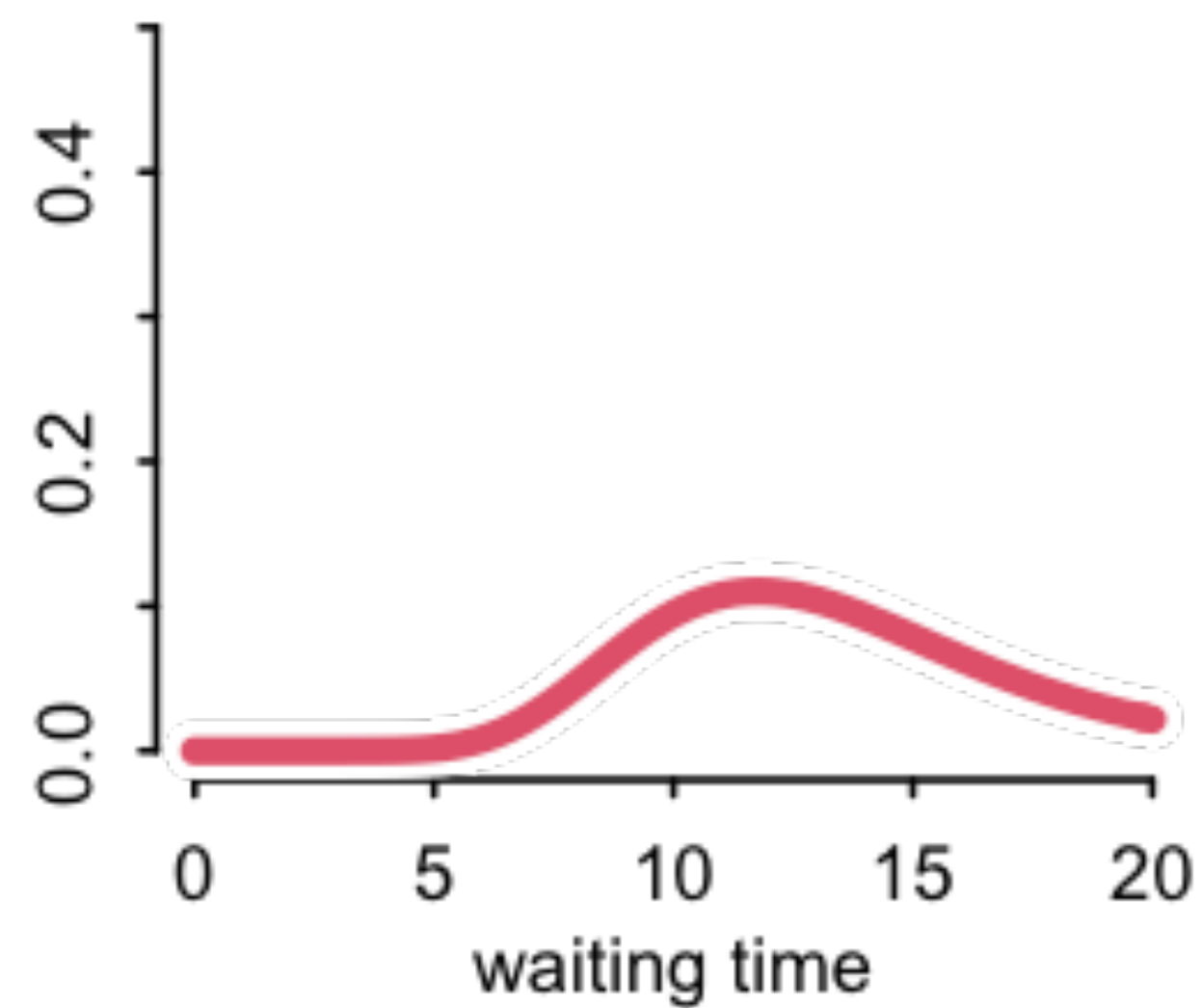
Population of cafes



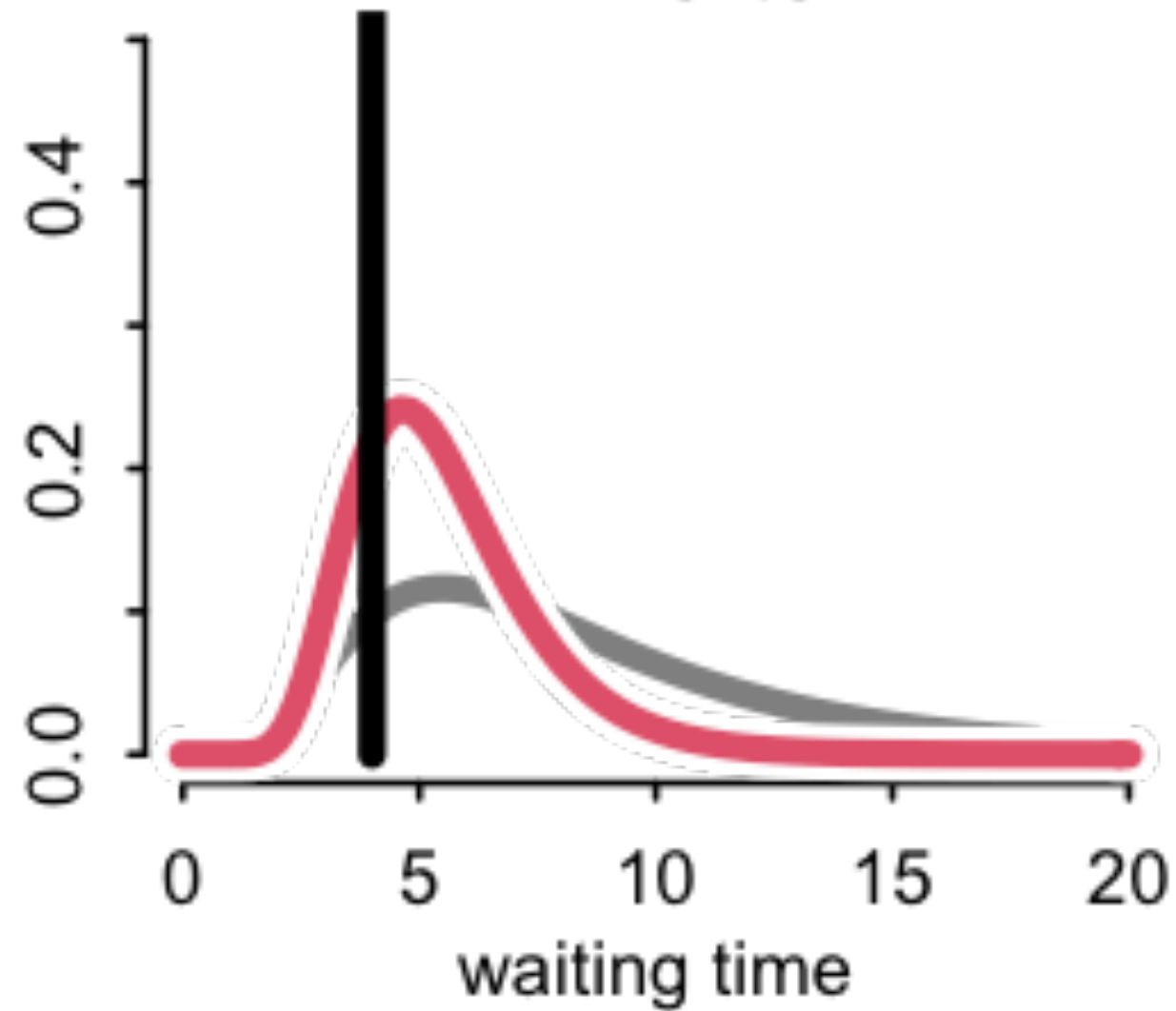
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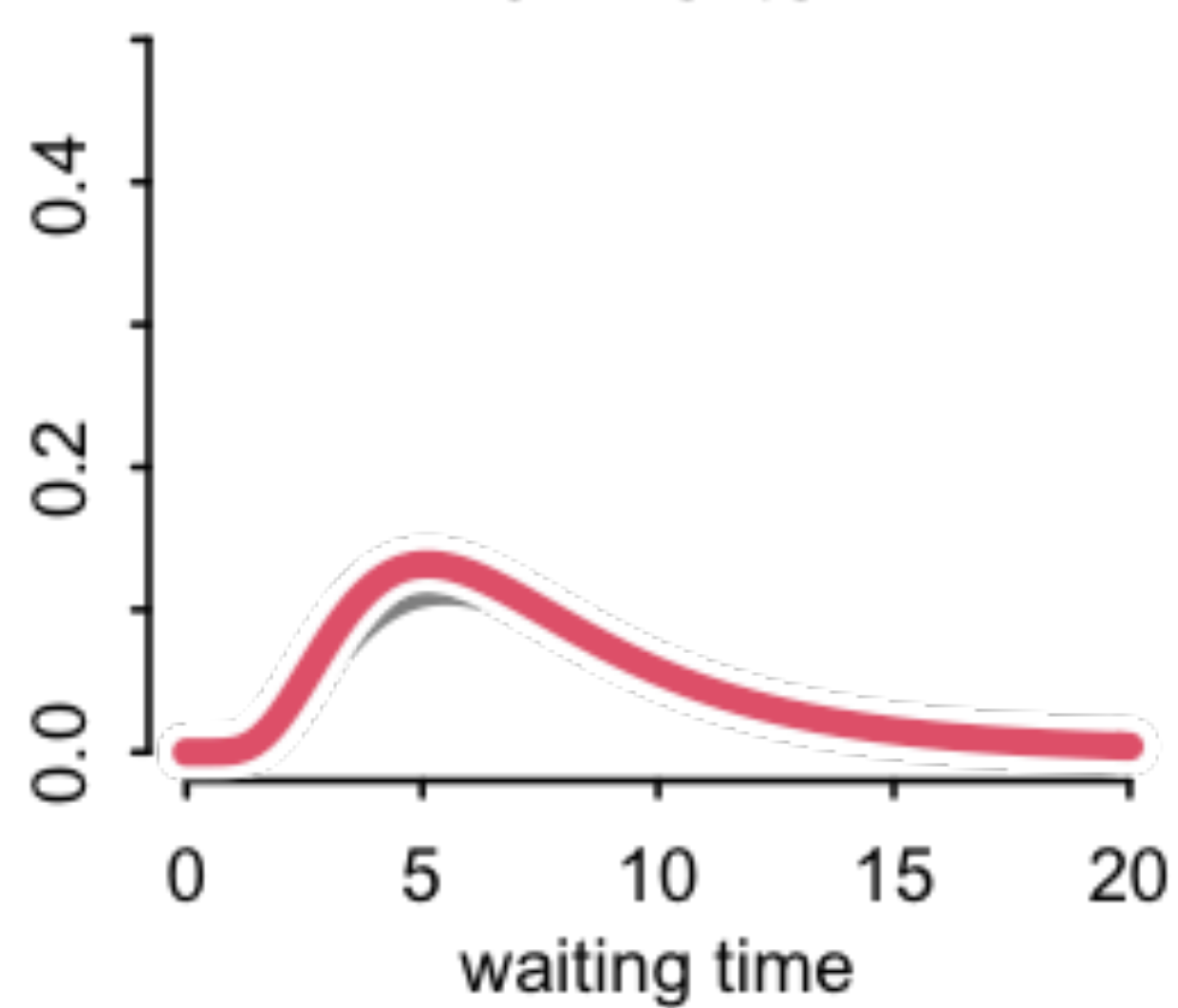
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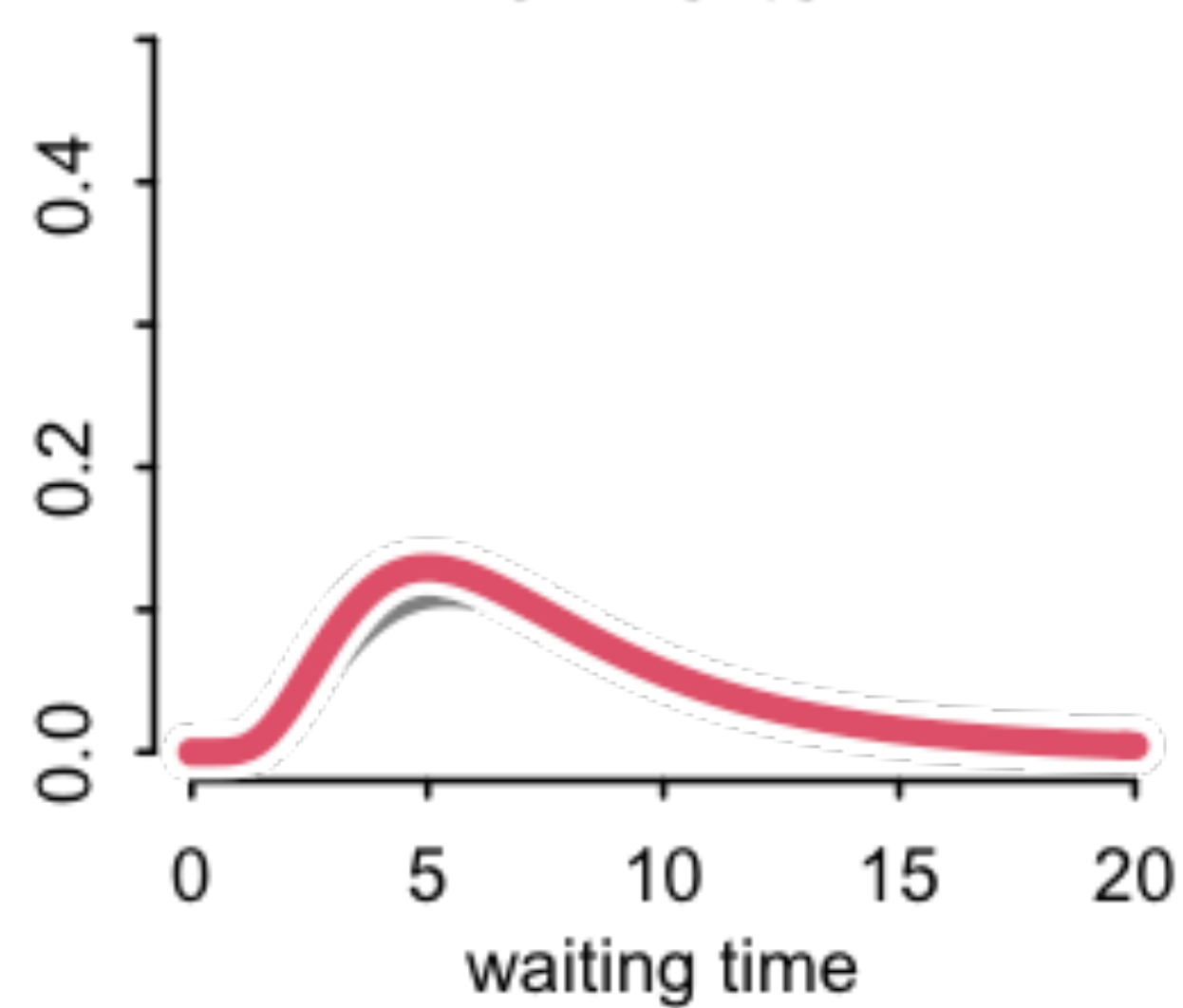
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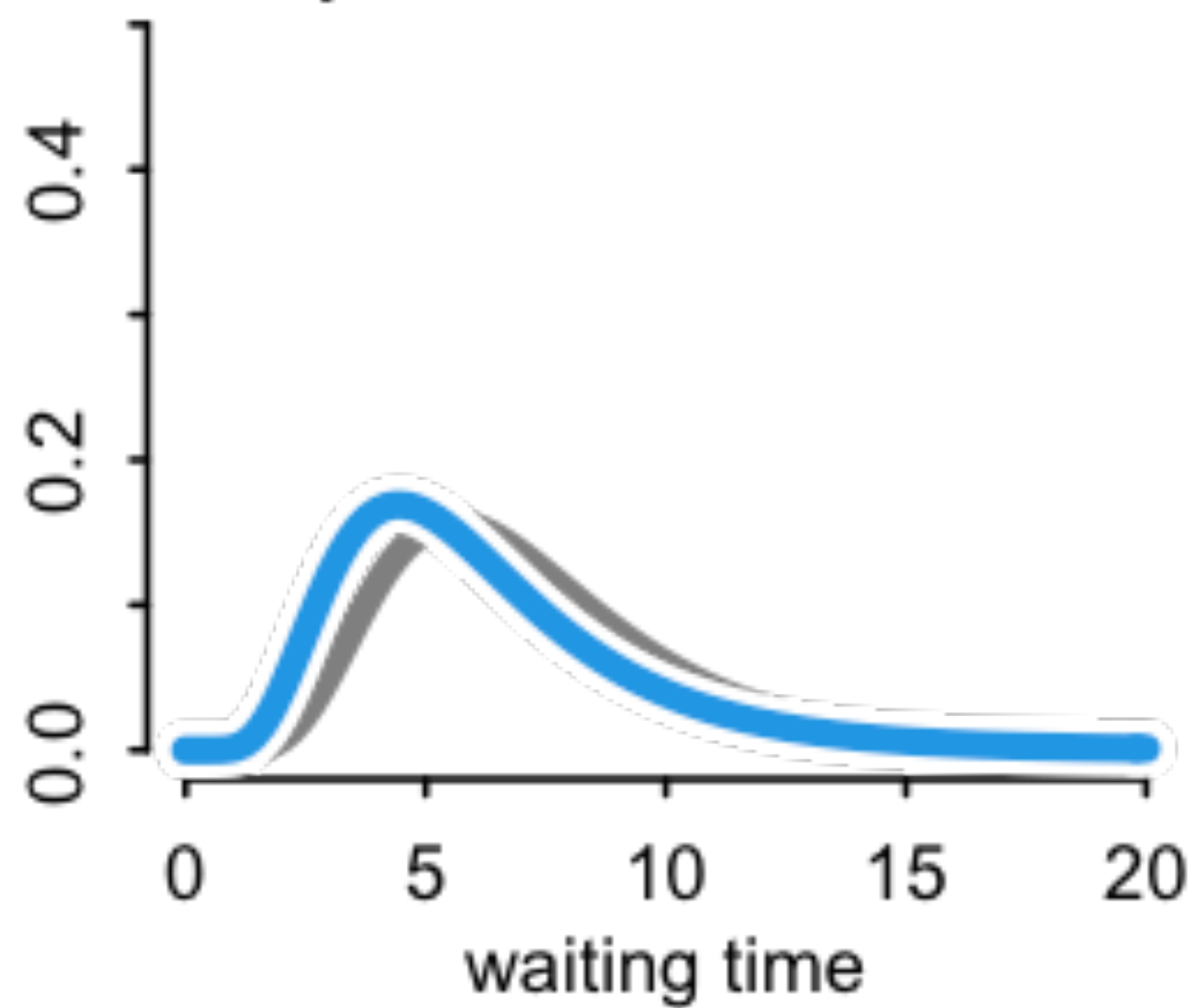
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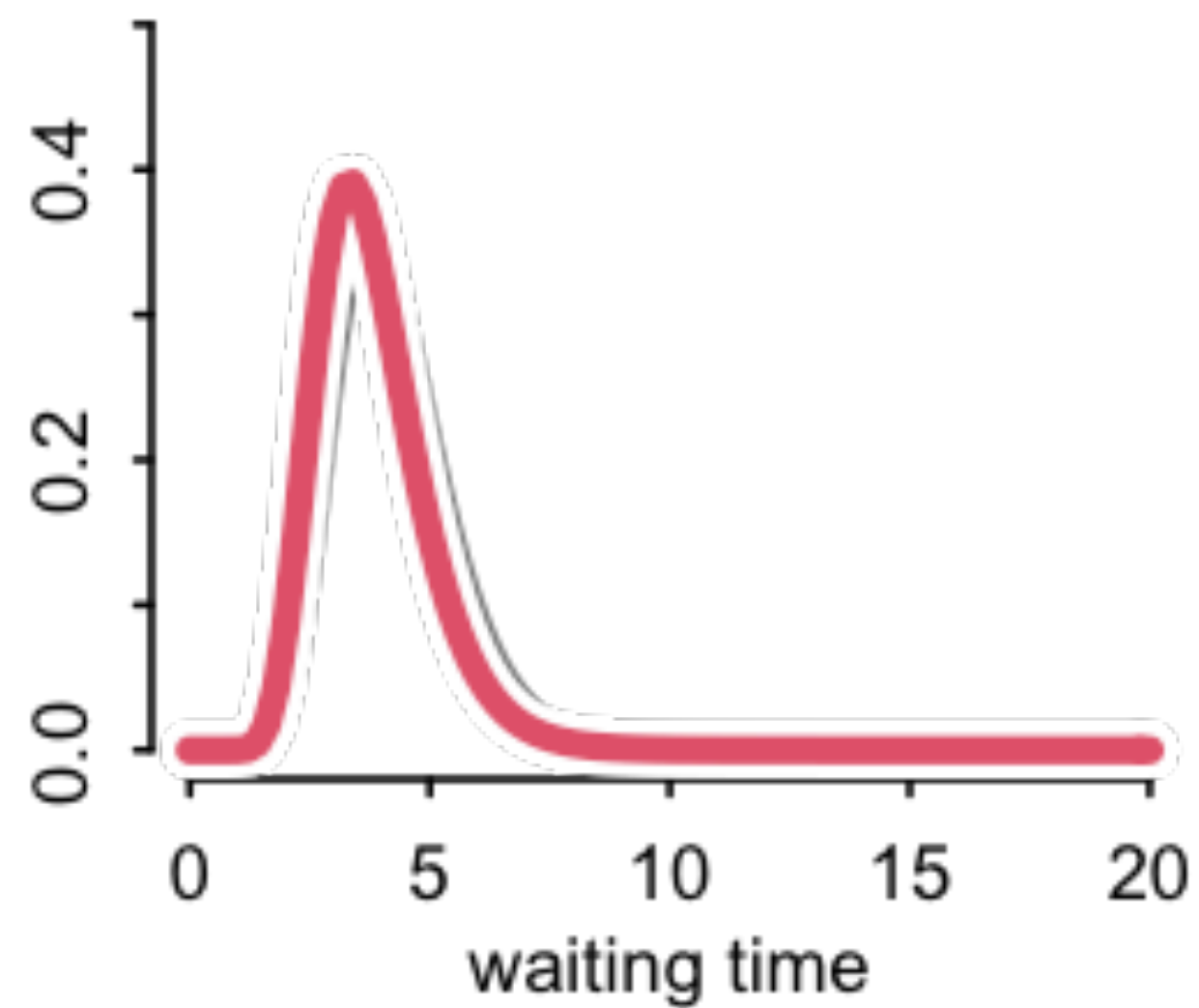
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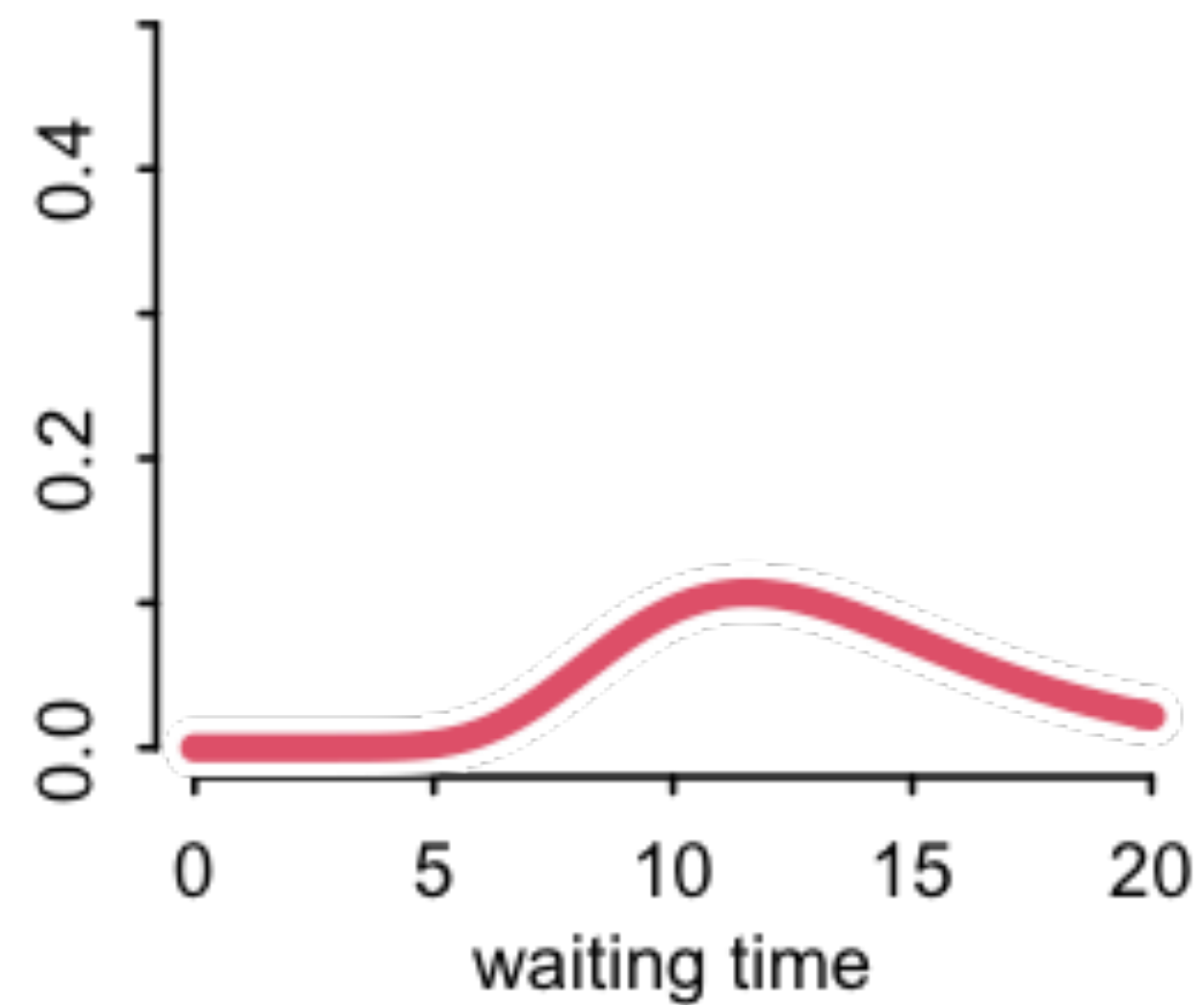
Population of cafes



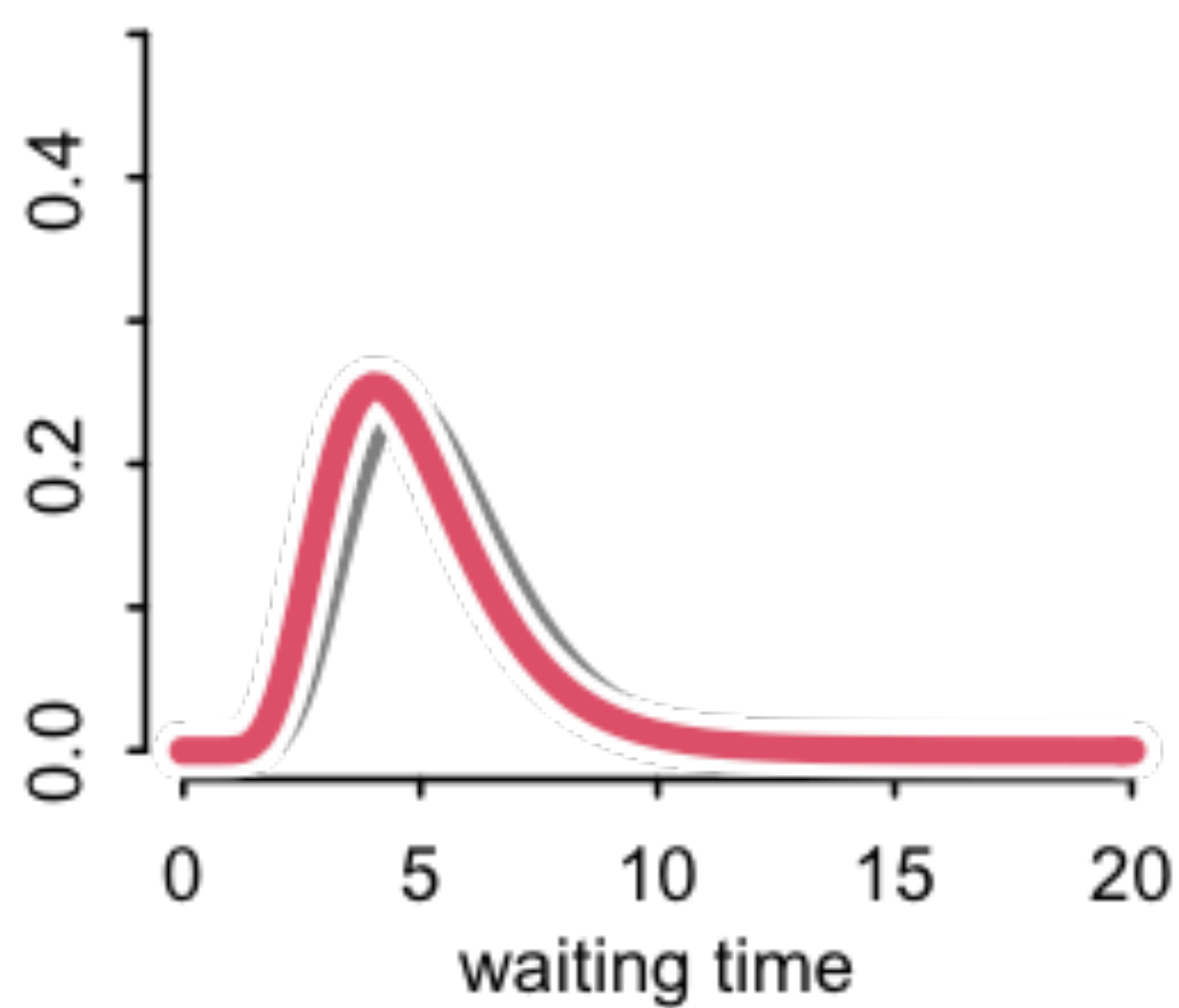
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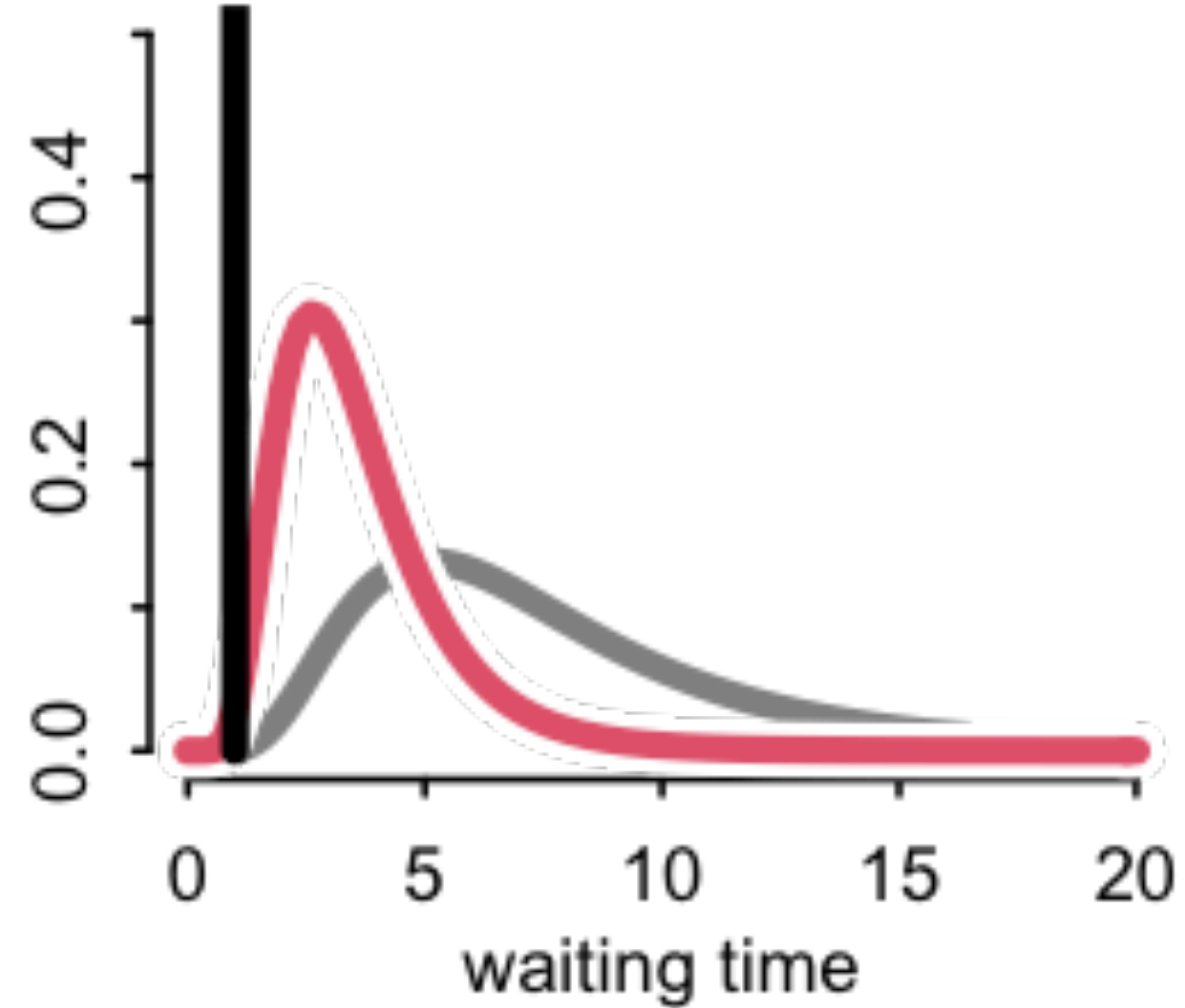
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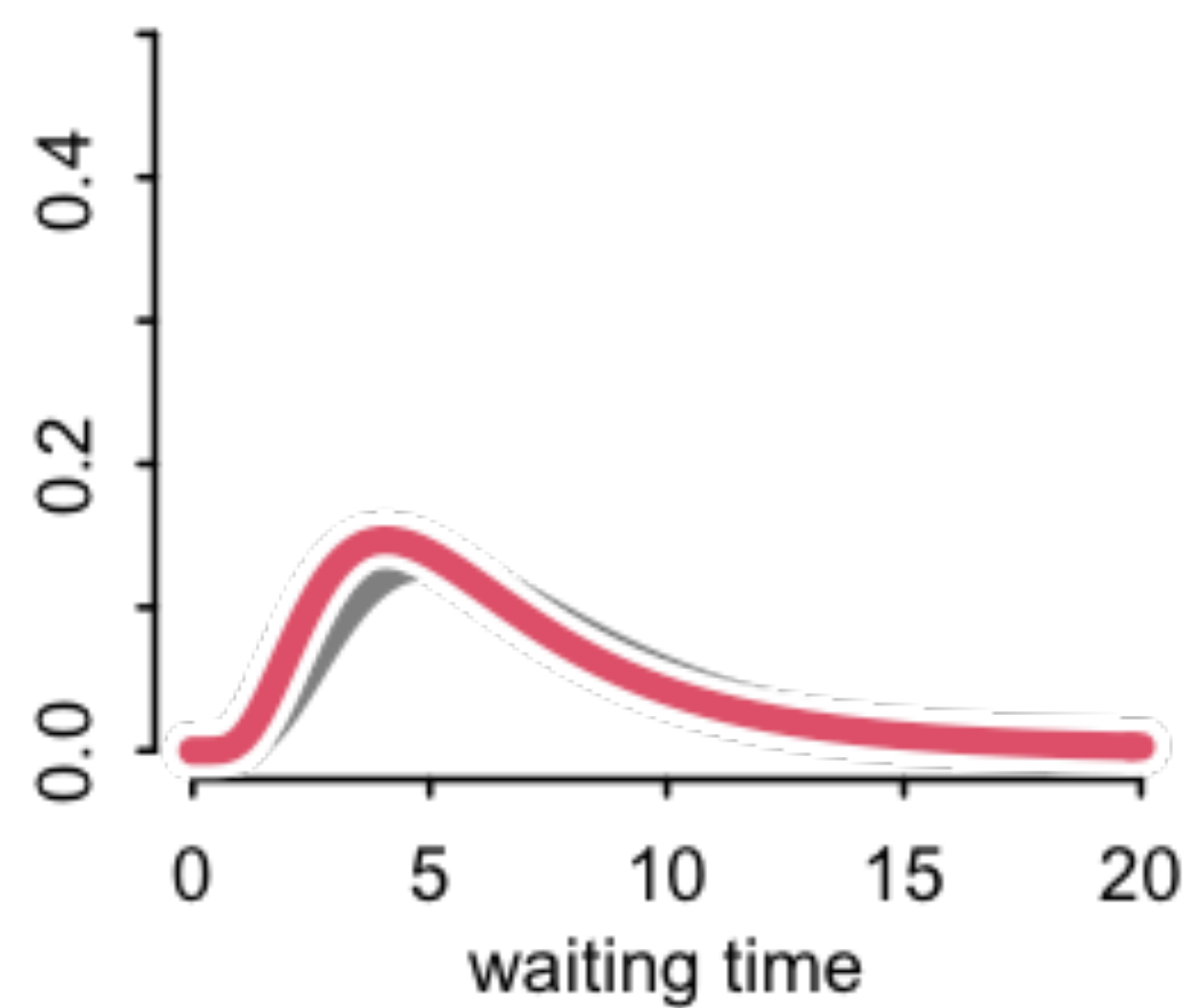
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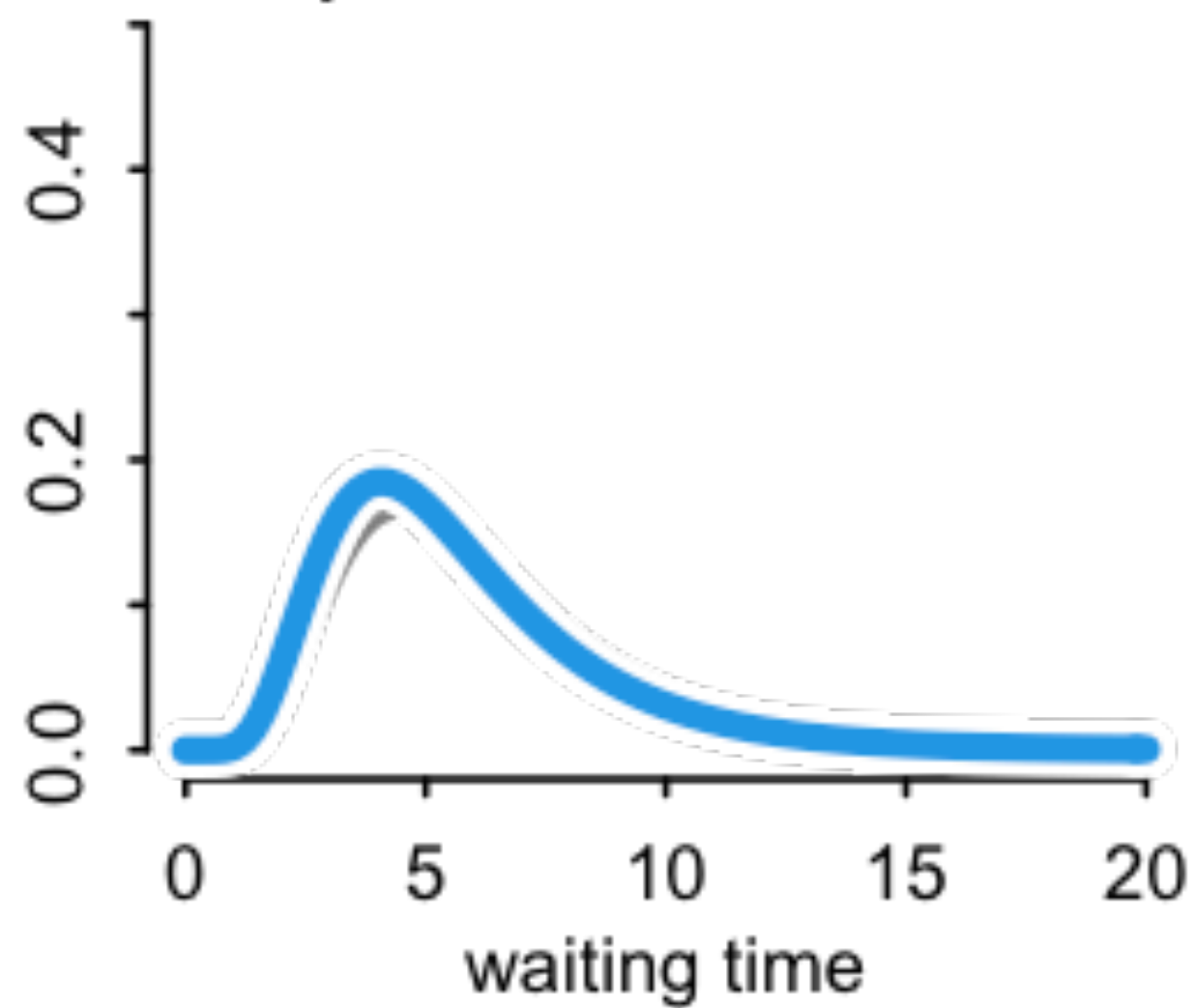
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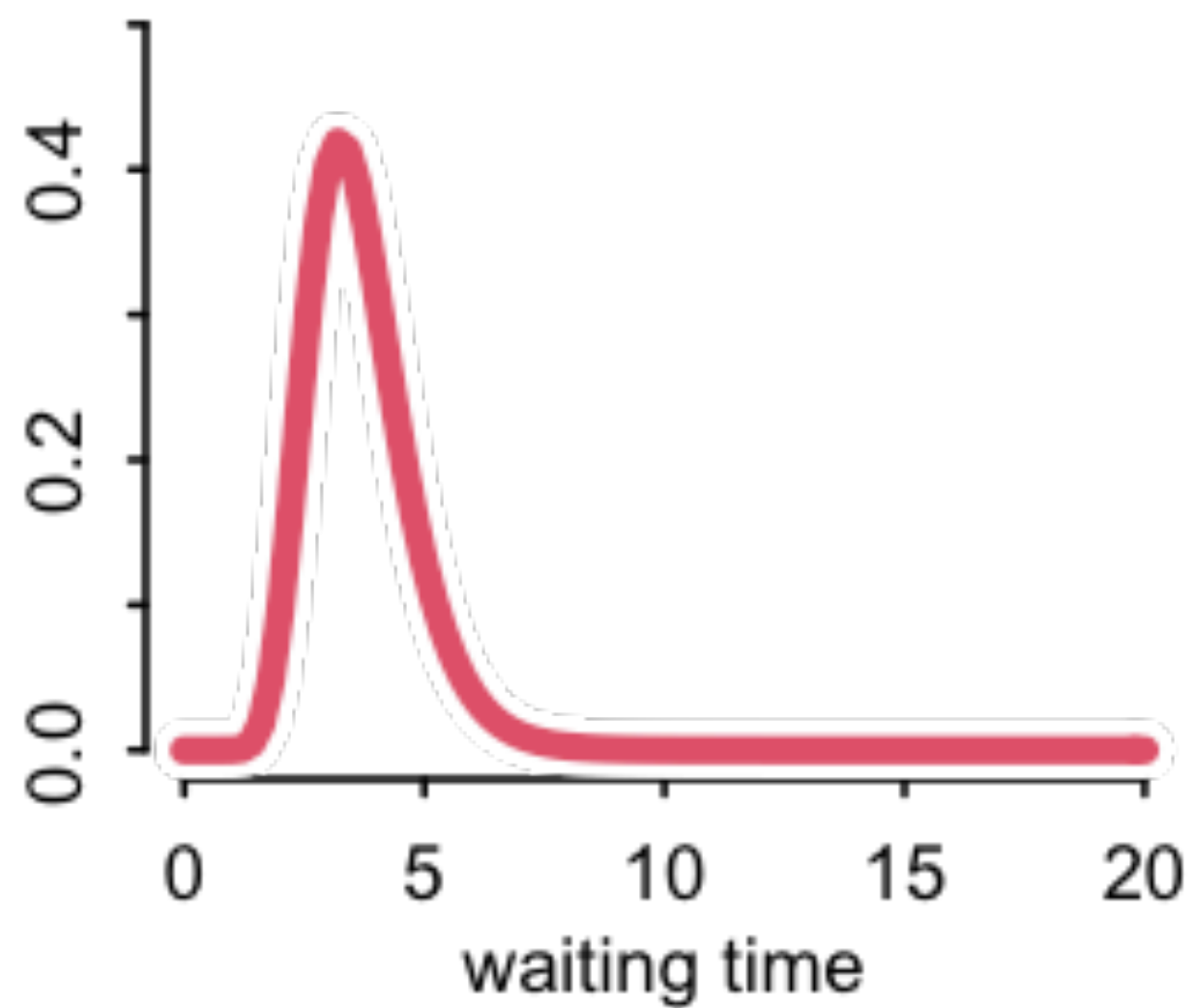
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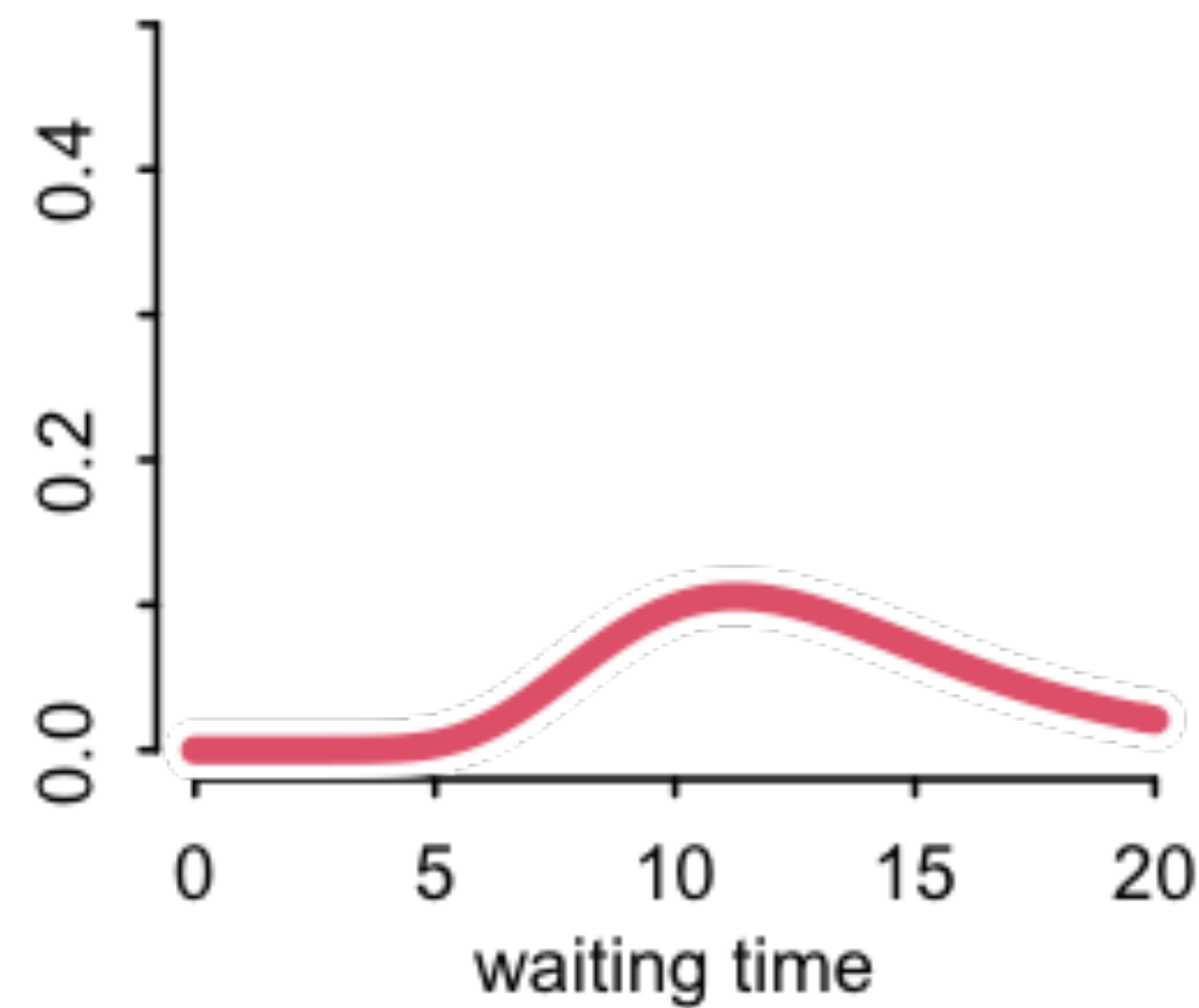
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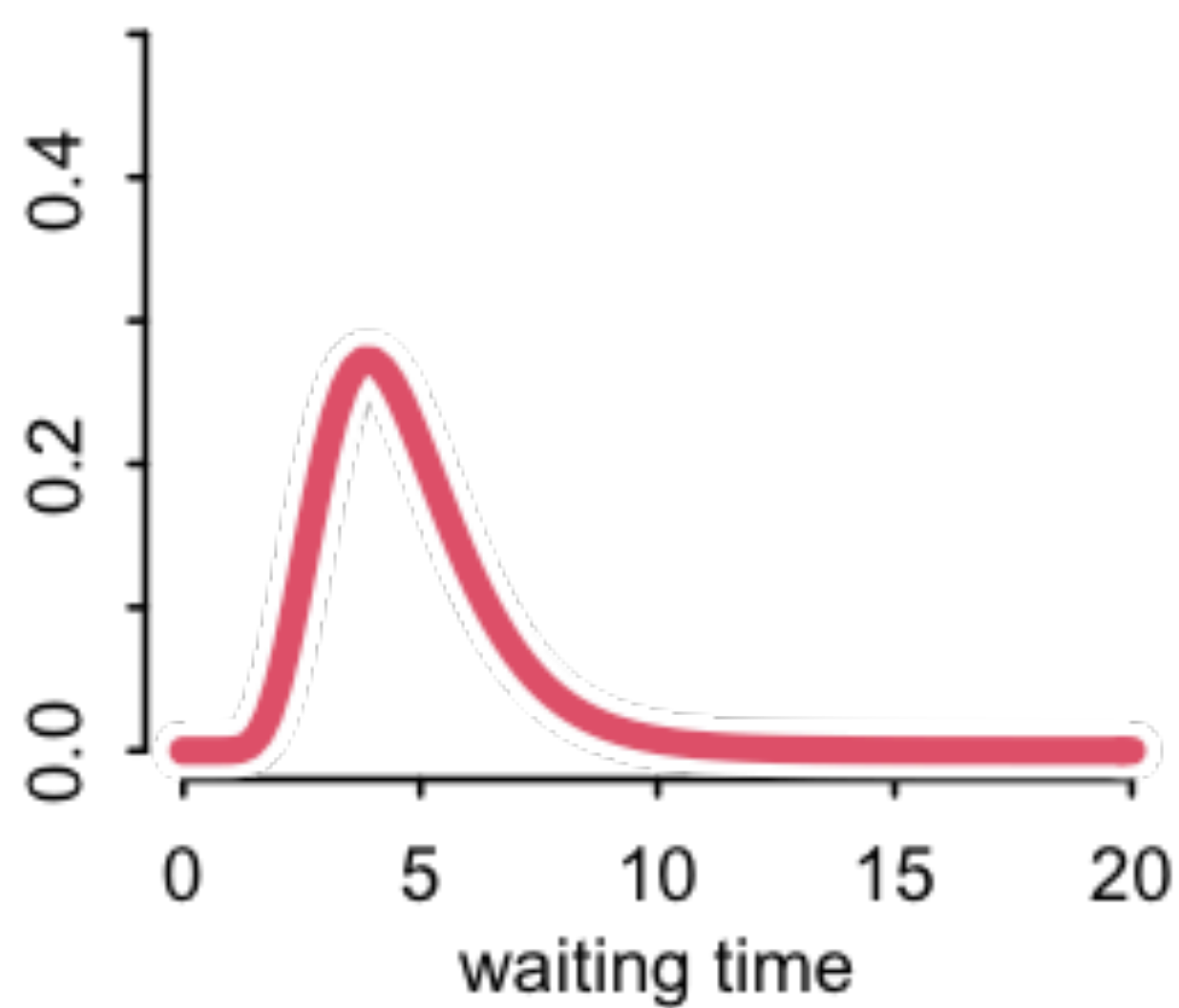
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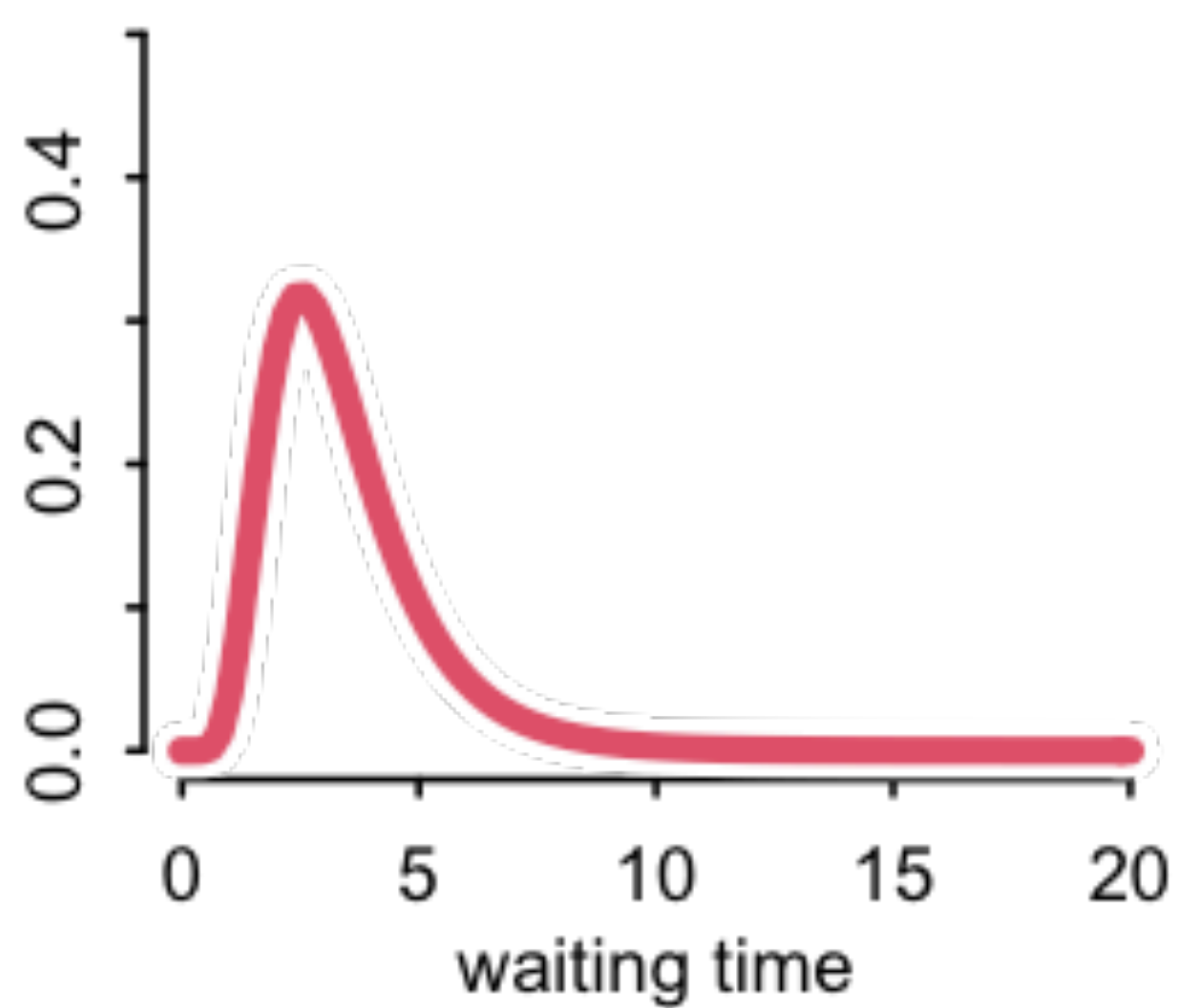
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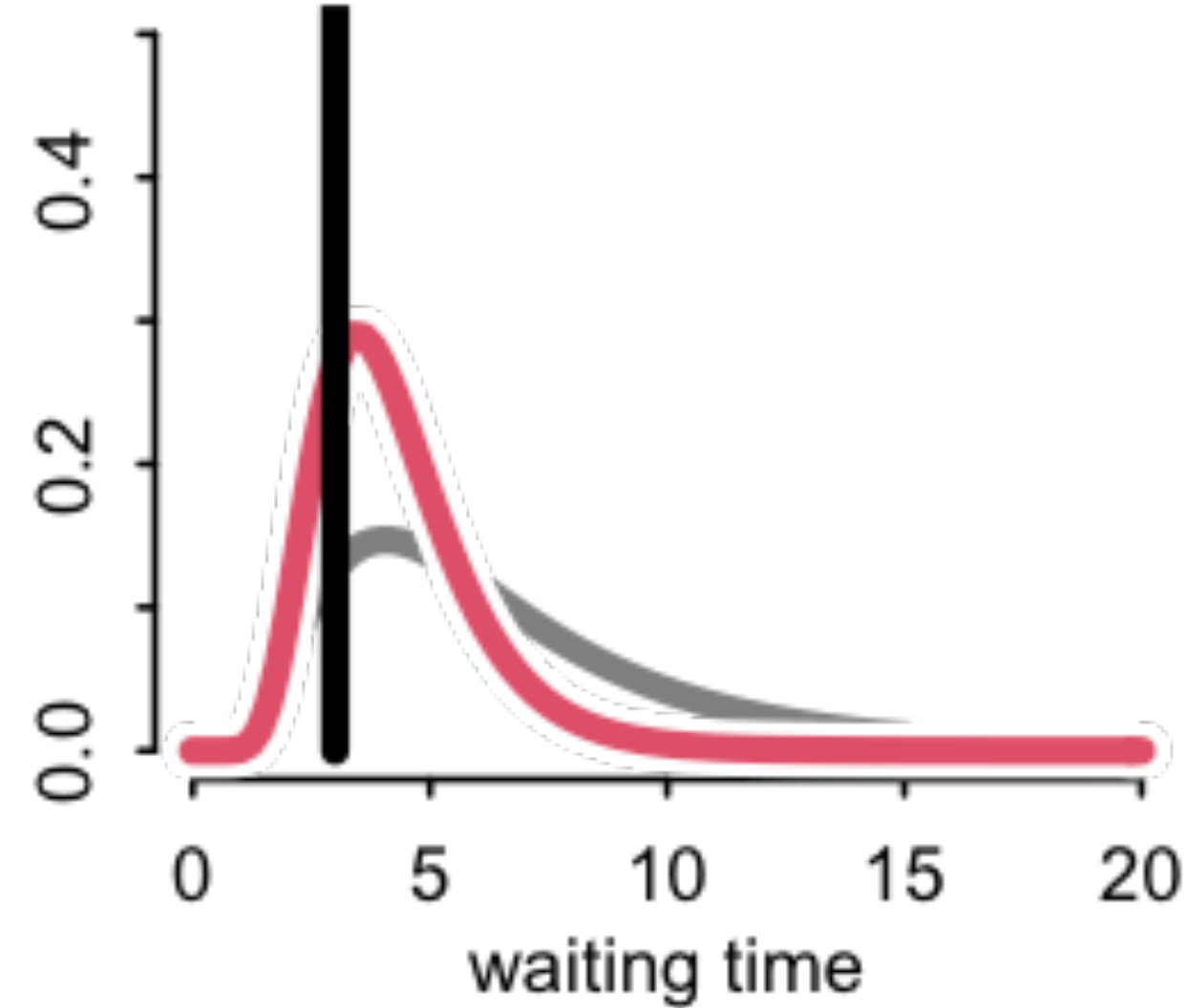
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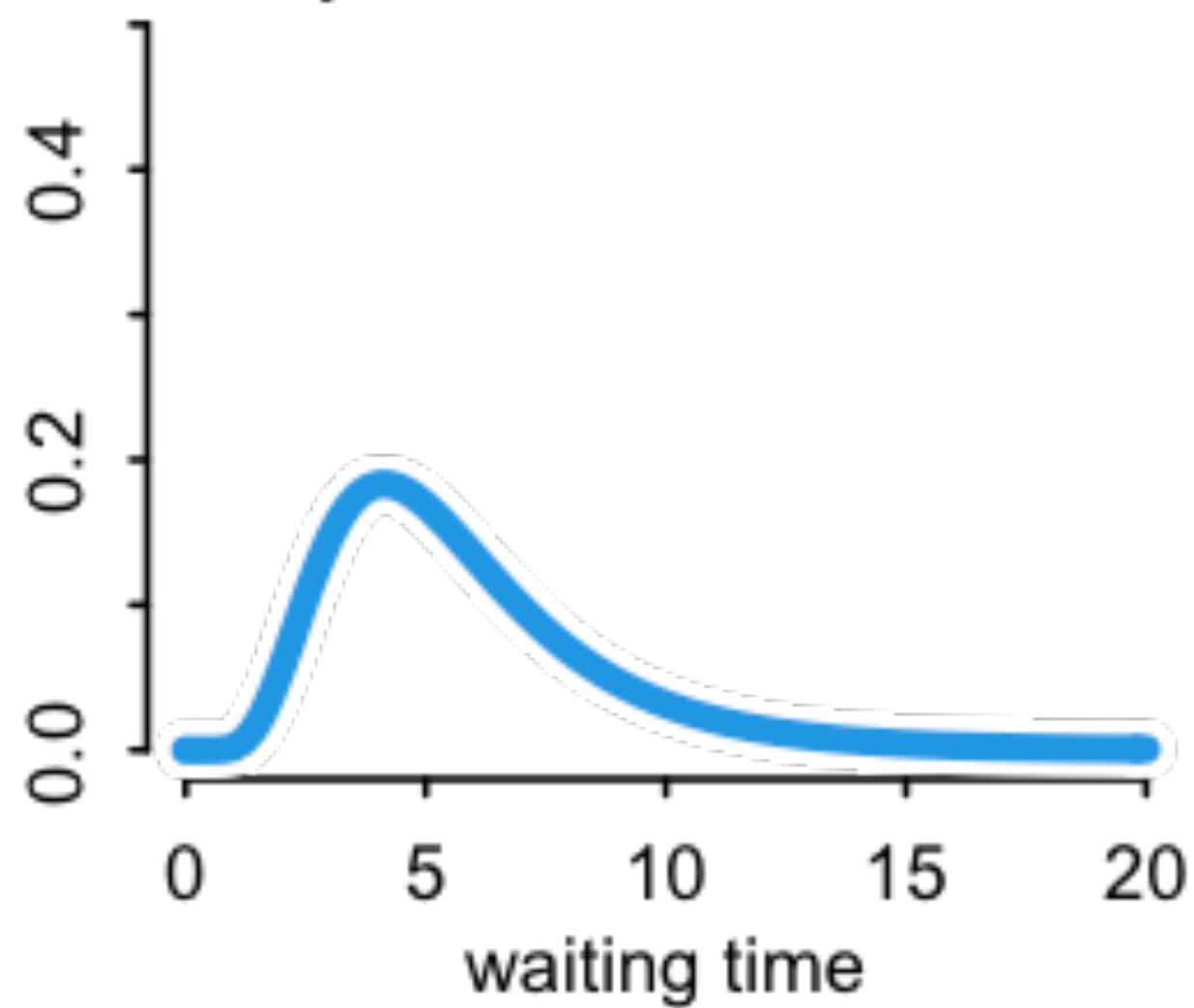
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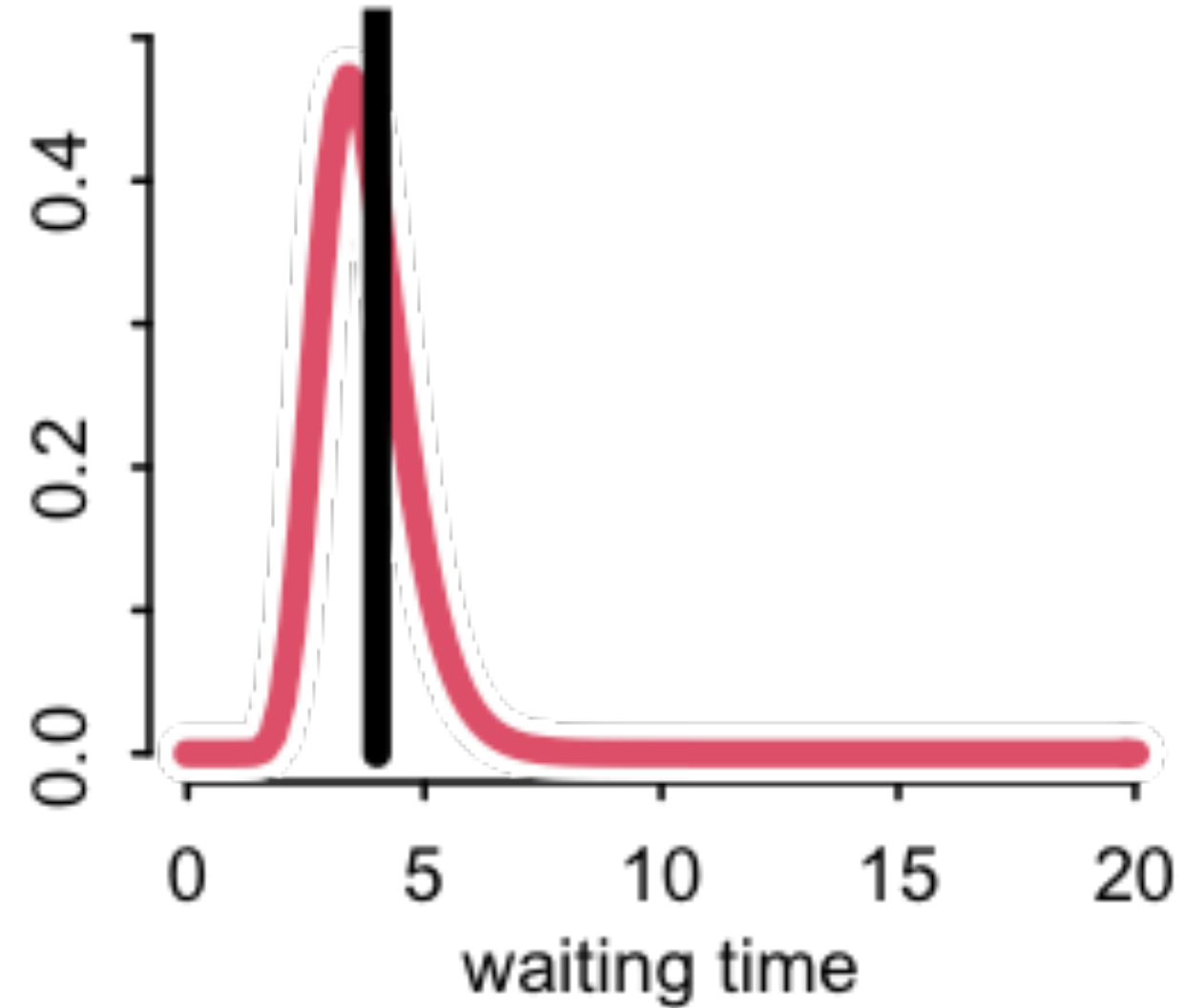
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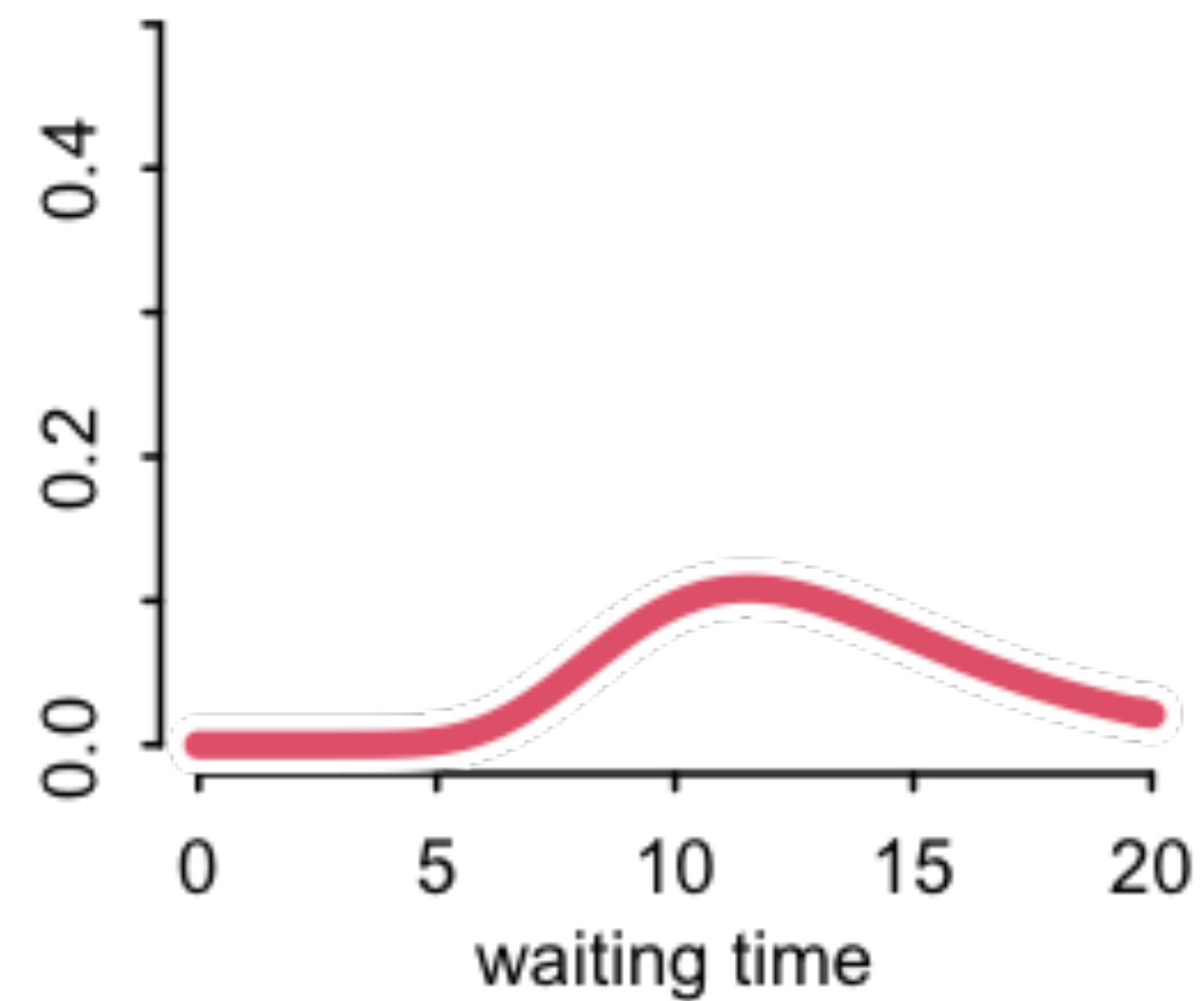
Population of cafes



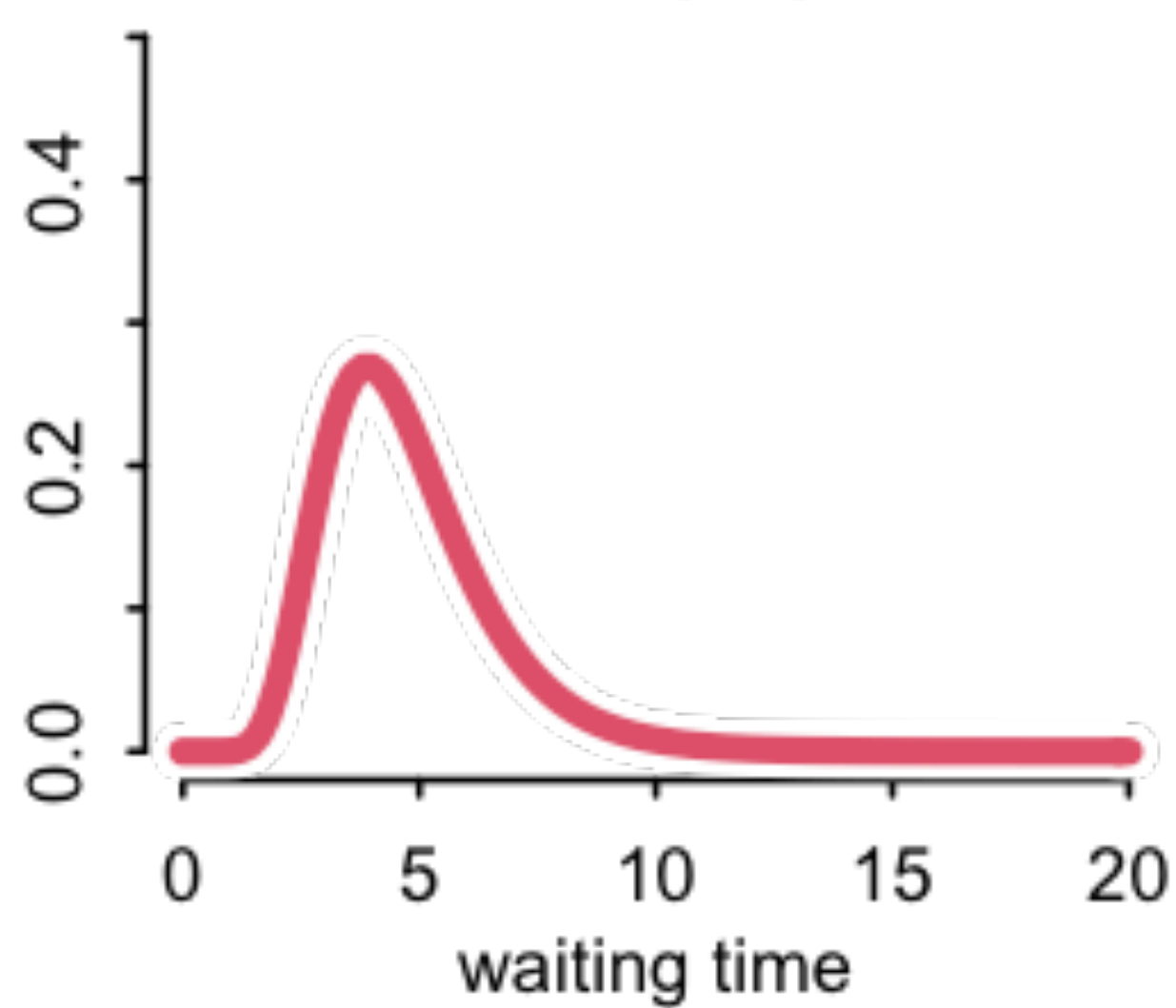
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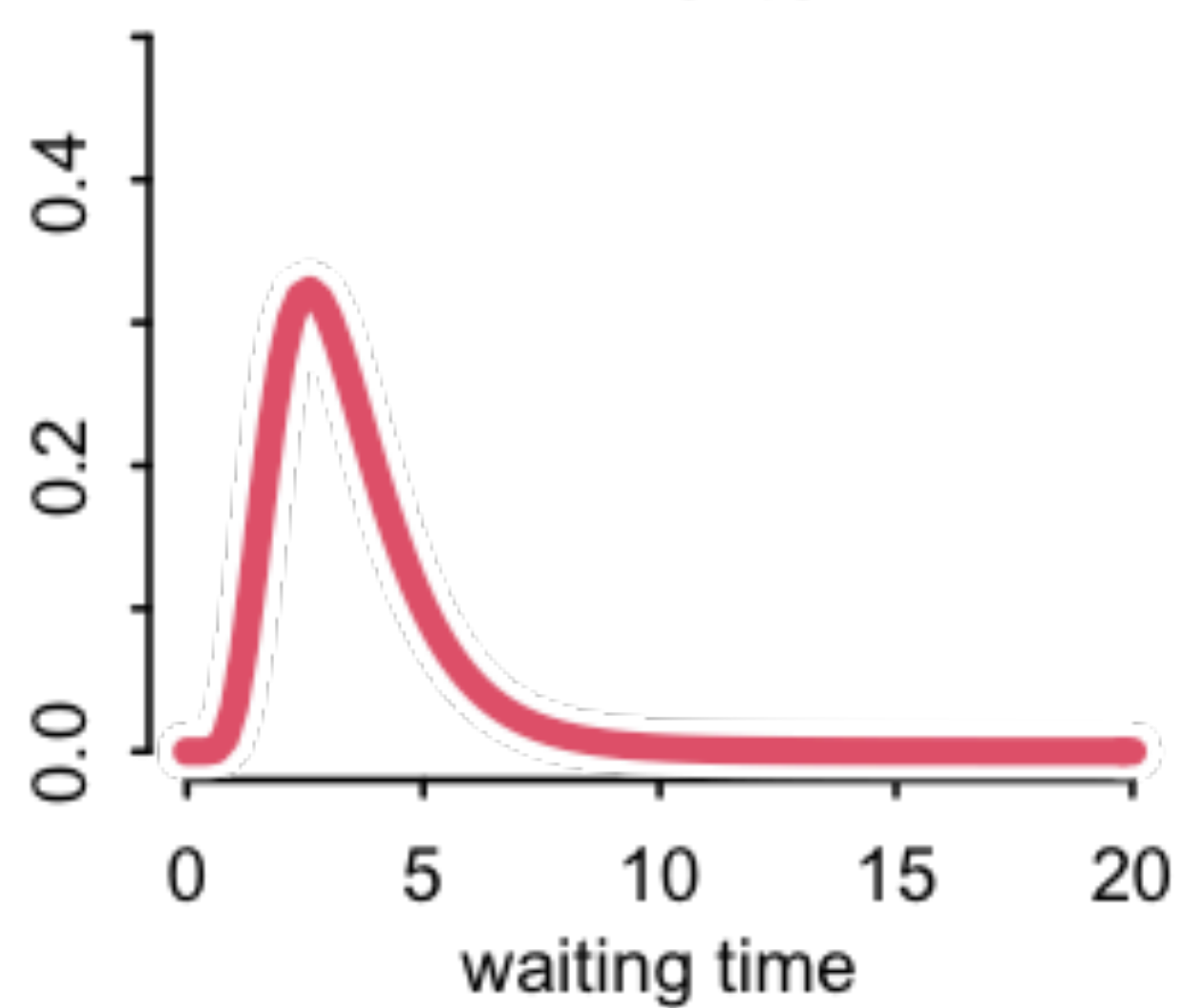
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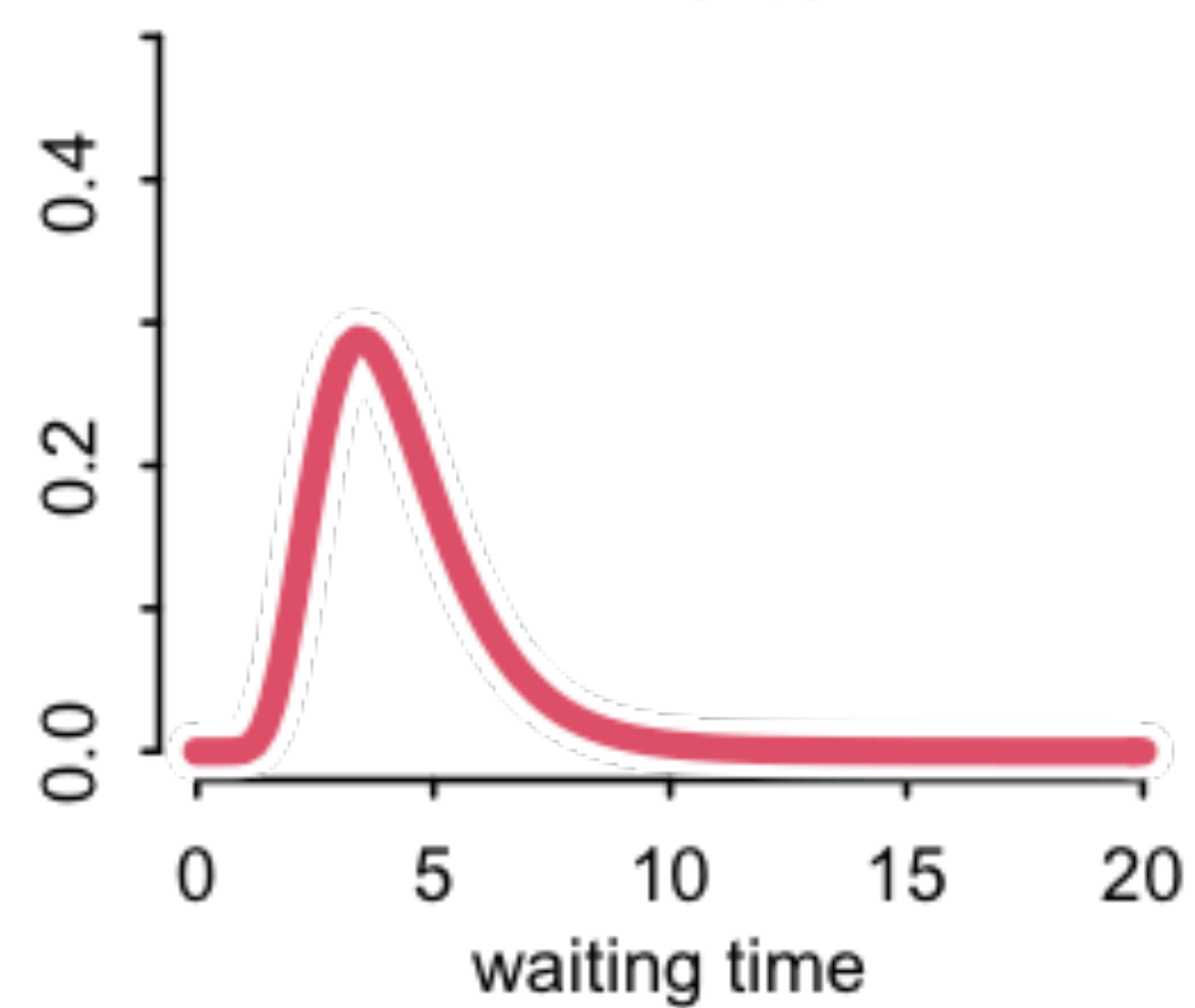
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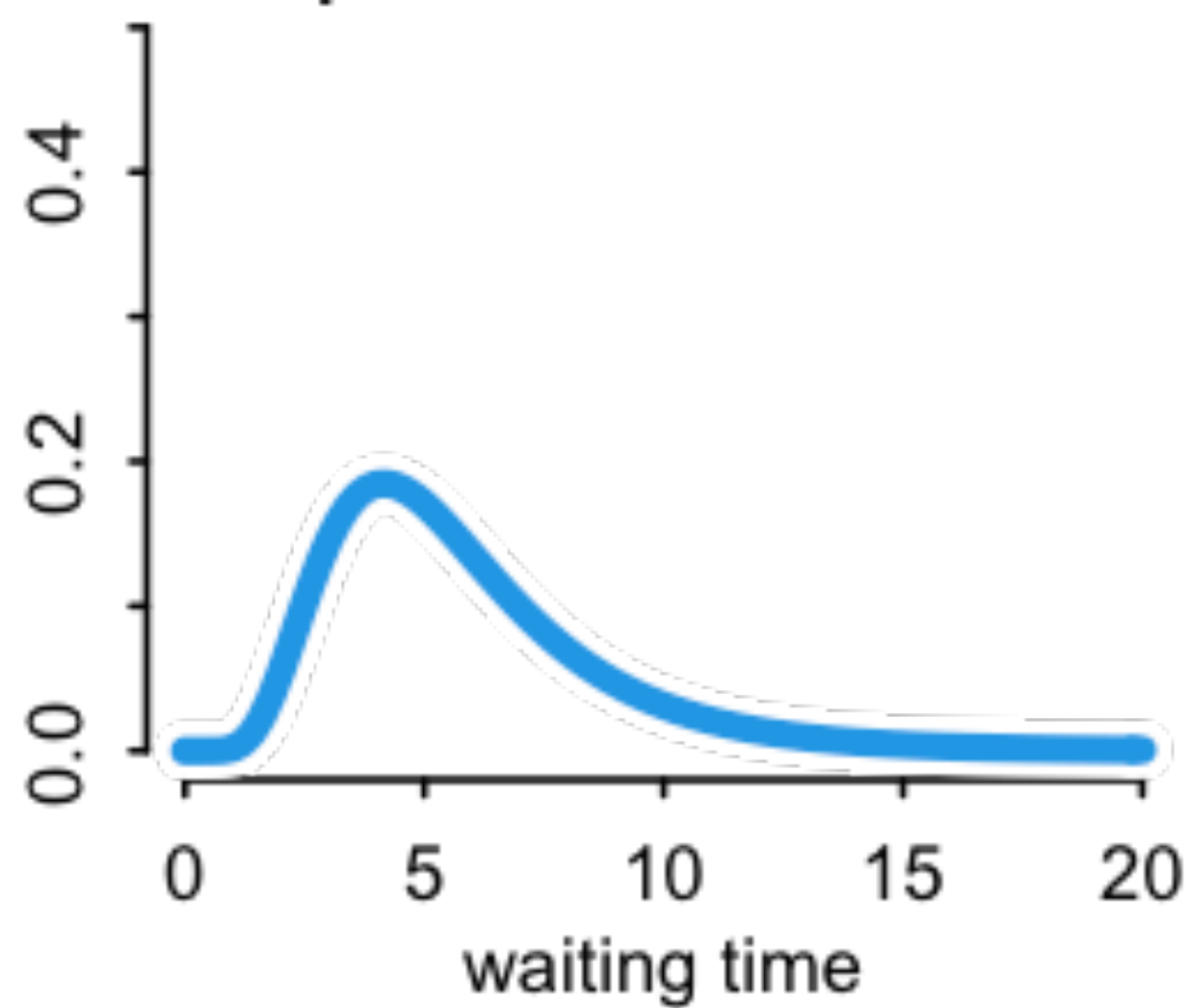
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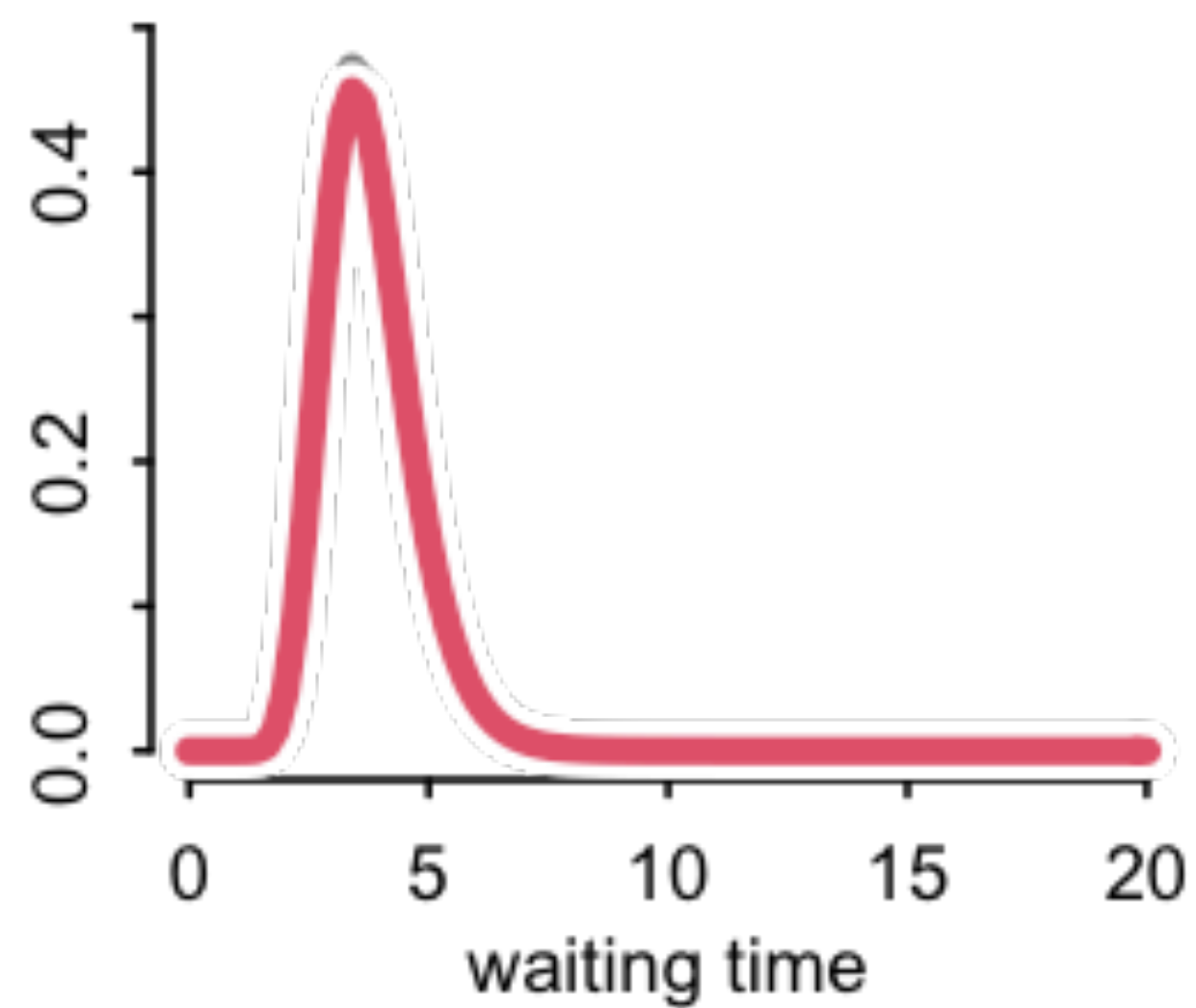
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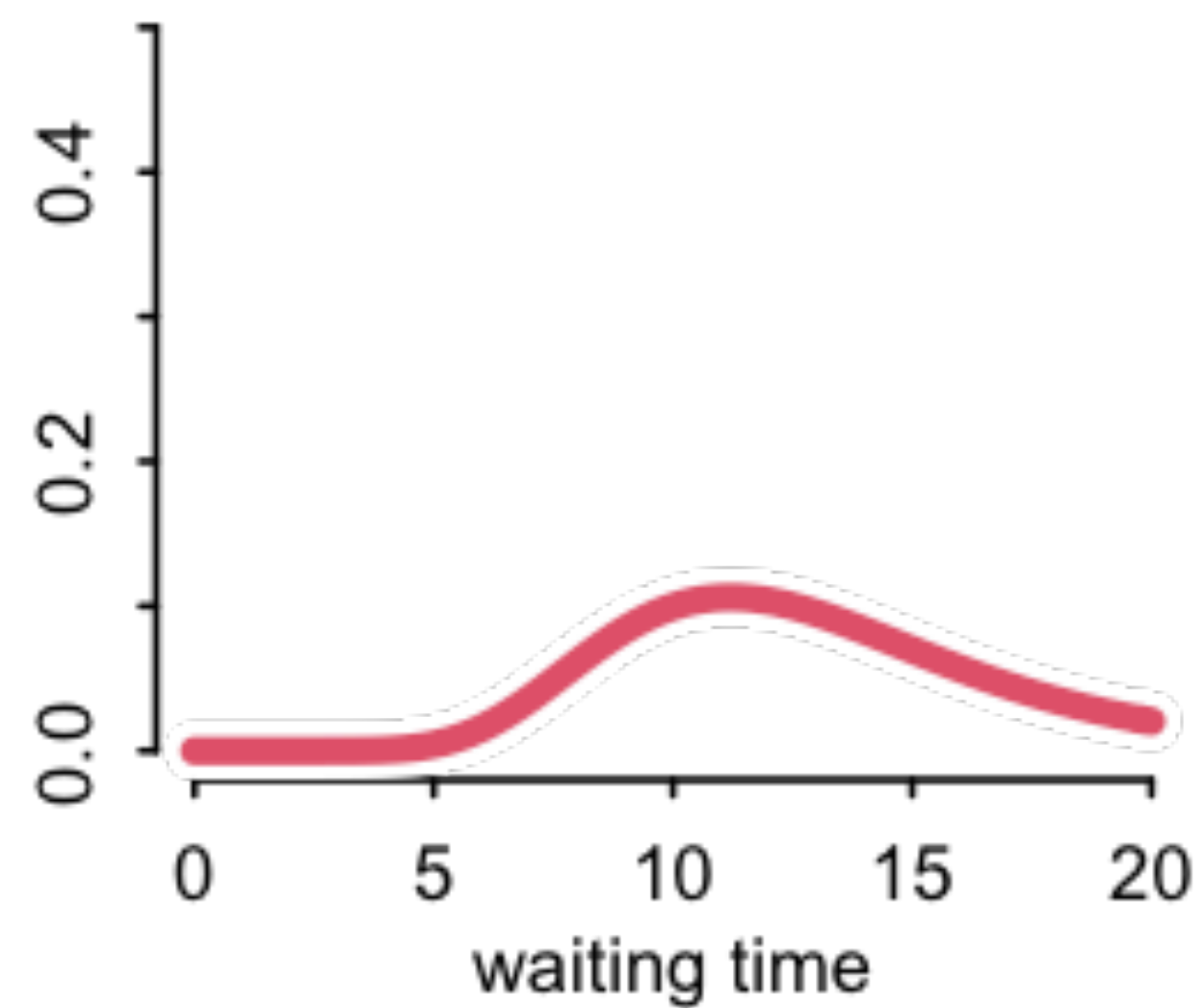
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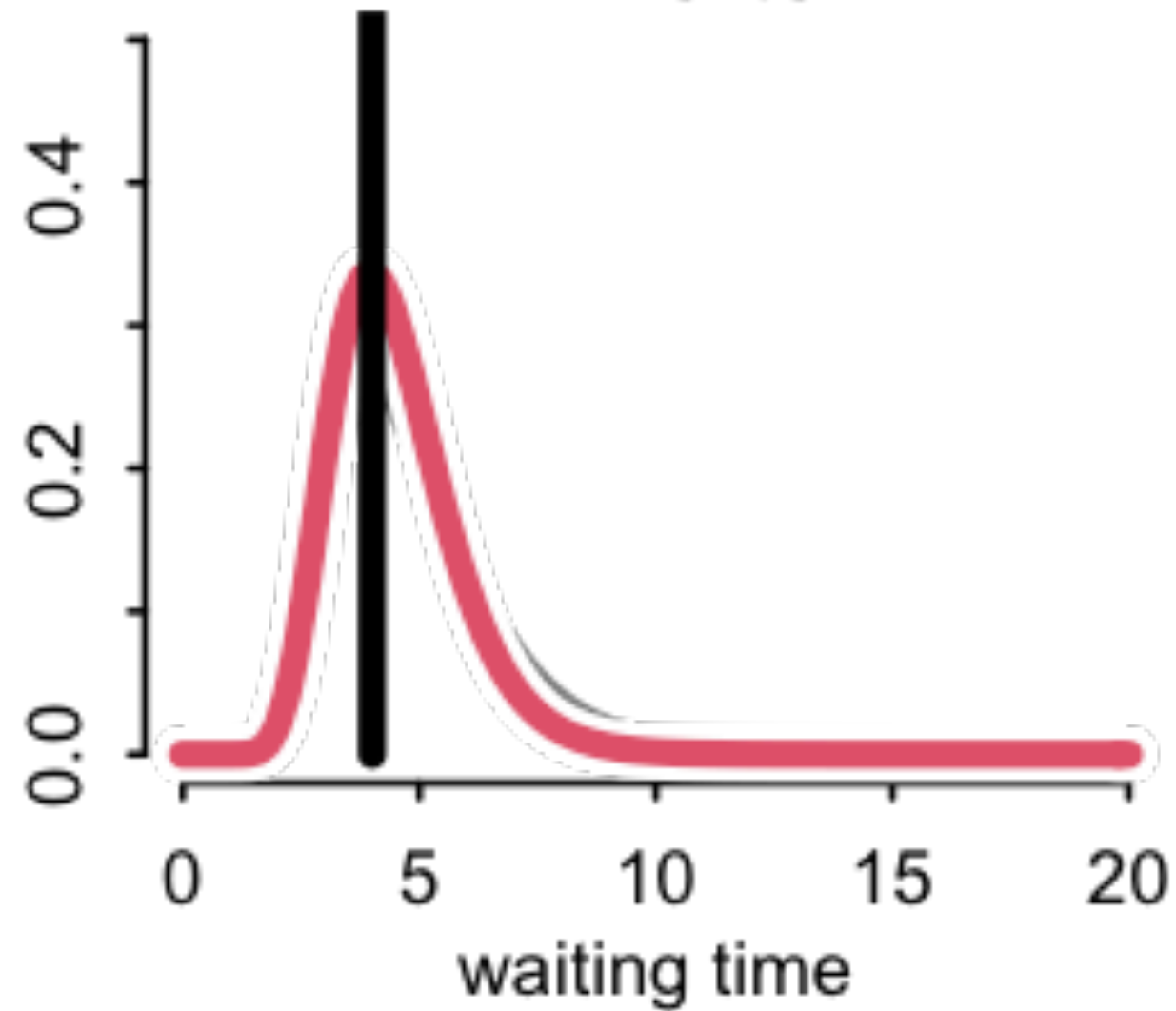
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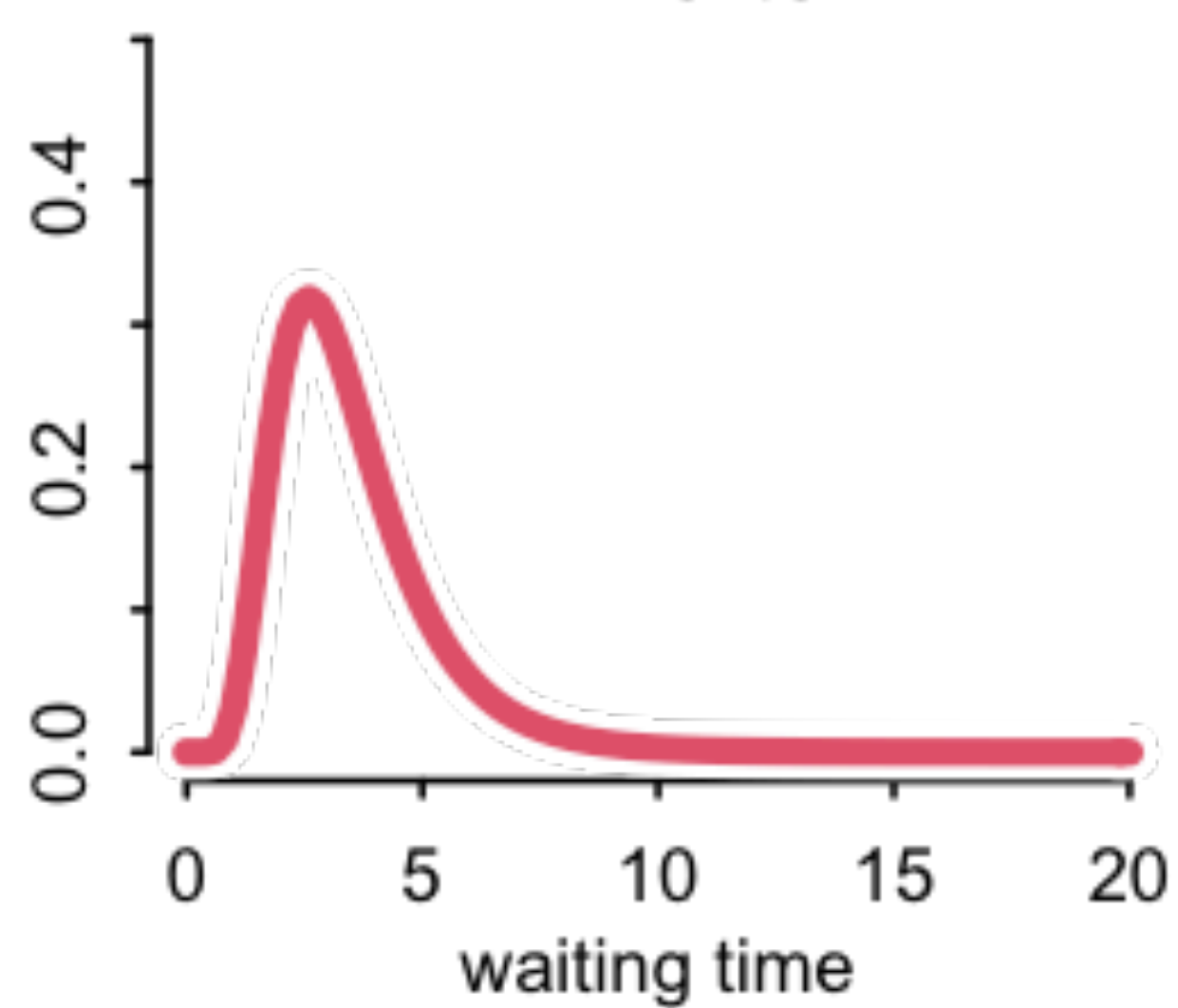
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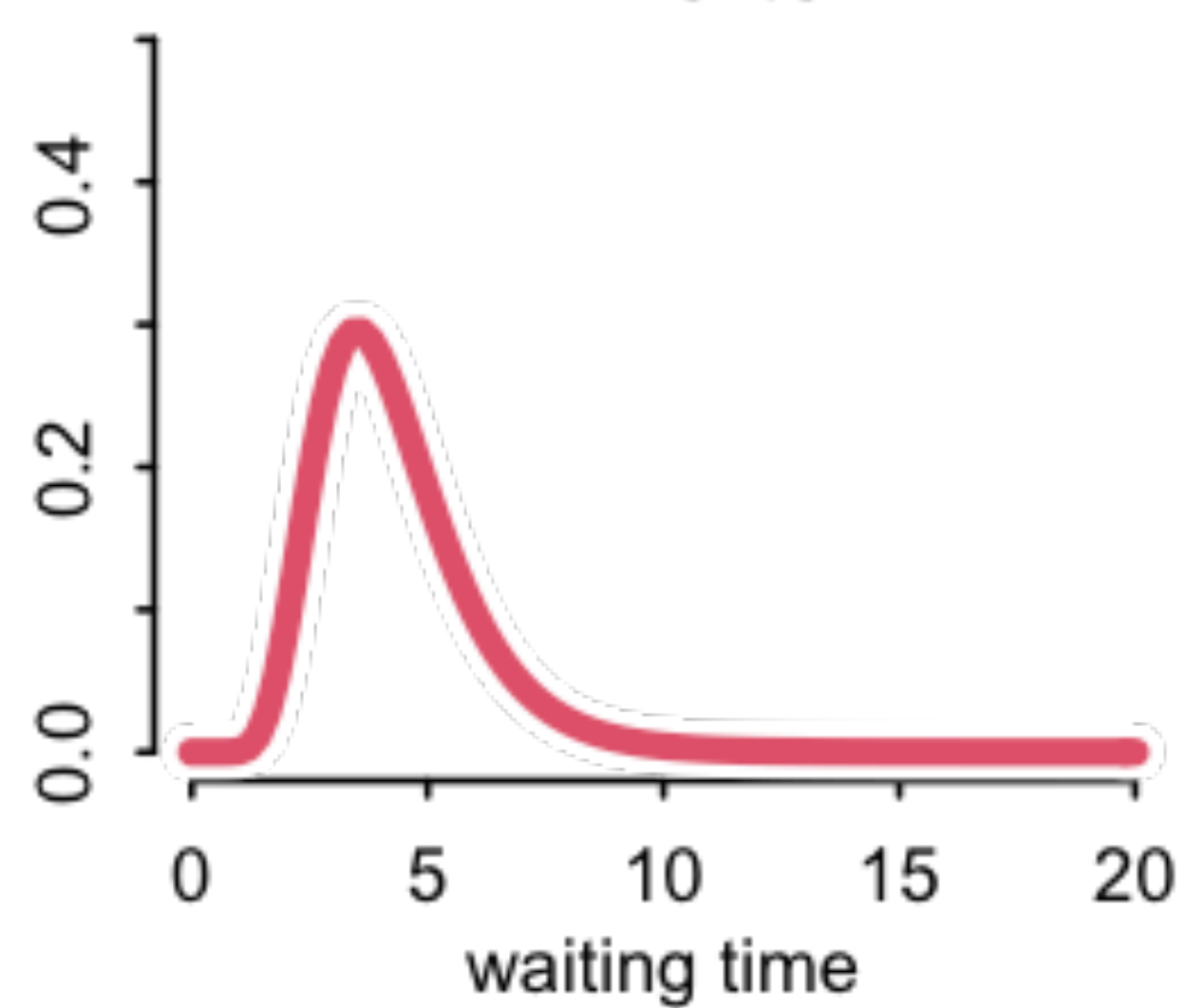
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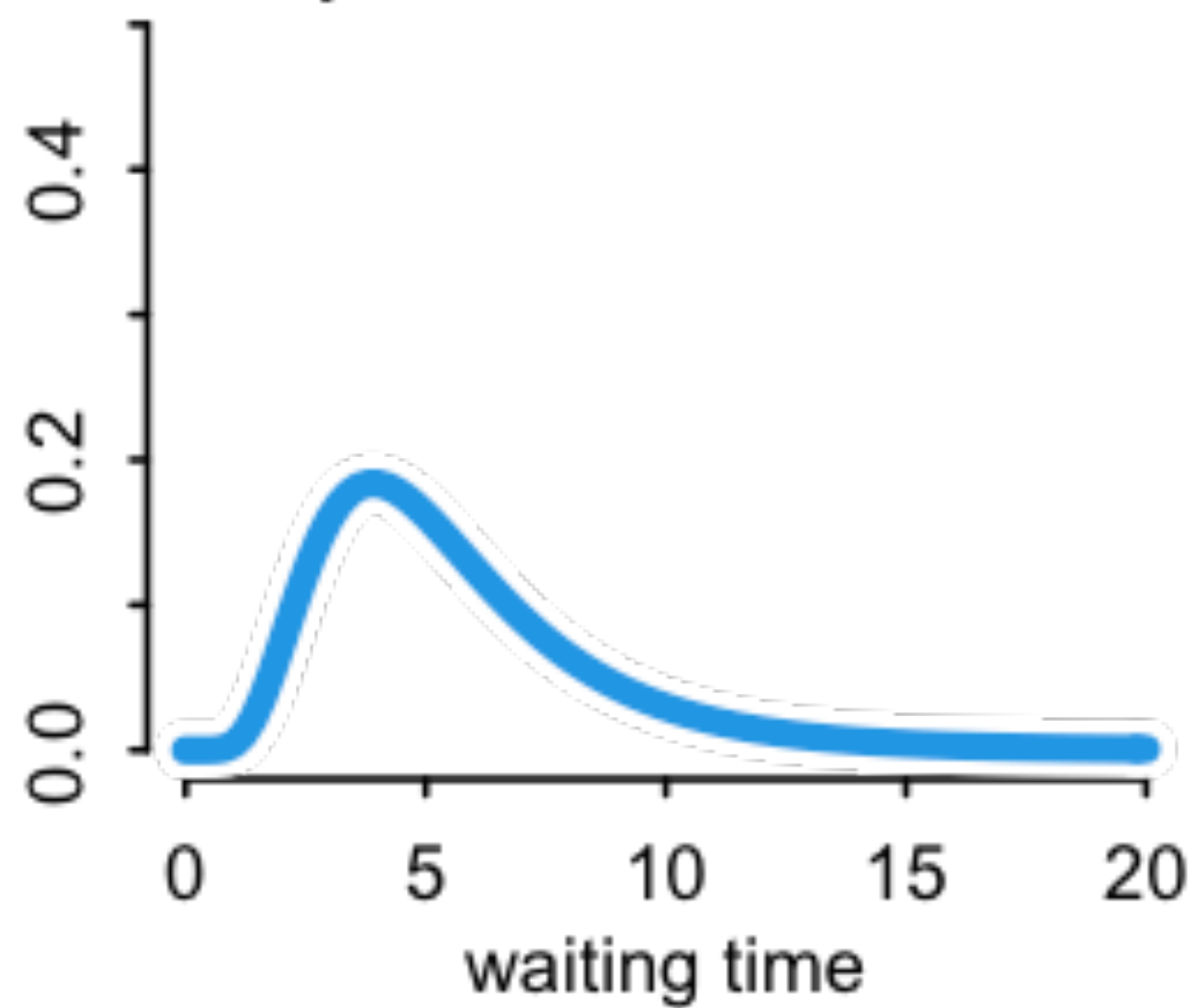
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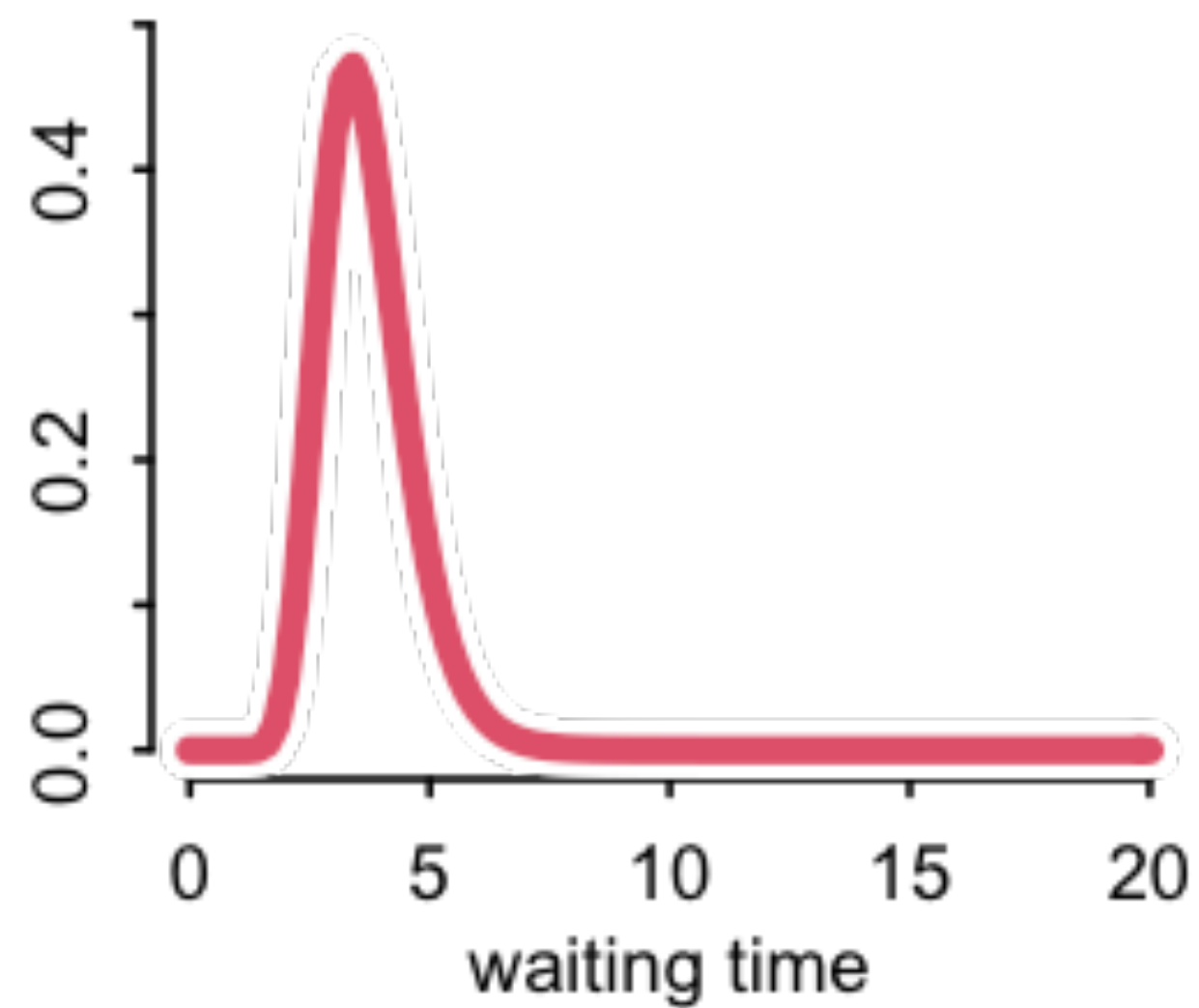
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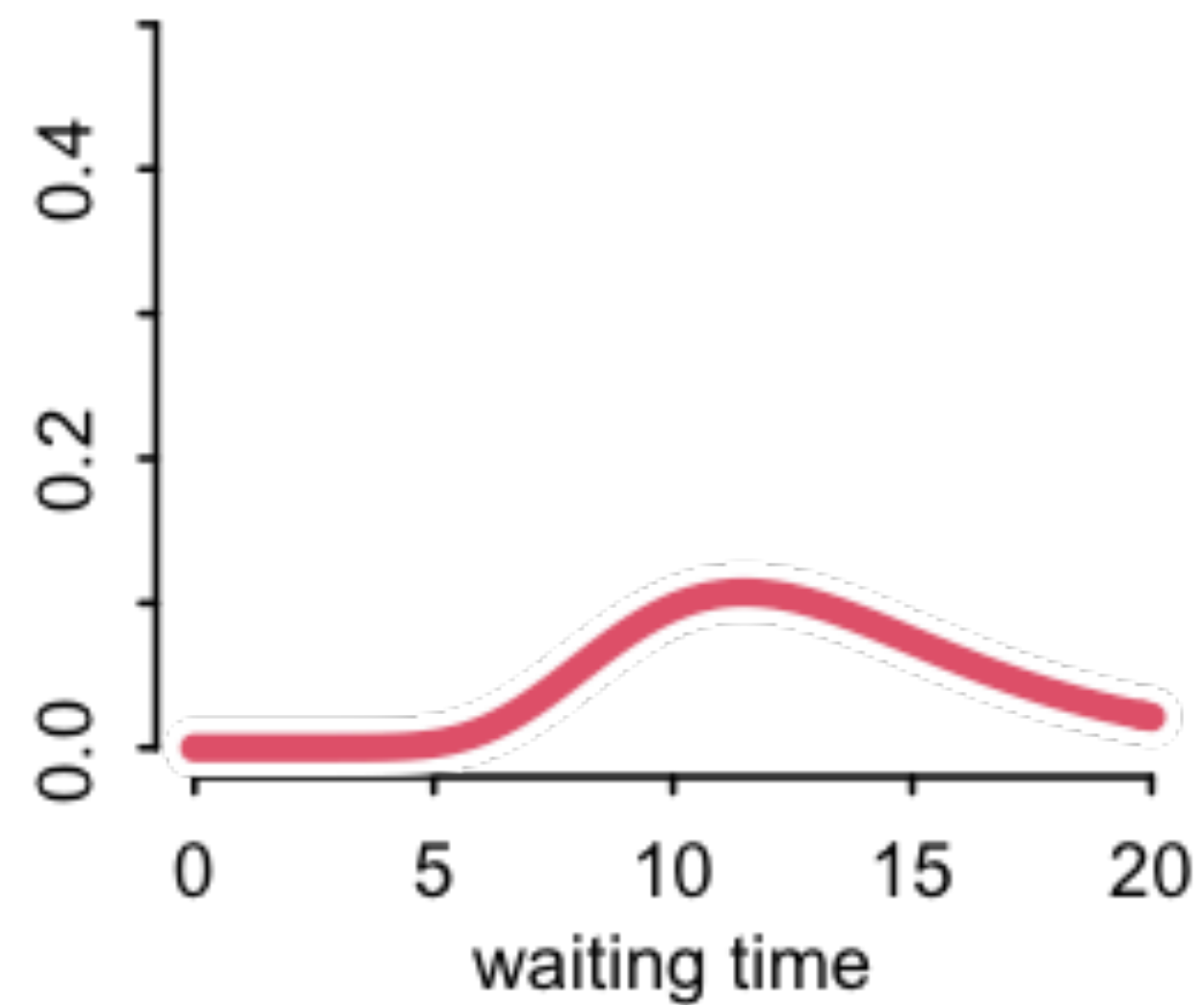
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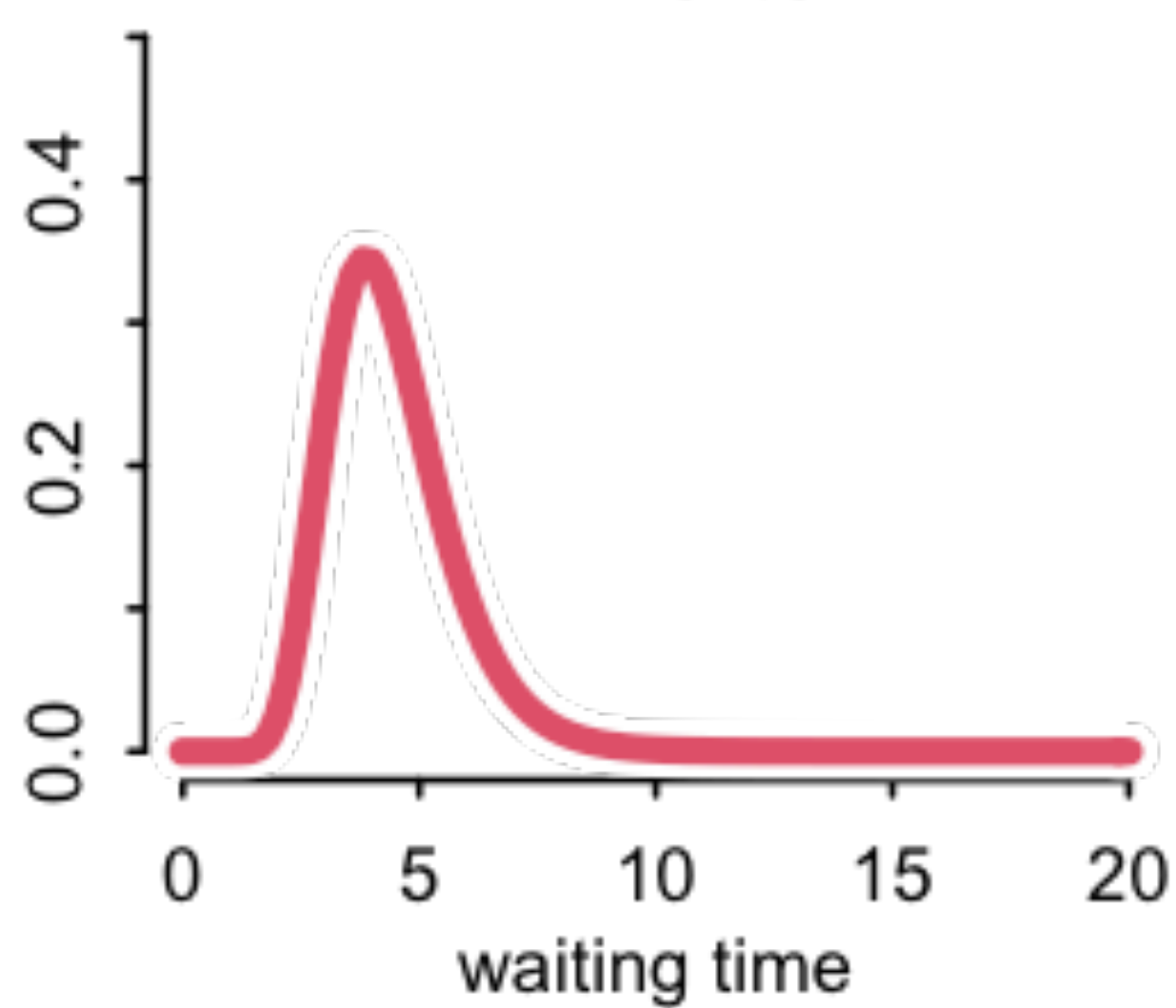
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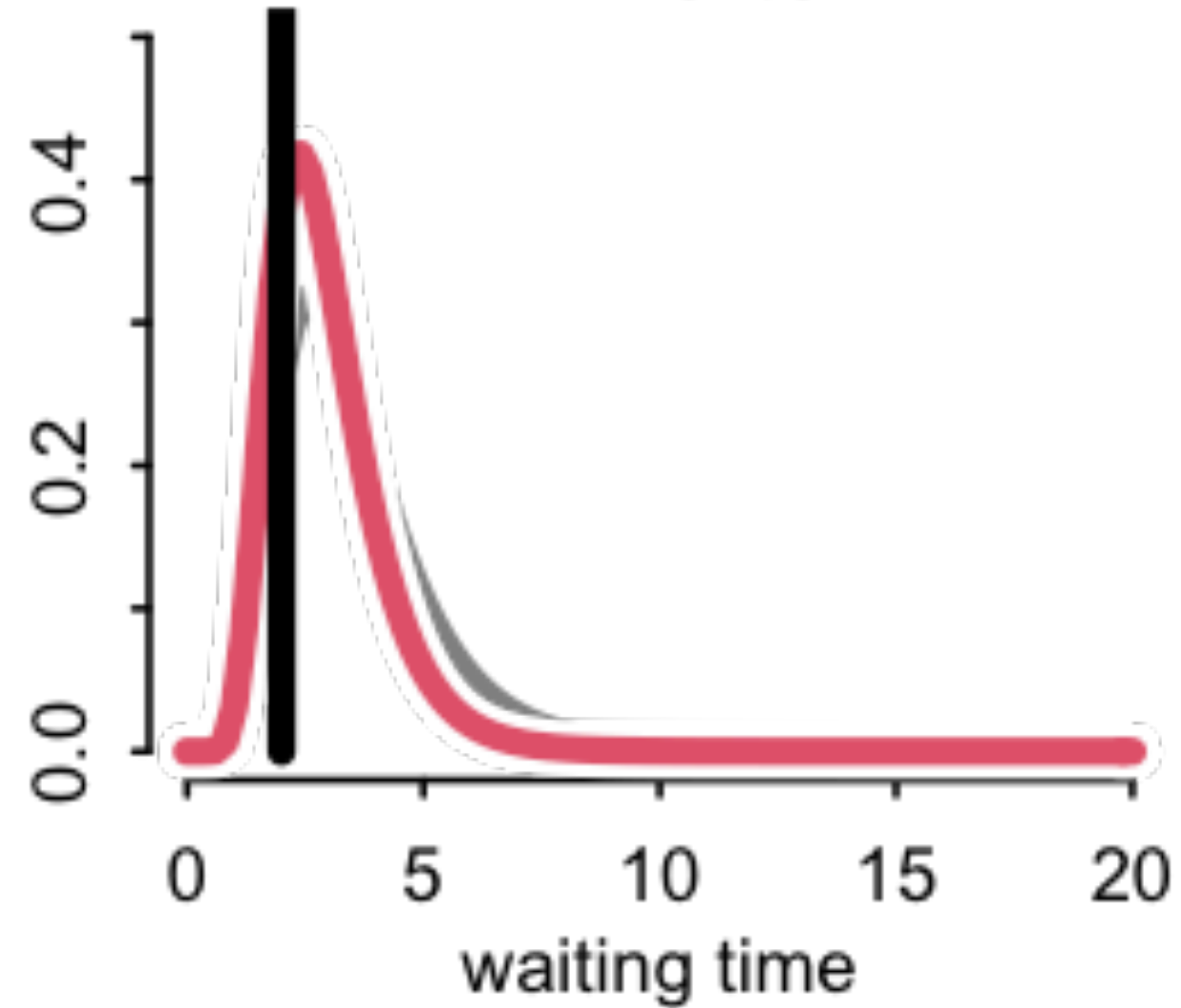
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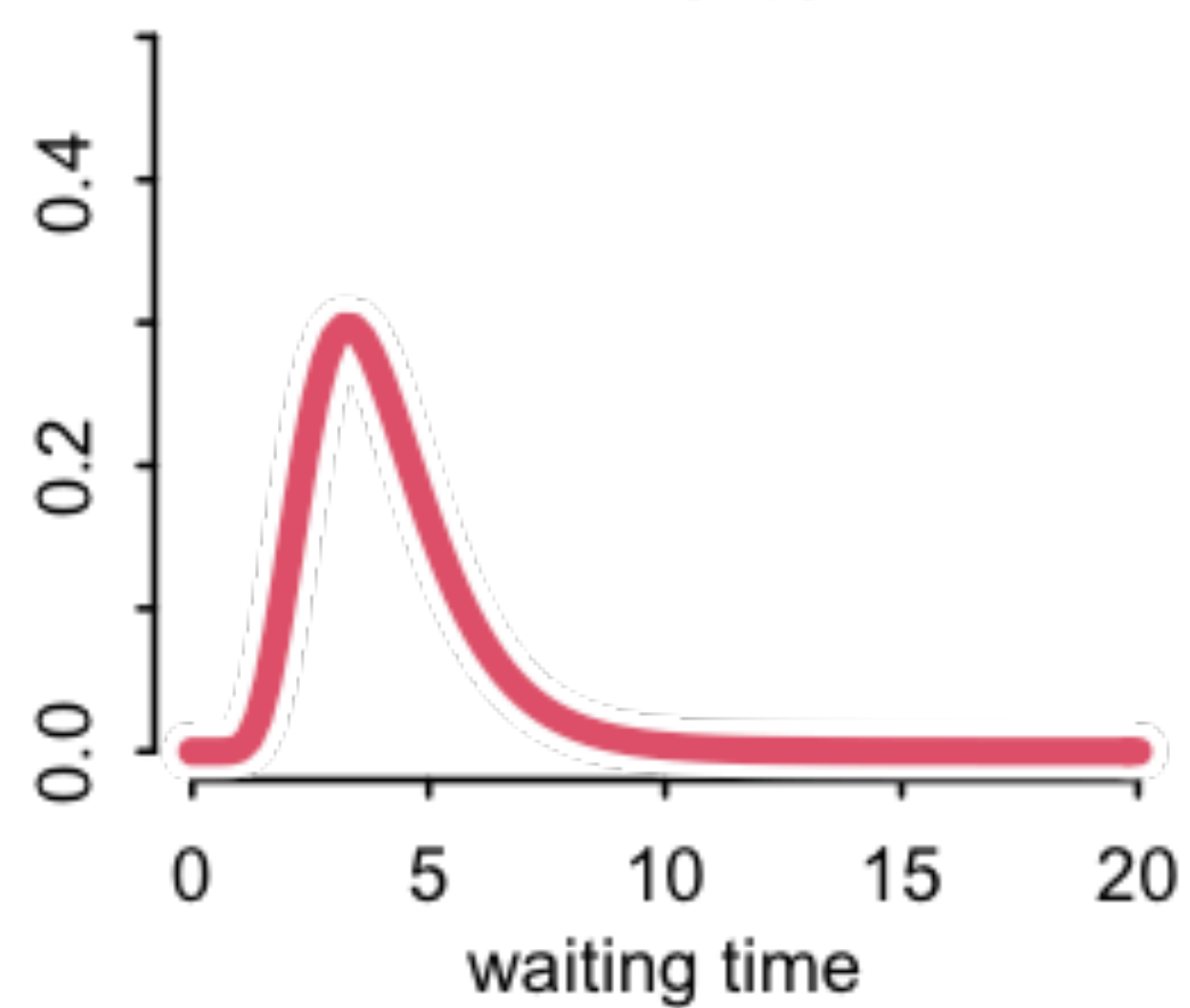
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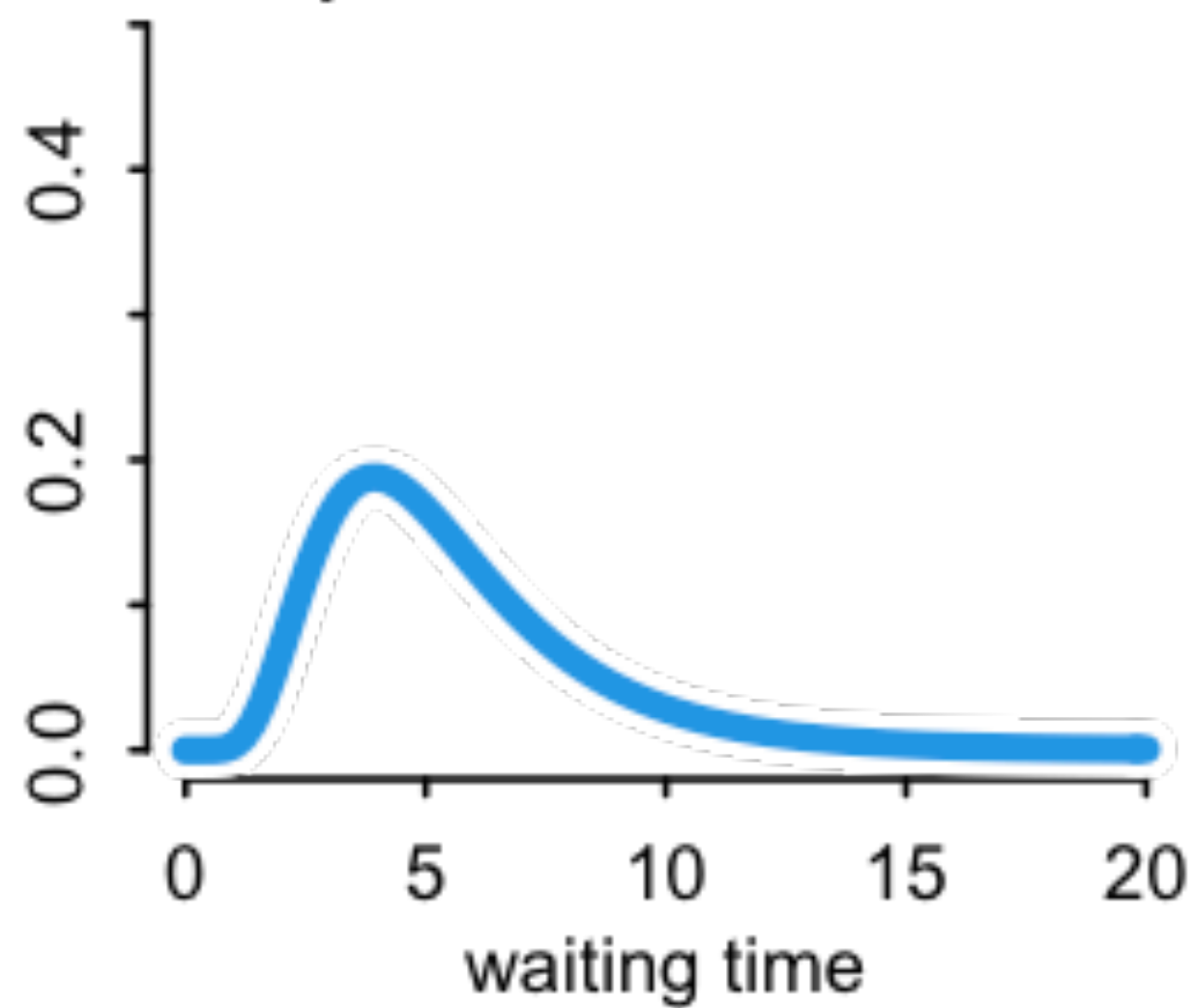
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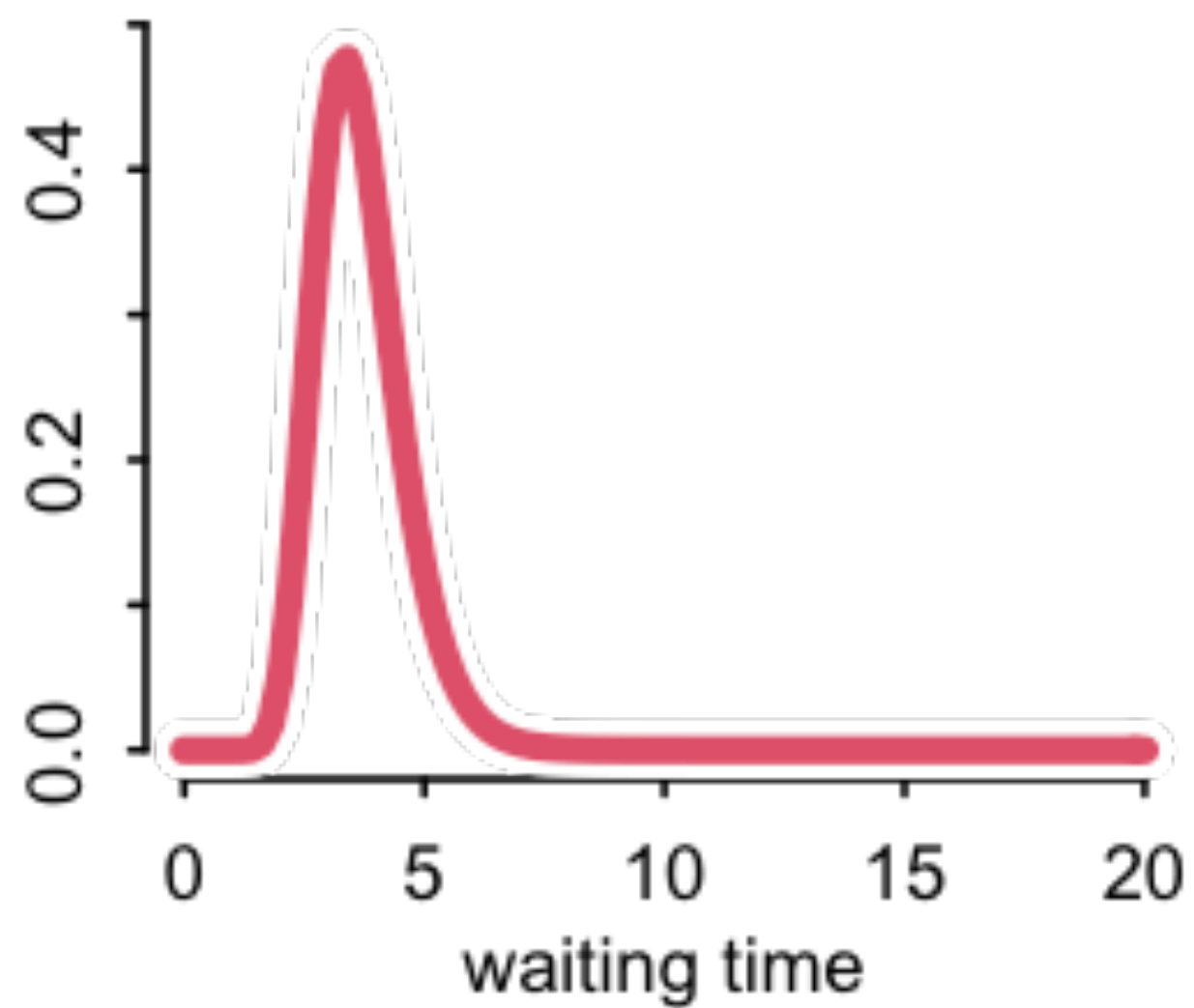
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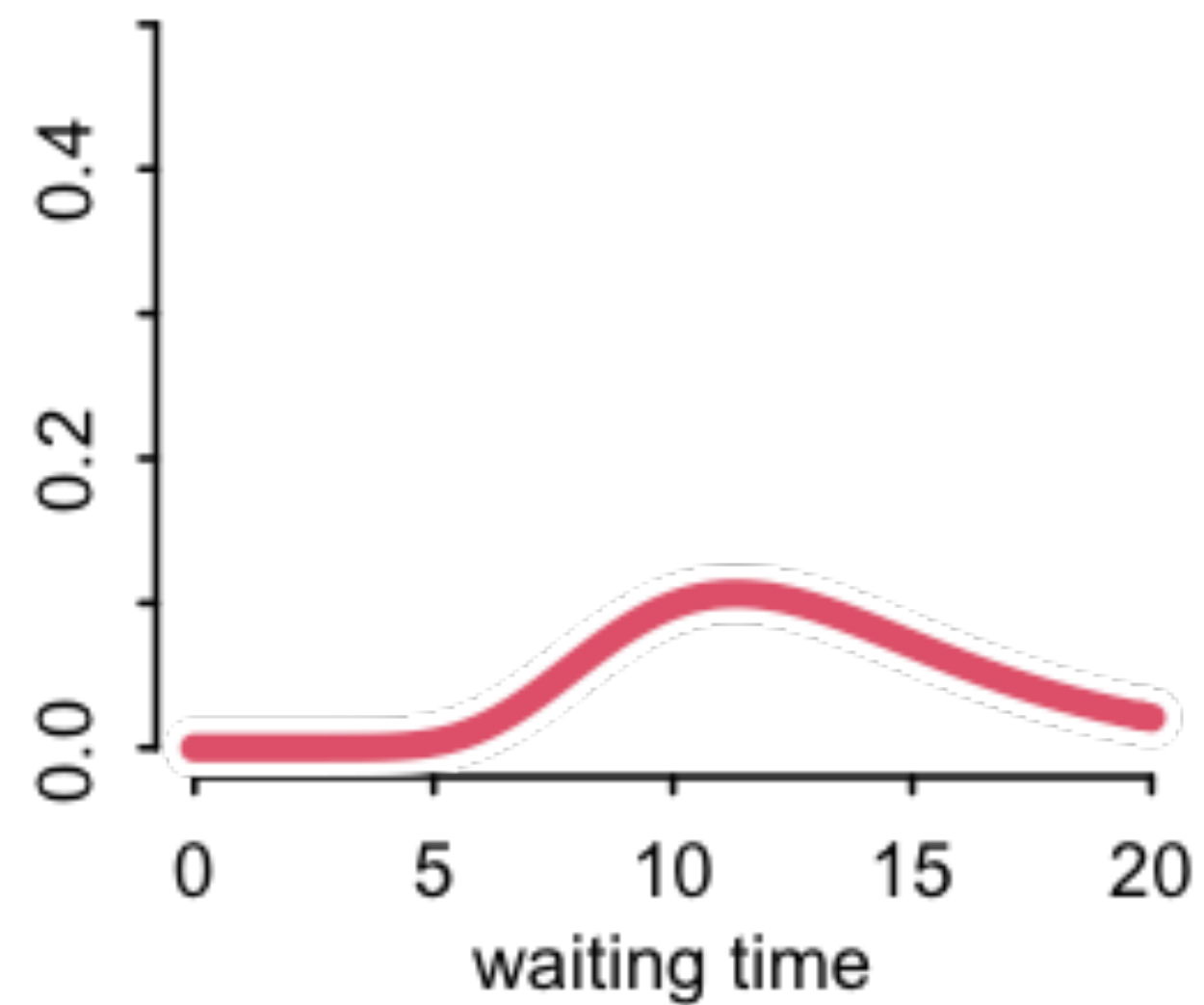
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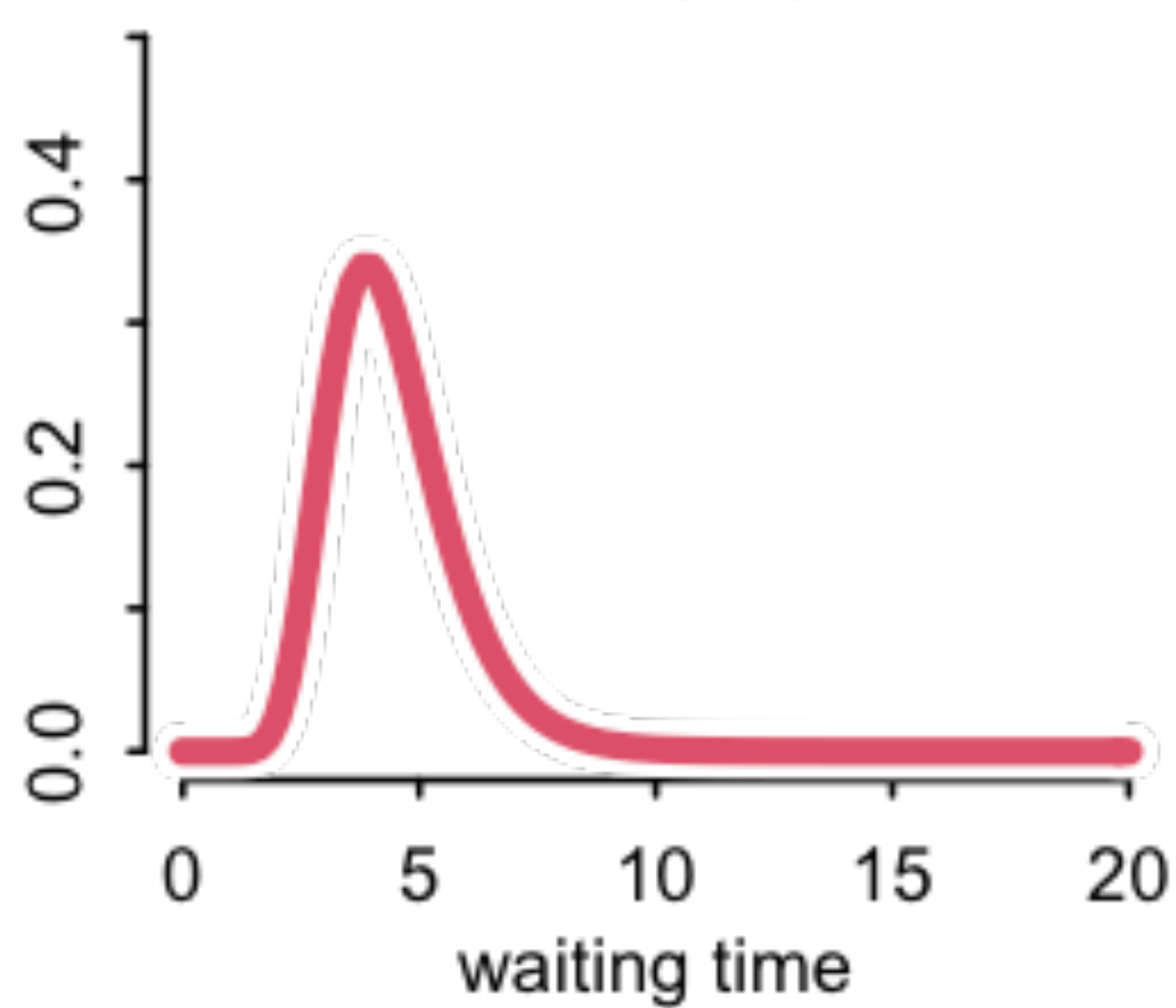
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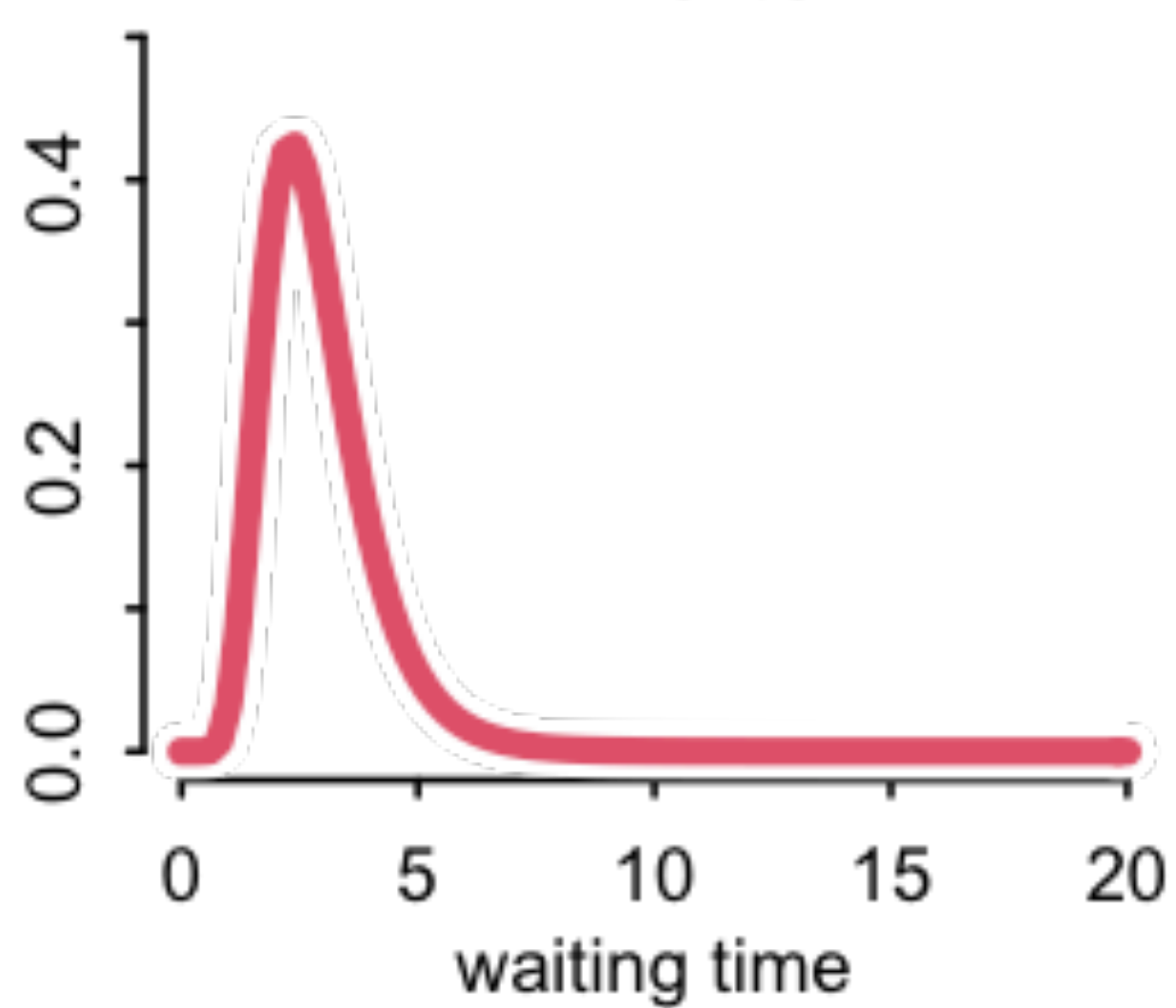
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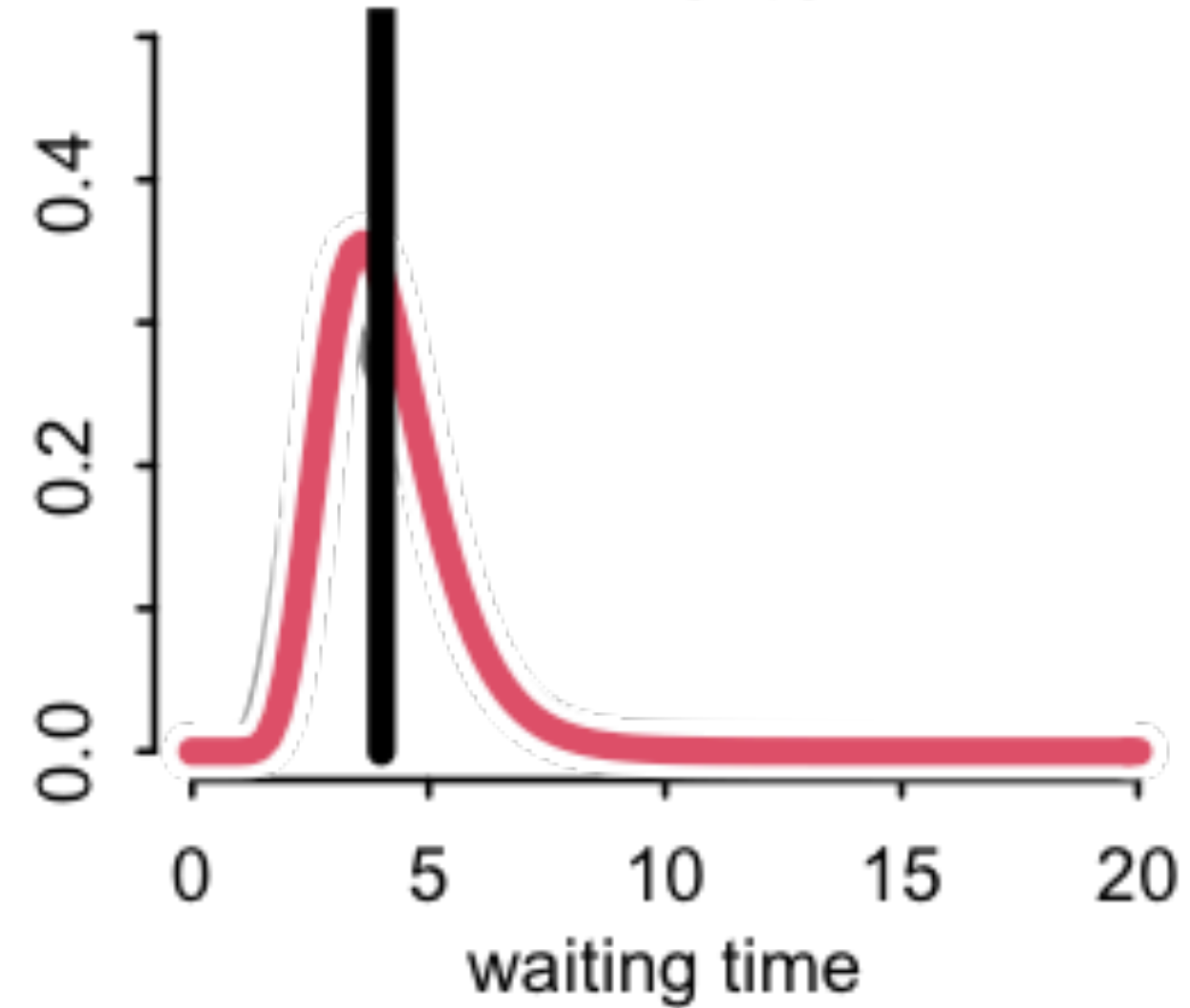
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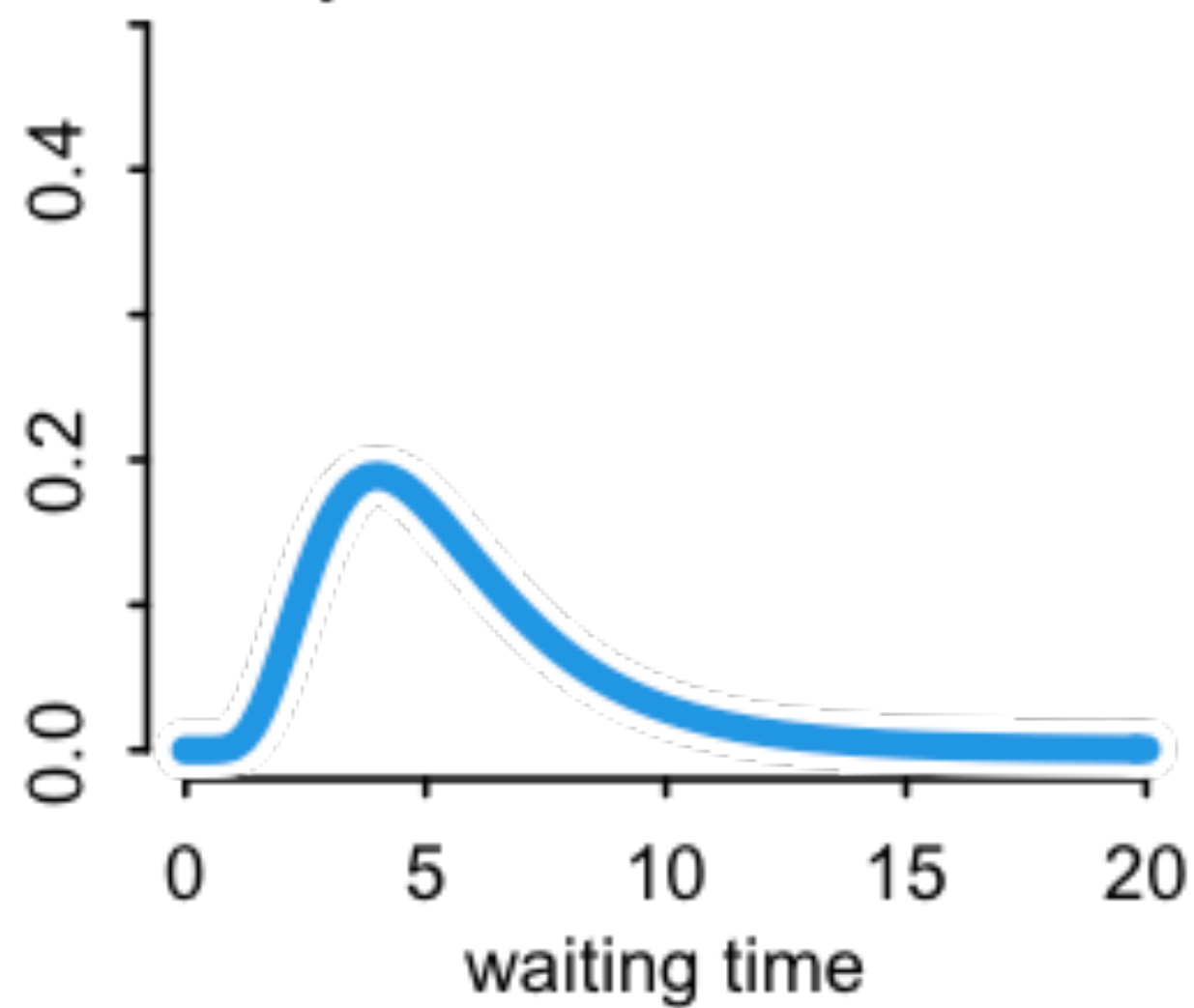
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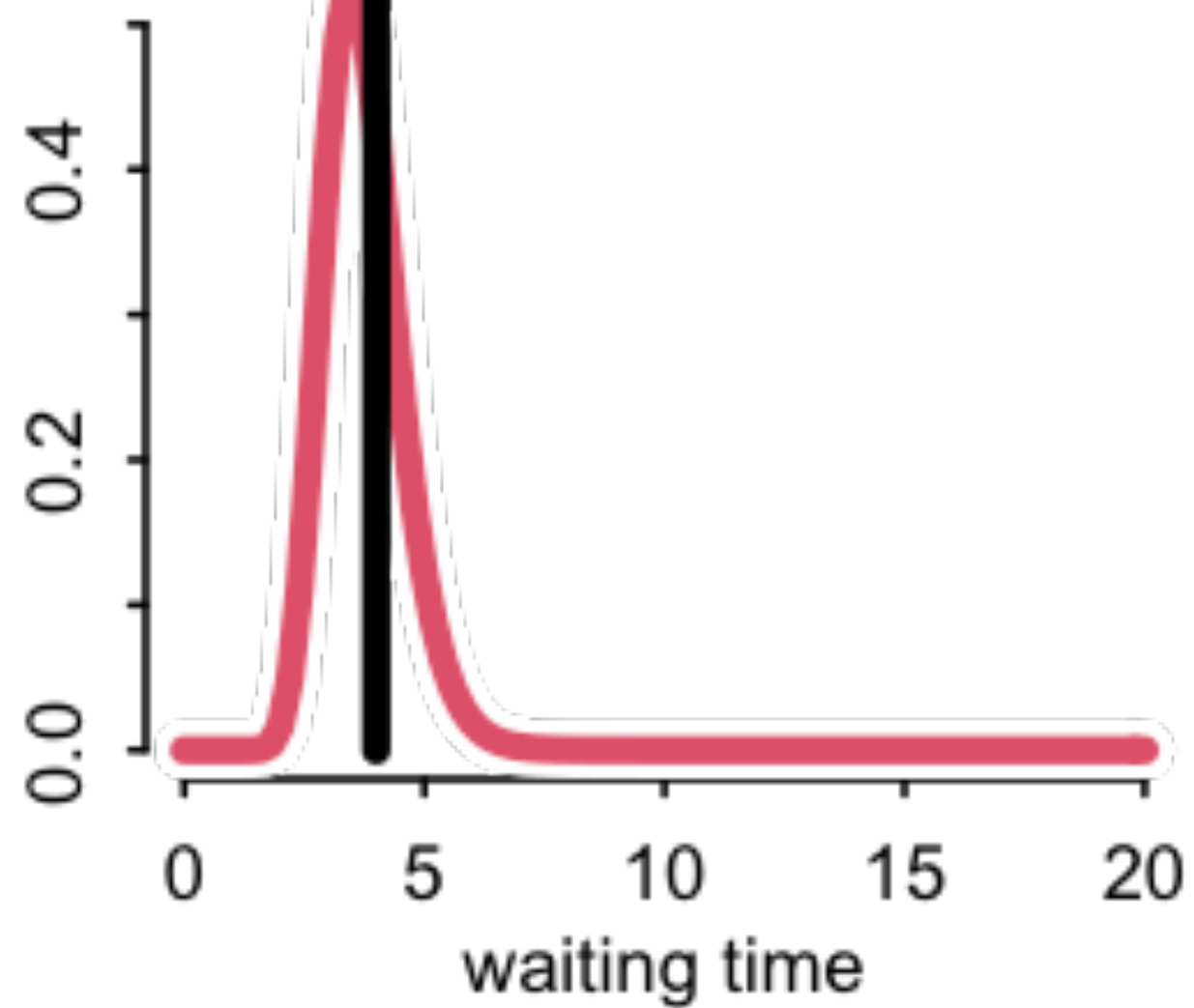
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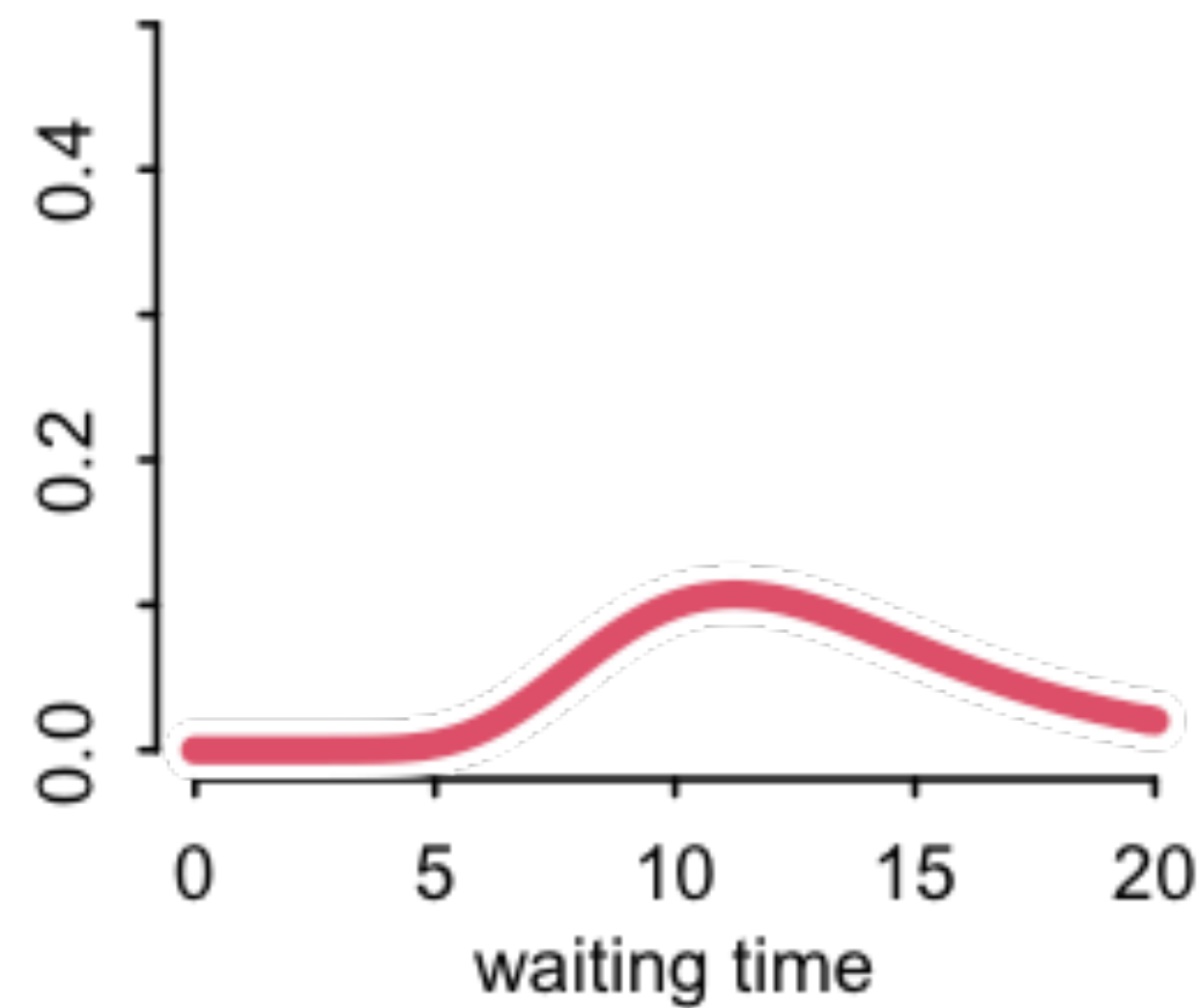
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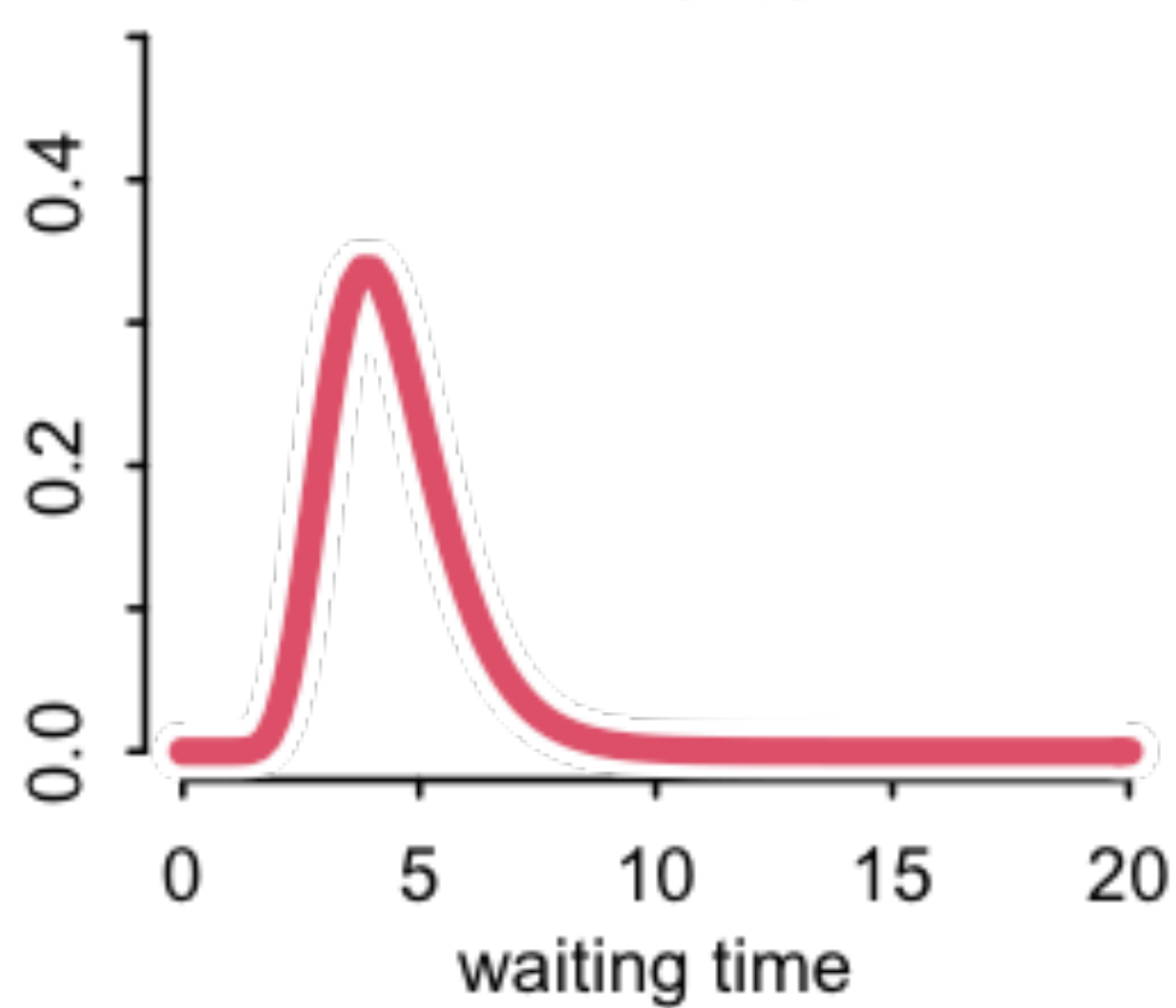
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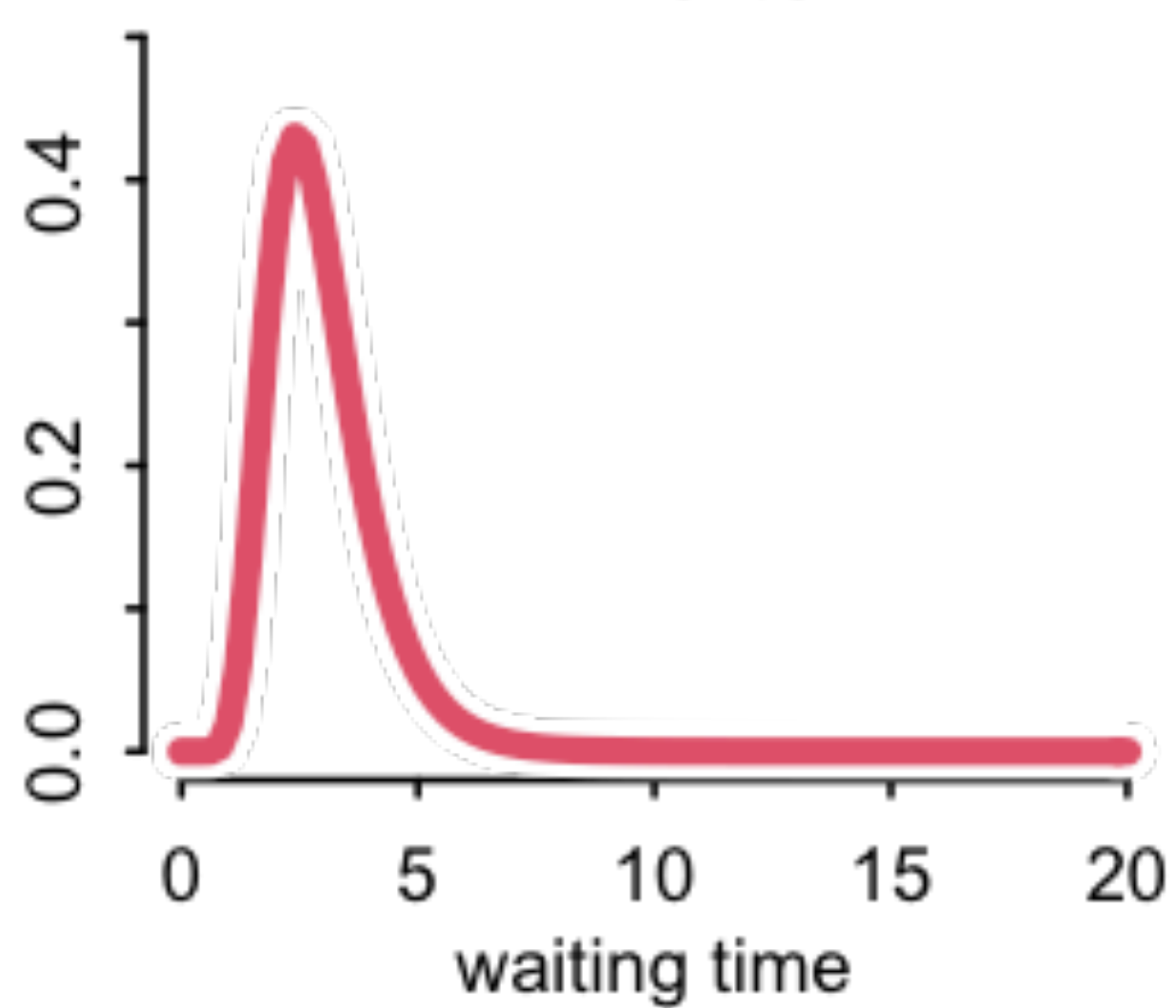
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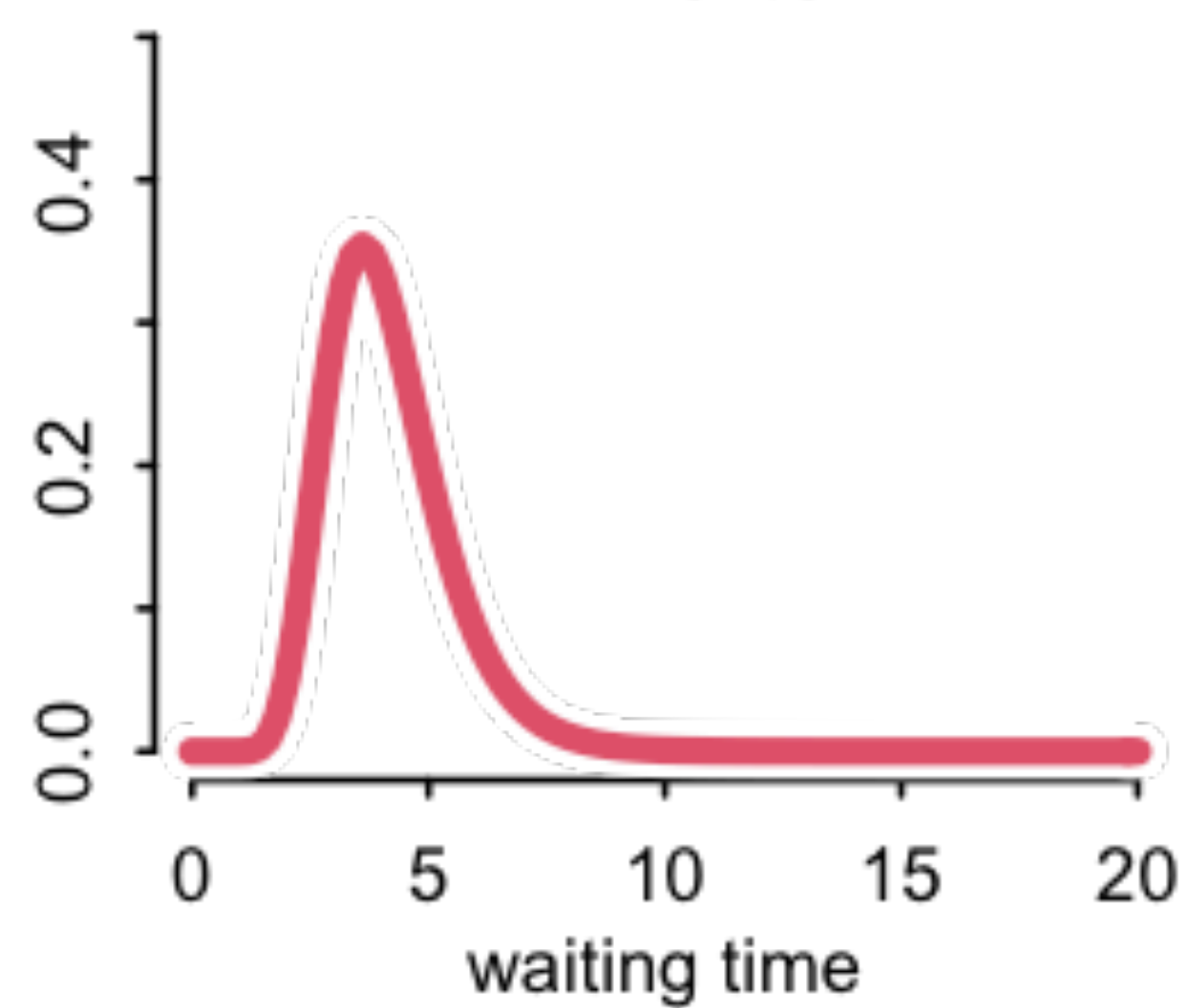
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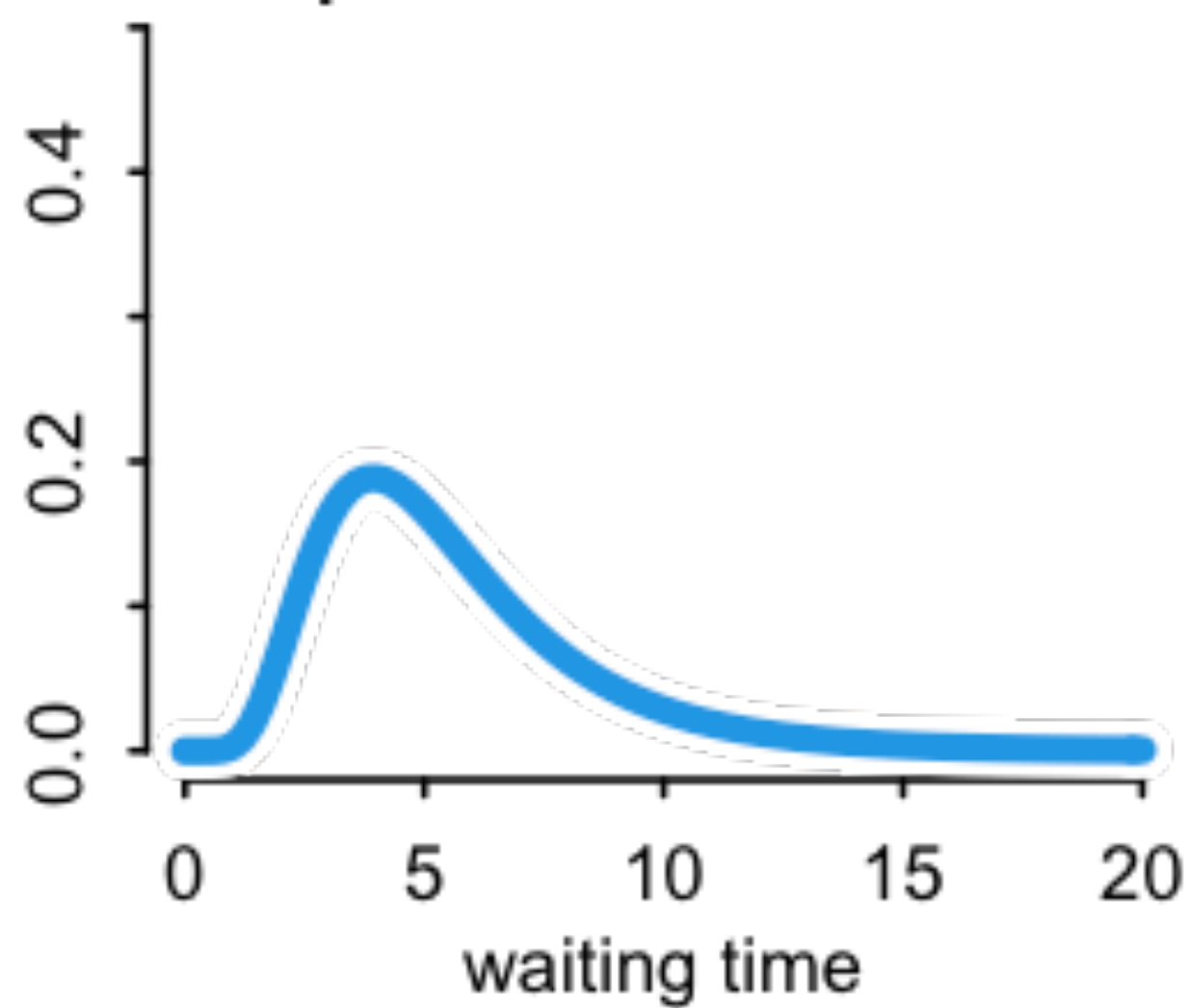
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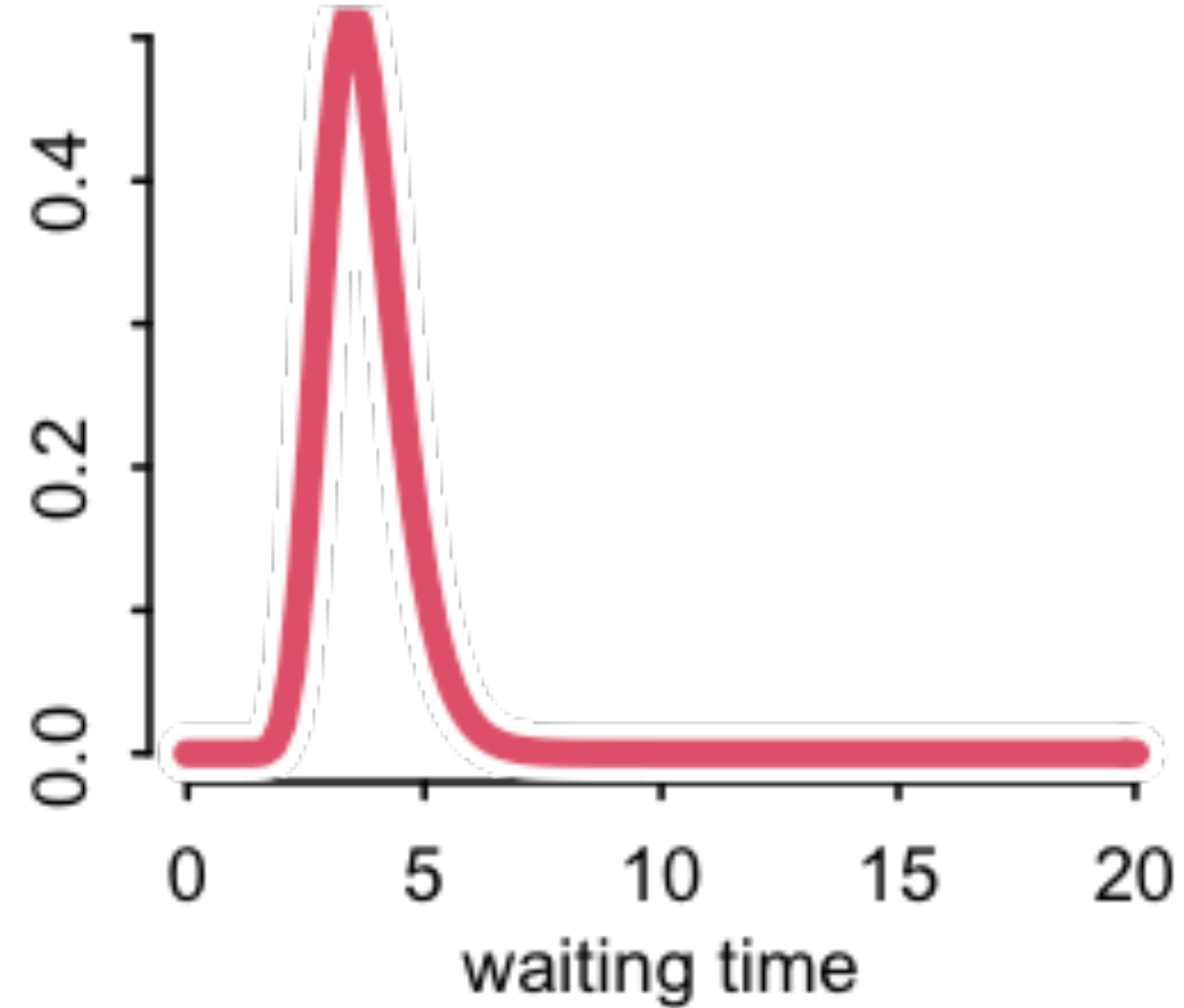
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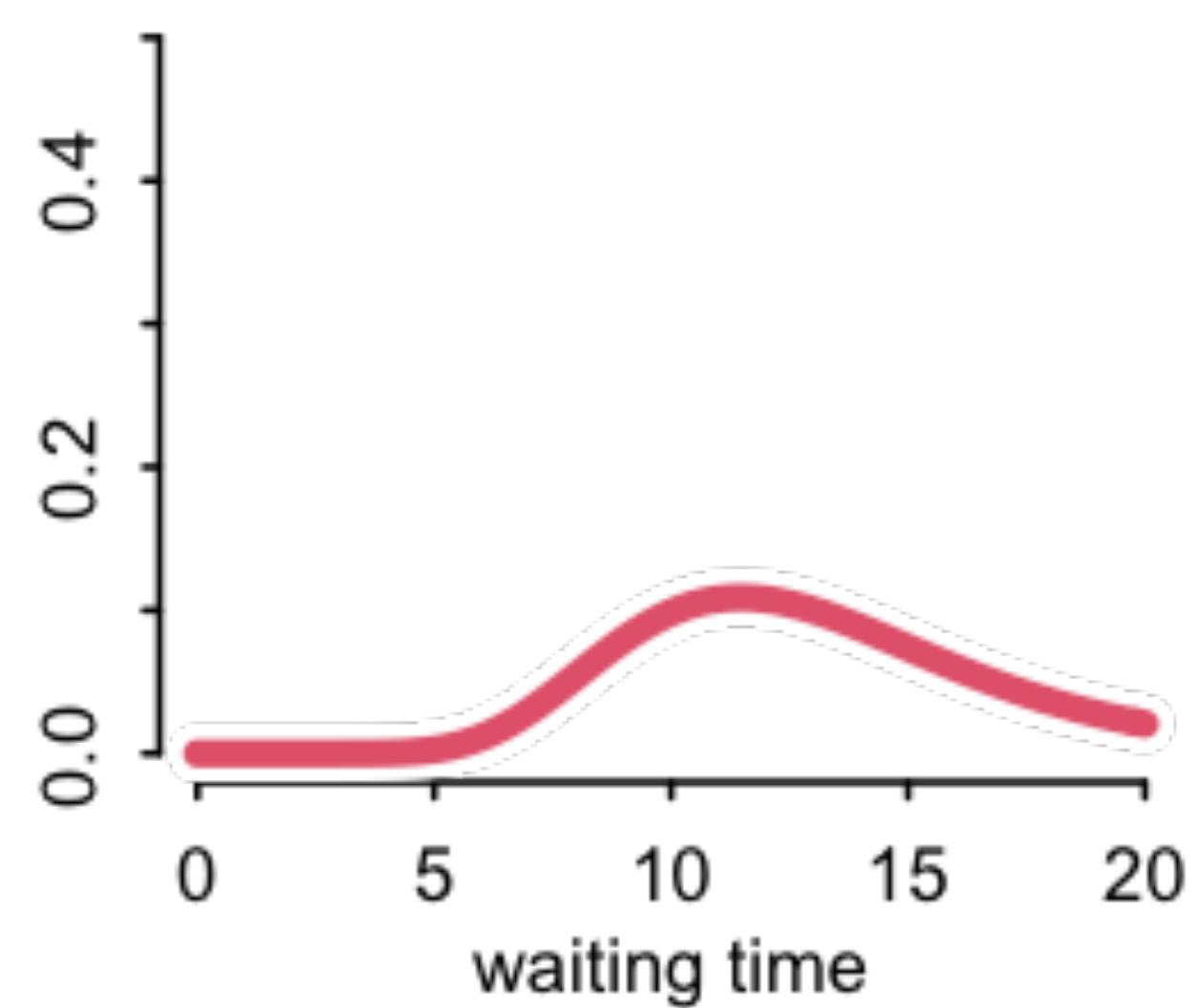
Population of cafes



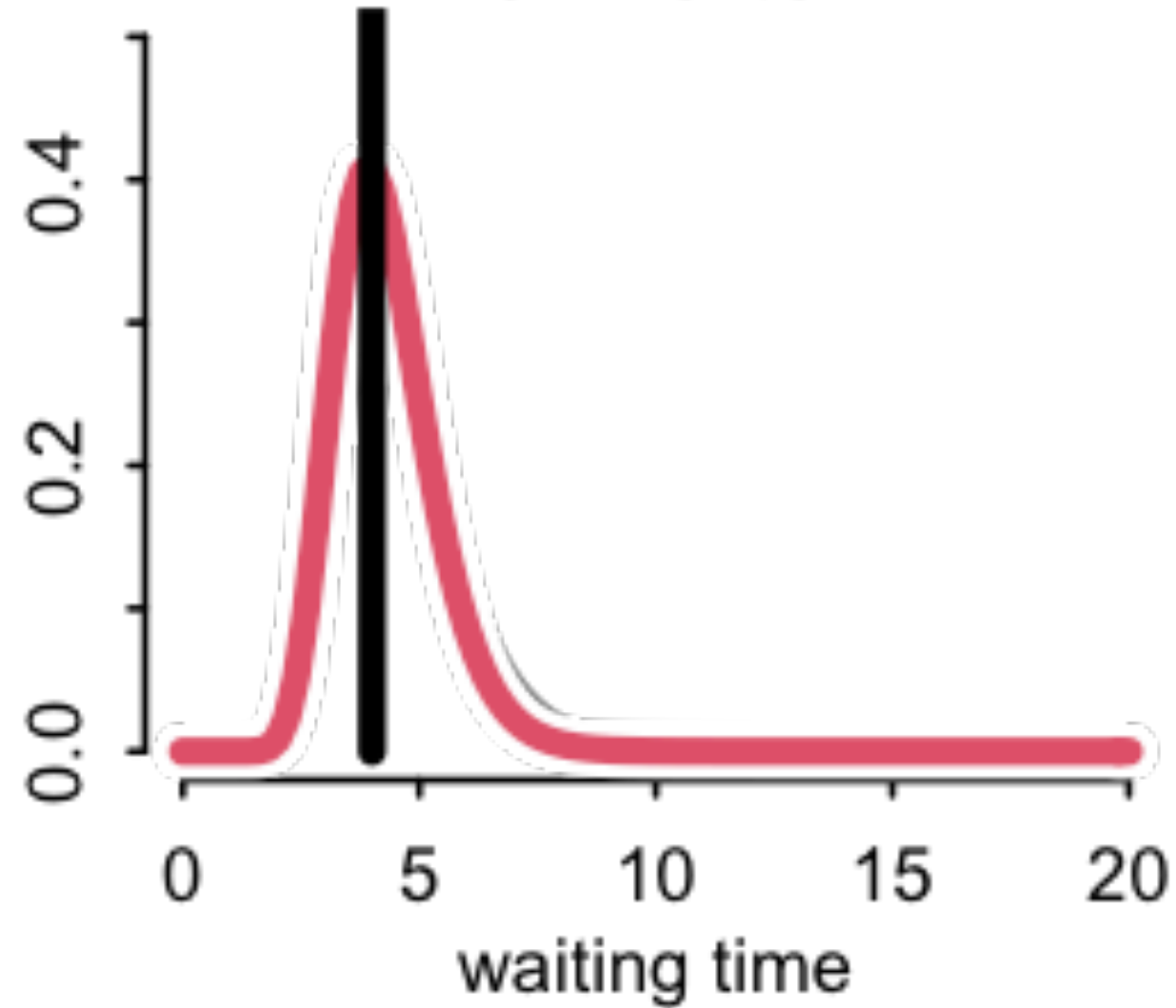
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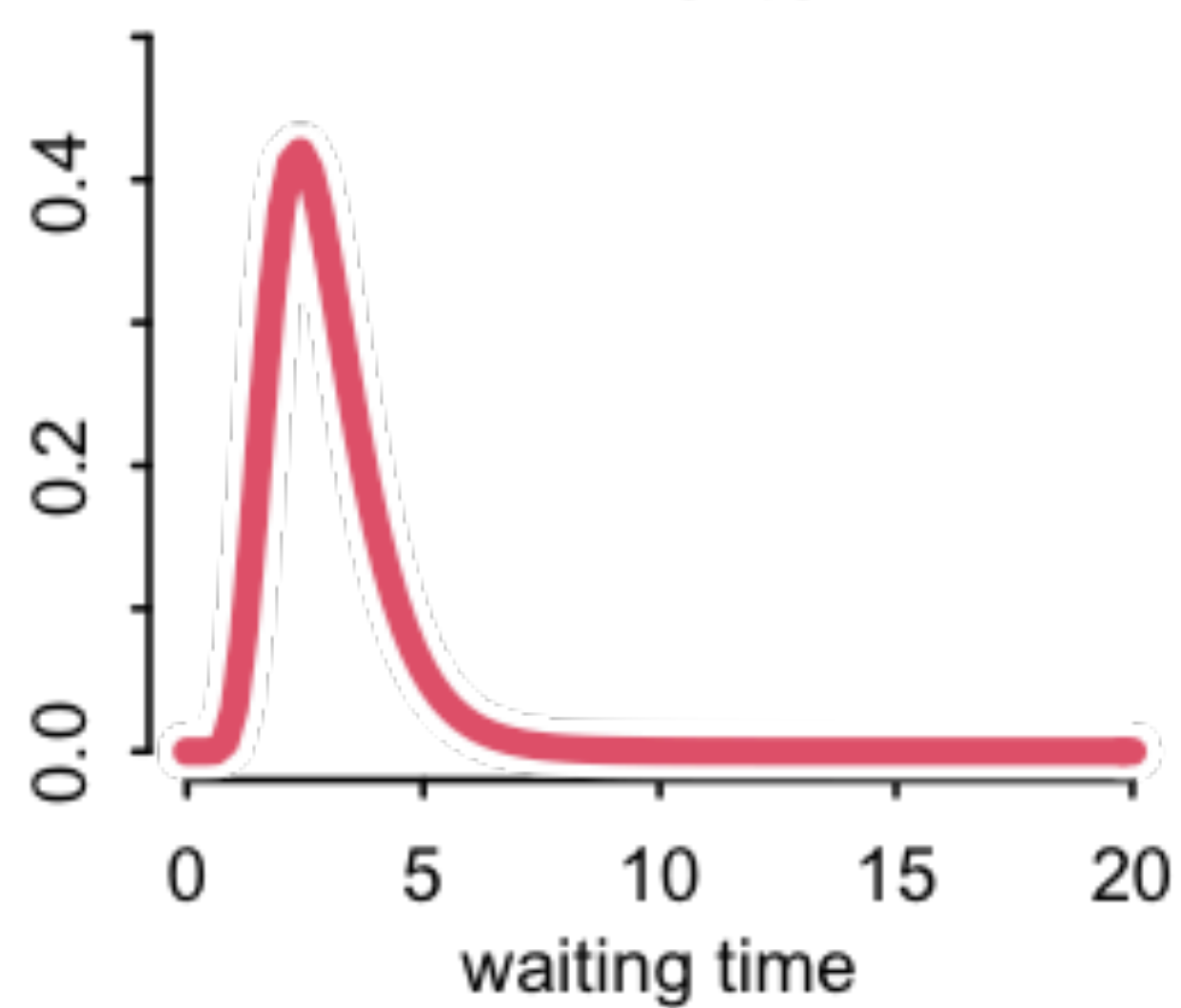
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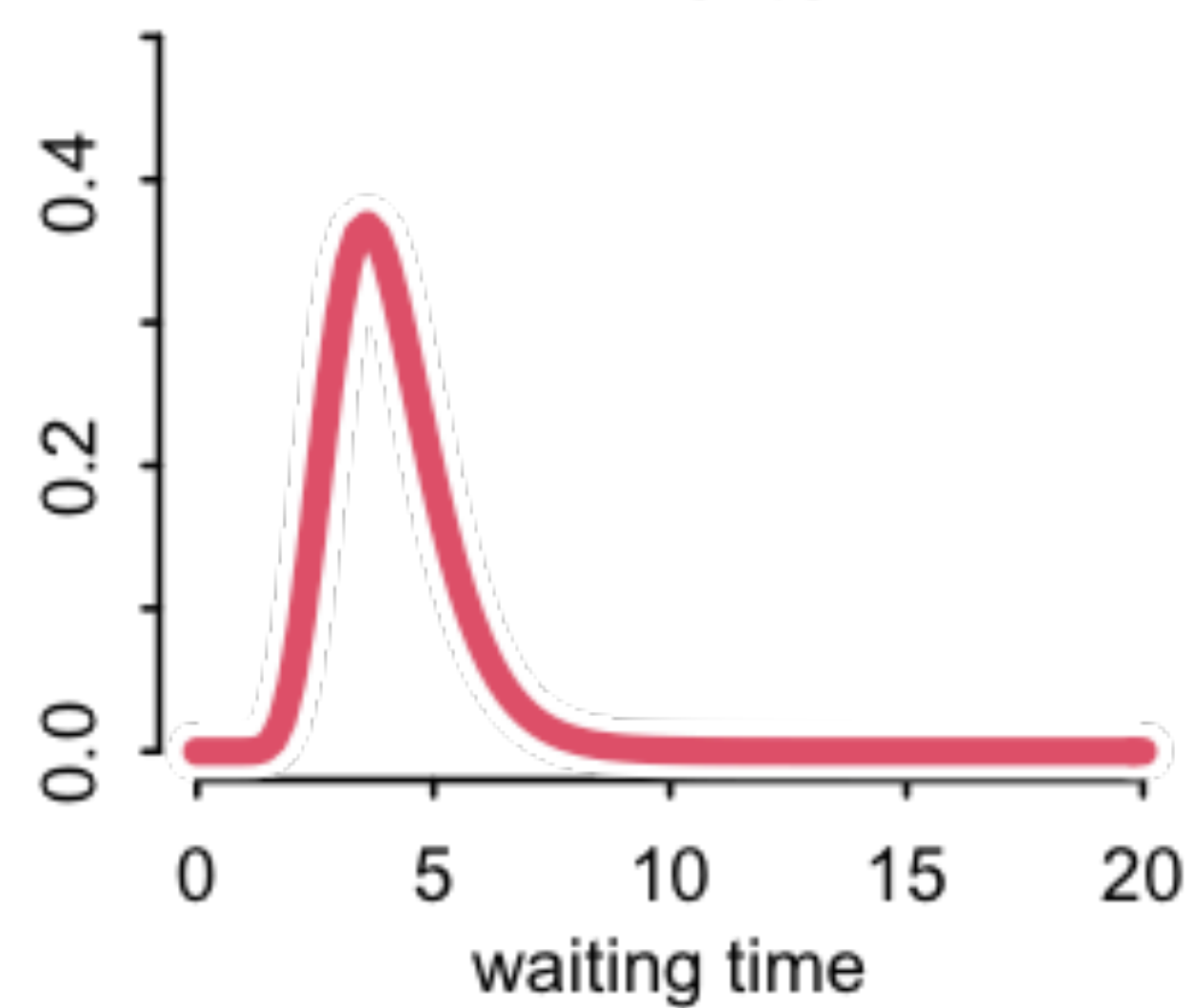
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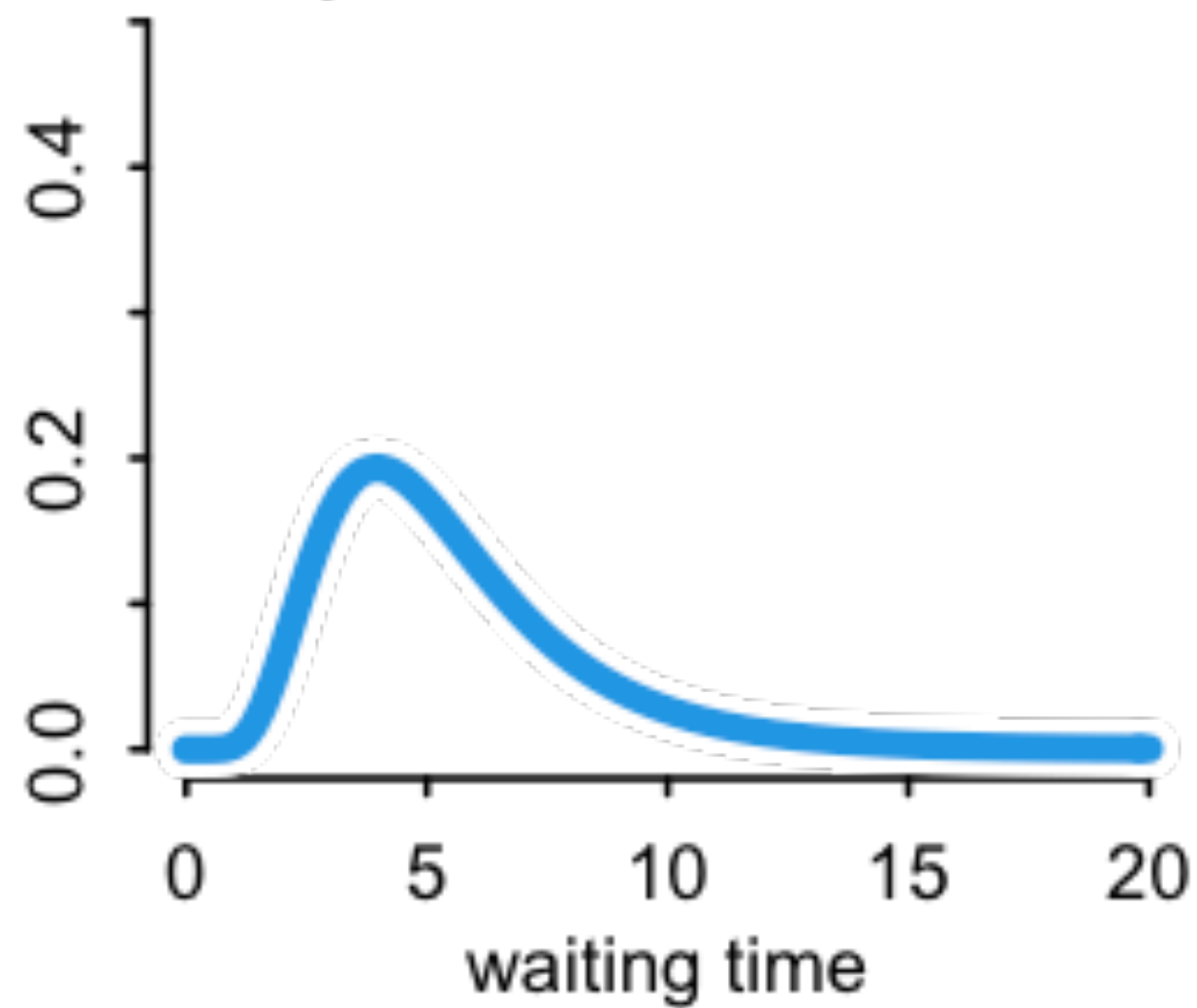
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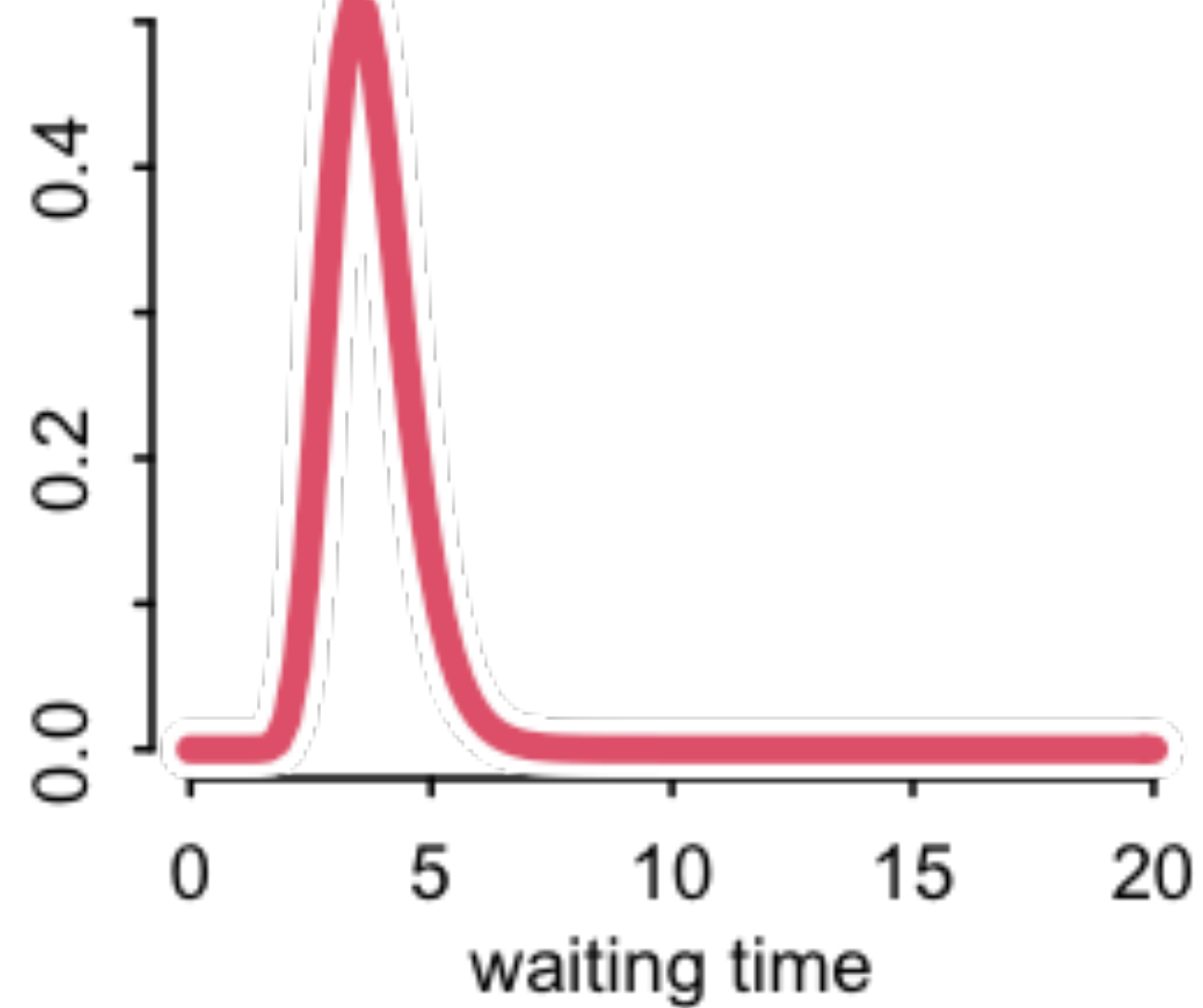
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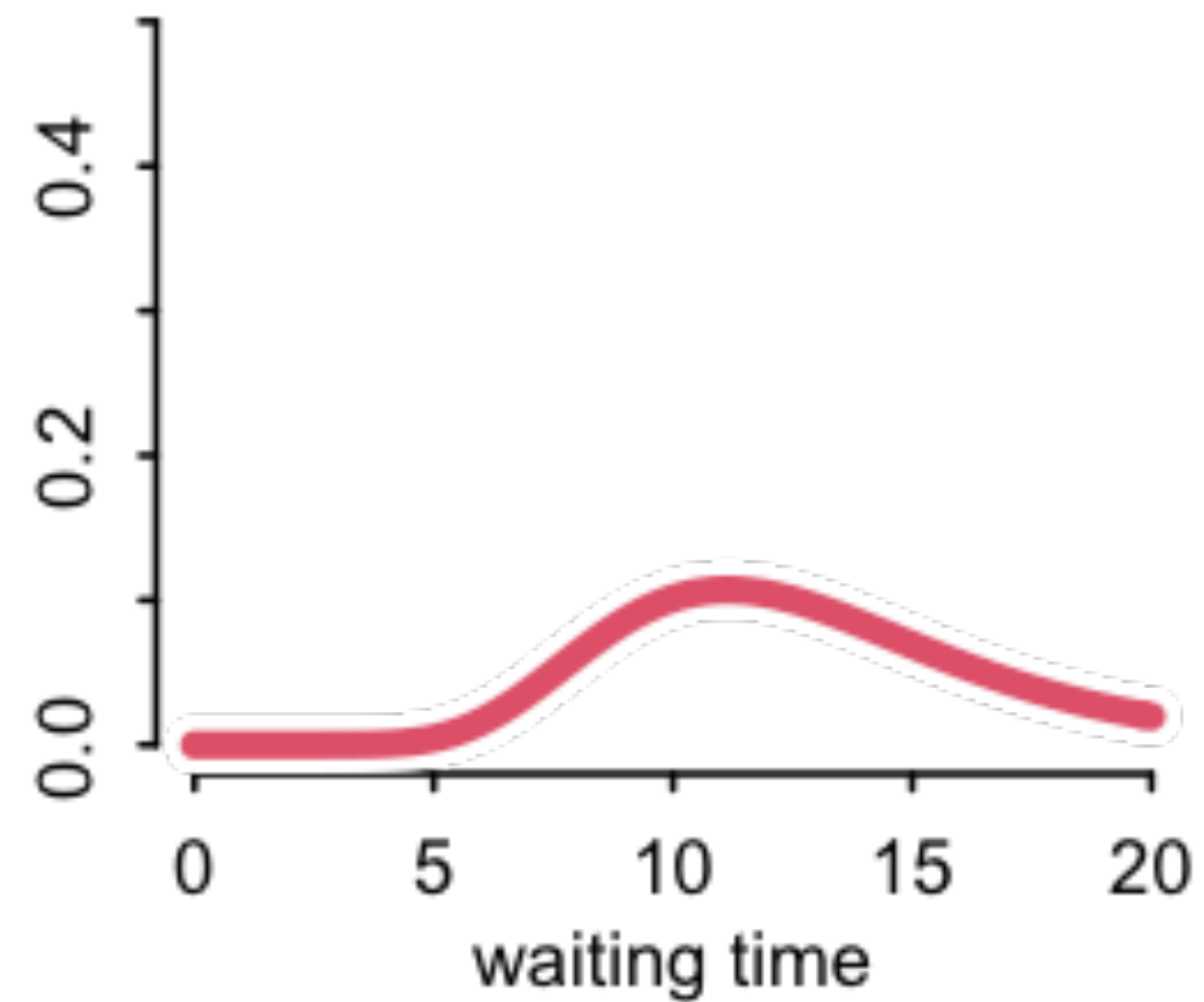
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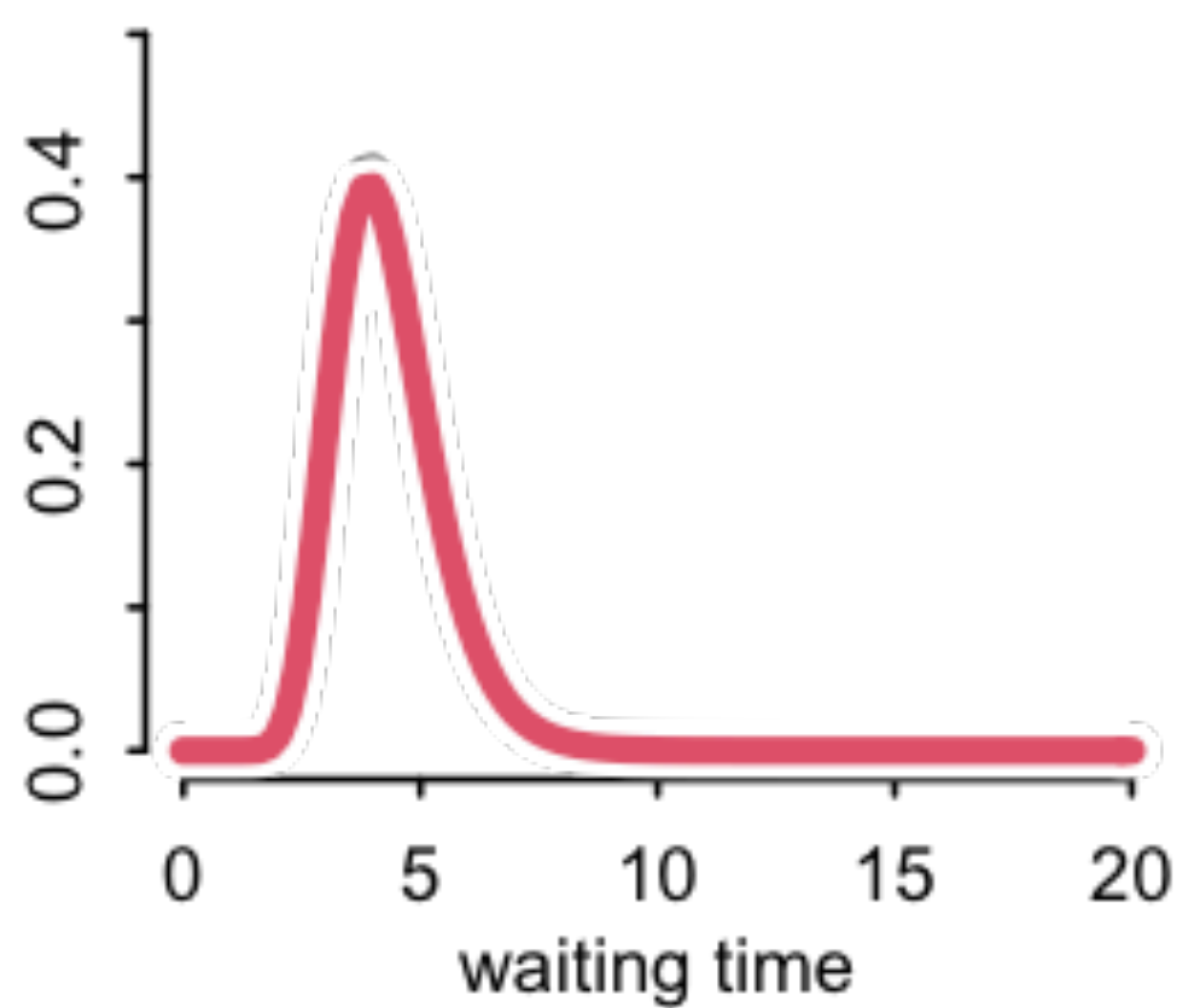
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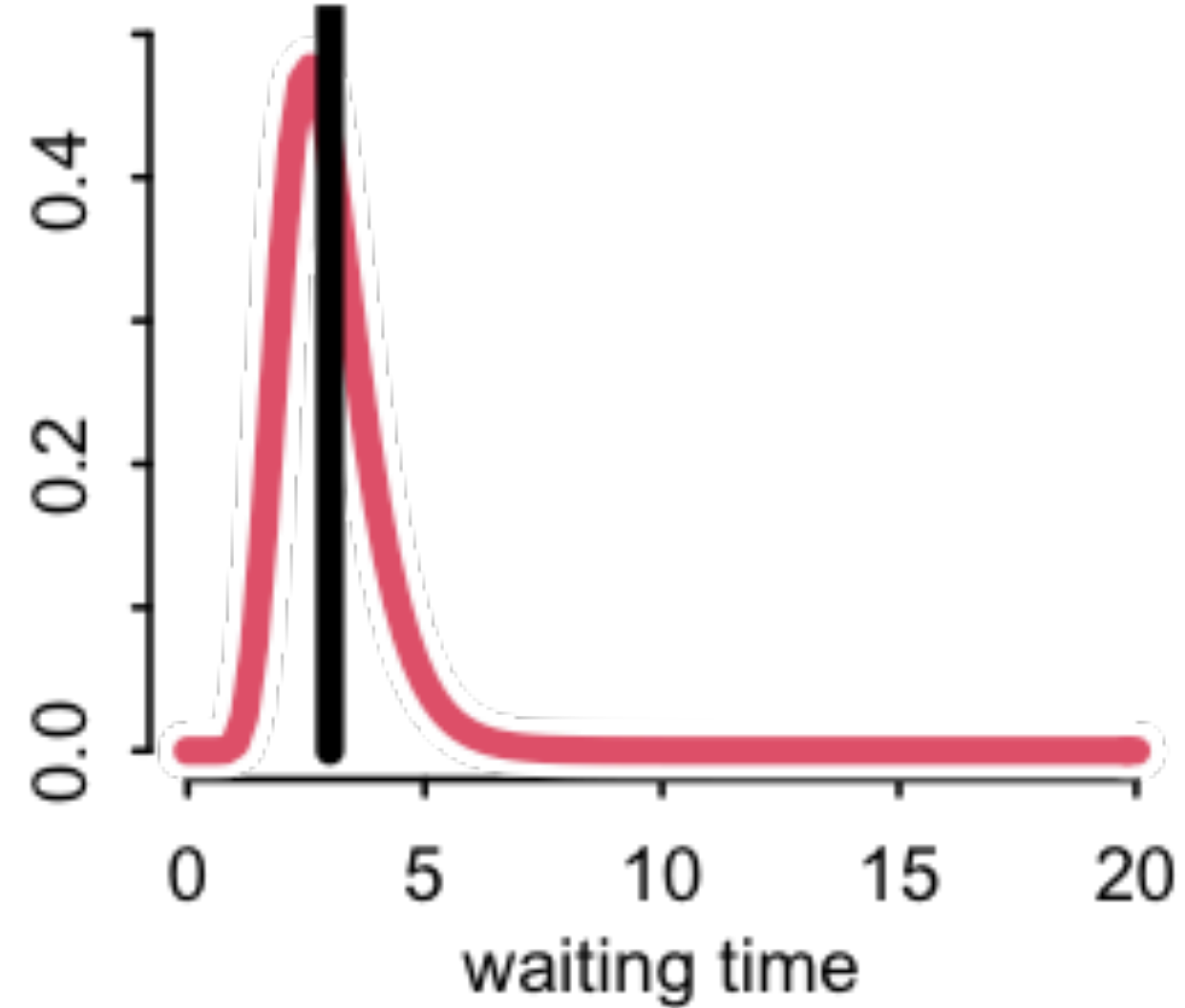
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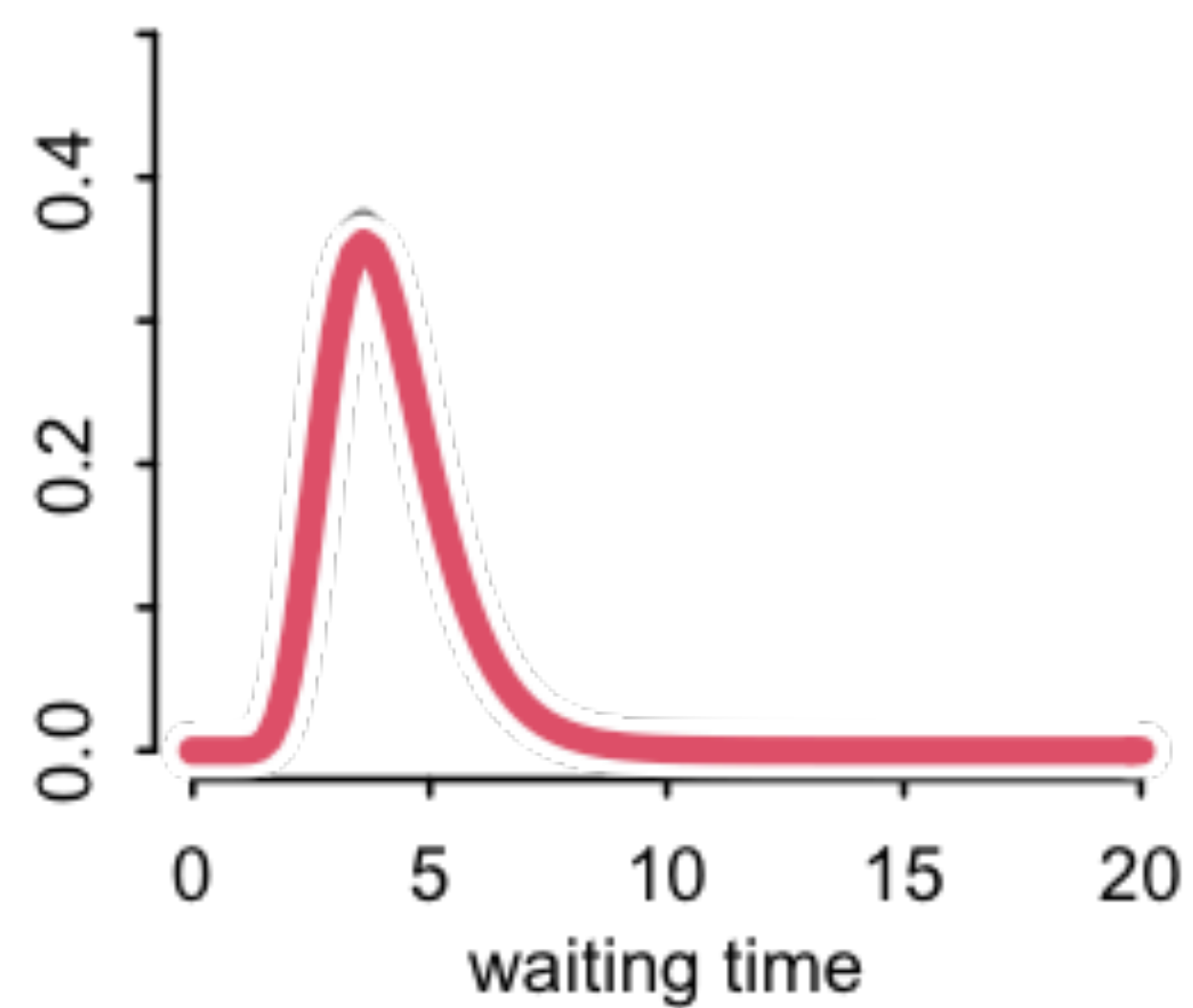
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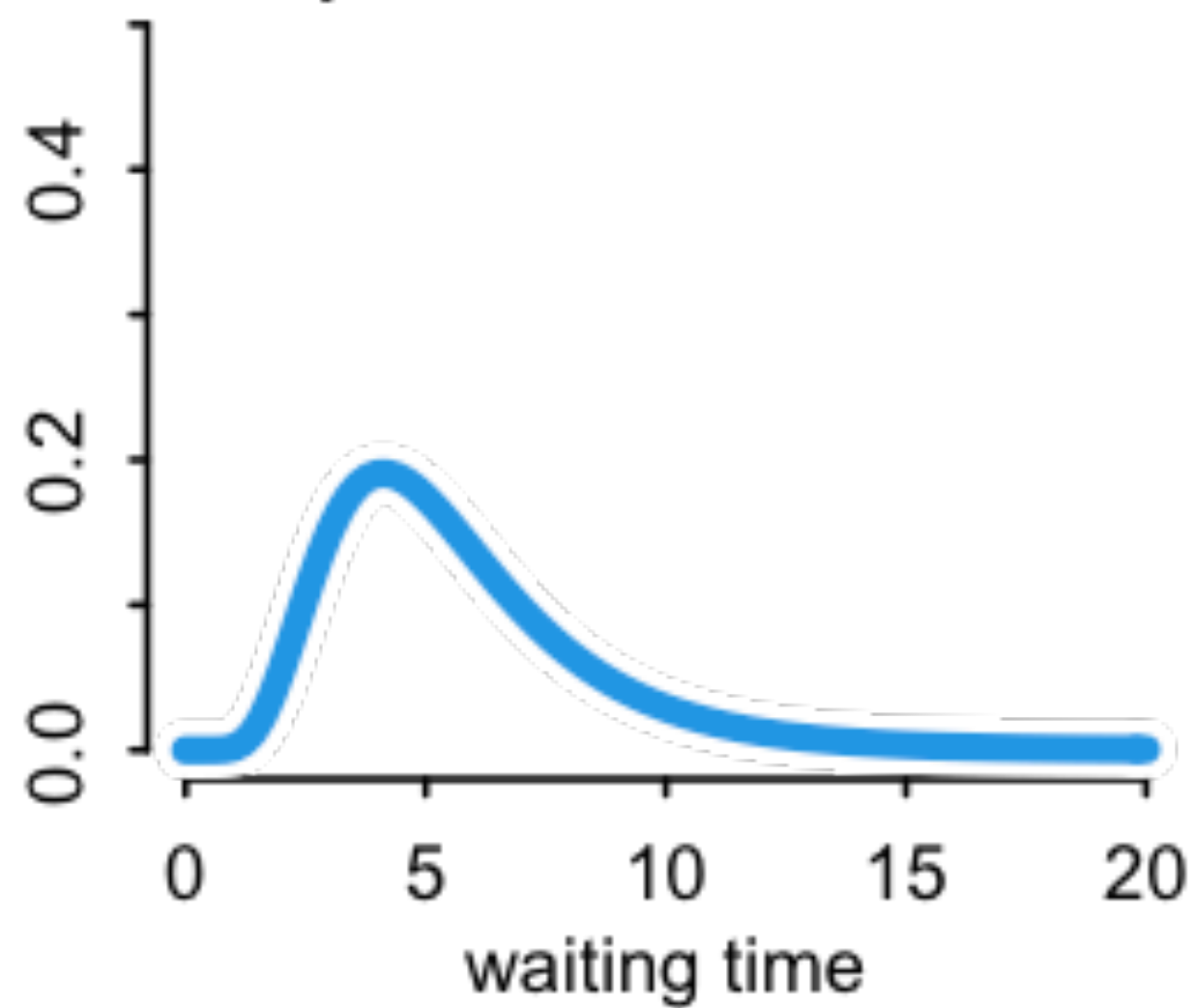
3 visits



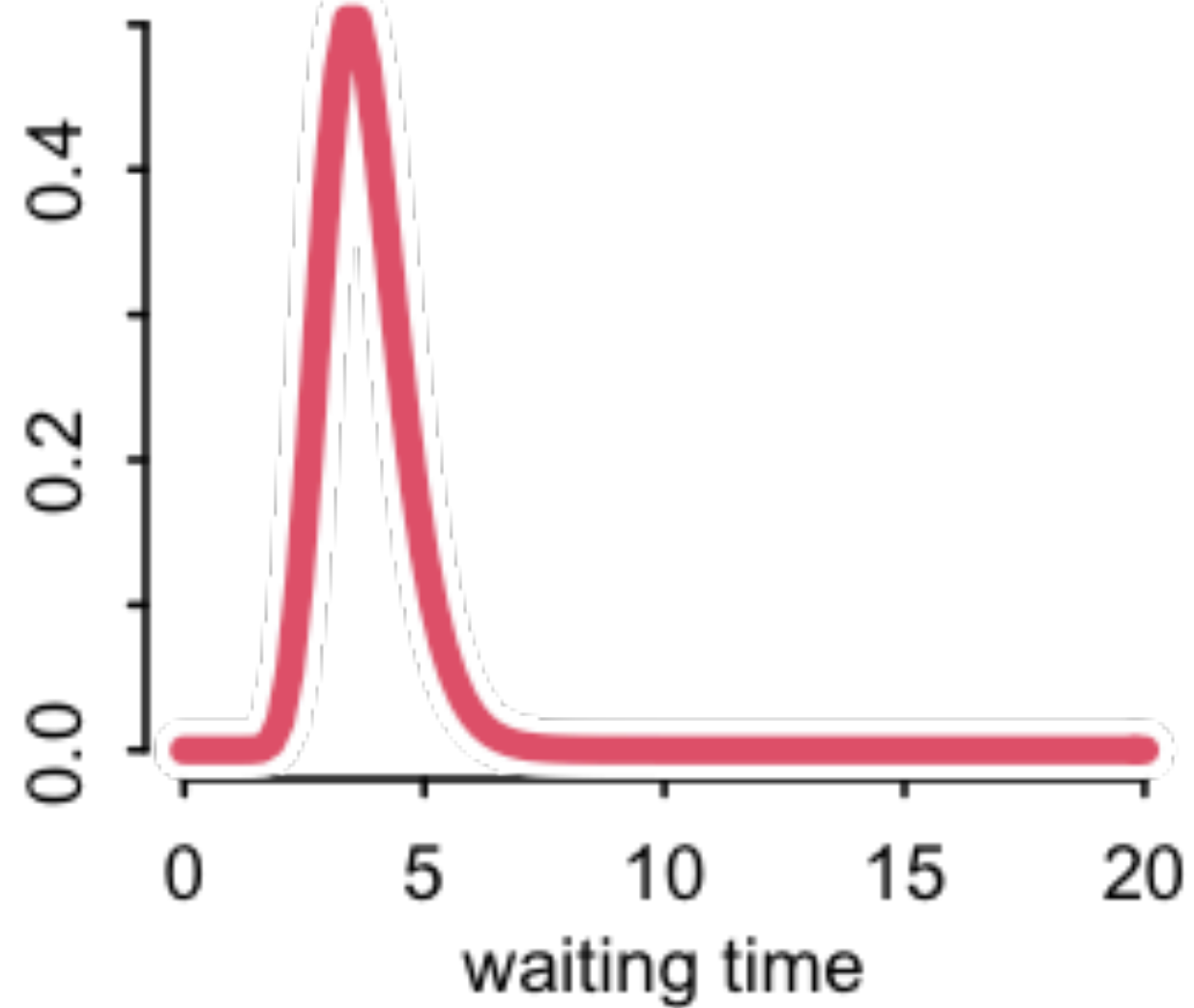
2 visits



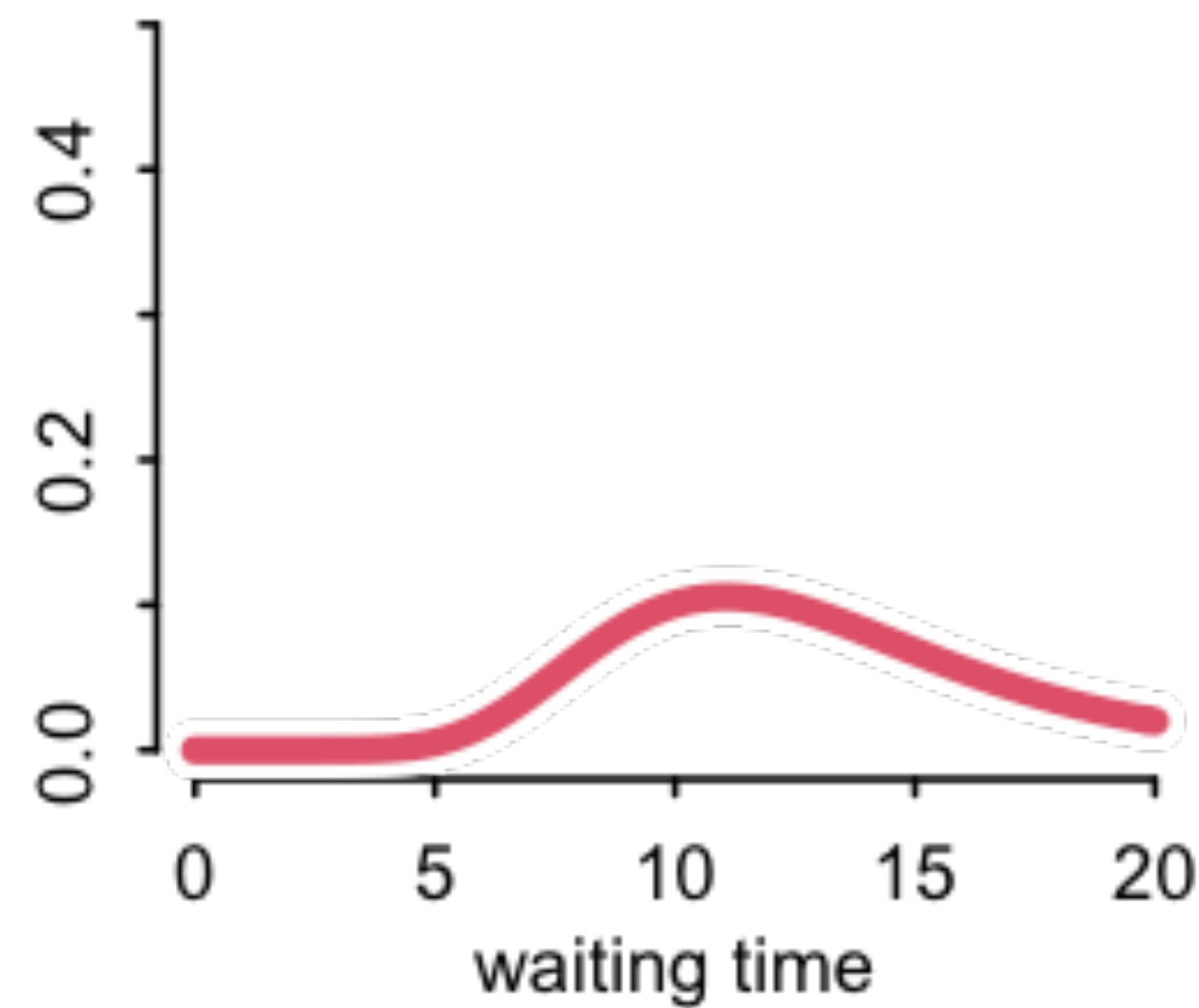
Population of cafes



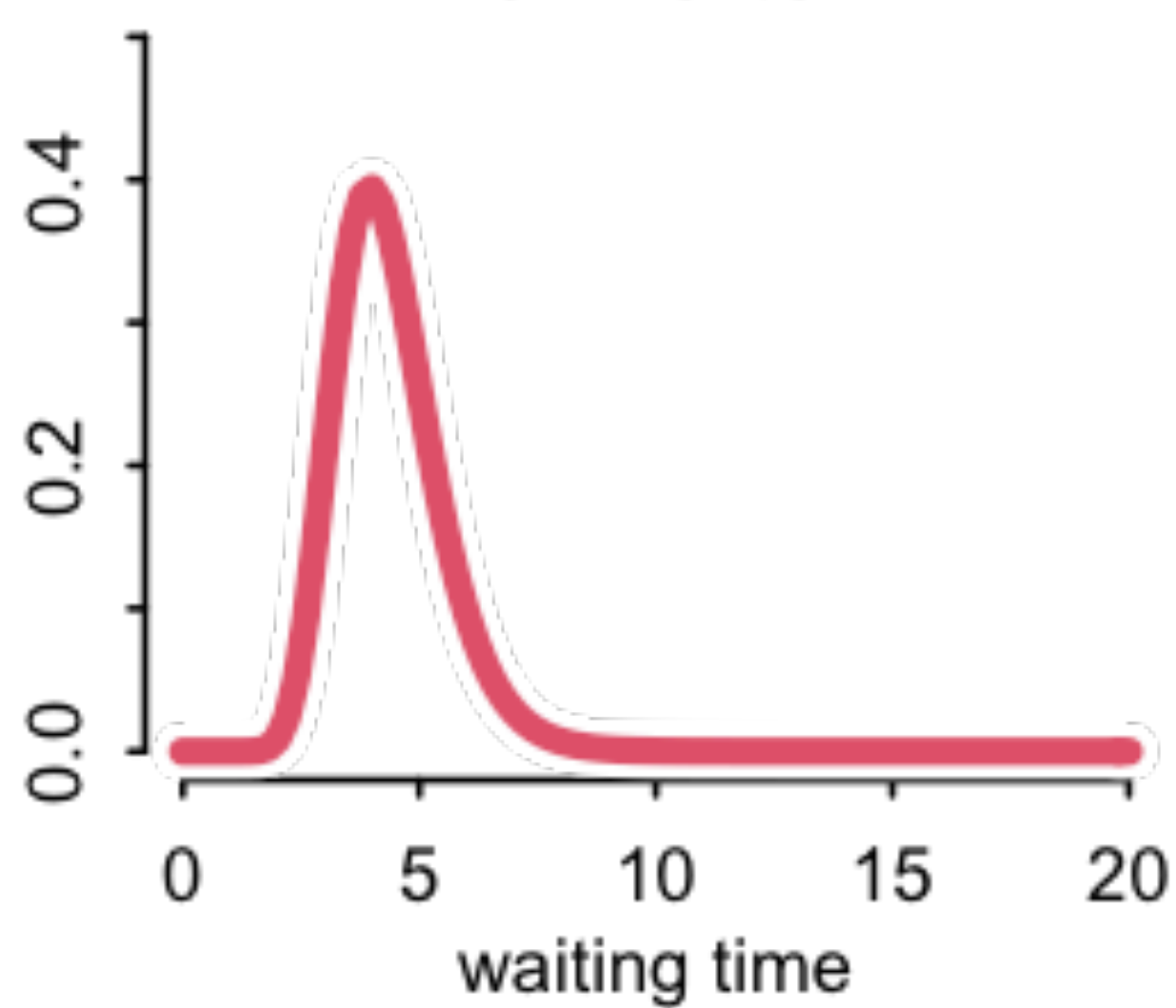
5 visits



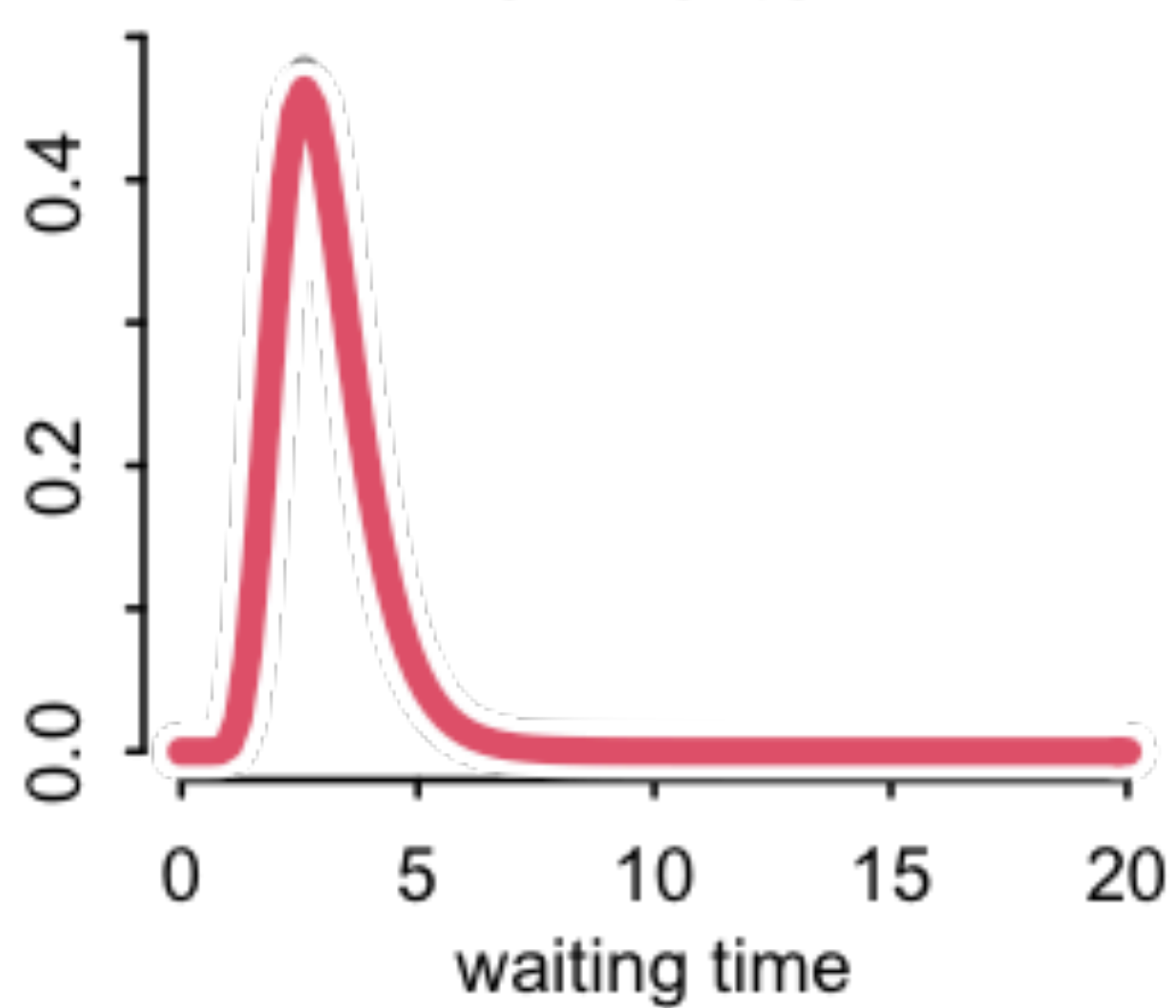
1 visits



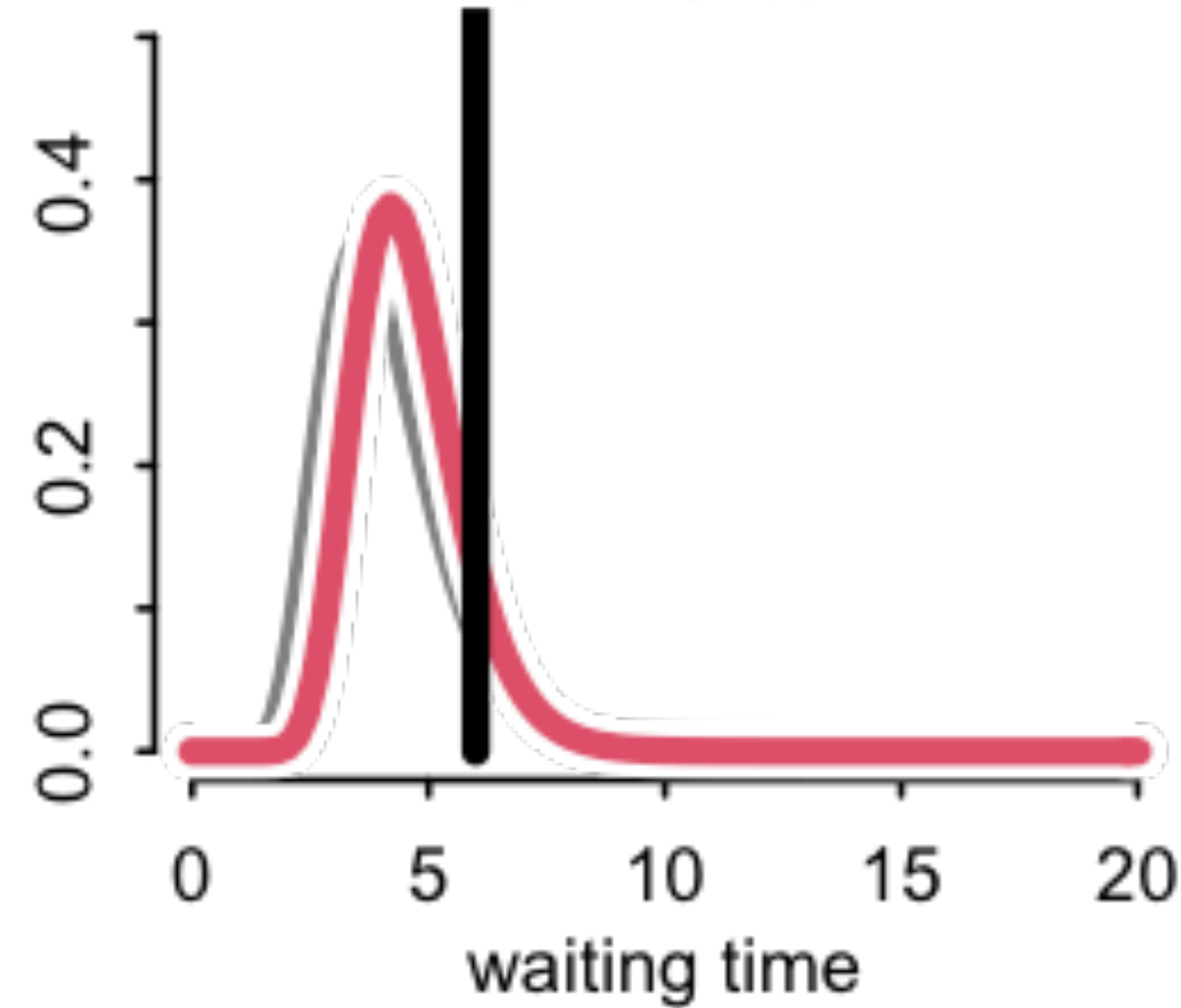
3 visits



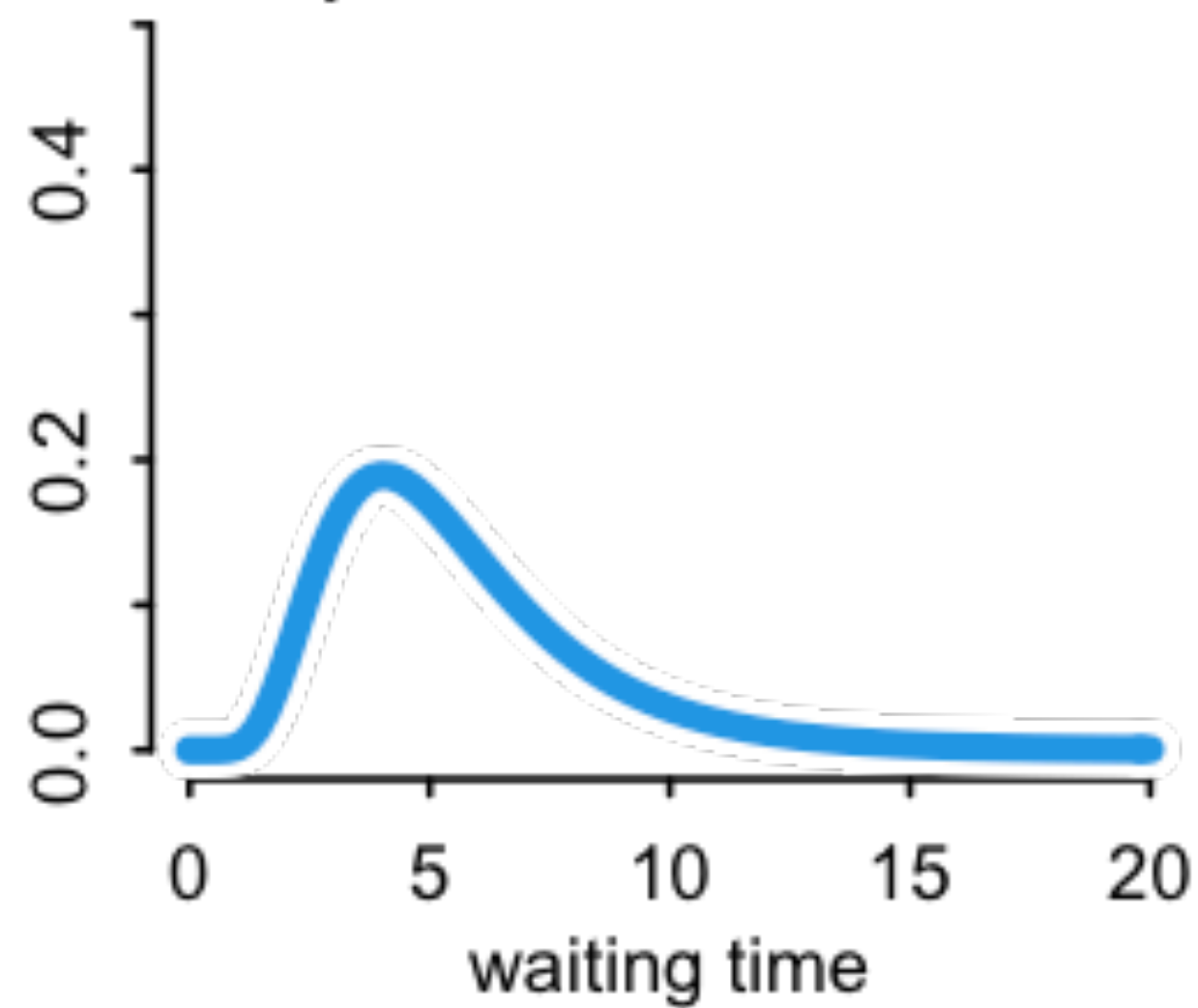
3 visits



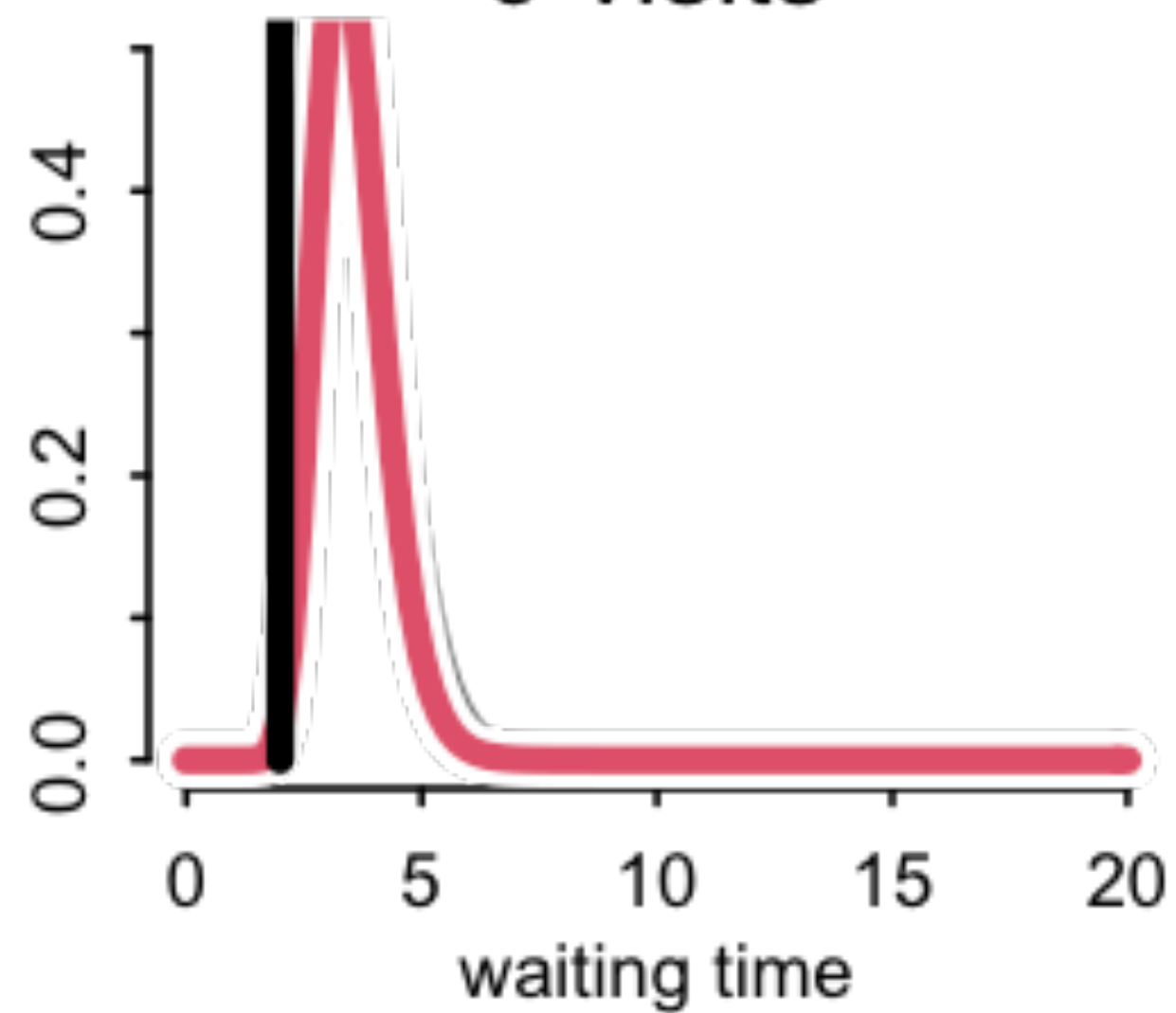
3 visits



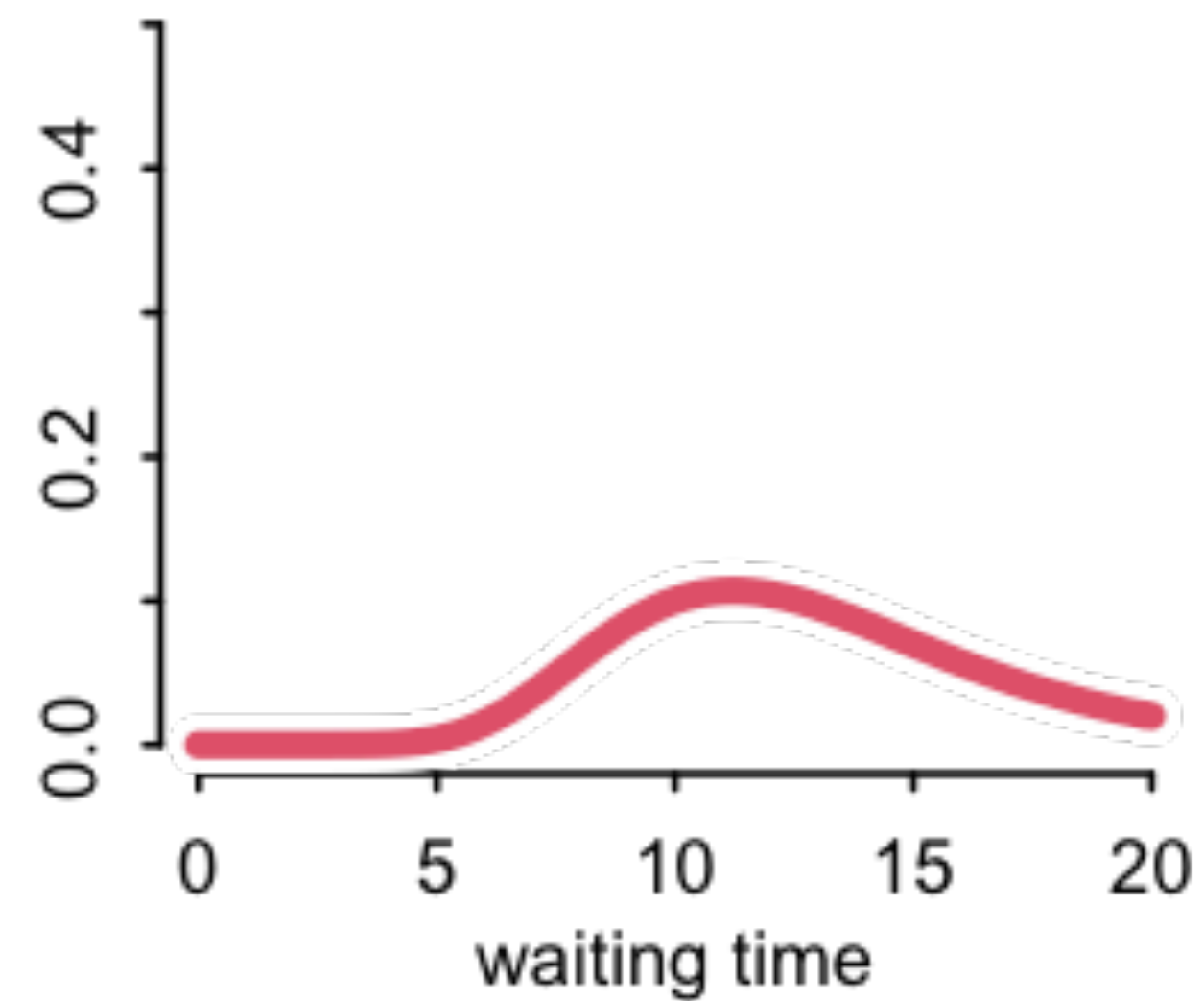
Population of cafes



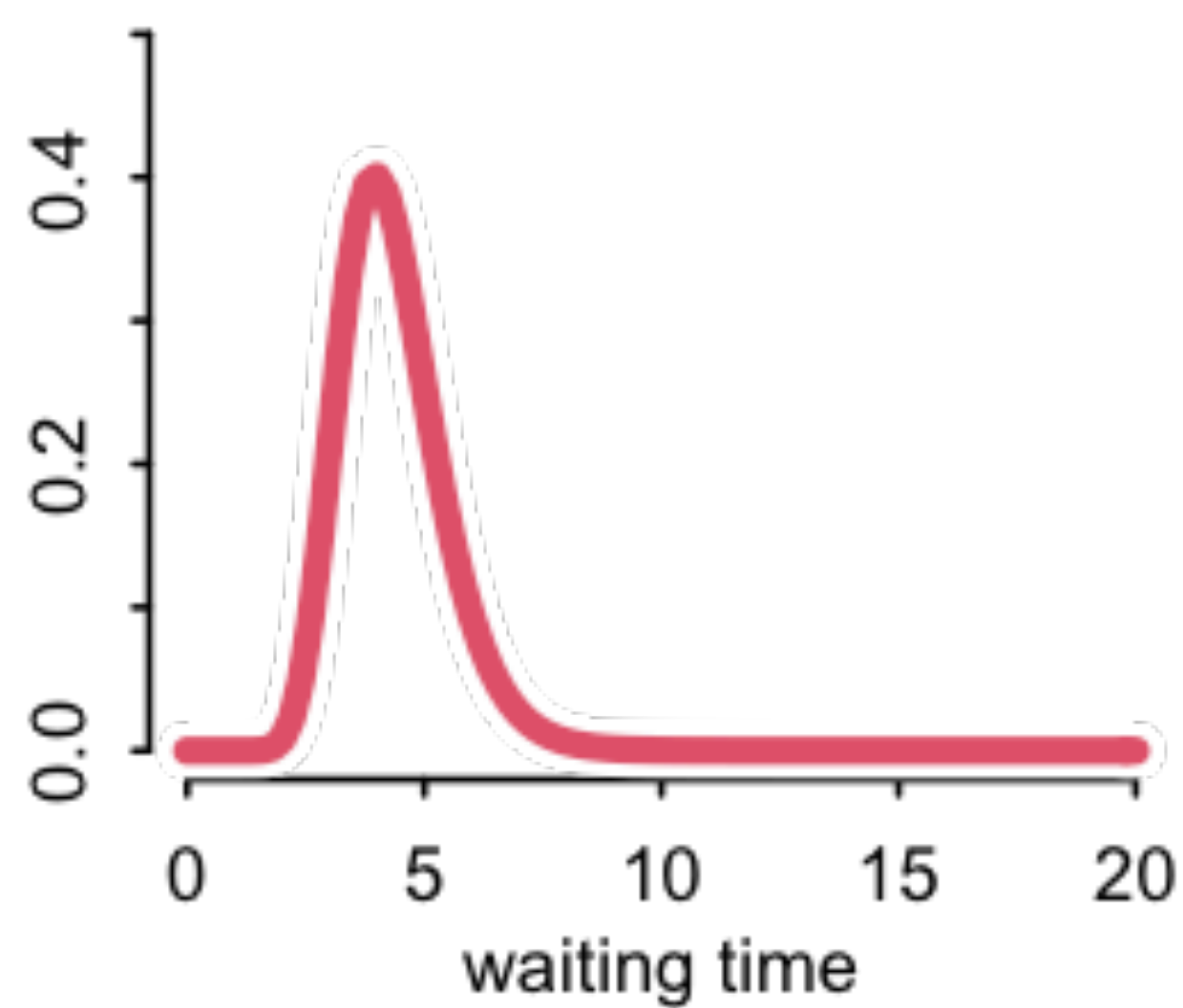
6 visits



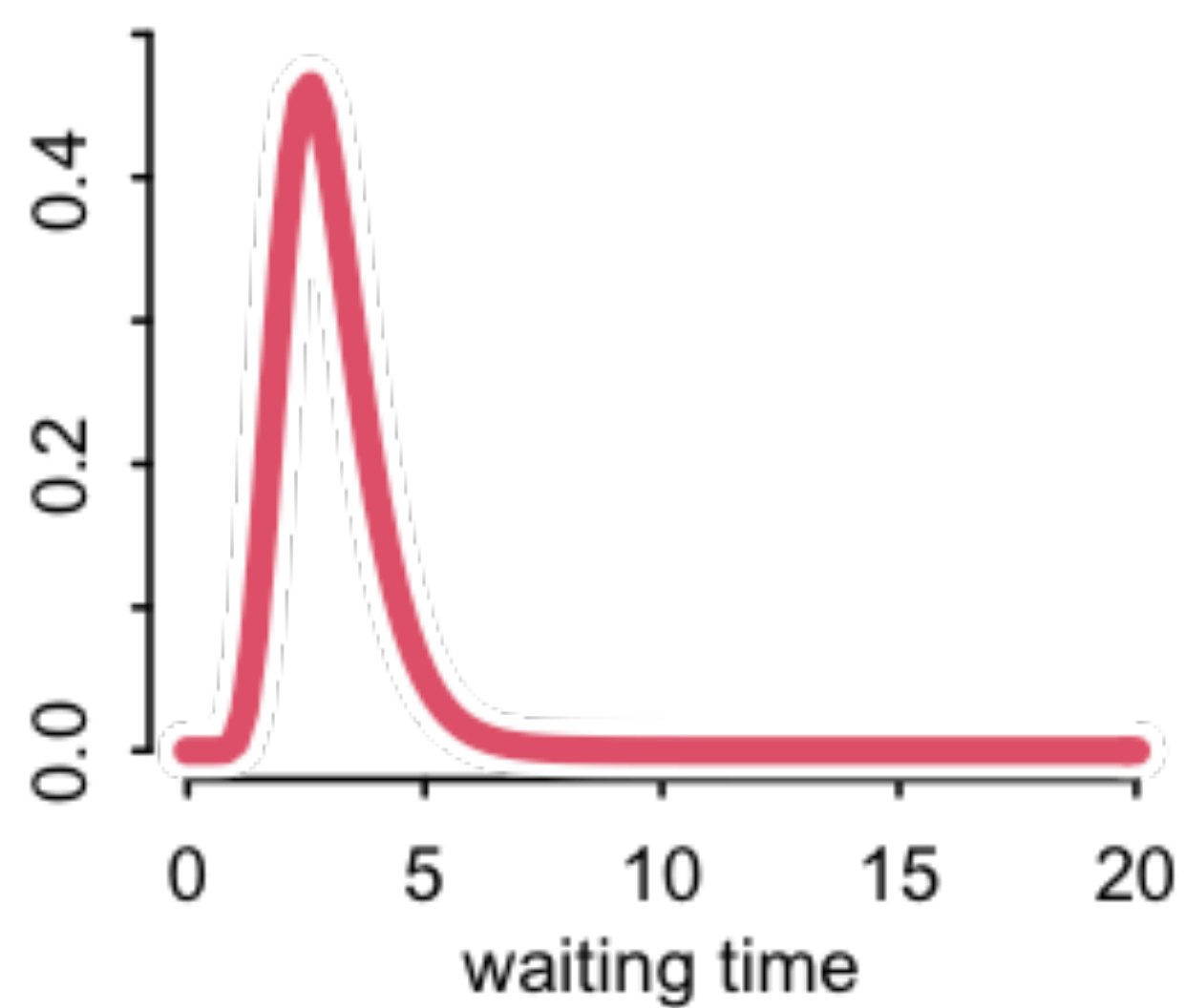
1 visits



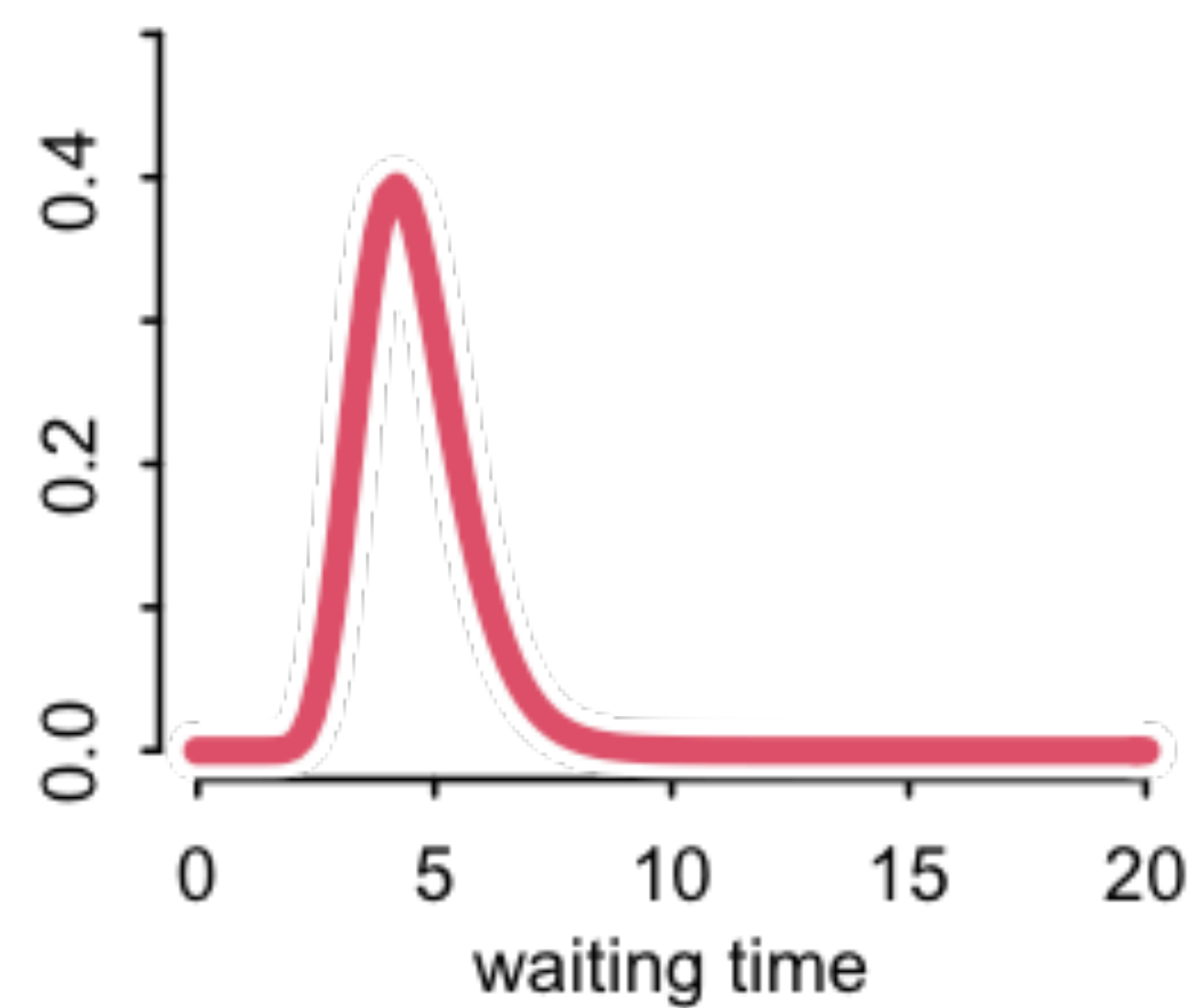
3 visits



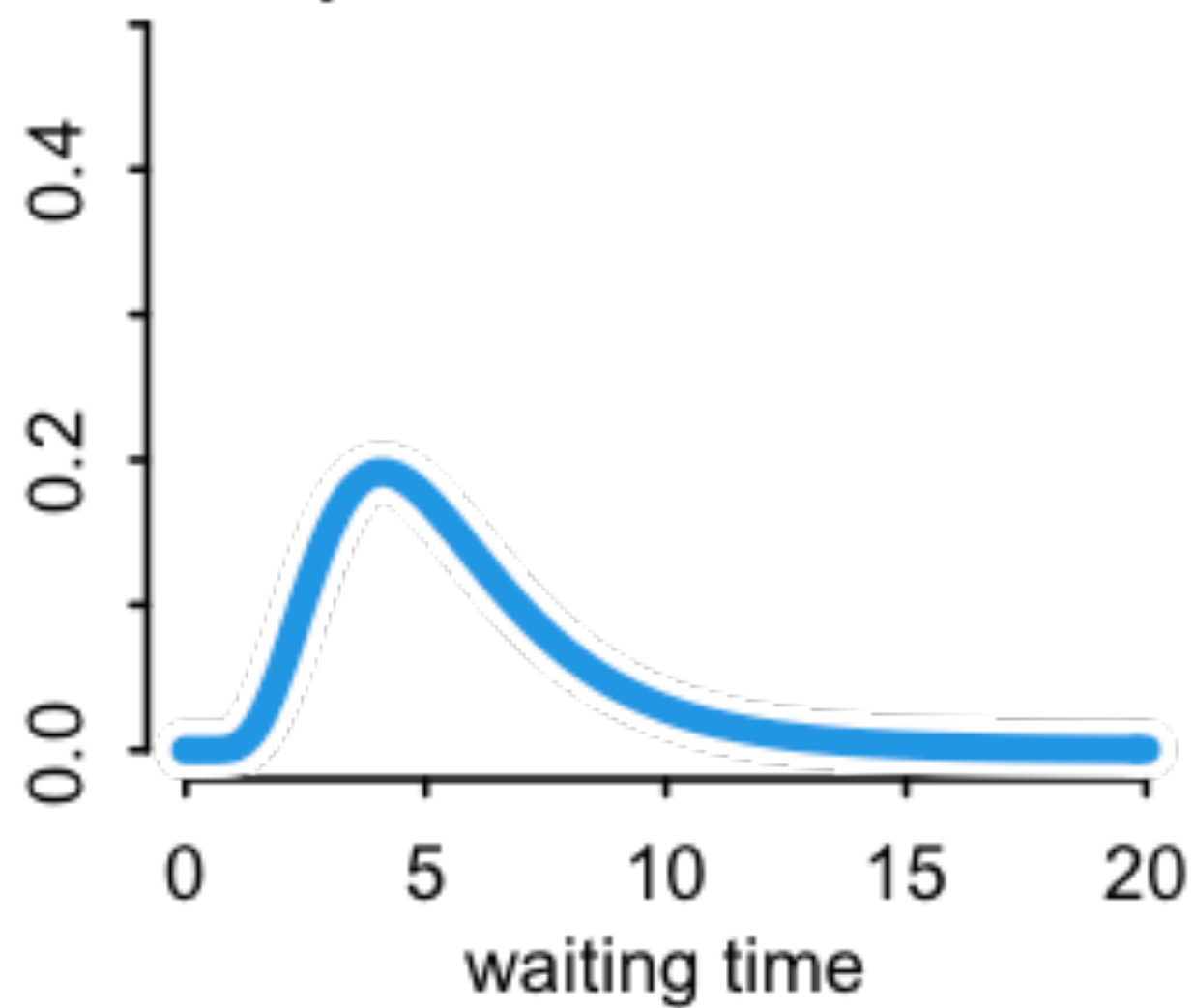
3 visits



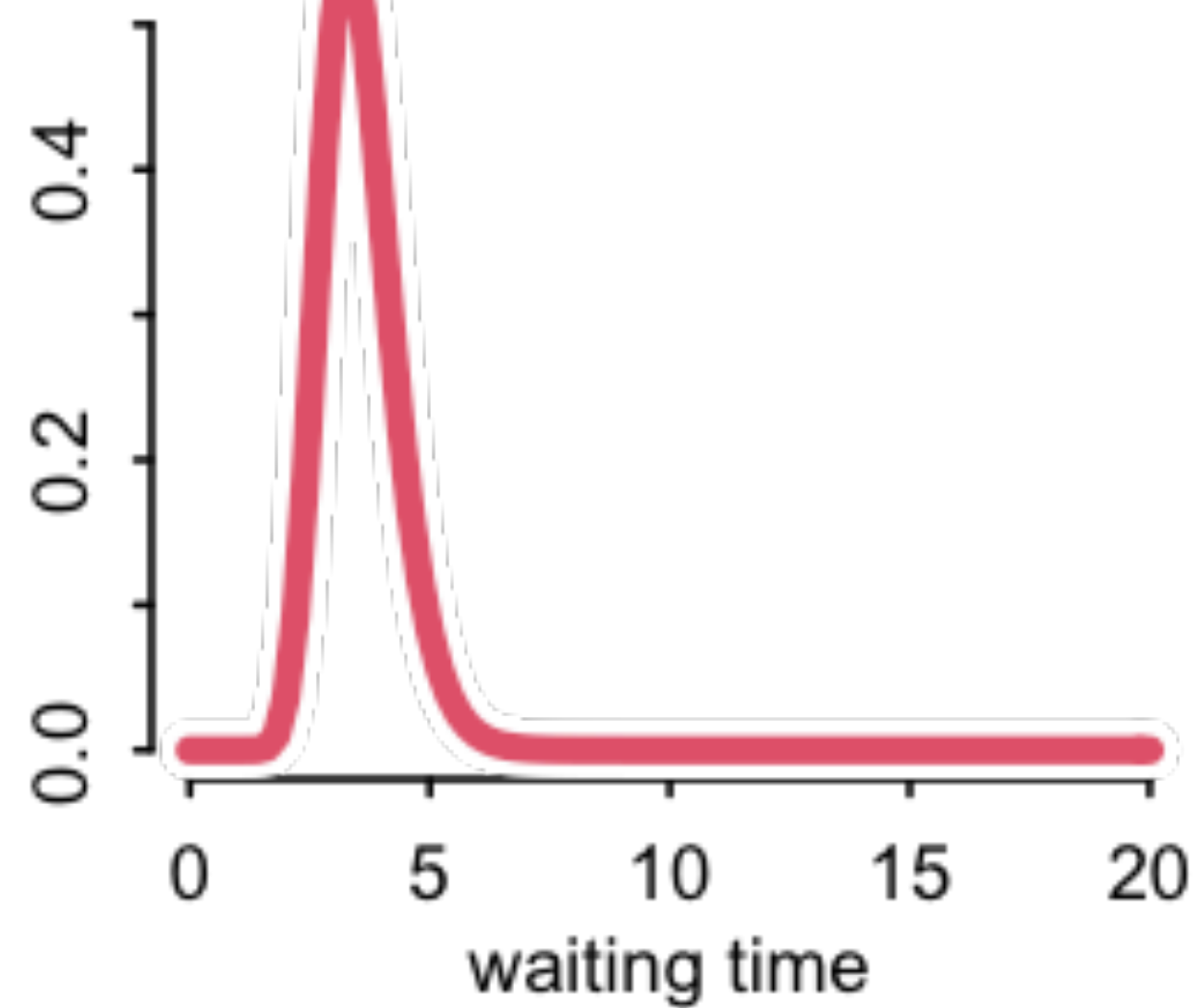
3 visits



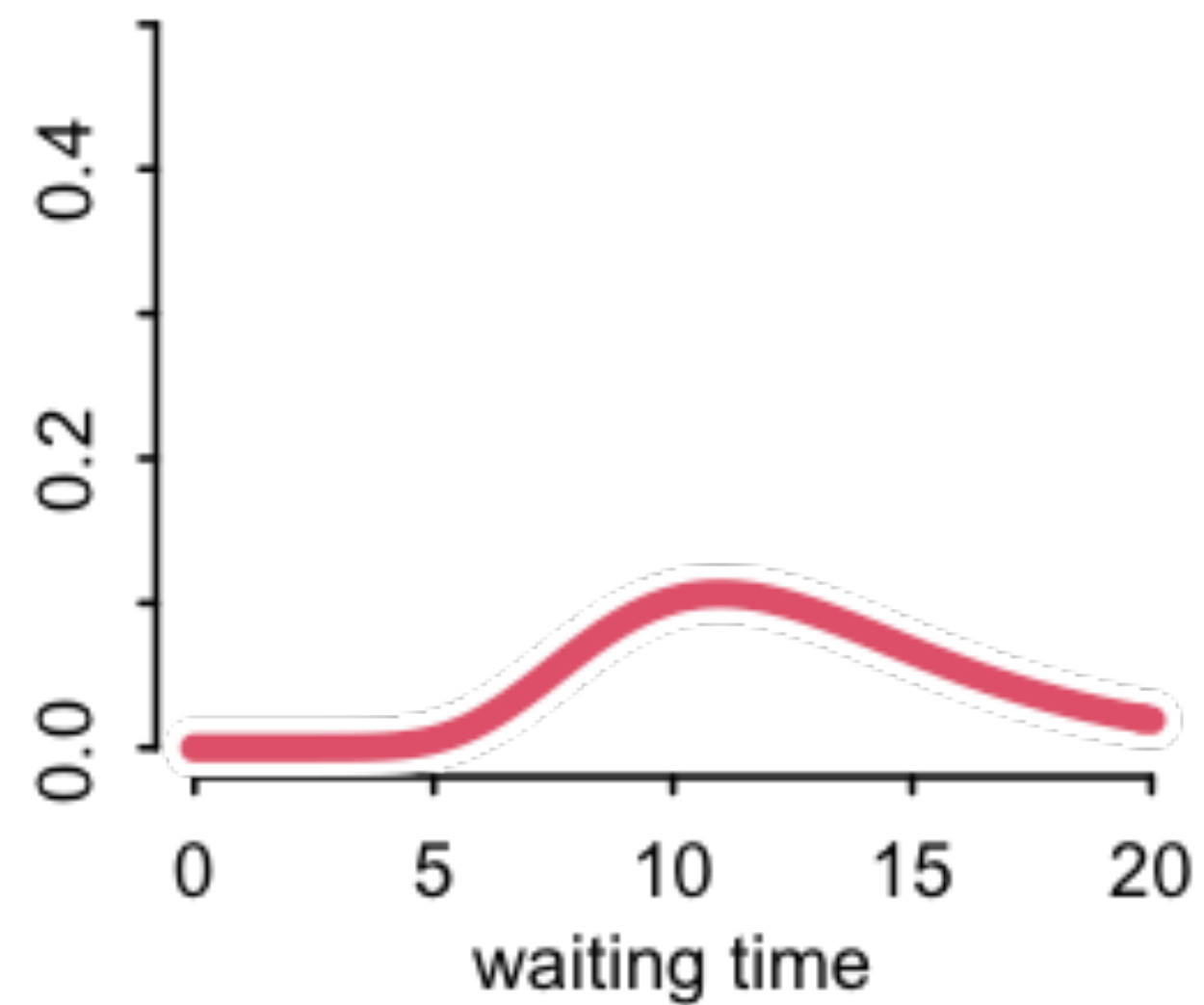
Population of cafes



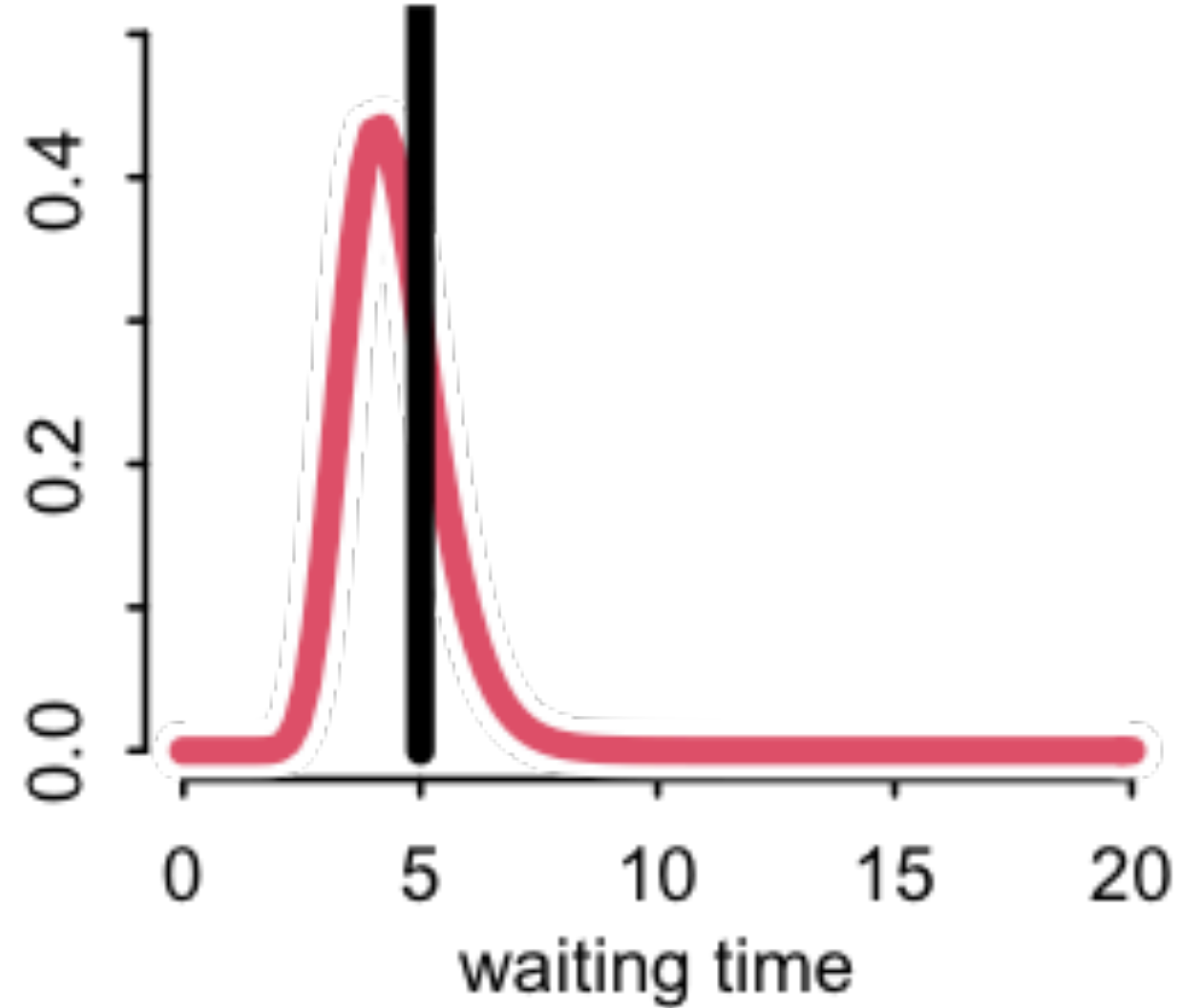
6 visits



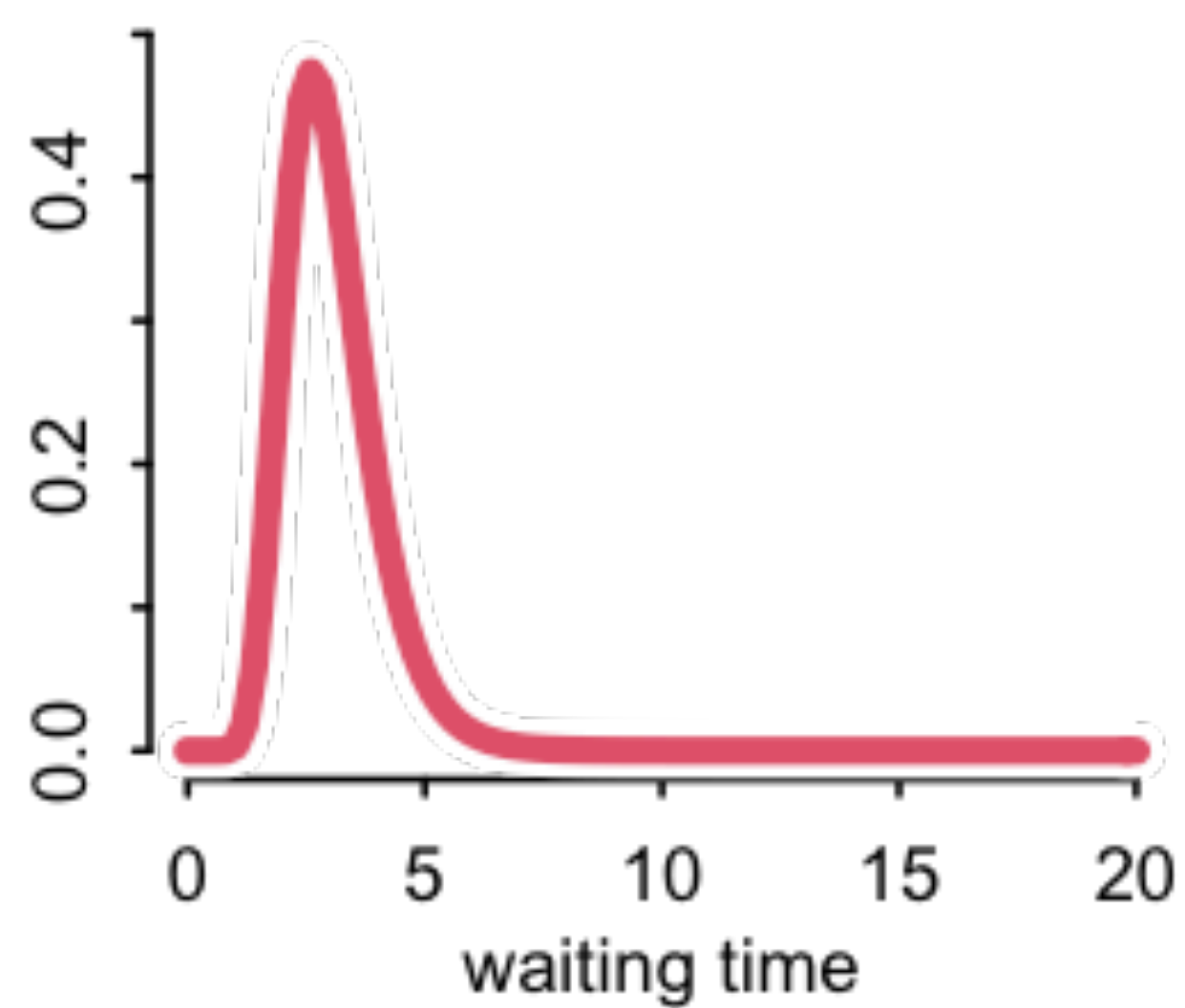
1 visits



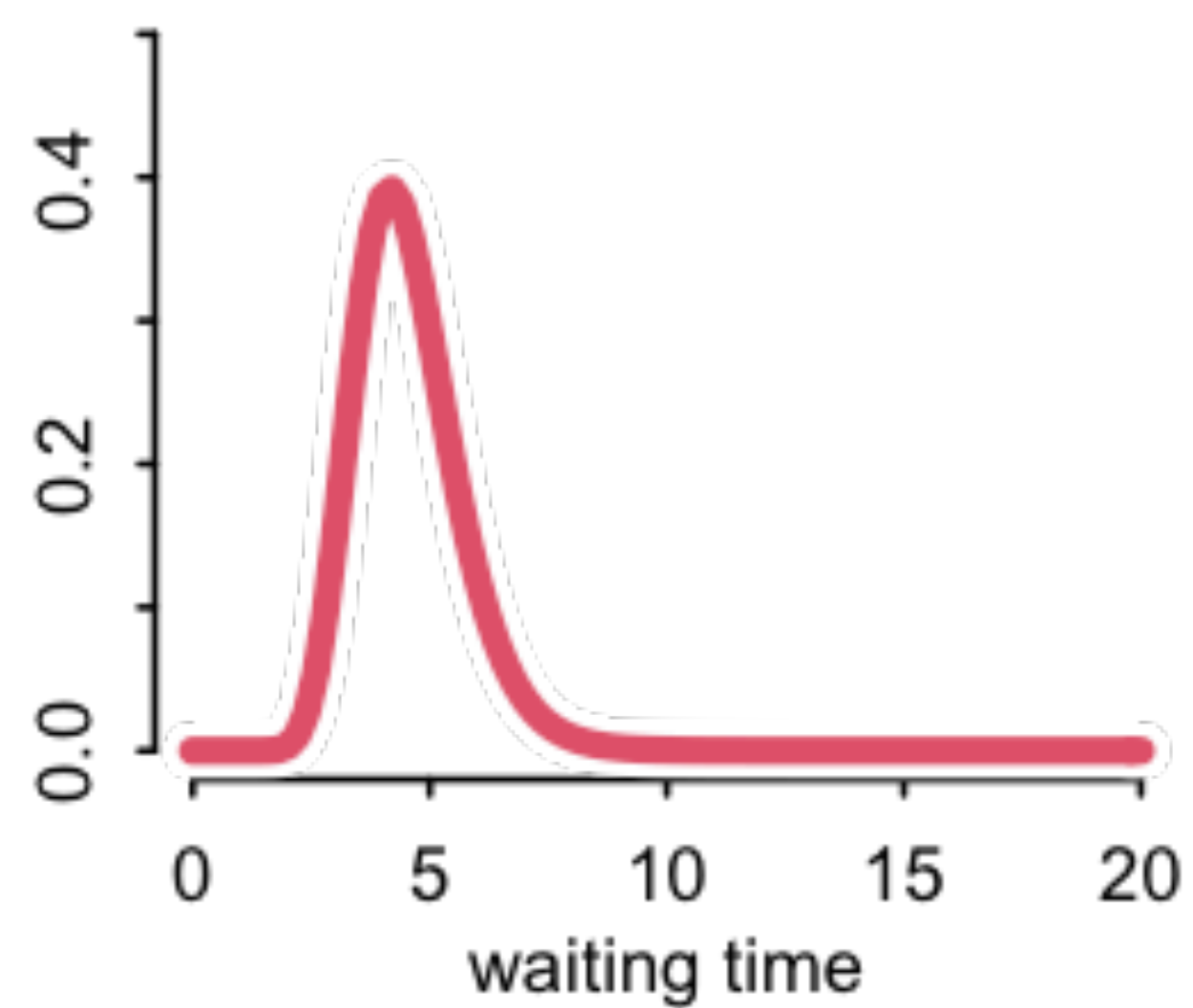
4 visits



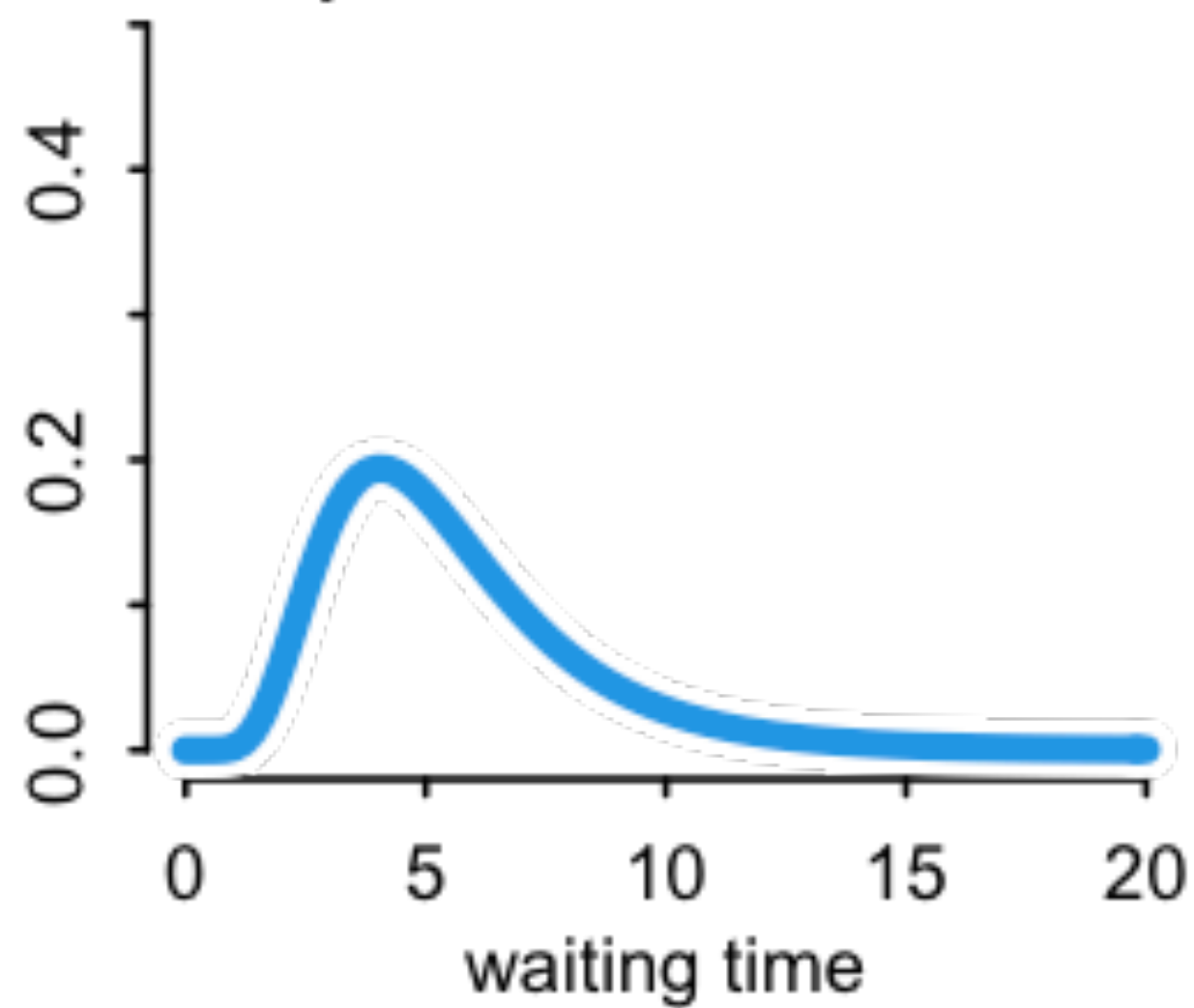
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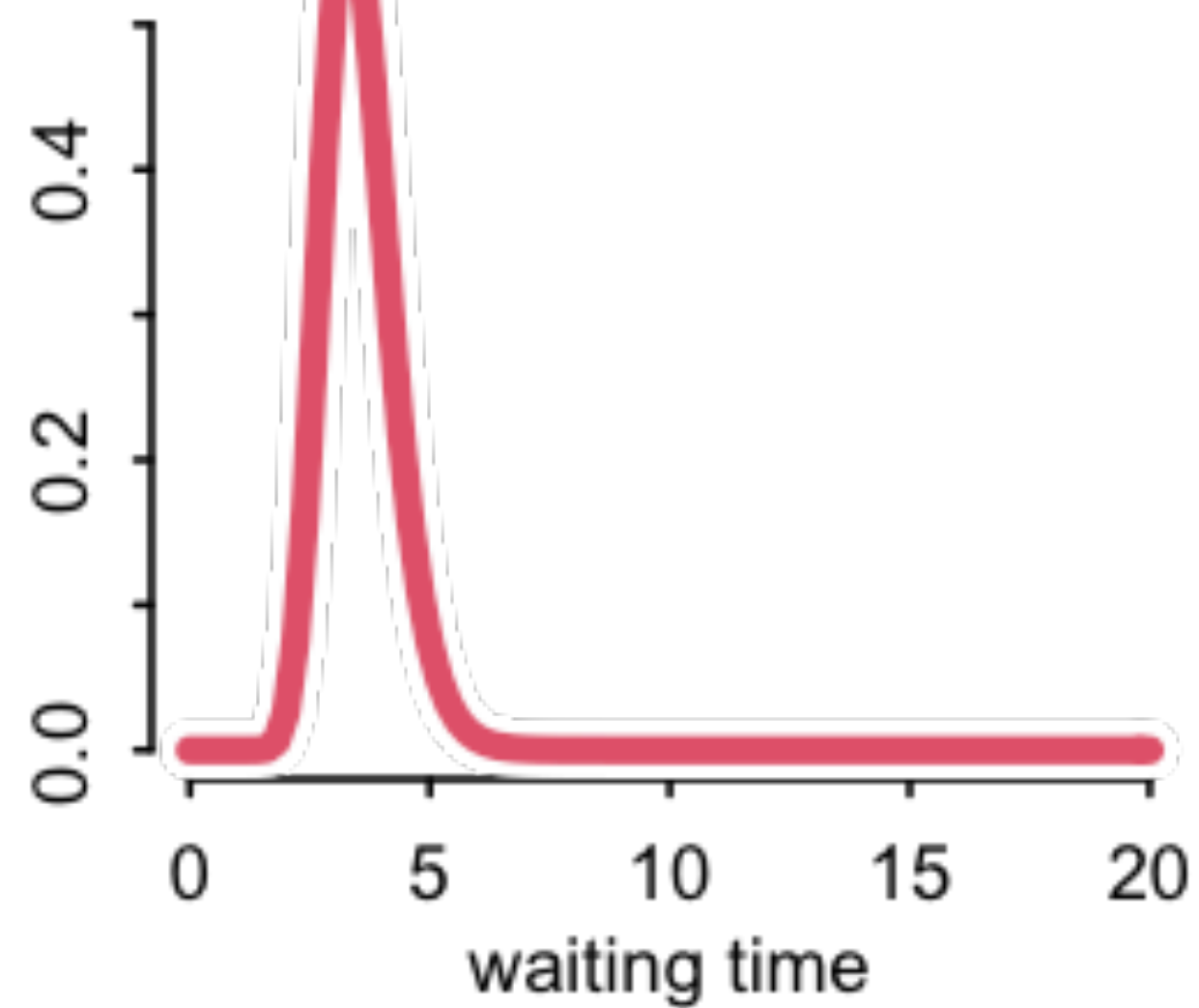
3 visits



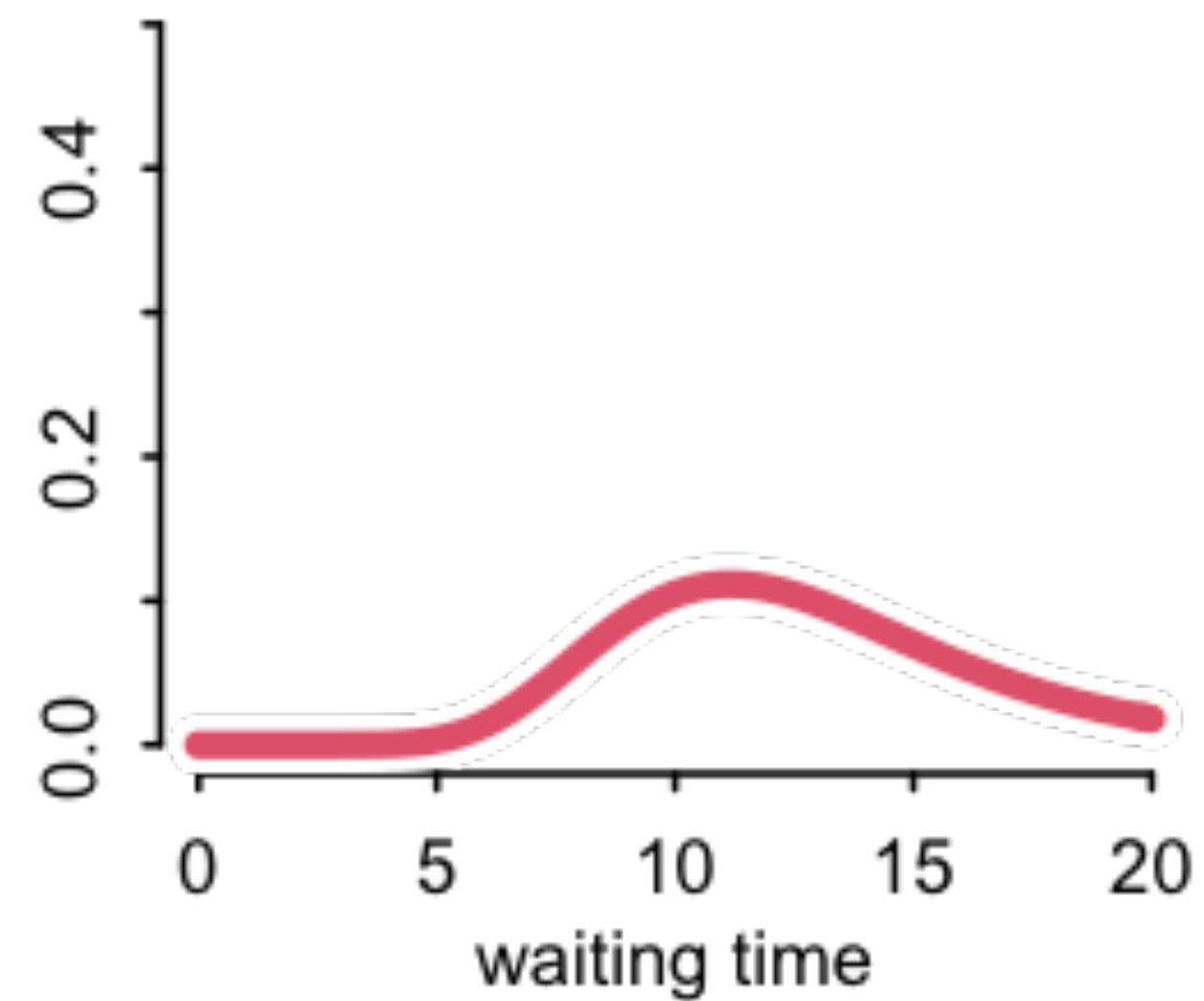
Population of cafes



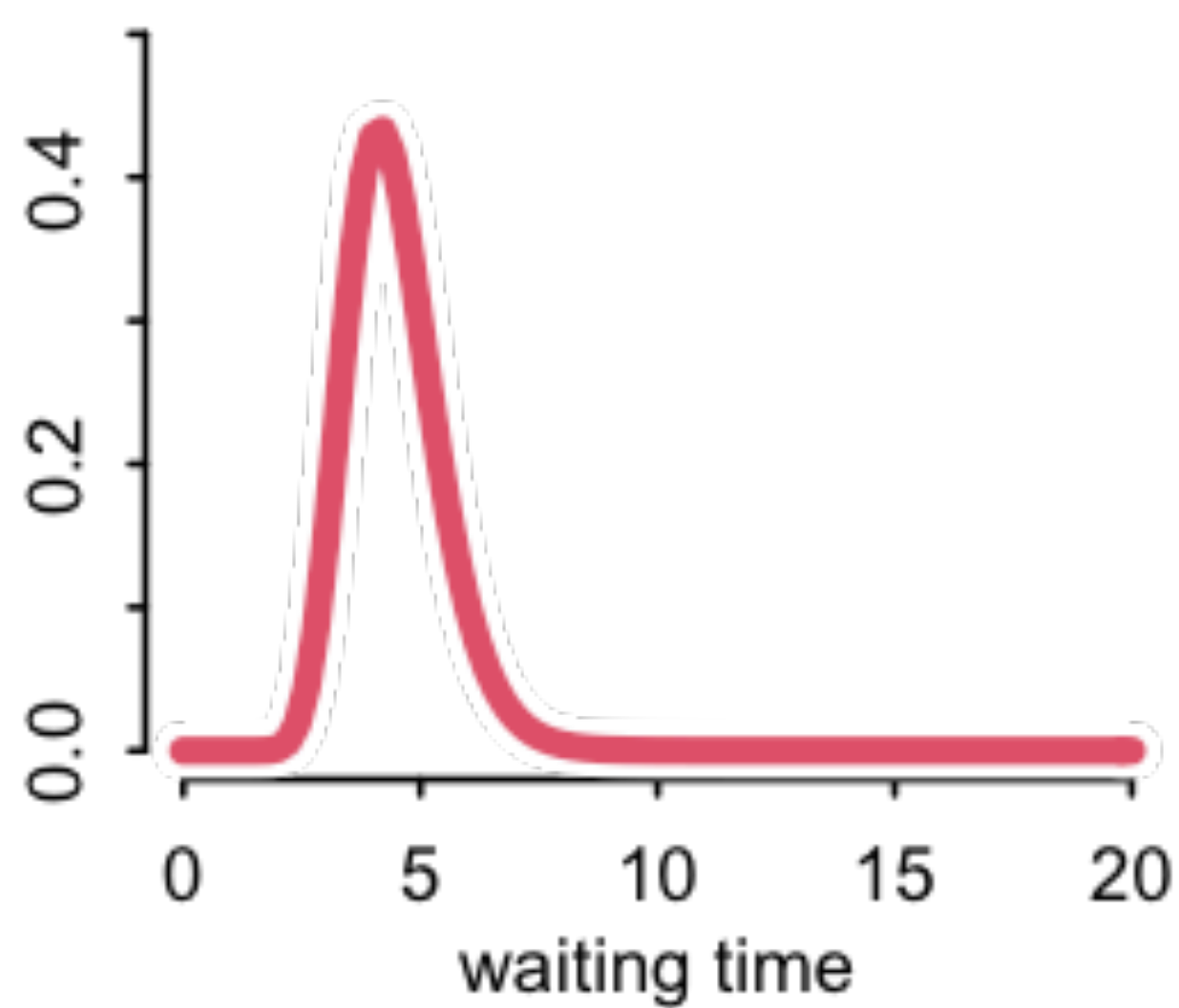
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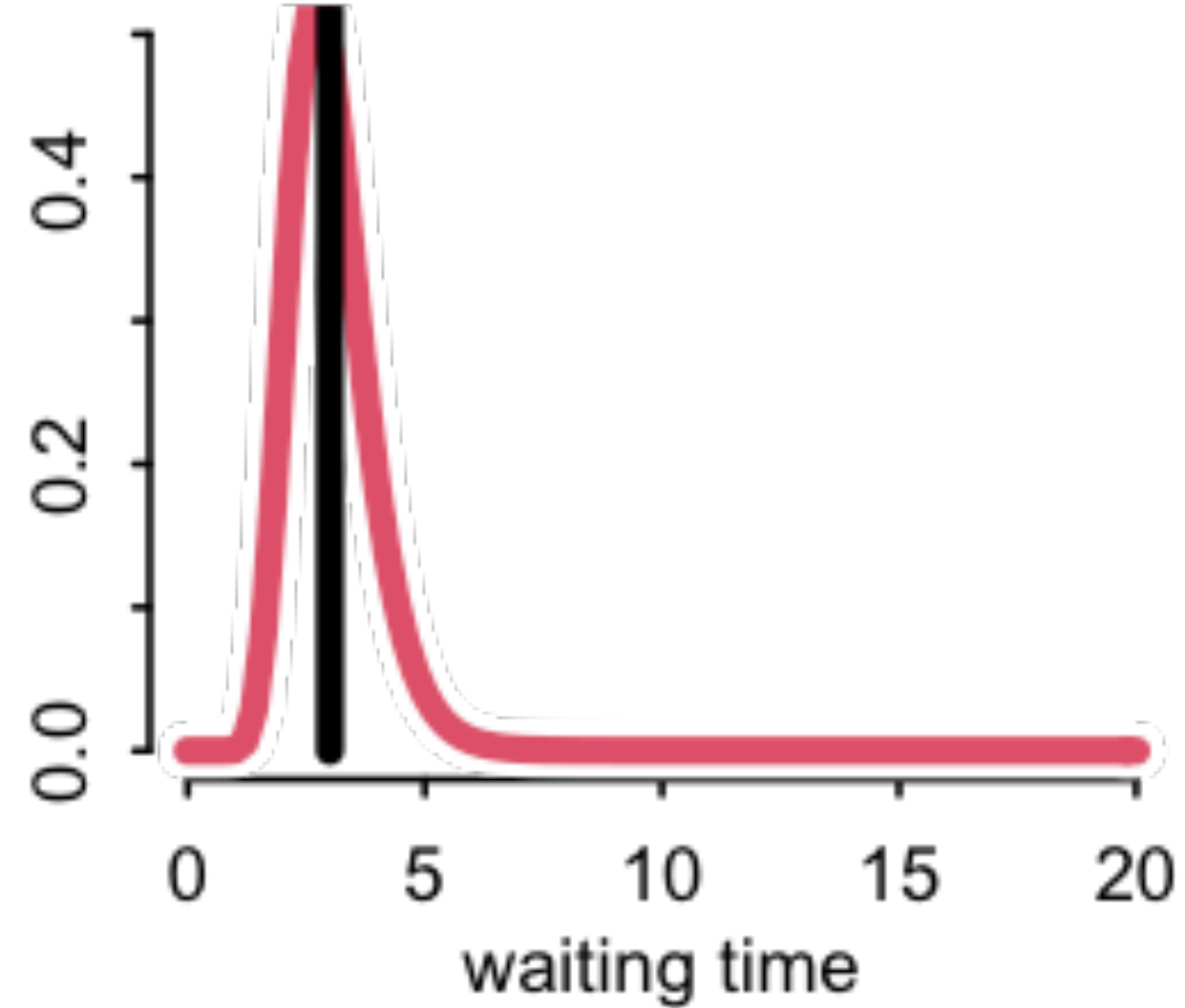
1 visits



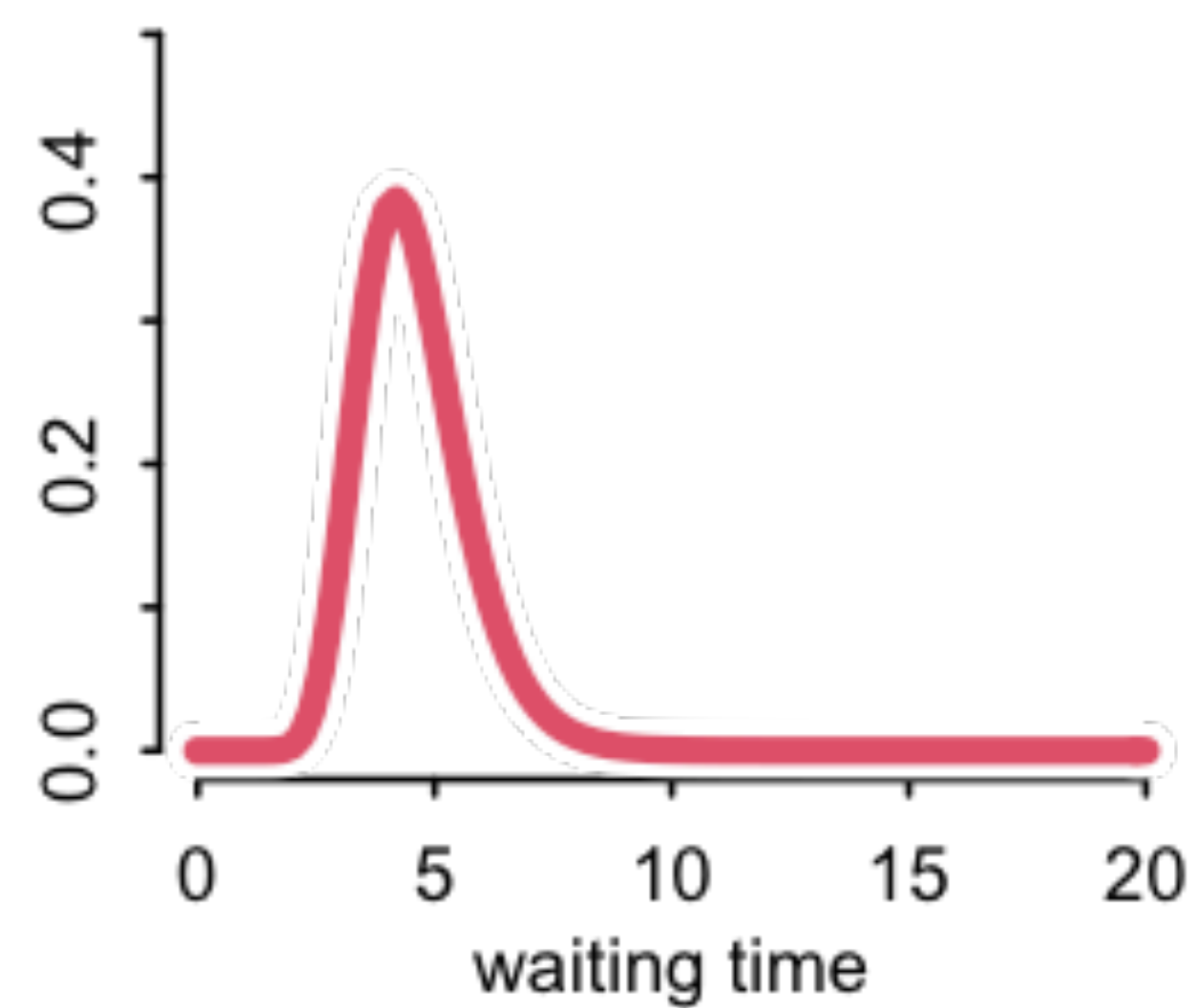
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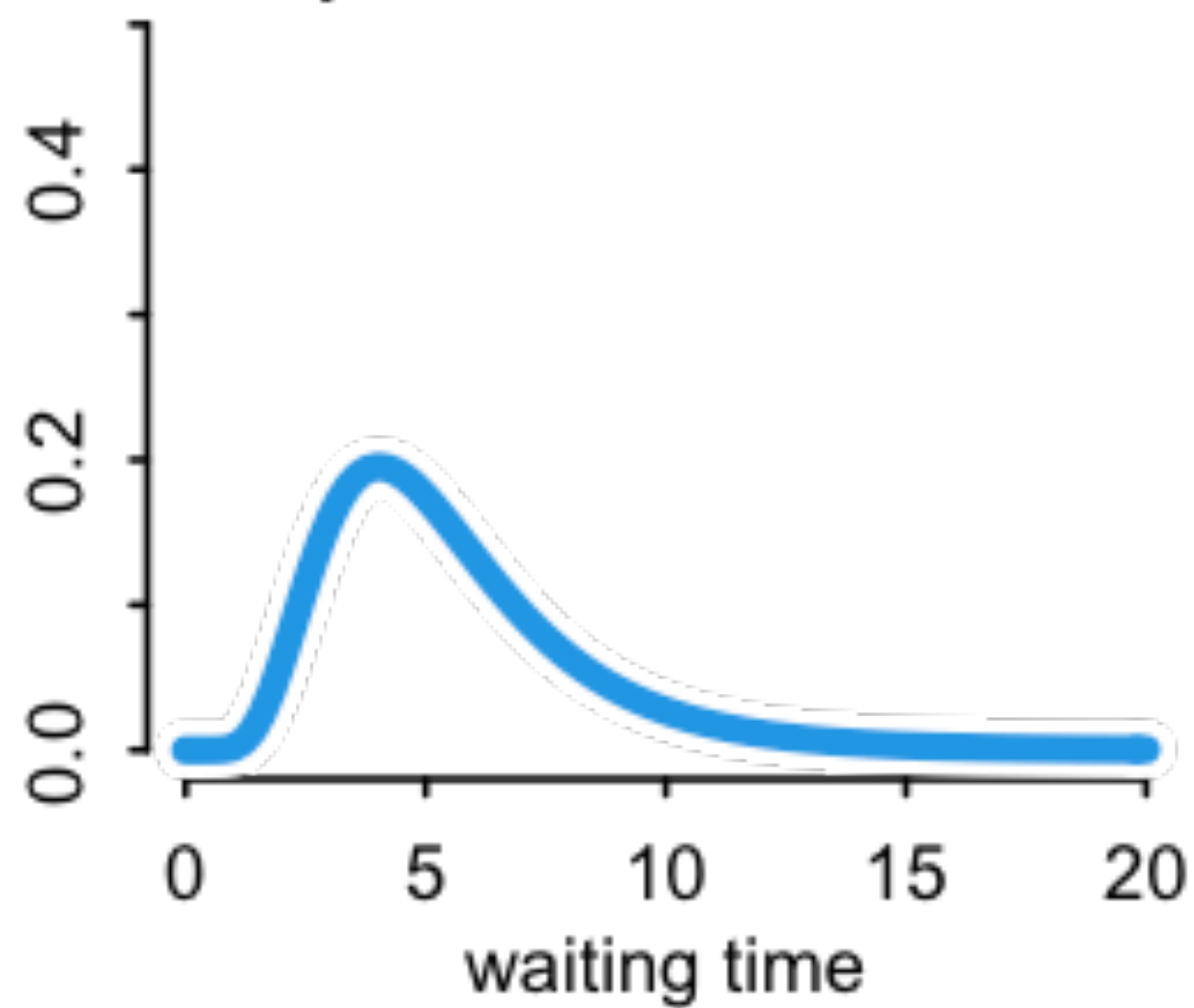
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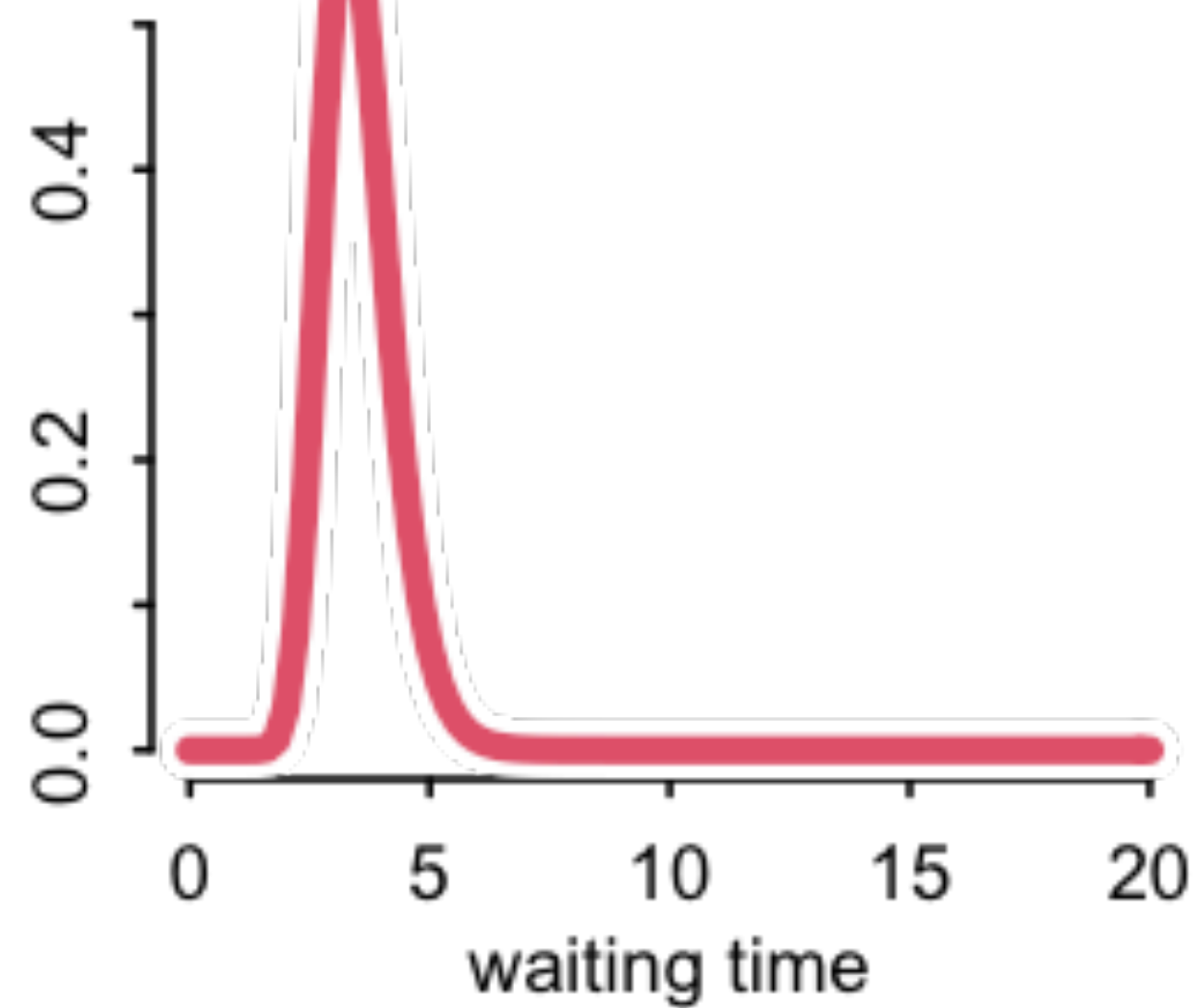
3 visits



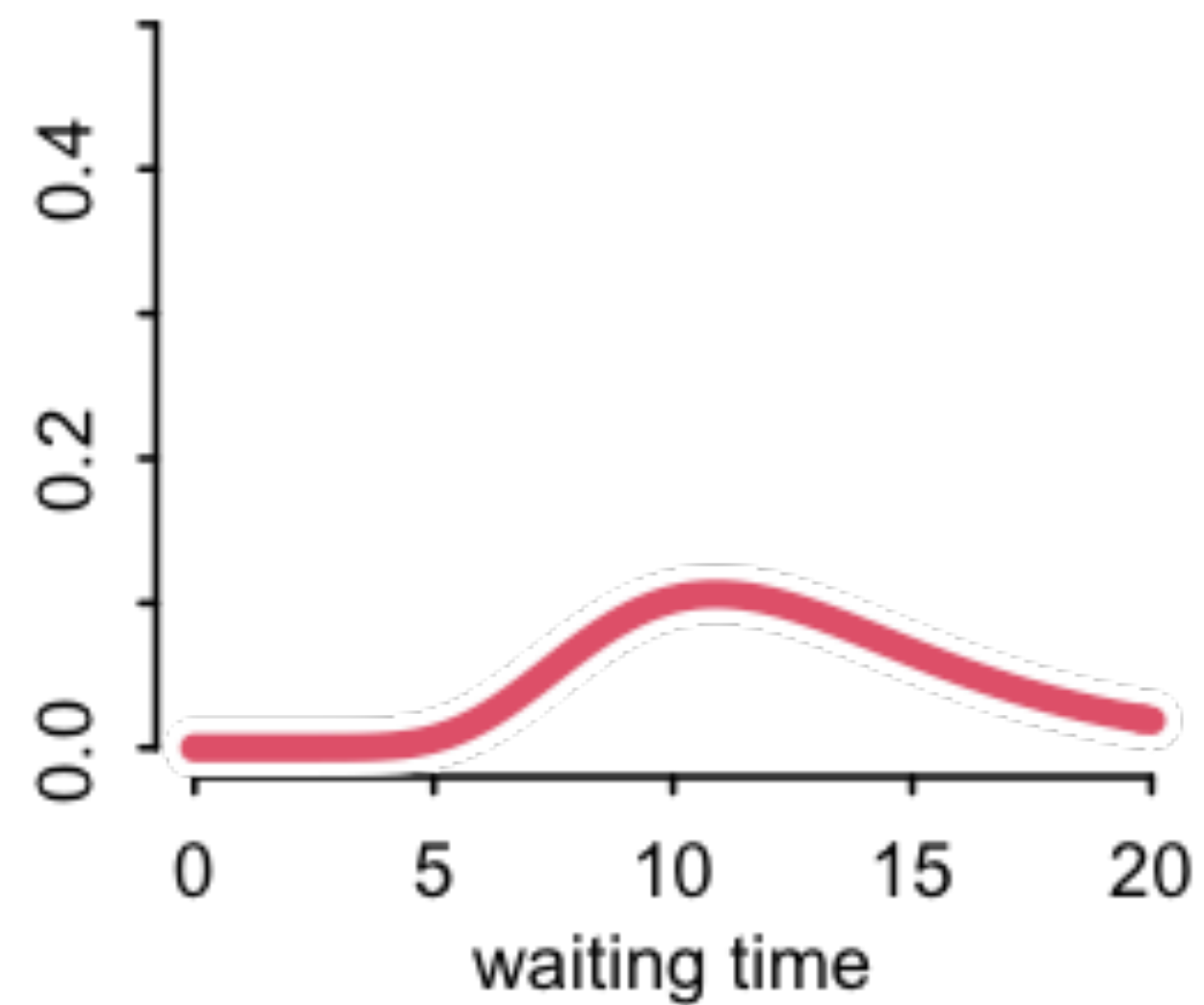
Population of cafes



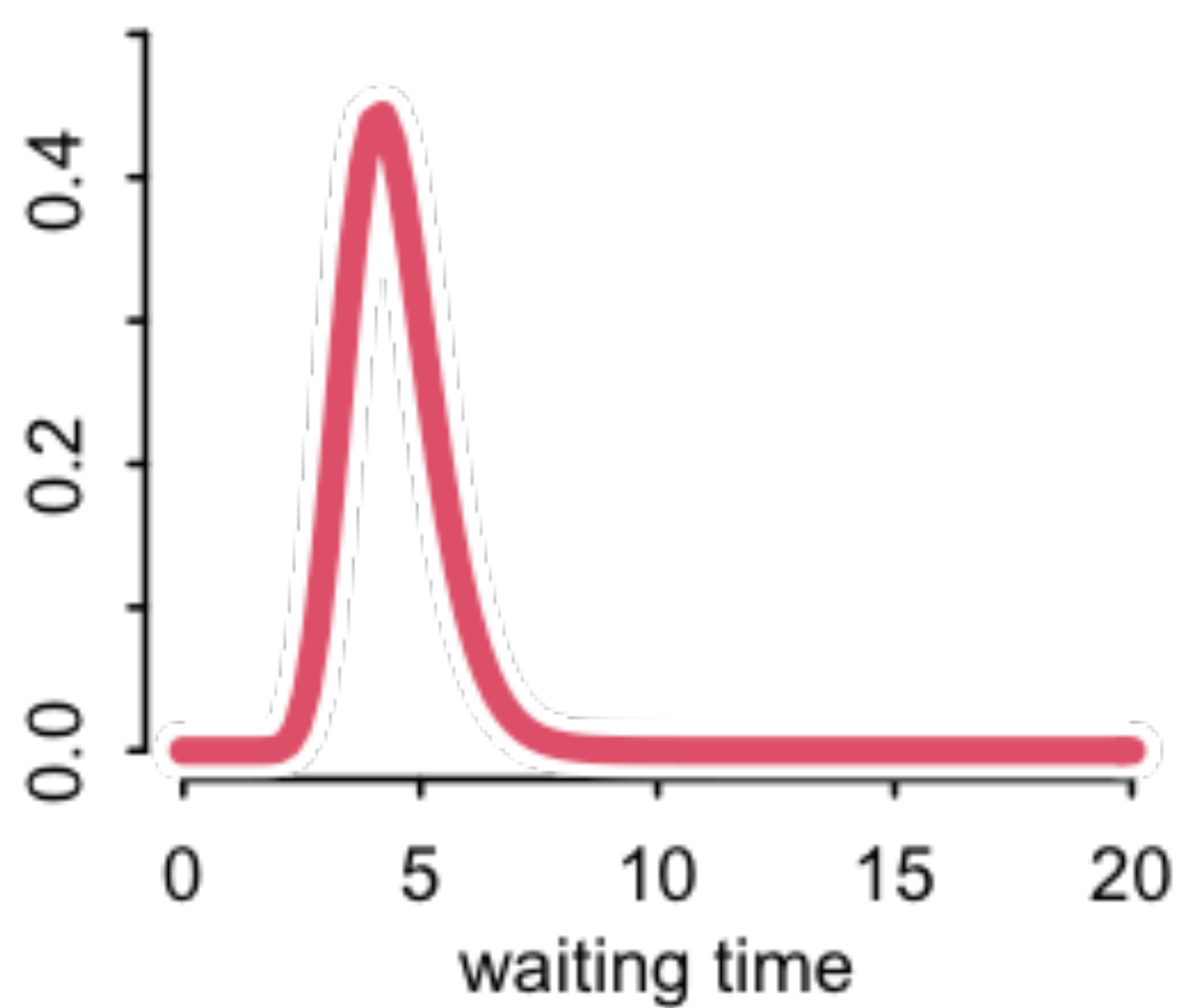
6 visits



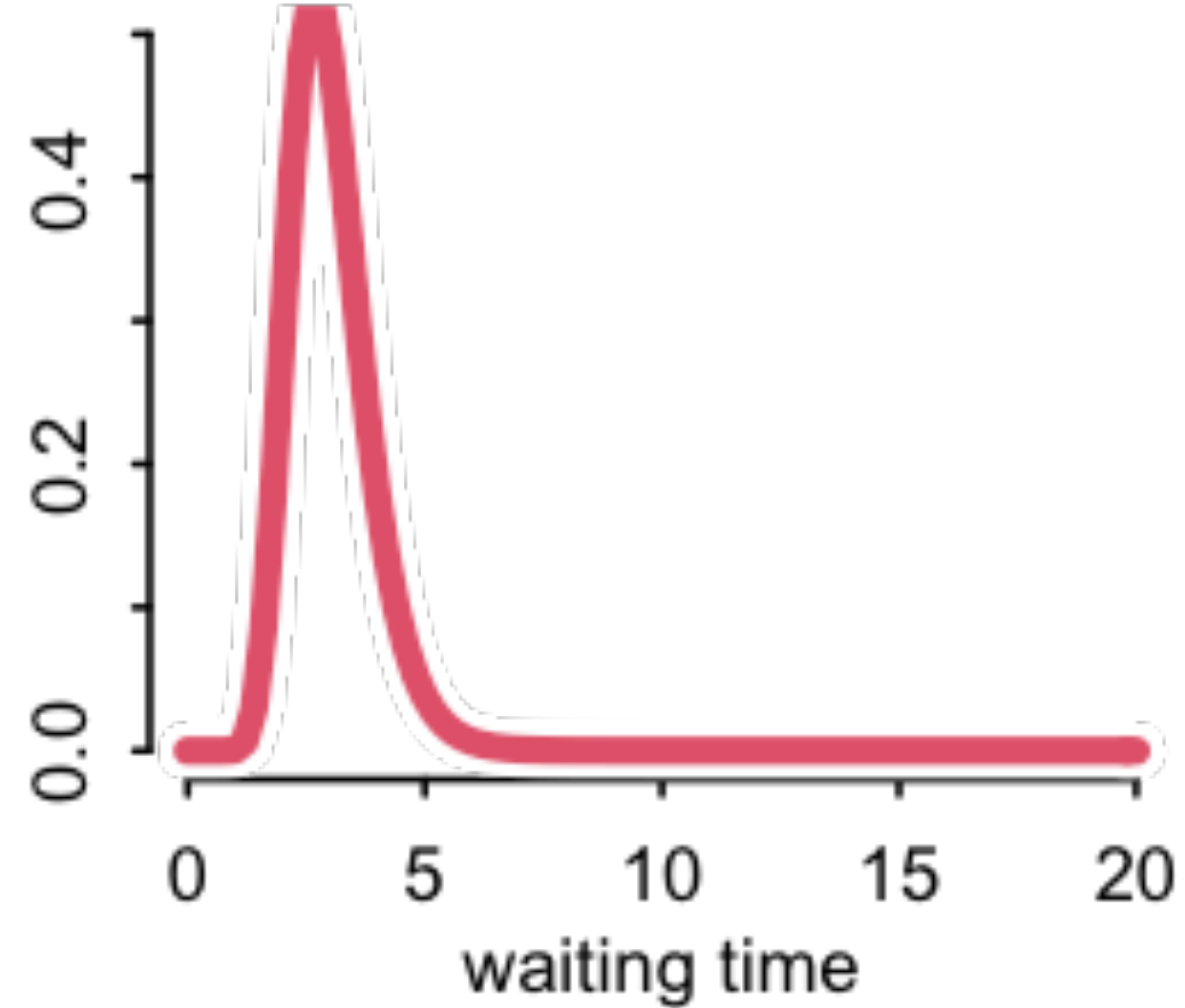
1 visits



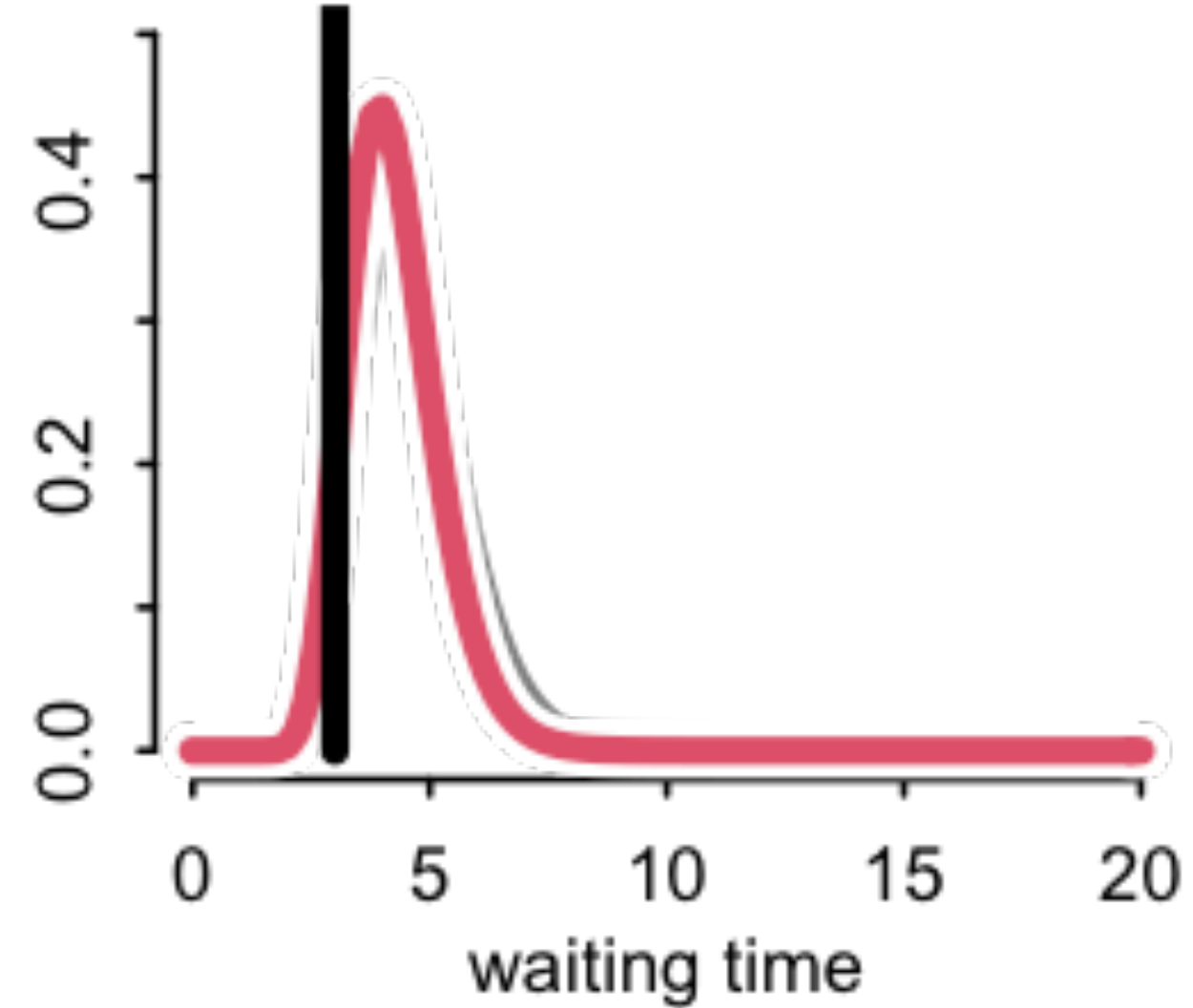
4 visits



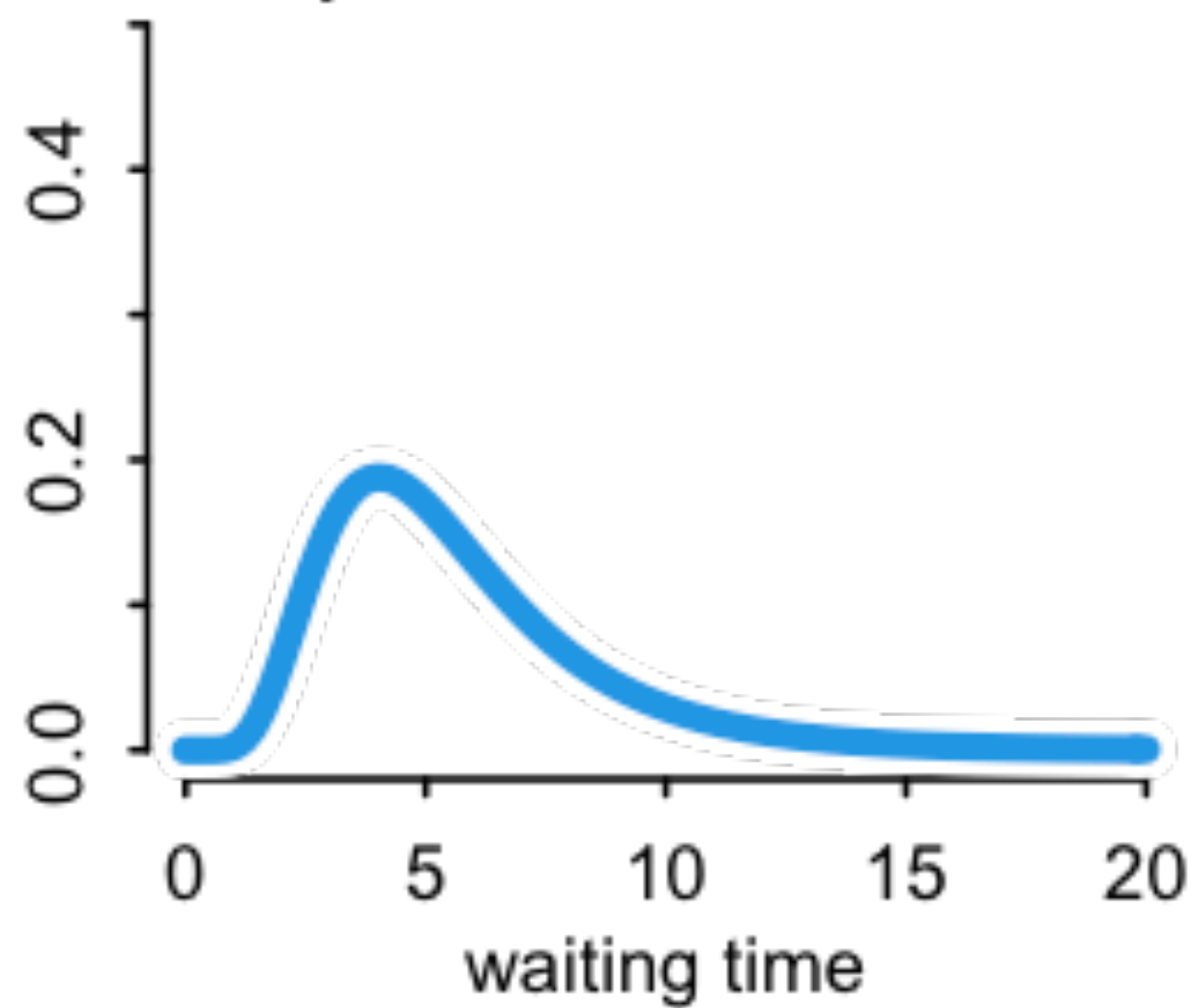
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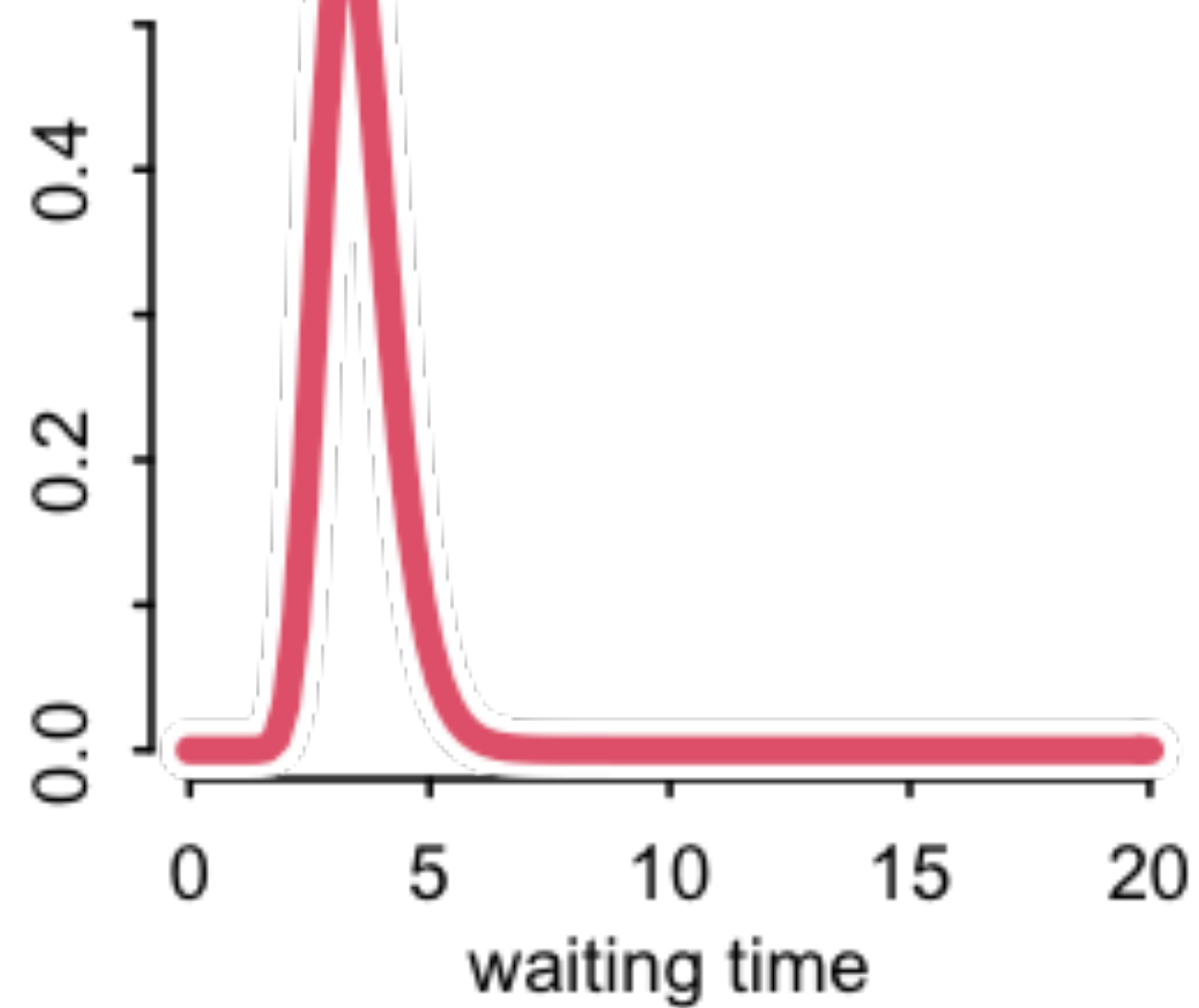
4 visits



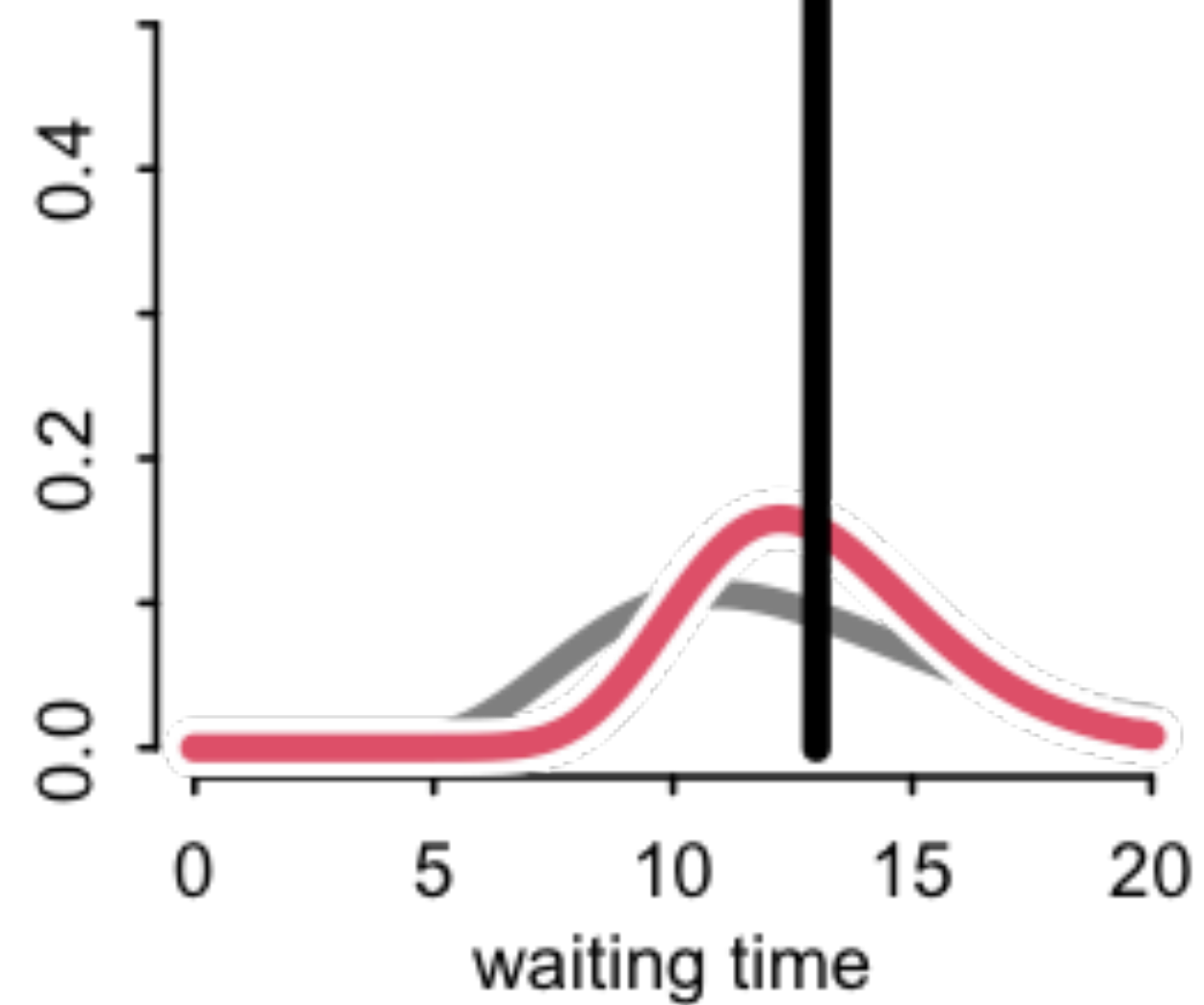
Population of cafes



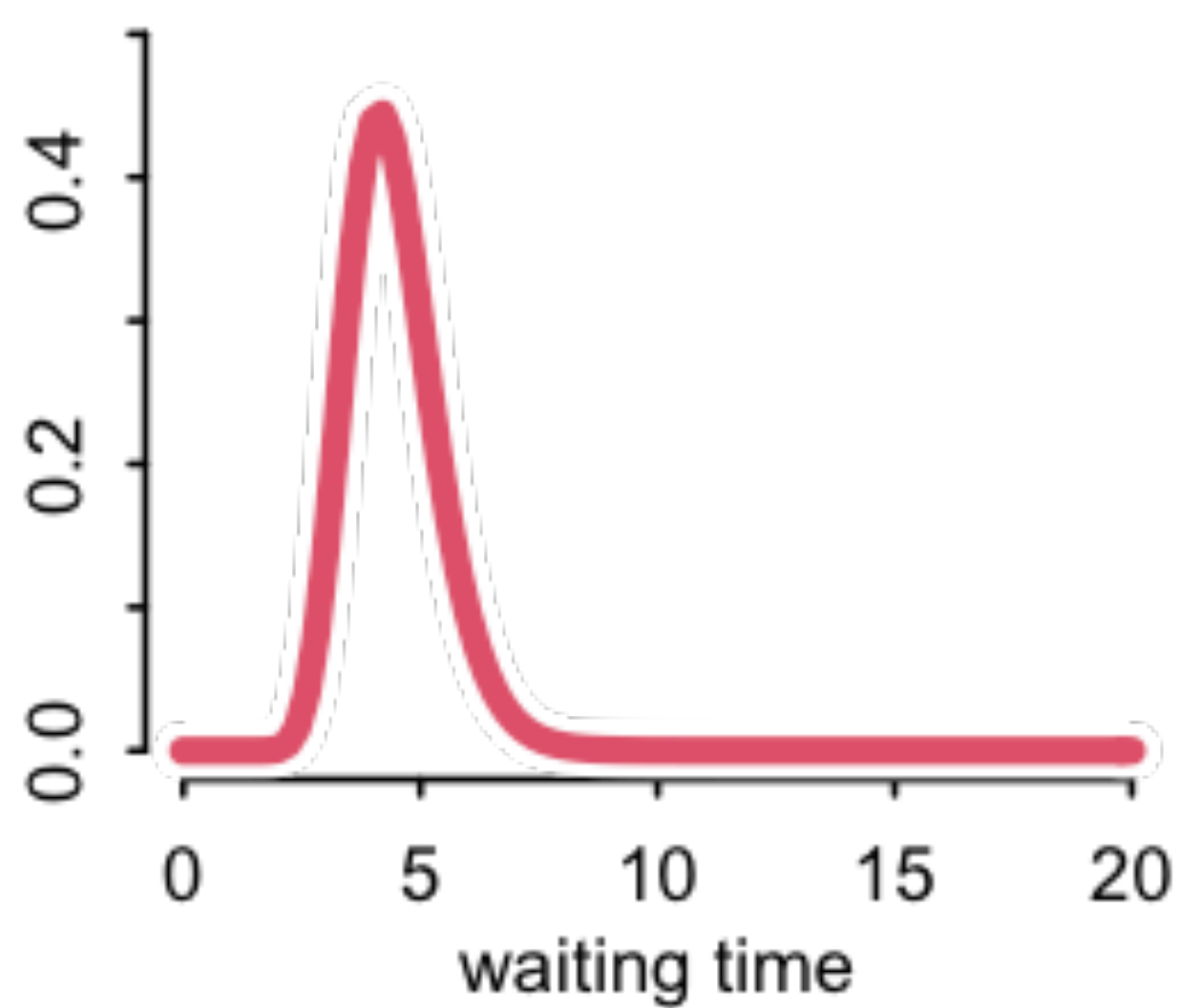
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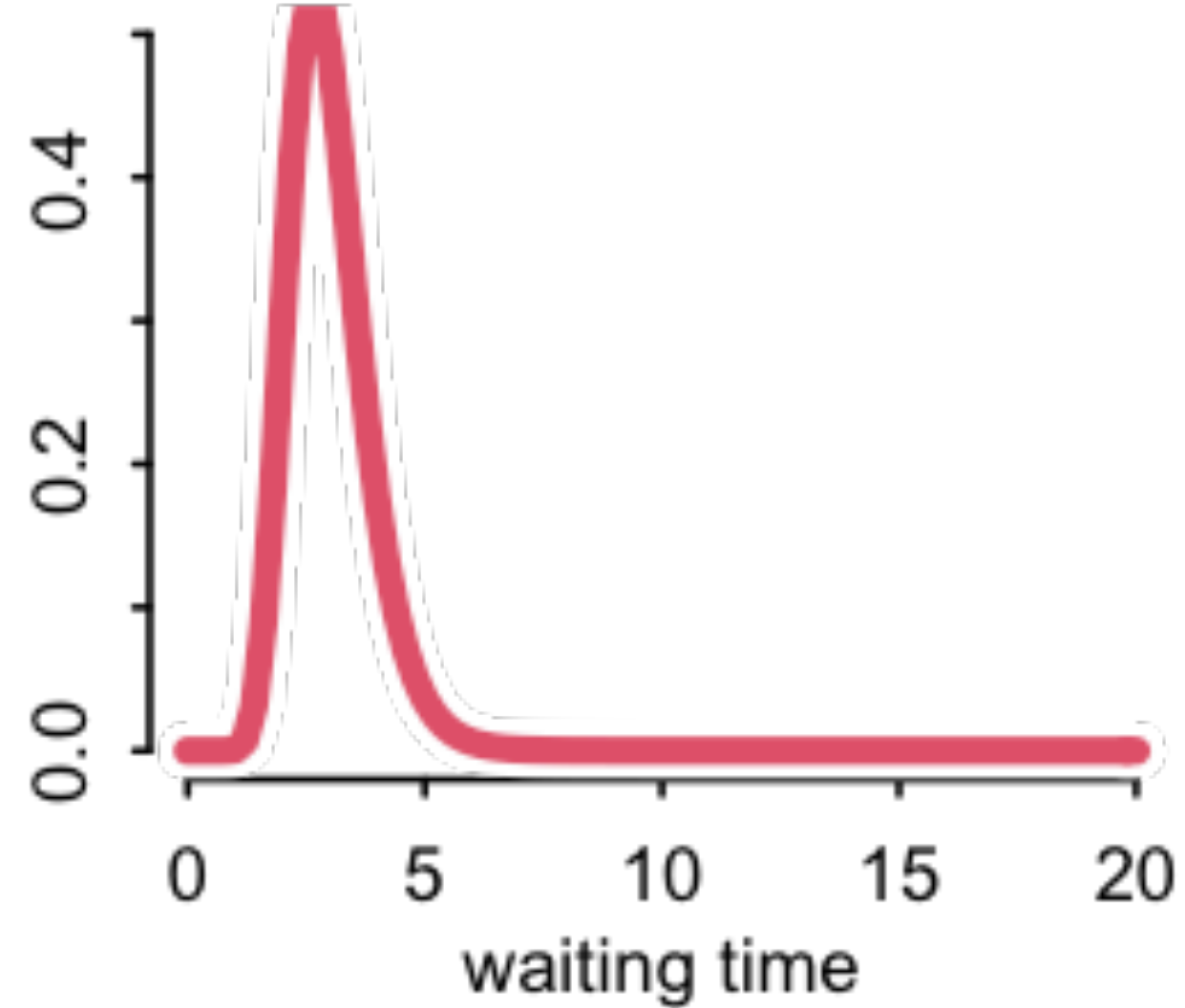
2 visits



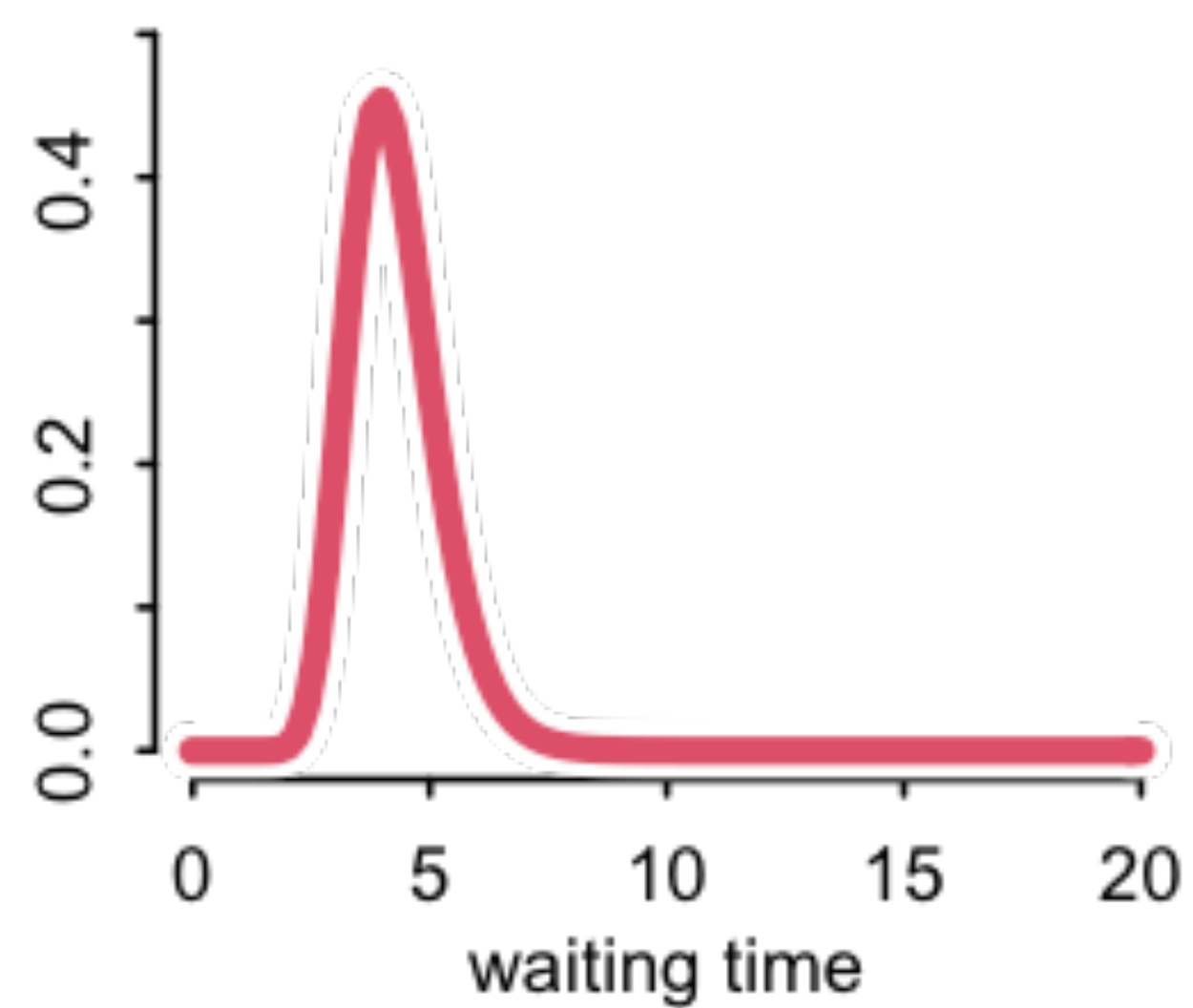
4 visits



4 visits



4 visits



Regularization

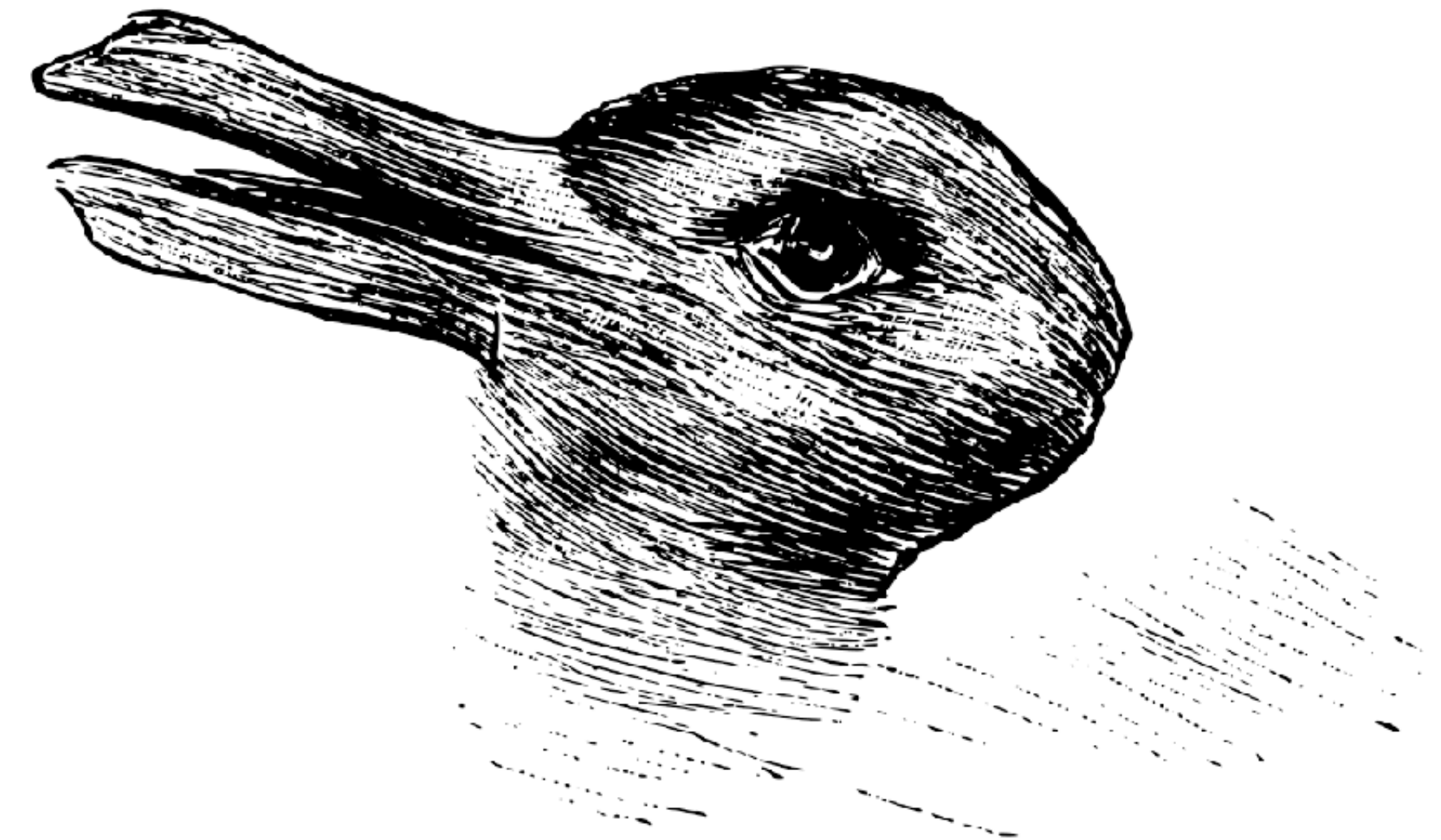
Another reason for multilevel models is that they adaptively regularize

Complete pooling: Treat all clusters as identical => underfitting

No pooling: Treat all clusters as unrelated => overfitting

Partial pooling: Adaptive compromise

Welche Tiere gleichen ein-
ander am meisten?



Kaninchen und Ente.

Reedfrogs in peril

data(reedfrogs)

48 groups (“tanks”) of tadpoles

Treatments: density, size, predation

Outcome: survival



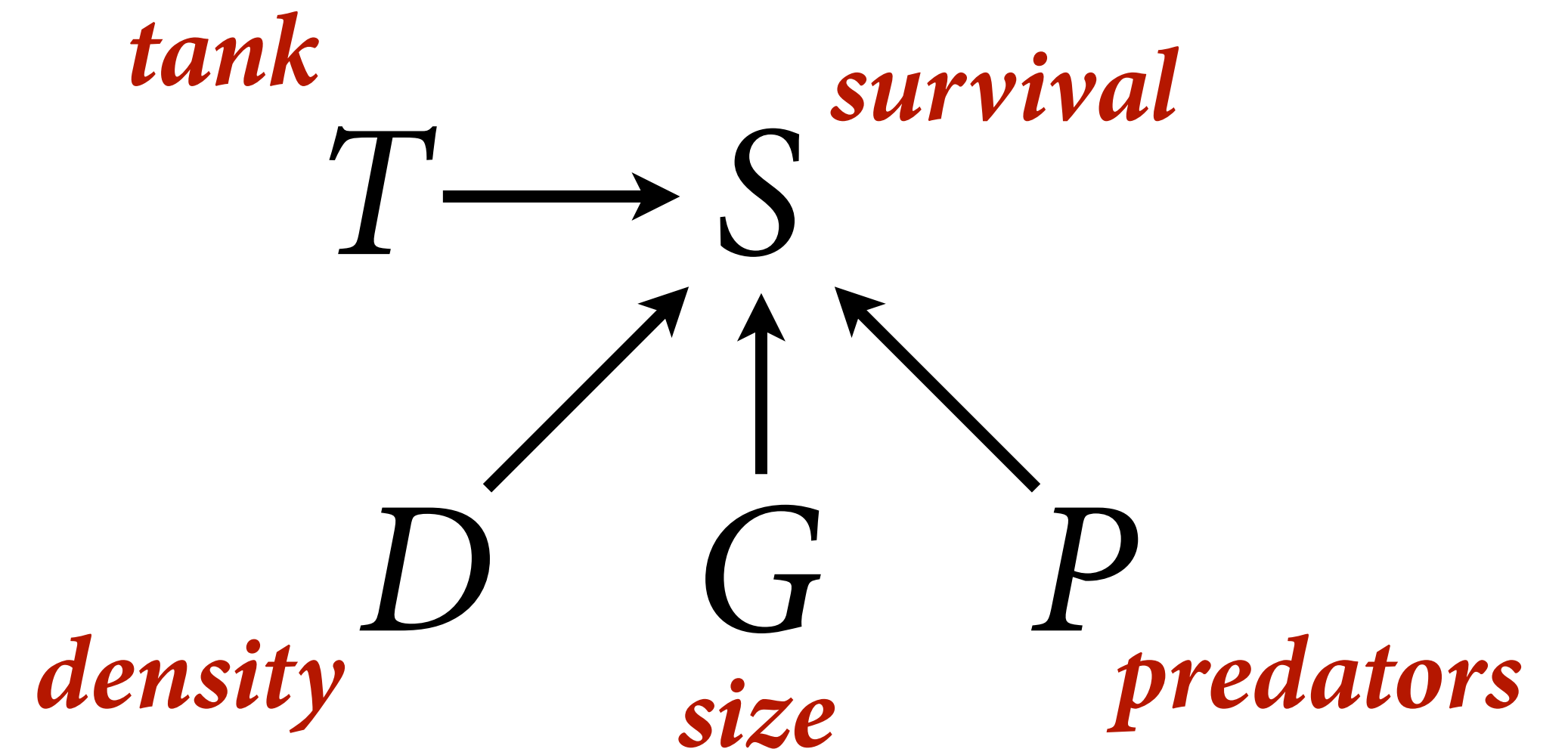
Reedfrogs in peril

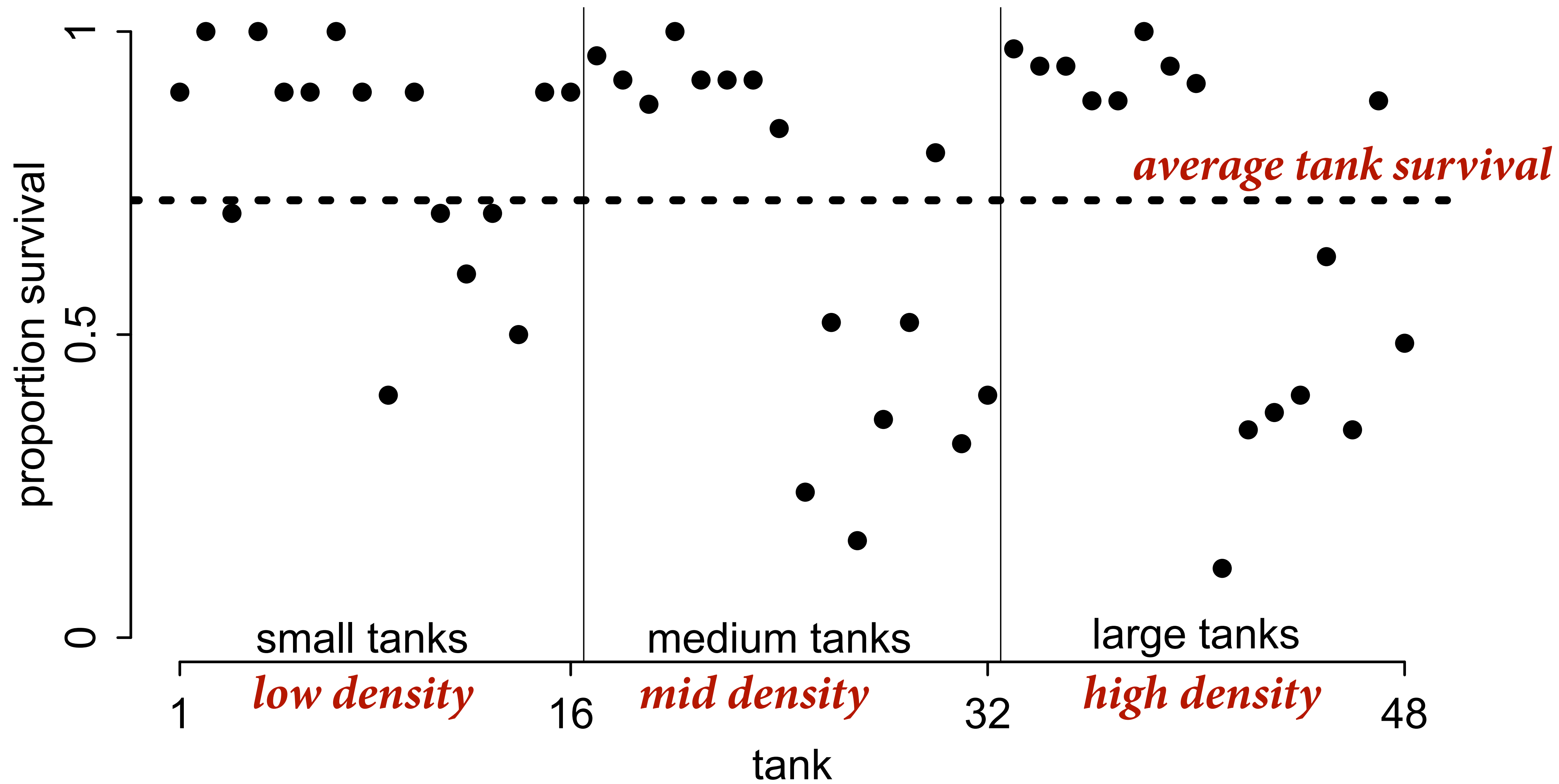
data(reedfrogs)

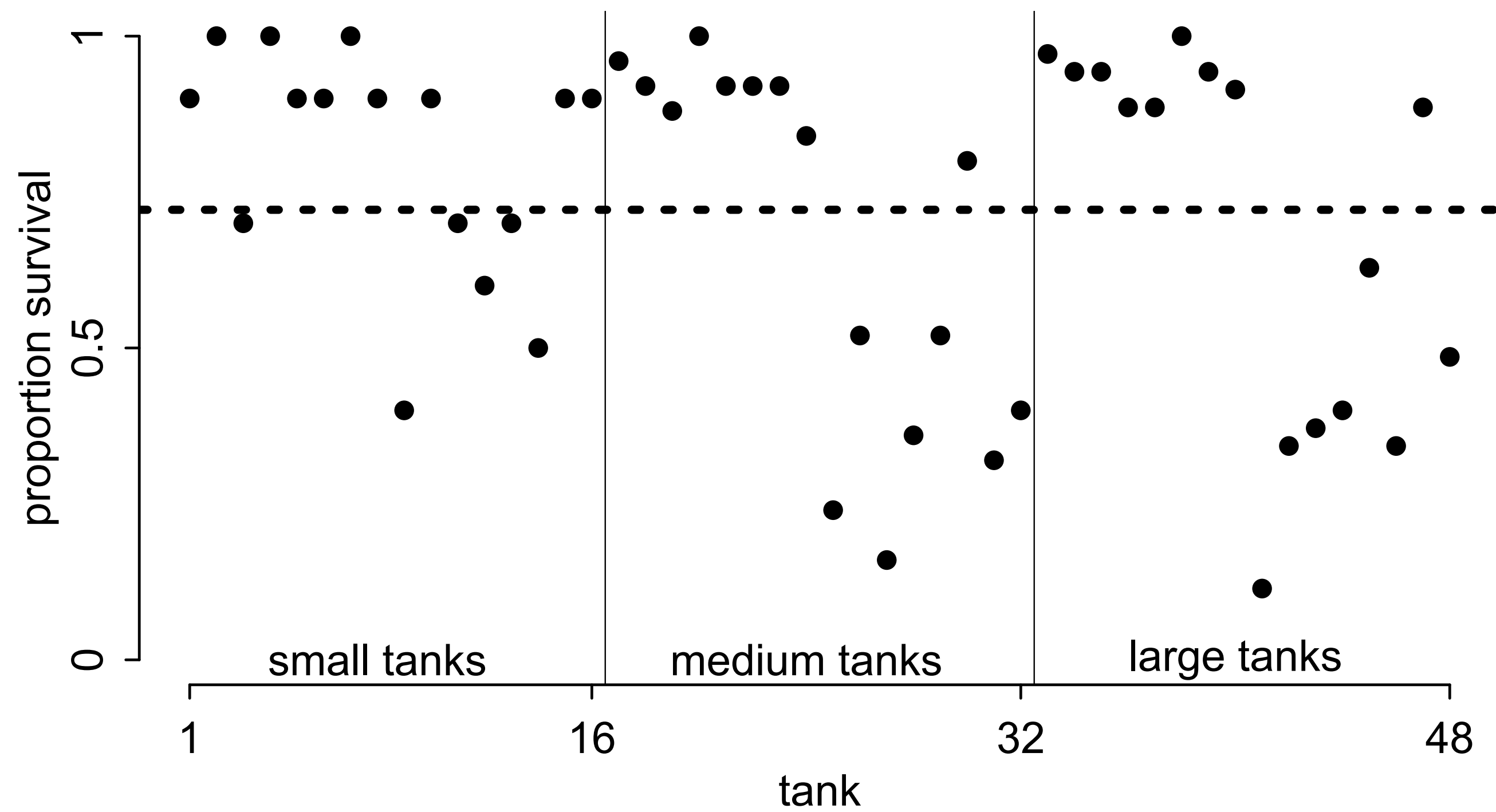
48 groups (“tanks”) of tadpoles

Treatments: density, size, predation

Outcome: survival





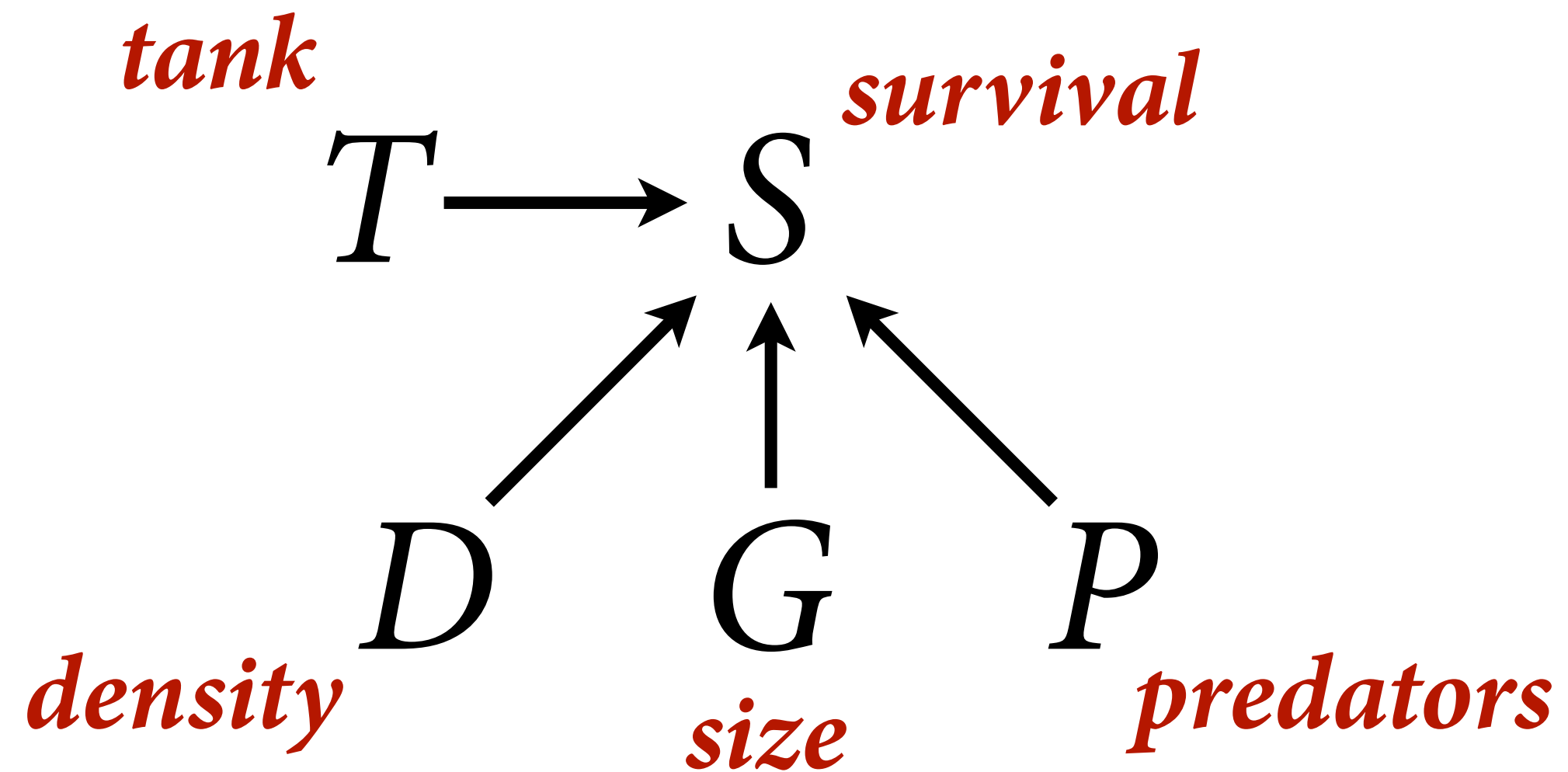


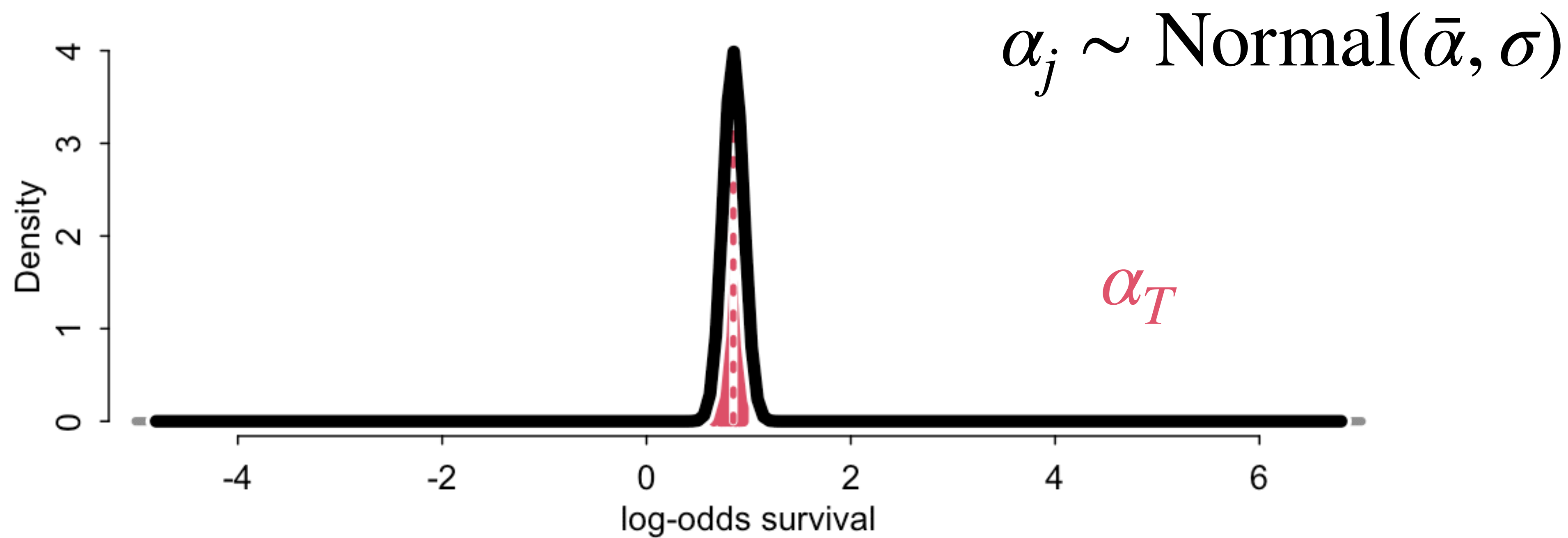
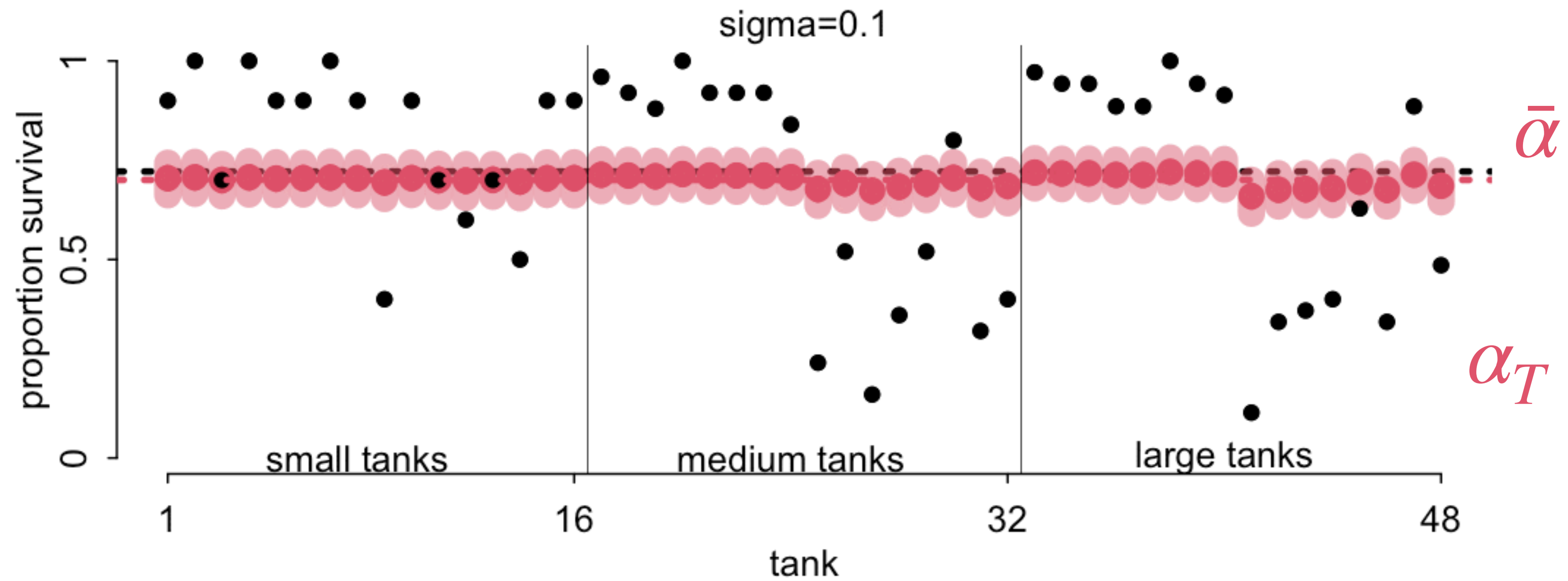
$$S_i \sim \text{Binomial}(D_i, p_i)$$

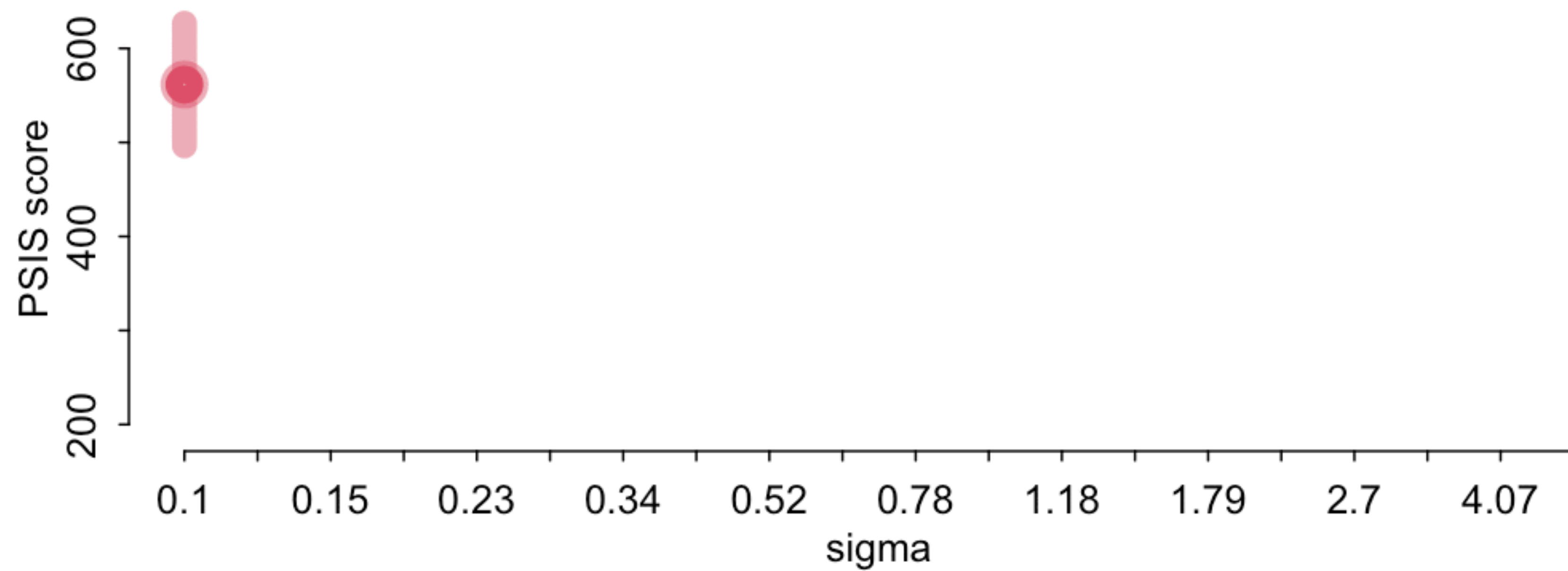
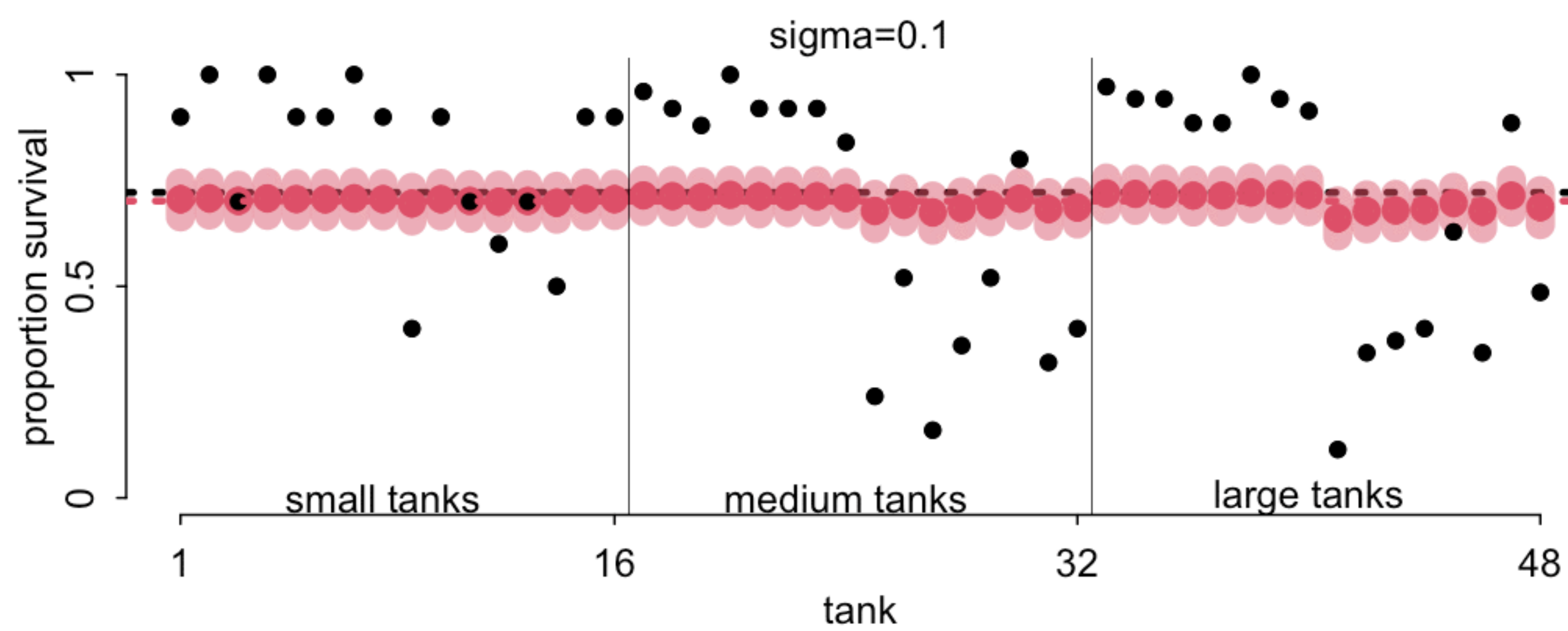
$$\text{logit}(p_i) = \alpha_{T[i]}$$

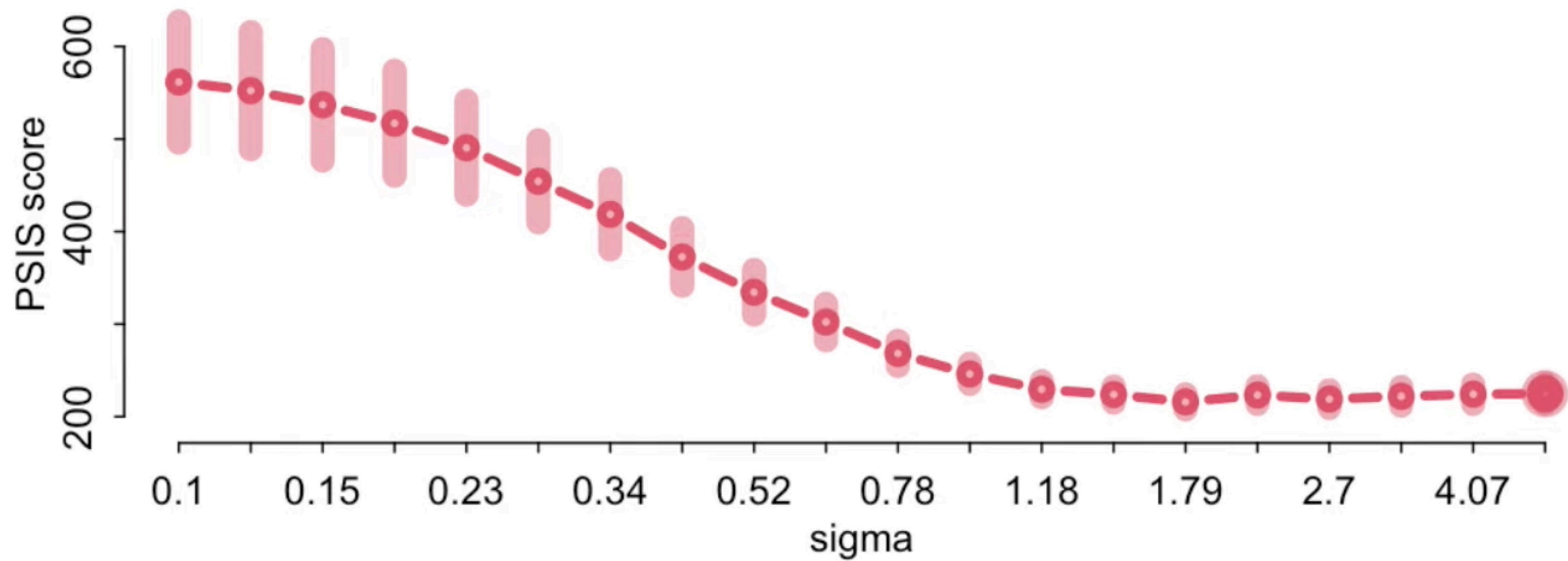
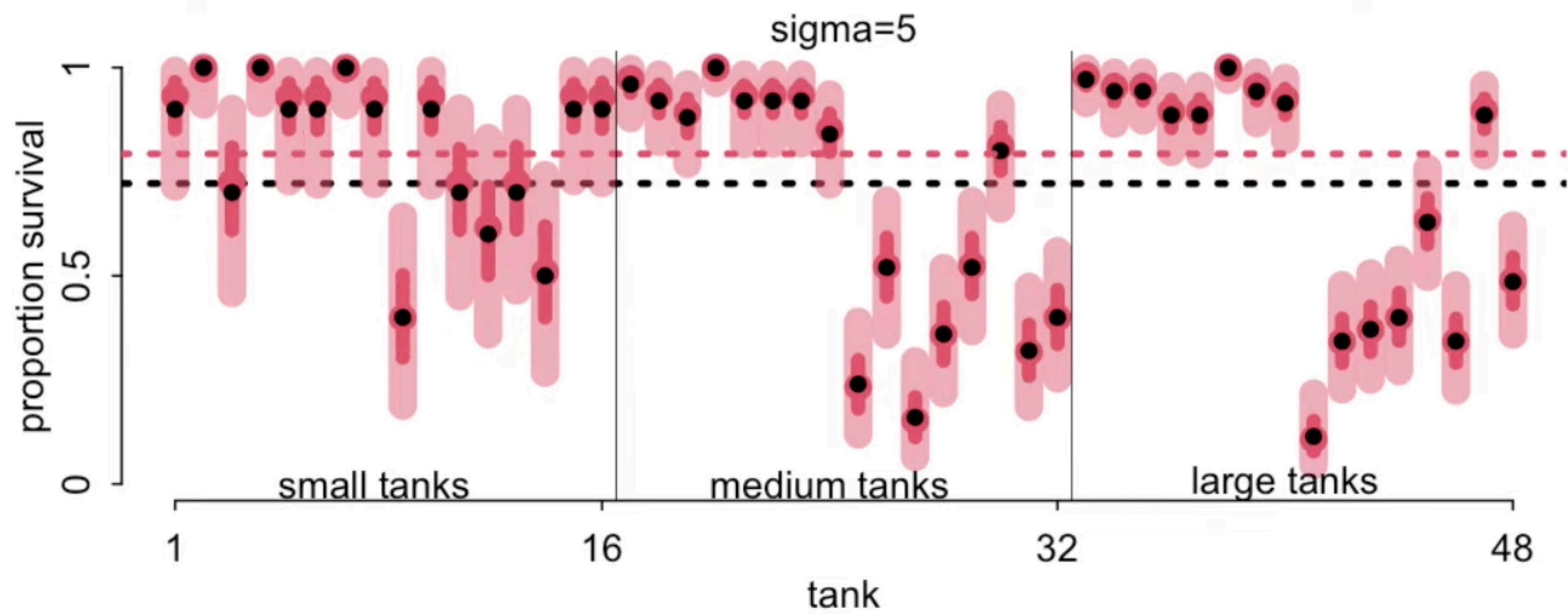
$$\alpha_j \sim \text{Normal}(\bar{\alpha}, ?)$$

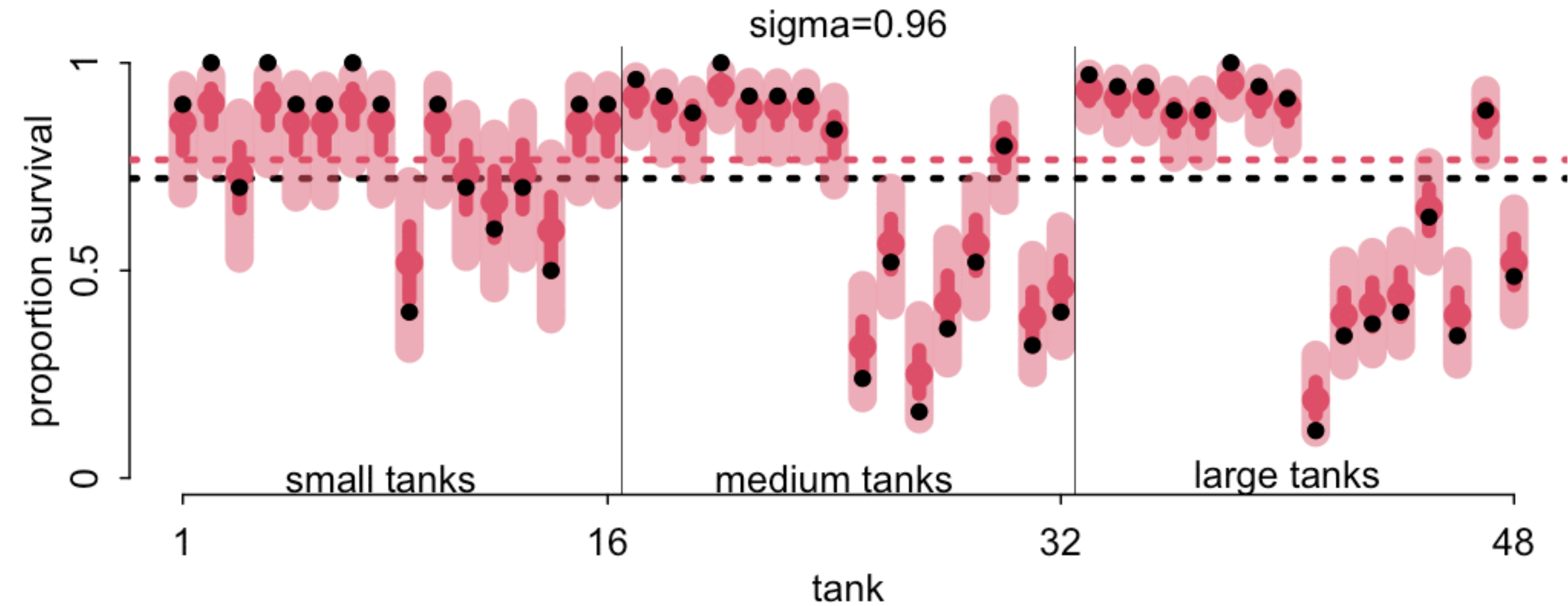
$$\bar{\alpha} \sim \text{Normal}(0, 1.5)$$







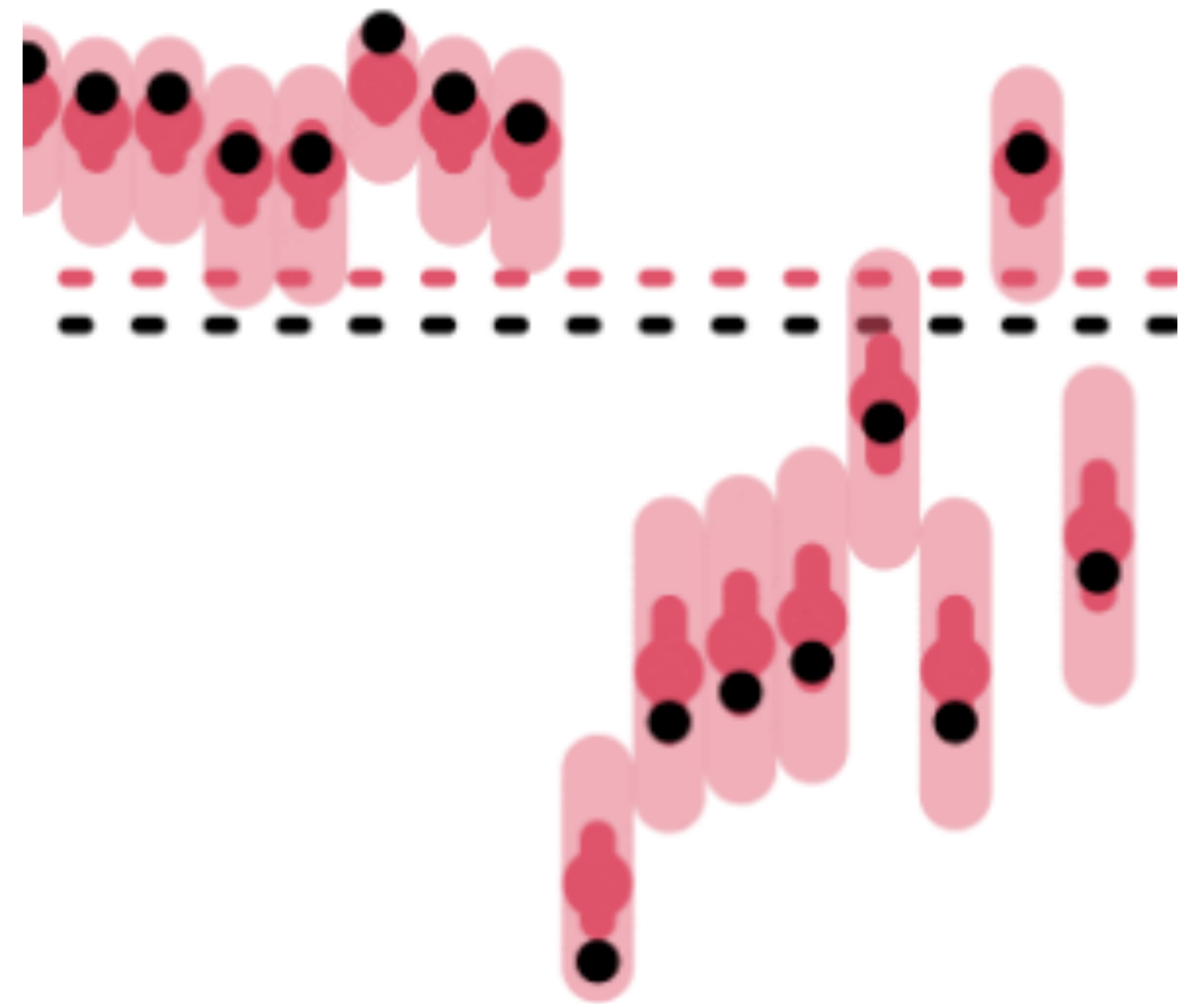


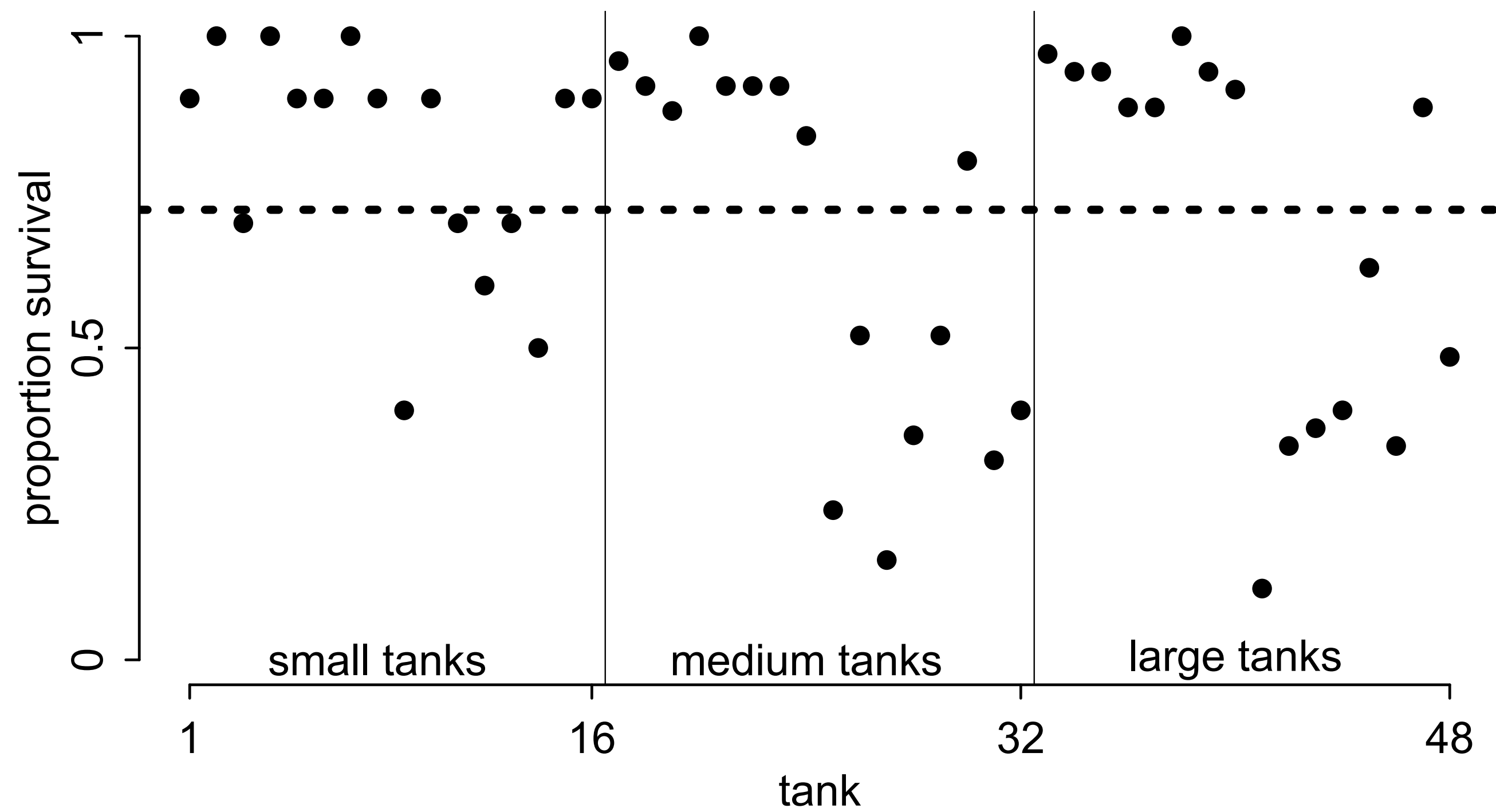


Automatic regularization

Wouldn't it be nice if we could find a good sigma without running so many models?

Maybe we could learn it from the data?





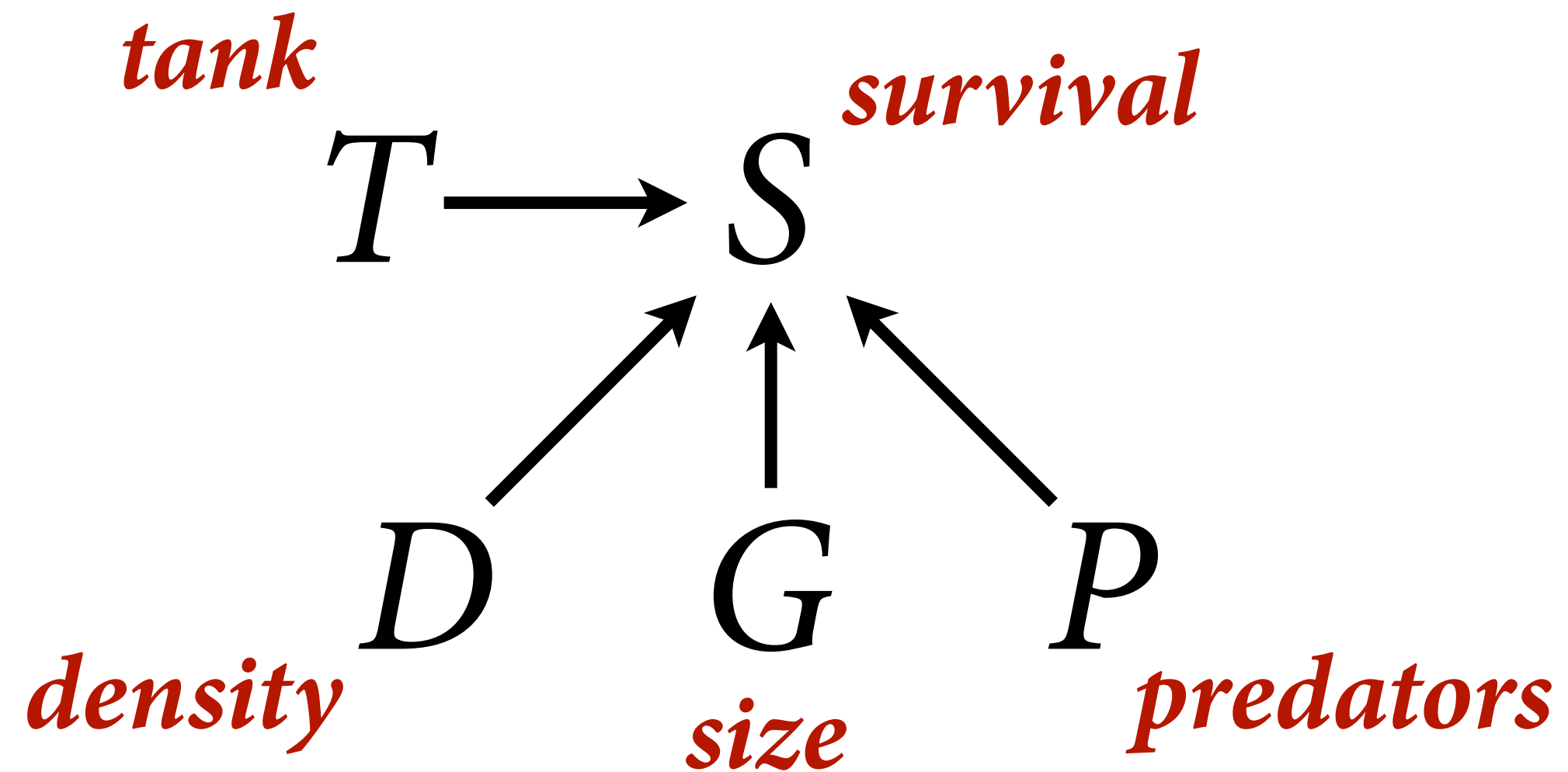
$$S_i \sim \text{Binomial}(D_i, p_i)$$

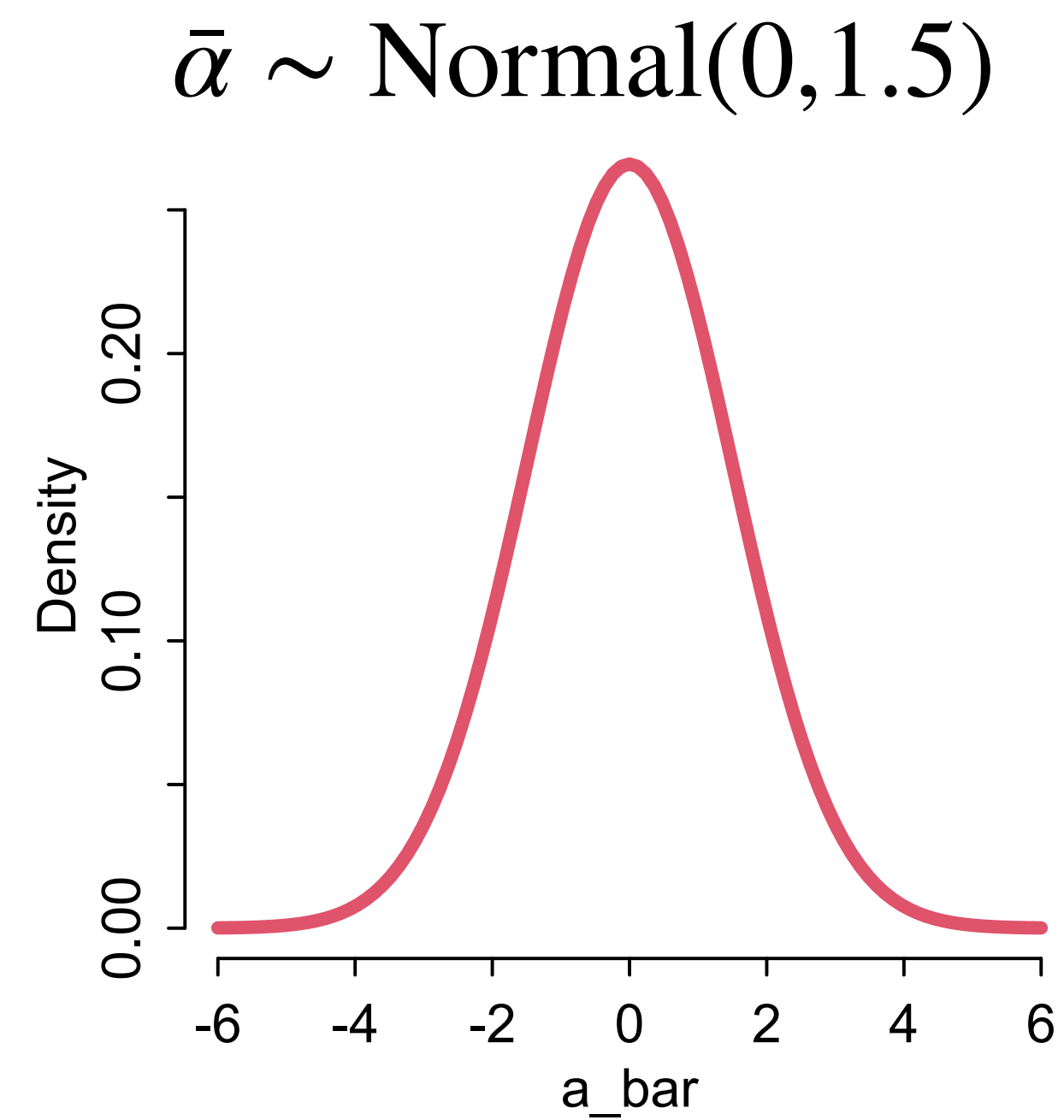
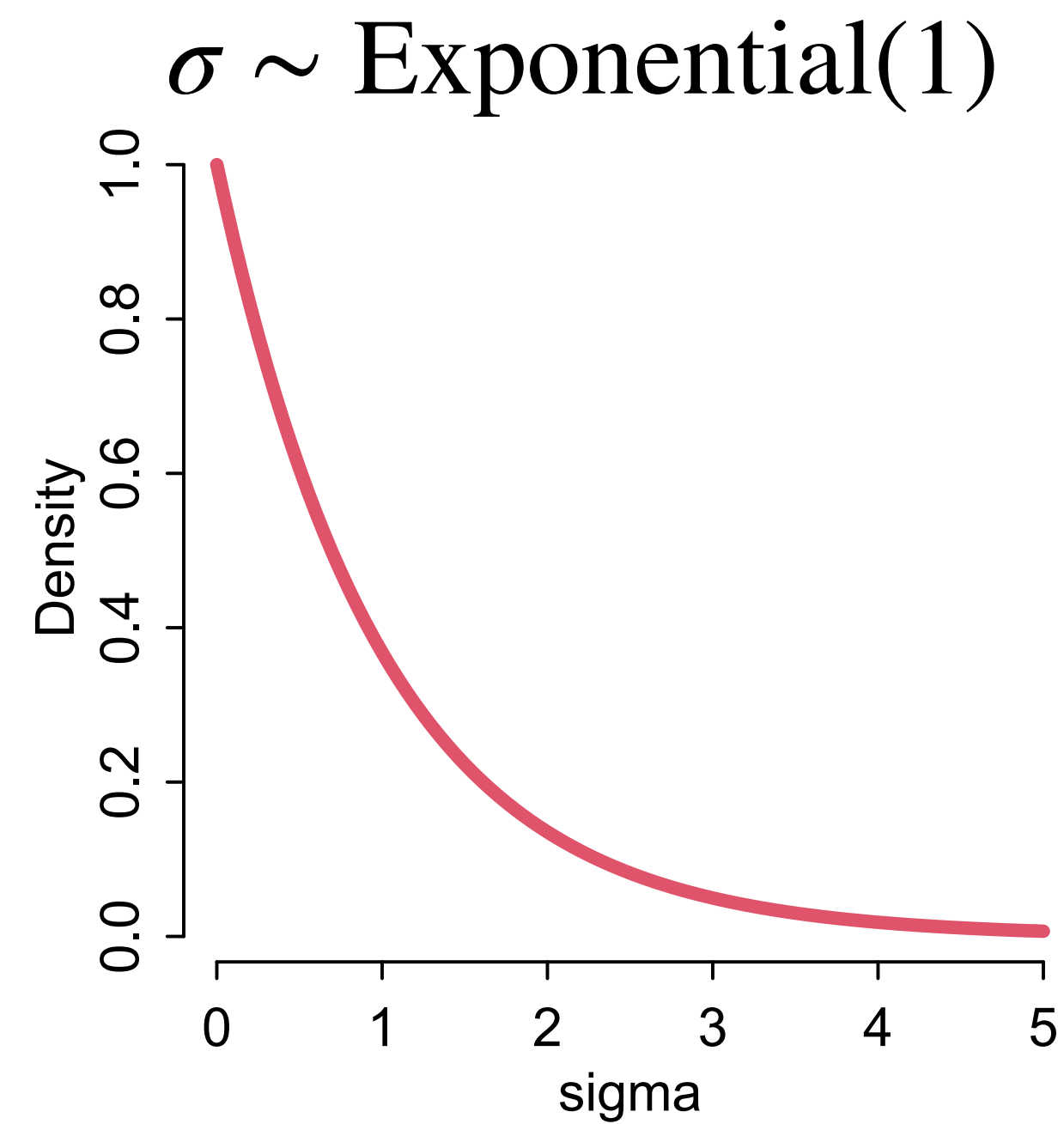
$$\text{logit}(p_i) = \alpha_{T[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\bar{\alpha} \sim \text{Normal}(0, 1.5)$$

$$\sigma \sim \text{Exponential}(1)$$





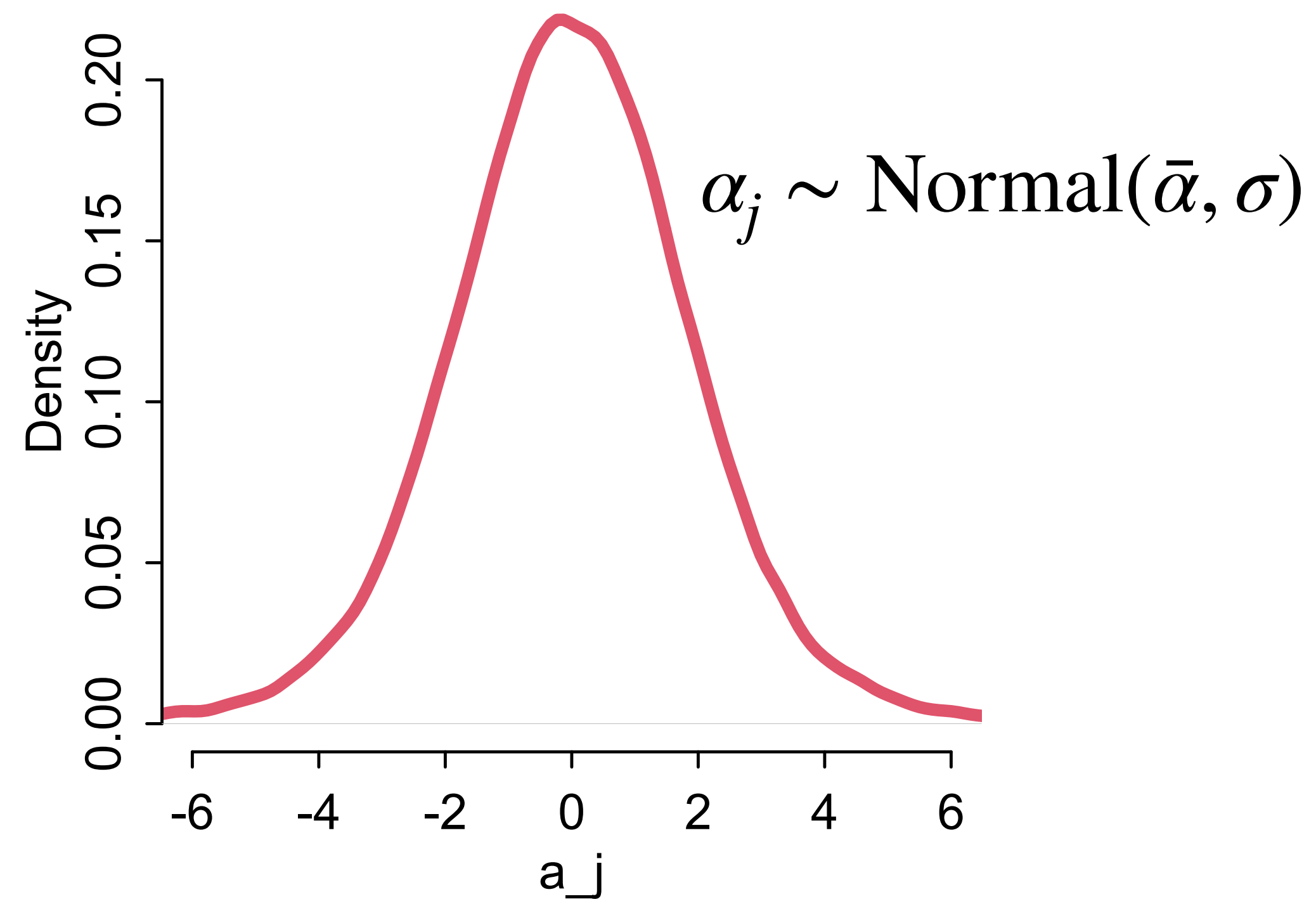
$$S_i \sim \text{Binomial}(D_i, p_i)$$

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```

library(rethinking)
data(reedfrogs)
d <- reedfrogs
d$tank <- 1:nrow(d)
dat <- list(
  S = d$surv,
  D = d$density,
  T = d$tank )

mST <- ulam(
  alist(

    S ~ dbinom( D , p ) ,
    logit(p) <- a[T] ,
    a[T] ~ dnorm( a_bar , sigma ) ,
    a_bar ~ dnorm( 0 , 1.5 ) ,
    sigma ~ dexp( 1 )

  ), data=dat , chains=4 , log_lik=TRUE )

```

$$S_i \sim \text{Binomial}(D_i, p_i)$$

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    sigma ~ dexp( 1 )

  ), data=dat , chains=4 , log_lik=TRUE )

```

```

> precis(mST,depth=2)
      mean  sd  5.5% 94.5% n_eff Rhat4
a[1]  2.13 0.85  0.89  3.54 2992    1
a[2]  3.06 1.04  1.57  4.82 2716    1
a[3]  1.01 0.67 -0.01  2.11 5635    1
a[4]  3.08 1.07  1.53  4.88 2441    1
a[5]  2.14 0.87  0.85  3.61 3460    1
a[6]  2.11 0.85  0.88  3.61 3628    1
a[7]  3.05 1.08  1.54  4.90 3603    1
a[8]  2.14 0.89  0.83  3.69 3190    1
a[9] -0.17 0.64 -1.20  0.88 5424    1
a[10] 2.15 0.90  0.83  3.72 2559    1
a[11] 1.00 0.66 -0.03  2.09 3265    1
a[12] 0.57 0.63 -0.44  1.60 6602    1
a[13] 1.01 0.67 -0.02  2.13 3618    1
a[14] 0.21 0.62 -0.75  1.21 4147    1
a[15] 2.10 0.85  0.84  3.51 4563    1
a[16] 2.12 0.85  0.89  3.58 3030    1
a[17] 2.88 0.77  1.82  4.22 3888    1
a[18] 2.38 0.65  1.42  3.46 3645    1
a[19] 2.01 0.58  1.16  2.95 4029    1
a[20] 3.65 1.04  2.17  5.47 2750    1
a[21] 2.39 0.65  1.43  3.47 3585    1
a[22] 2.39 0.66  1.41  3.51 3607    1
a[23] 2.40 0.66  1.45  3.49 3312    1
a[24] 1.71 0.53  0.92  2.58 3395    1
a[25] -0.99 0.43 -1.69 -0.32 3187    1
a[26]  0.16 0.39 -0.47  0.80 4611    1
a[27] -1.43 0.49 -2.23 -0.69 3289    1
a[28] -0.47 0.41 -1.15  0.15 5525    1
a[29]  0.17 0.40 -0.46  0.82 5628    1
a[30]  1.44 0.50  0.68  2.26 4925    1
a[31] -0.62 0.41 -1.29  0.03 5449    1
a[32] -0.30 0.39 -0.95  0.32 4039    1
a[33]  3.19 0.80  2.06  4.60 2357    1
a[34]  2.71 0.63  1.79  3.80 3117    1
a[35]  2.71 0.62  1.82  3.71 3185    1
a[36]  2.07 0.53  1.28  2.95 3616    1
a[37]  2.05 0.47  1.34  2.82 4705    1
a[38]  3.87 0.91  2.56  5.42 3022    1
a[39]  2.71 0.66  1.76  3.80 3479    1
a[40]  2.37 0.59  1.47  3.38 2981    1
a[41] -1.79 0.45 -2.54 -1.13 4340    1
a[42] -0.58 0.35 -1.14  0.00 4354    1
a[43] -0.45 0.36 -1.01  0.10 5360    1
a[44] -0.33 0.34 -0.89  0.22 4541    1
a[45]  0.58 0.35  0.02  1.15 5803    1
a[46] -0.56 0.37 -1.14  0.02 4696    1
a[47]  2.06 0.51  1.27  2.91 3955    1
a[48]  0.01 0.35 -0.55  0.56 5353    1
a_bar 1.35 0.26  0.93  1.78 2746    1
sigma 1.61 0.21  1.32  1.96 1557    1

```

~ Binomial(D_i, p_i)

= $\alpha_{T[i]}$

~ Normal($\bar{\alpha}, \sigma$)

~ Normal(0,1.5)

~ Exponential(1)

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  T = d$tank )
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  alist(
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```
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    a[T] ~ dnorm( a_bar , sigma ) ,
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```

```
      mean   sd  5.5% 94.5% n_eff Rhat4
a_bar  1.35 0.26  0.93  1.78 2746    1
sigma  1.61 0.21  1.32  1.96 1557    1
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```
a[29]  0.17 0.40 -0.46  0.82 5628    1
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a[45]  0.58 0.35  0.02  1.15 5803    1
a[46] -0.56 0.37 -1.14  0.02 4696    1
a[47]  2.06 0.51  1.27  2.91 3955    1
a[48]  0.01 0.35 -0.55  0.56 5353    1
a_bar  1.35 0.26  0.93  1.78 2746    1
sigma  1.61 0.21  1.32  1.96 1557    1
```

~ Binomial(D_i, p_i)

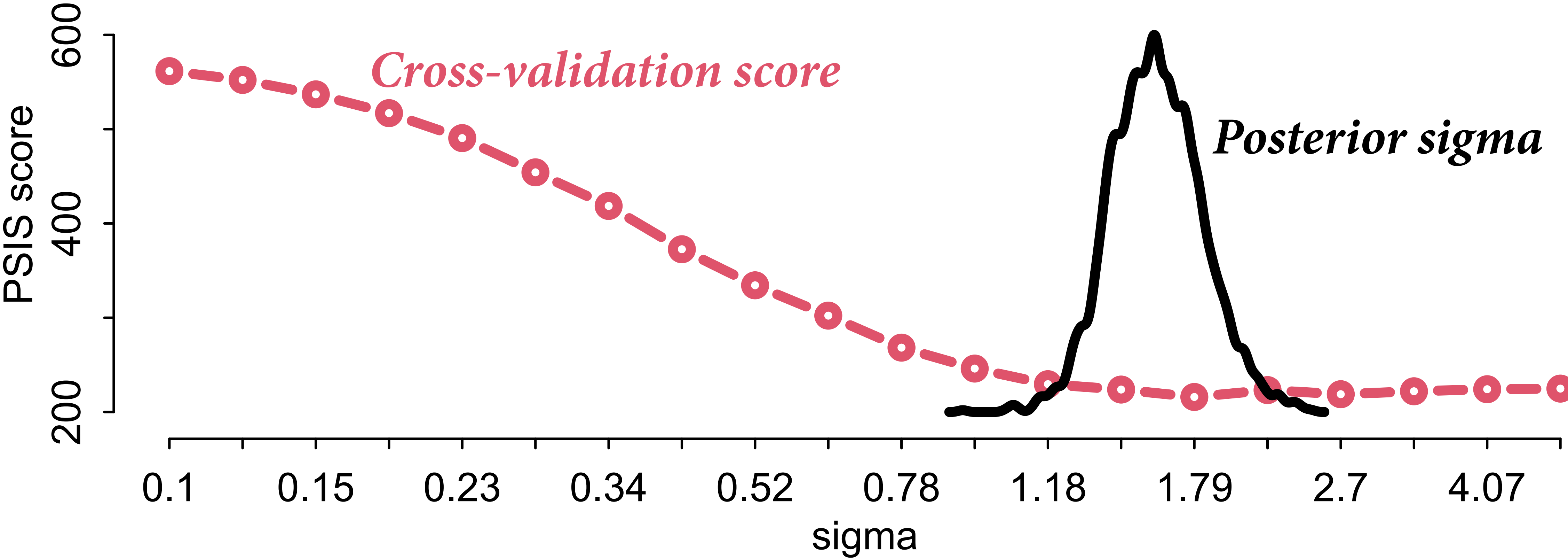
= $\alpha_{T[i]}$

~ Normal($\bar{\alpha}, \sigma$)

~ Normal(0,1.5)

~ Exponential(1)

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a_bar	1.35	0.26	0.93	1.78	2746	1
sigma	1.61	0.21	1.32	1.96	1557	1



```

mST <- ulam(
  alist(
    S ~ dbinom( D , p ) ,
    logit(p) <- a[T] ,
    a[T] ~ dnorm( a_bar , sigma ) ,
    a_bar ~ dnorm( 0 , 1.5 ) ,
    sigma ~ dexp( 1 )
  ), data=dat , chains=4 , log_lik=TRUE )

```

$$S_i \sim \text{Binomial}(D_i, p_i)$$

$$\text{logit}(p_i) = \alpha_{T[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\bar{\alpha} \sim \text{Normal}(0, 1.5)$$

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```

mSTnomem <- ulam(
  alist(
    S ~ dbinom( D , p ) ,
    logit(p) <- a[T] ,
    a[T] ~ dnorm( a_bar , 1 ) ,
    a_bar ~ dnorm( 0 , 1.5 )
  ), data=dat , chains=4 , log_lik=TRUE )

compare( mST , mSTnomem , func=WAIC )

```

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    sigma ~ dexp( 1 )
```

```
), data=dat , chains=4 )
> compare( mST , mSTnomem , func=WAIC )
```

	WAIC	SE	dWAIC	dSE	pWAIC	weight
mST	200.6	7.52	0.0	NA	21.1	1
mSTnomem	217.4	7.80	16.8	4.35	25.6	0

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mSTnomem <- ulam(
  alist(
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compare( mST , mSTnomem , func=WAIC )
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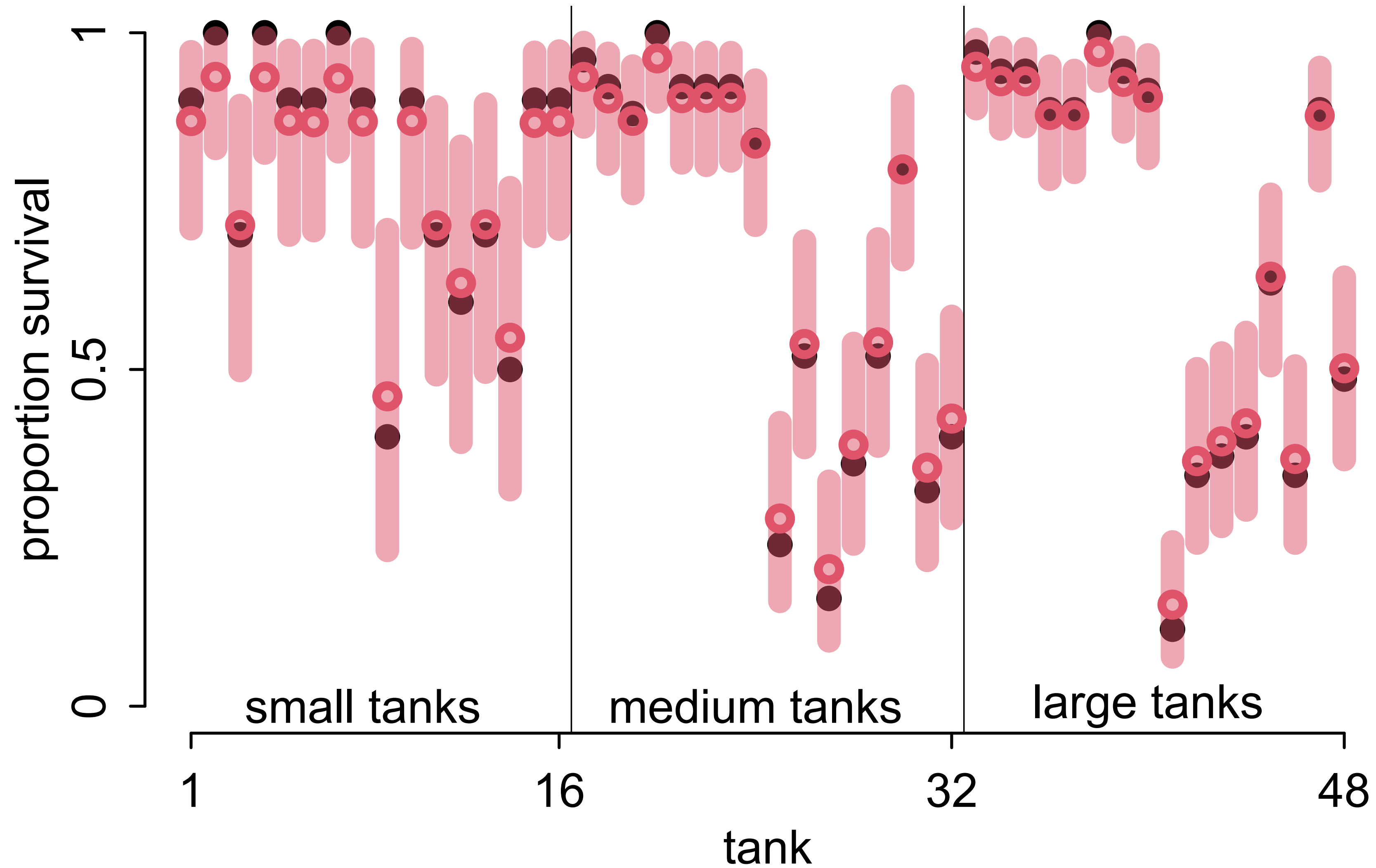
Adding parameters can reduce overfitting

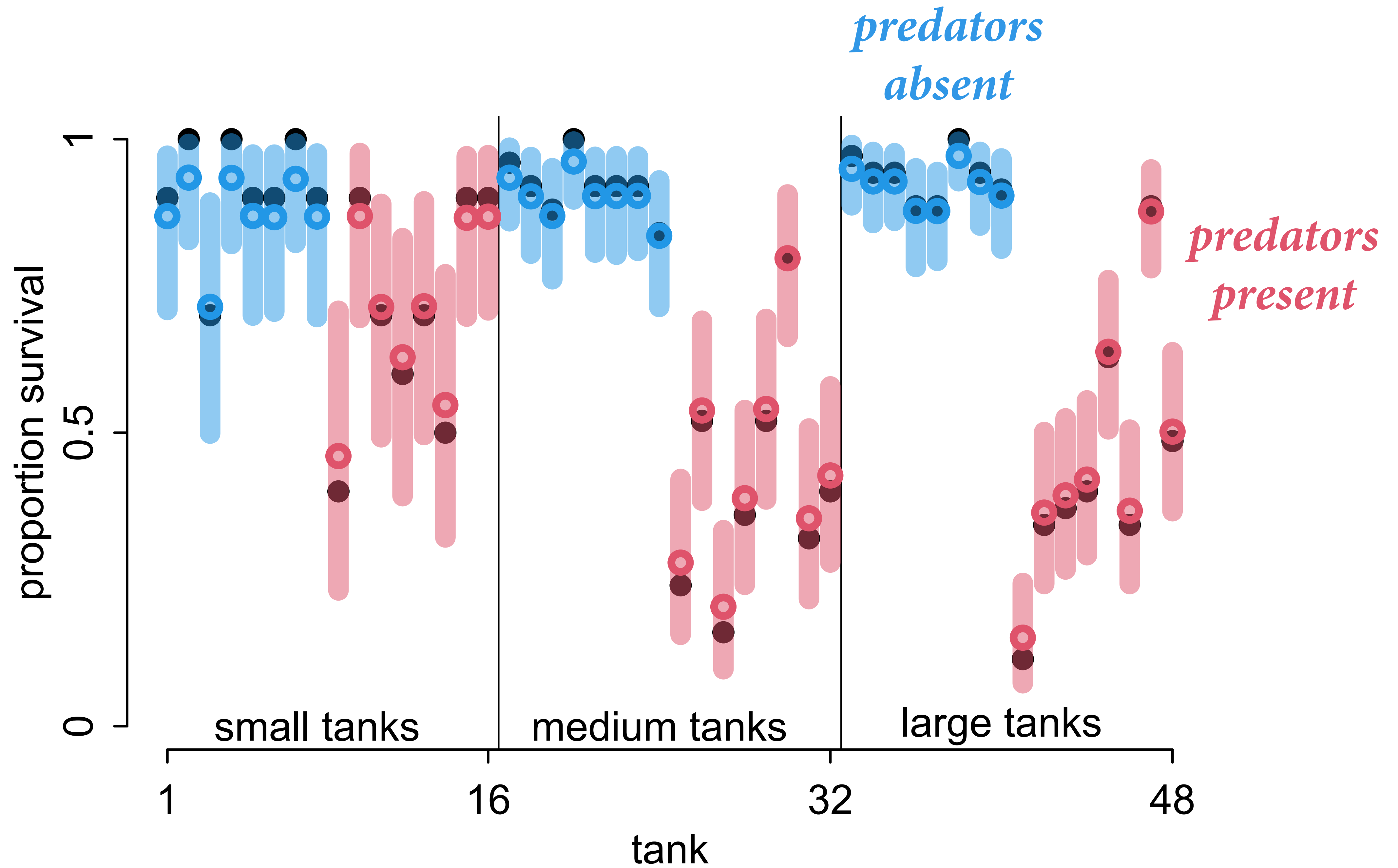
What matters is structure, not number


```
compare( mST , mSTnomem , func=WAIC )
```

*less evidence,
more conservative
estimates*

*more evidence,
less conservative
estimates*





Stratify mean by predators

$$S_i \sim \text{Binomial}(D_i, p_i)$$

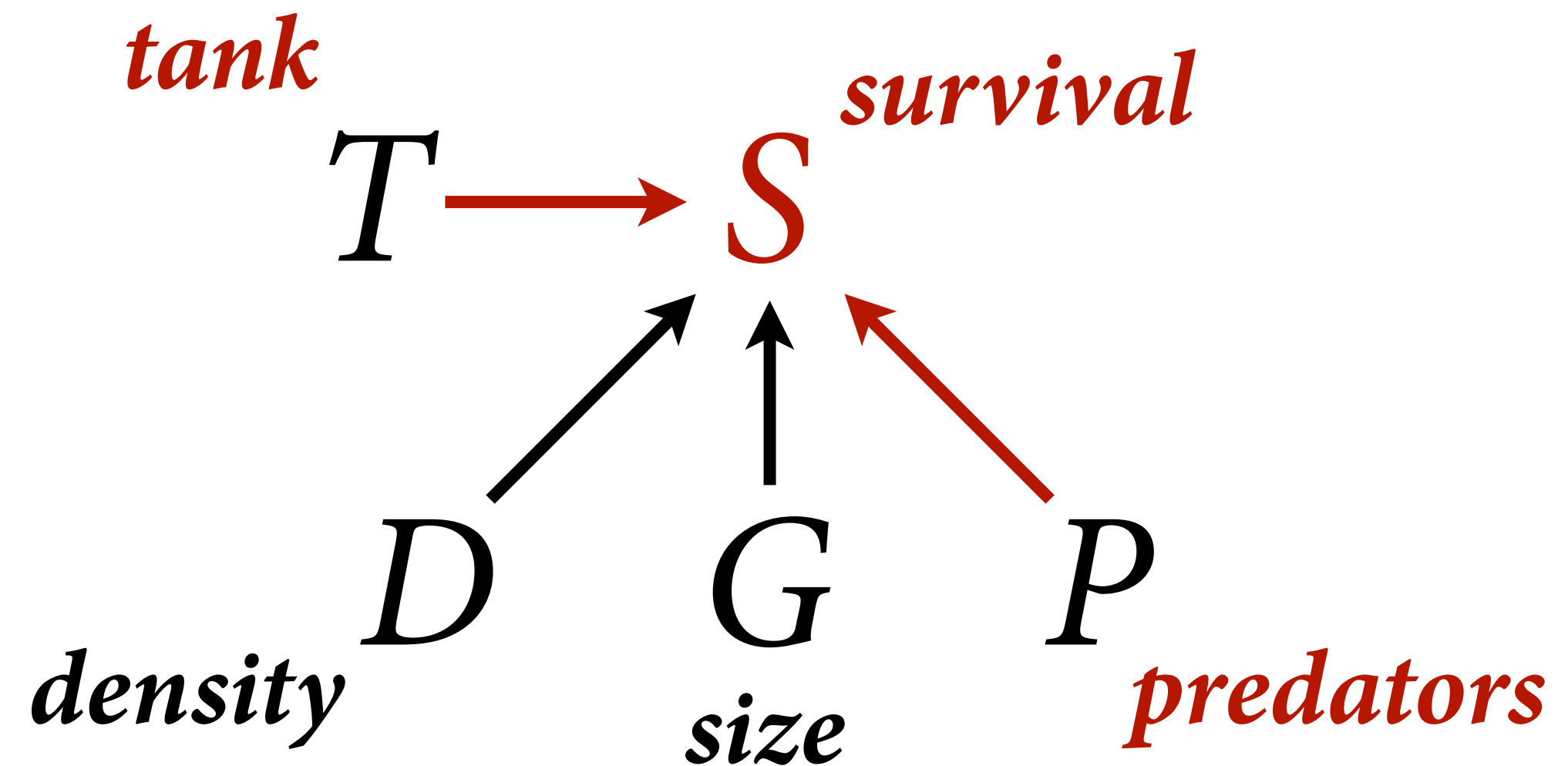
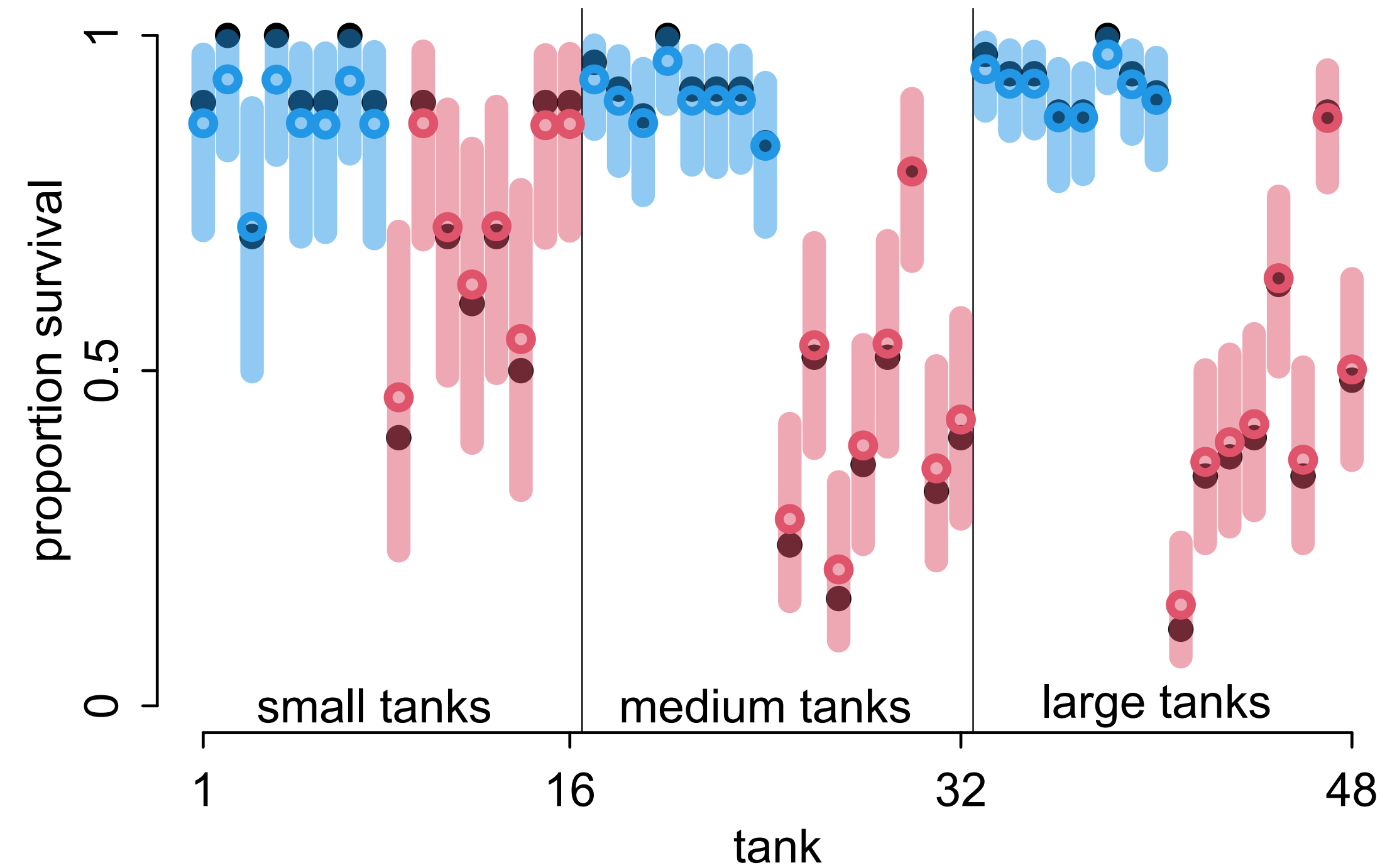
$$\text{logit}(p_i) = \alpha_{T[i]} + \beta_P P_i$$

$$\beta_P \sim \text{Normal}(0, 0.5)$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\bar{\alpha} \sim \text{Normal}(0, 1.5)$$

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$$\sigma \sim \text{Exponential}(1)$$

```
dat$P <- ifelse(d$pred=="pred",1,0)
mSTP <- ulam(
  alist(
    S ~ dbinom( D , p ) ,
    logit(p) <- a[T] + bP*P ,
    bP ~ dnorm( 0 , 0.5 ),
    a[T] ~ dnorm( a_bar , sigma ) ,
    a_bar ~ dnorm( 0 , 1.5 ) ,
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Stratify mean by predators

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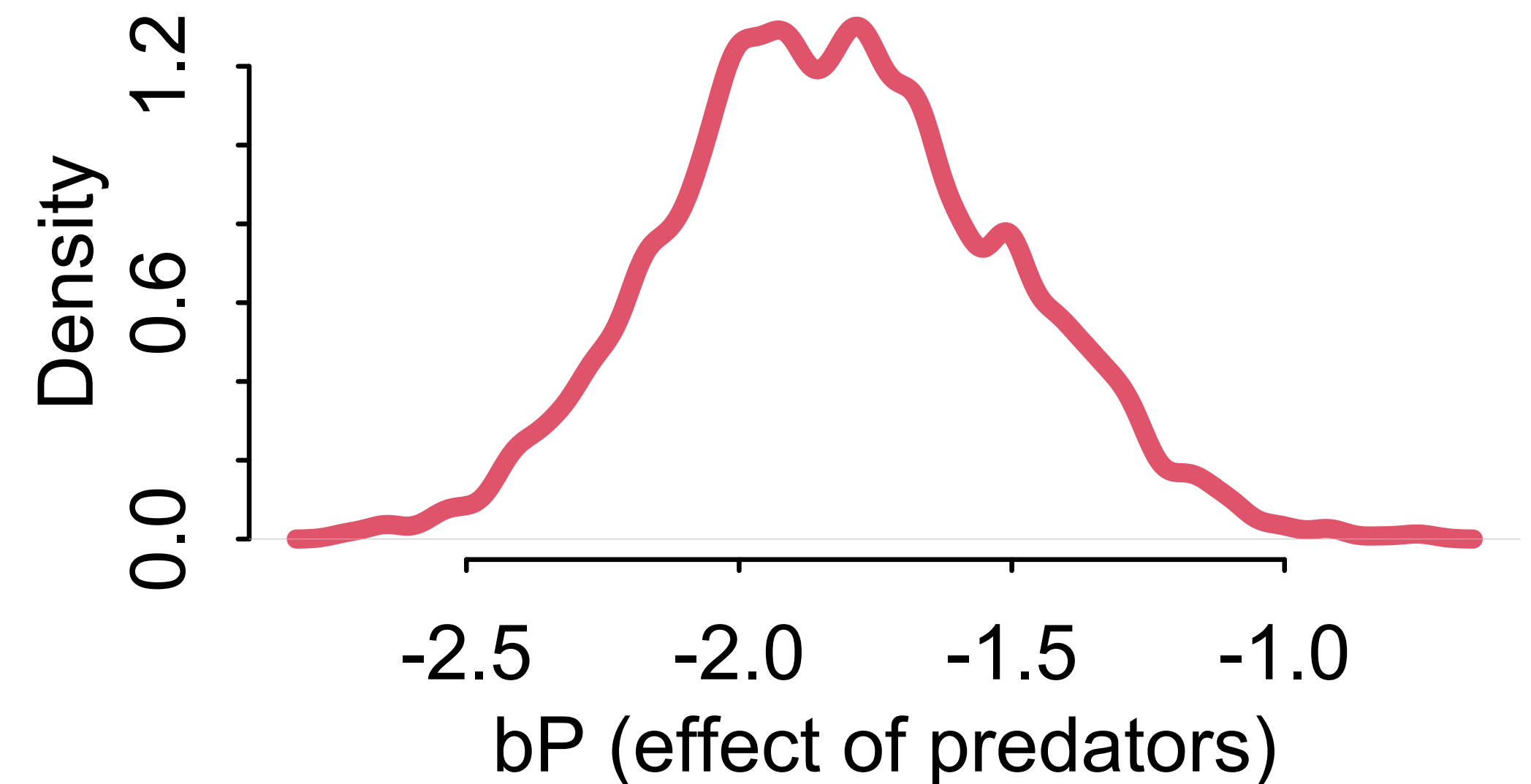
$$\beta_P \sim \text{Normal}(0, 0.5)$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

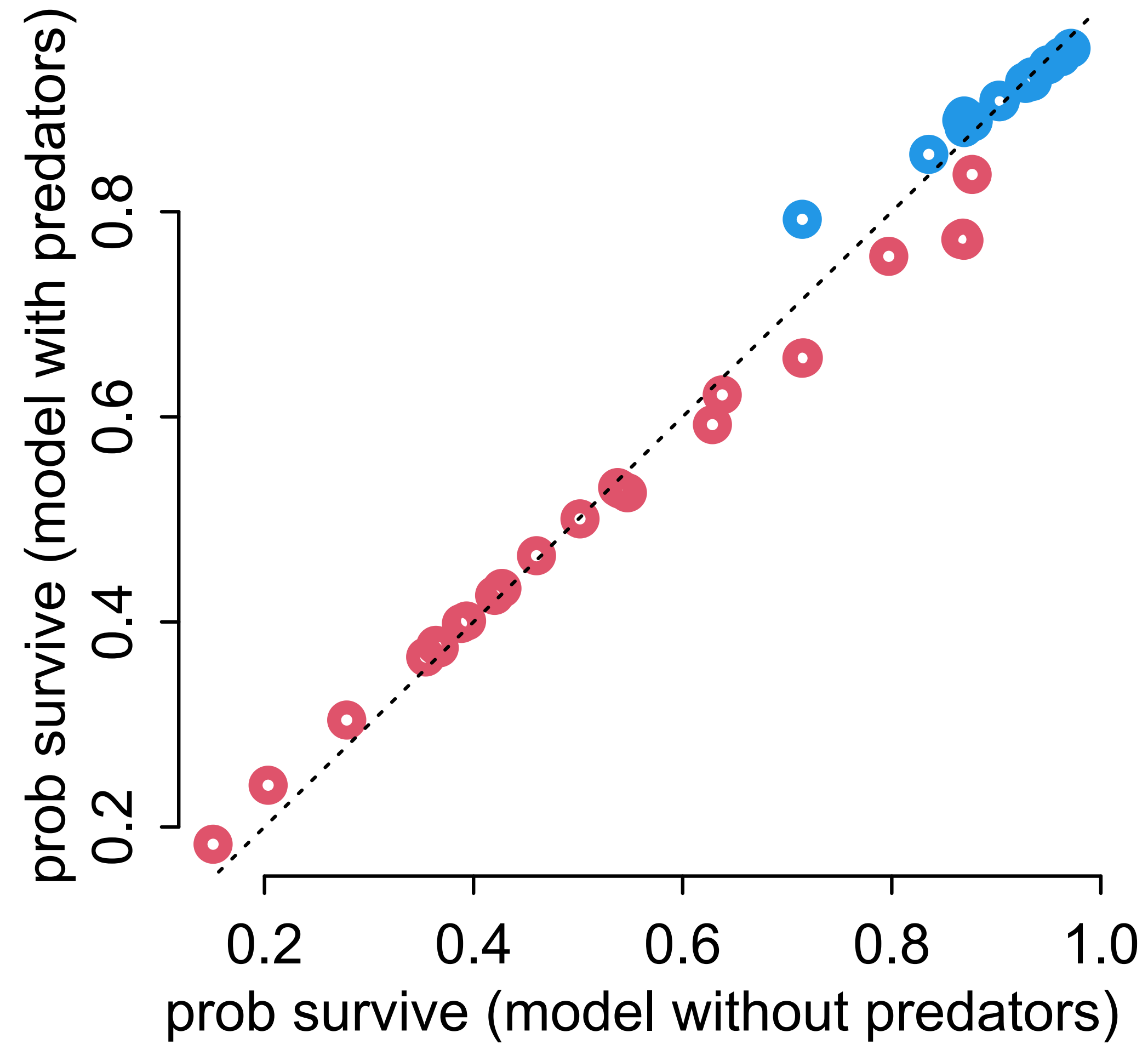
$$\bar{\alpha} \sim \text{Normal}(0, 1.5)$$

$$\sigma \sim \text{Exponential}(1)$$

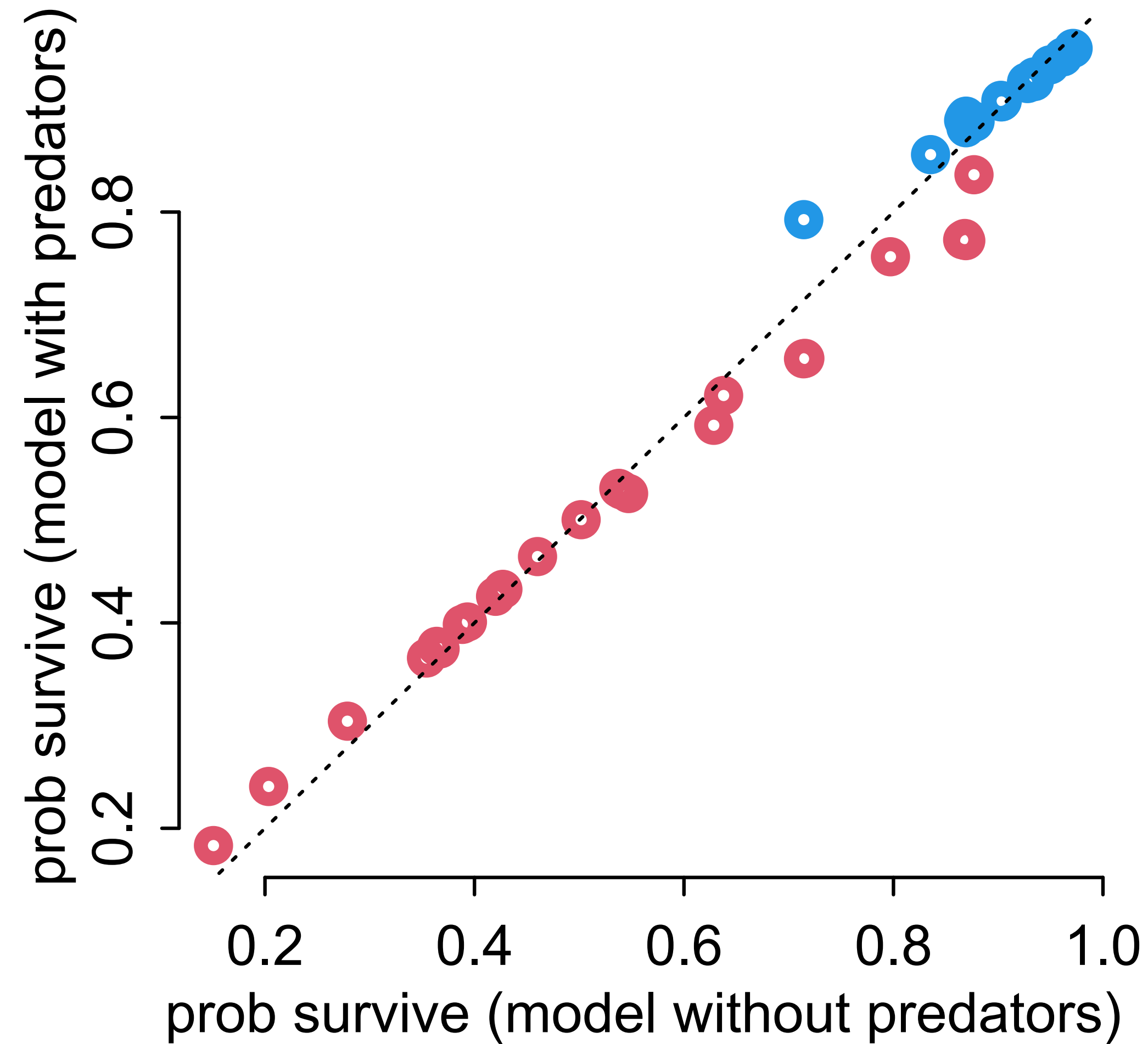
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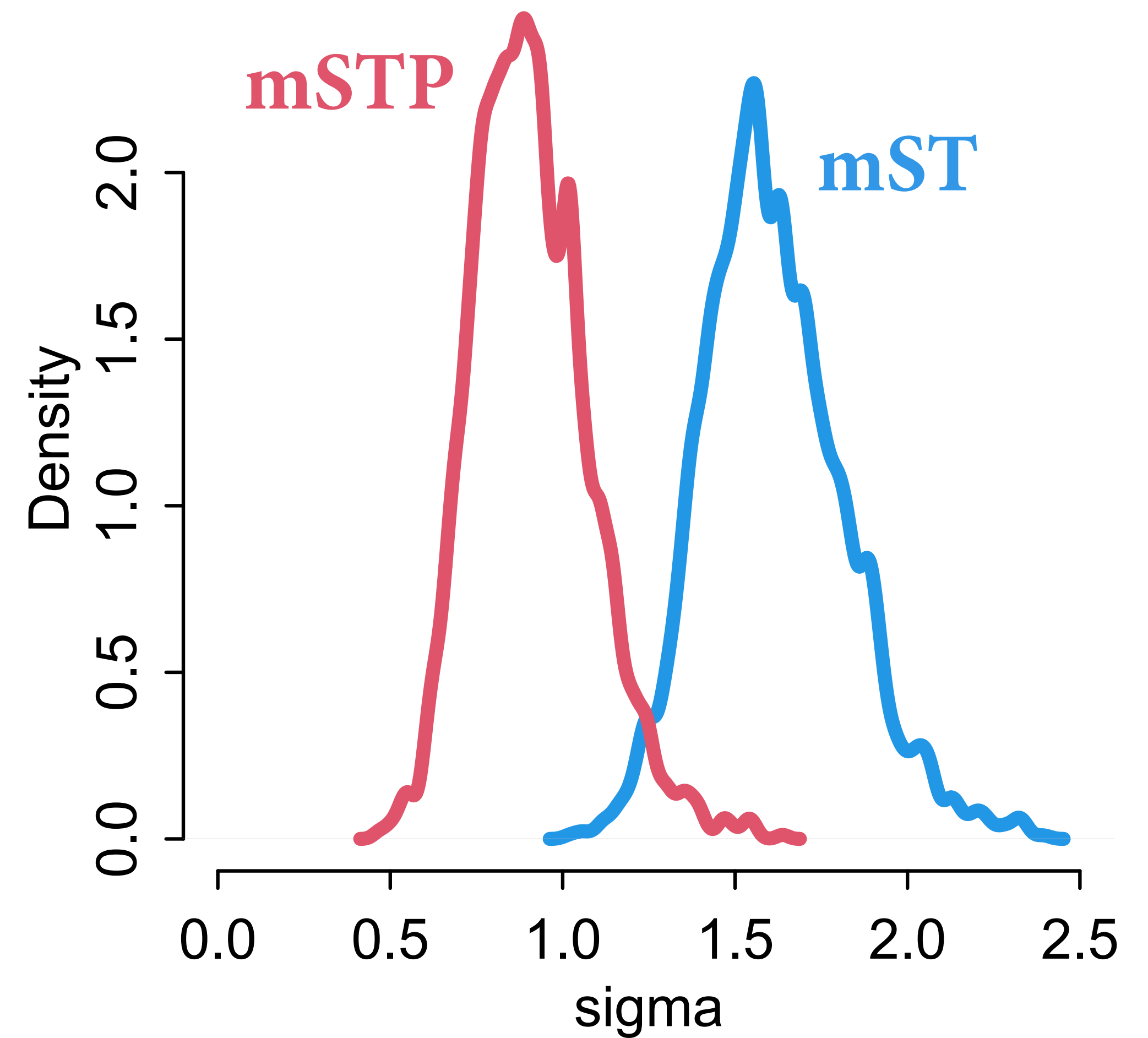
Extremely similar predictions



Extremely similar predictions



Very different sigma values



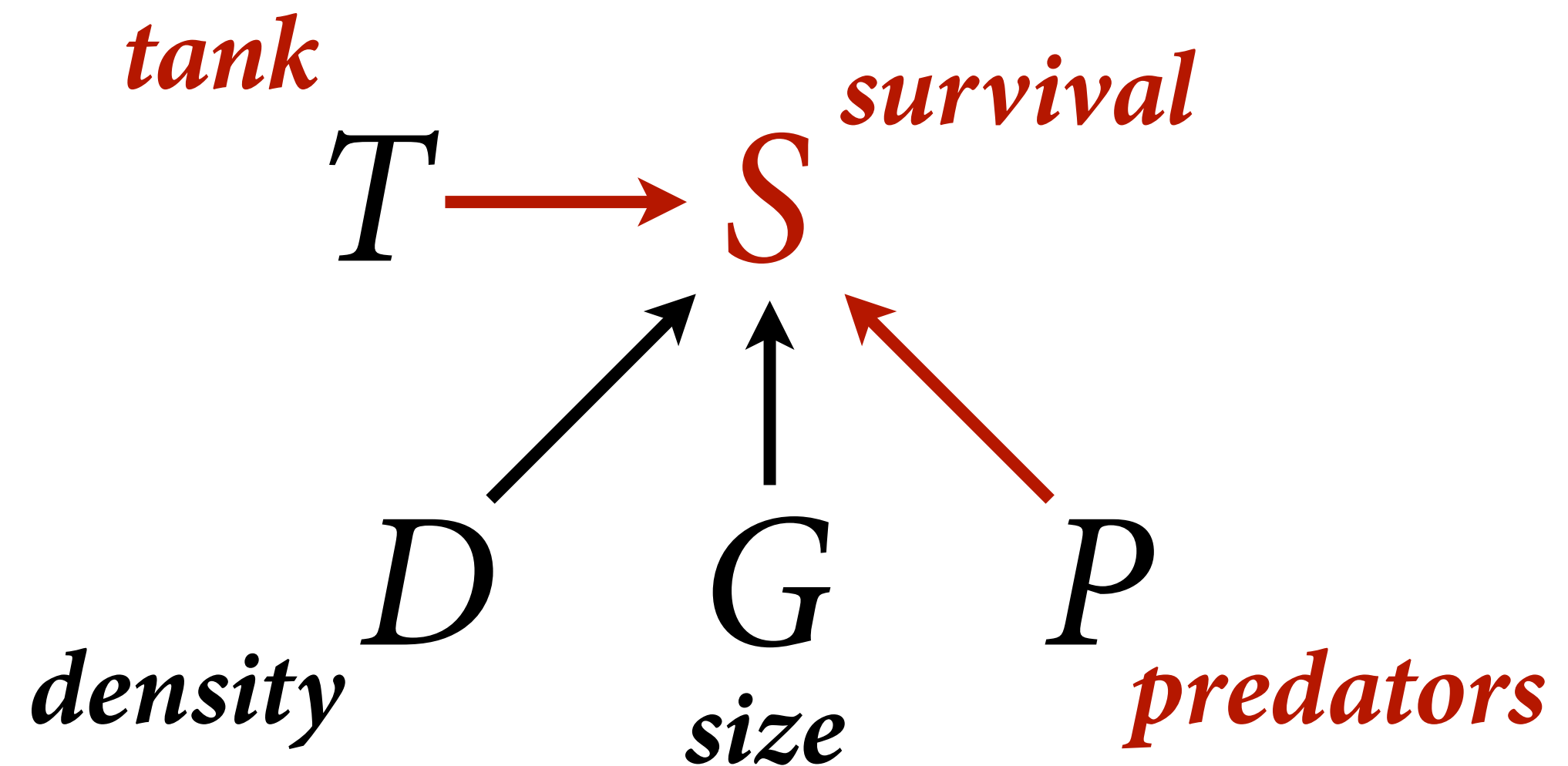
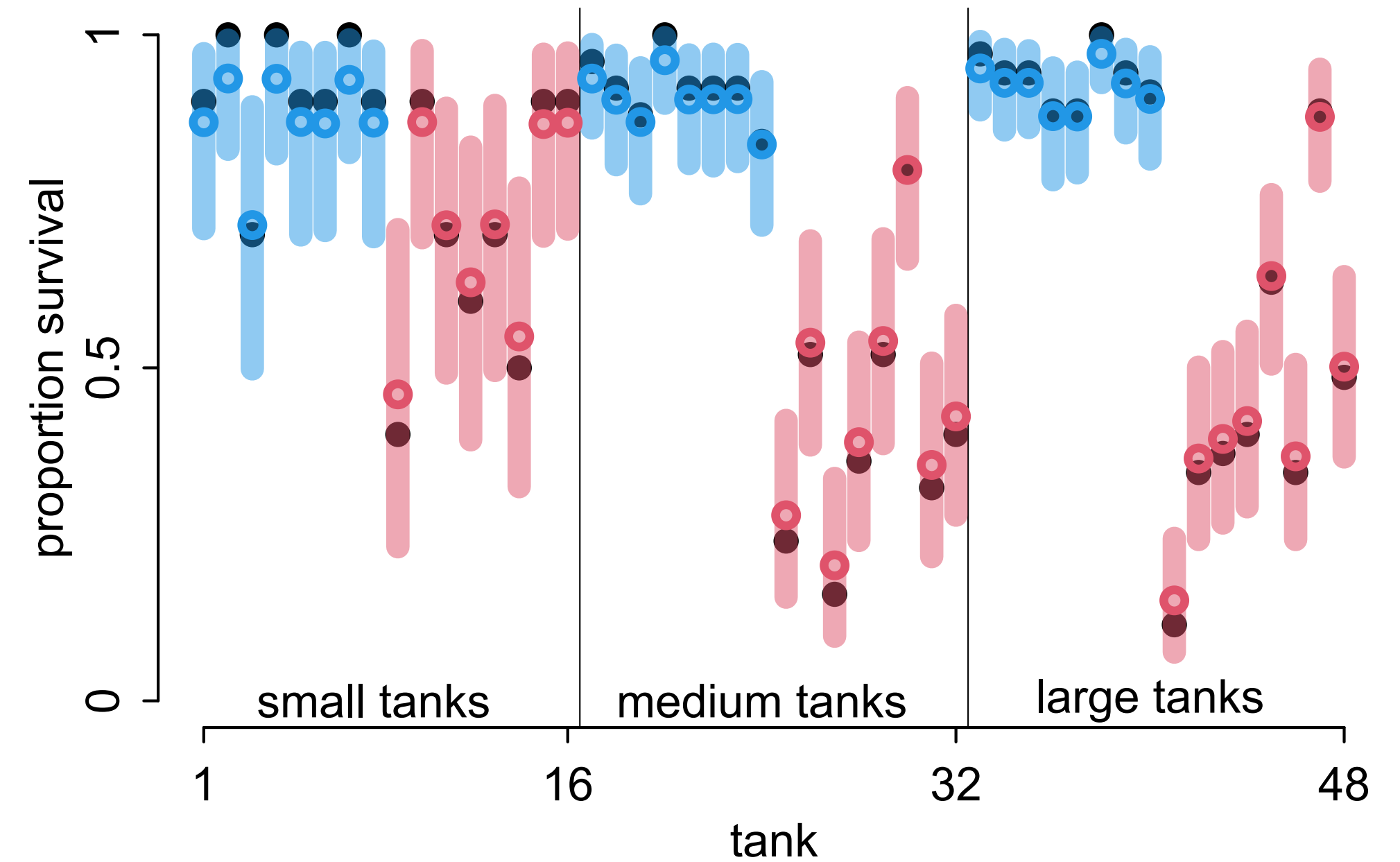
Multilevel Tadpoles

Model of unobserved population
helps learn about observed units

Use data efficiently, reduce overfitting

Varying effects: Unit-specific
partially pooled estimates

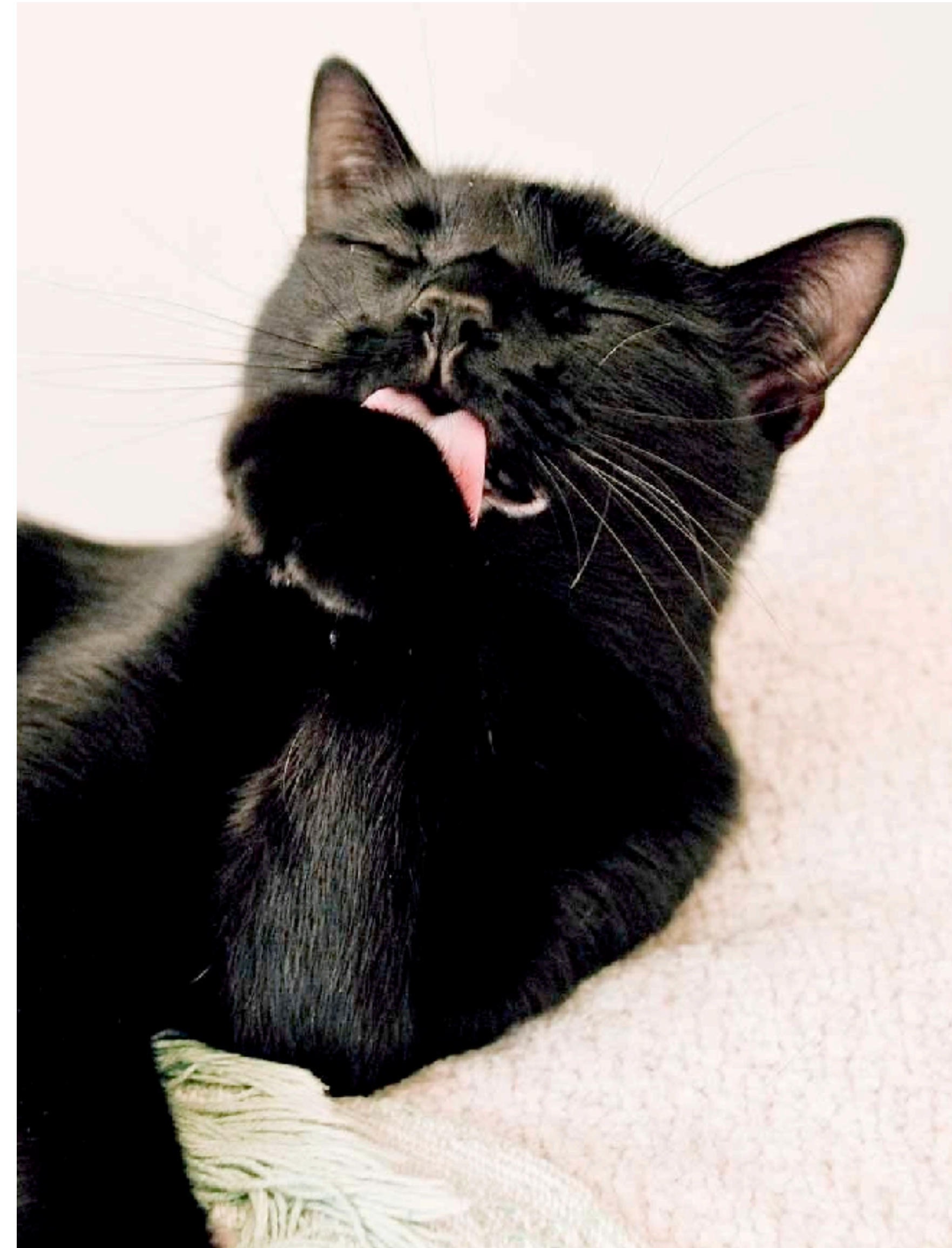
What about D and G ?



Varying Effect Superstitions

Varying effect models are plagued by superstition

- (1) Units must be sampled at random
- (2) Number of units must be large
- (3) Assumes Gaussian variation



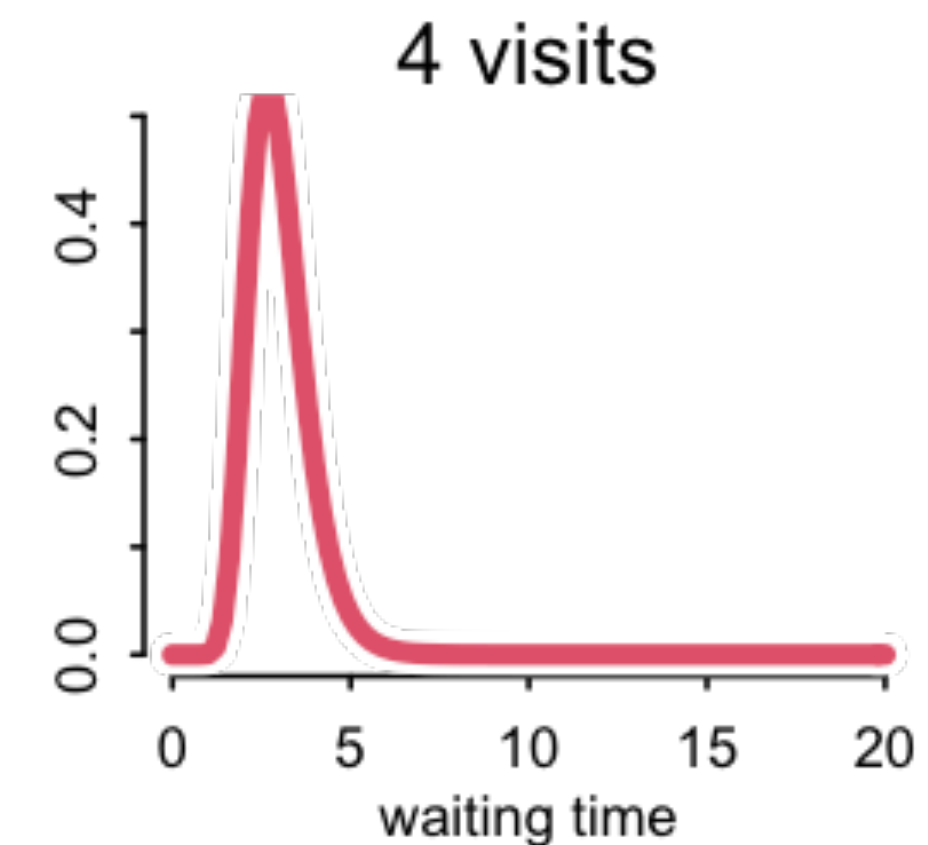
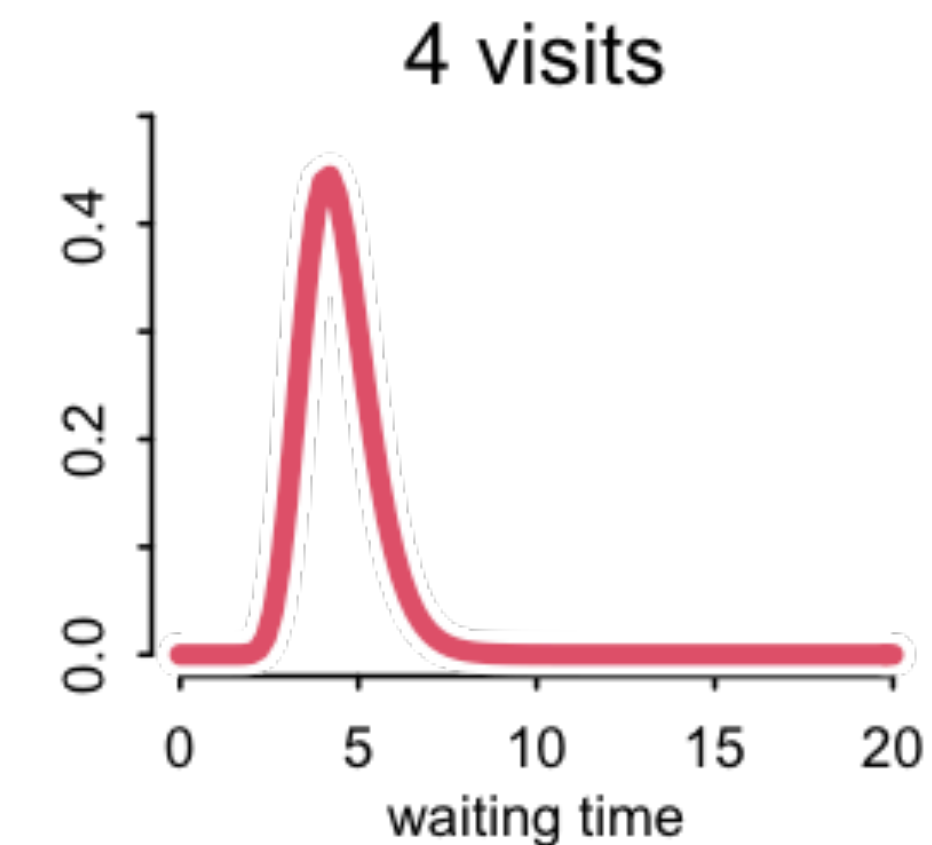
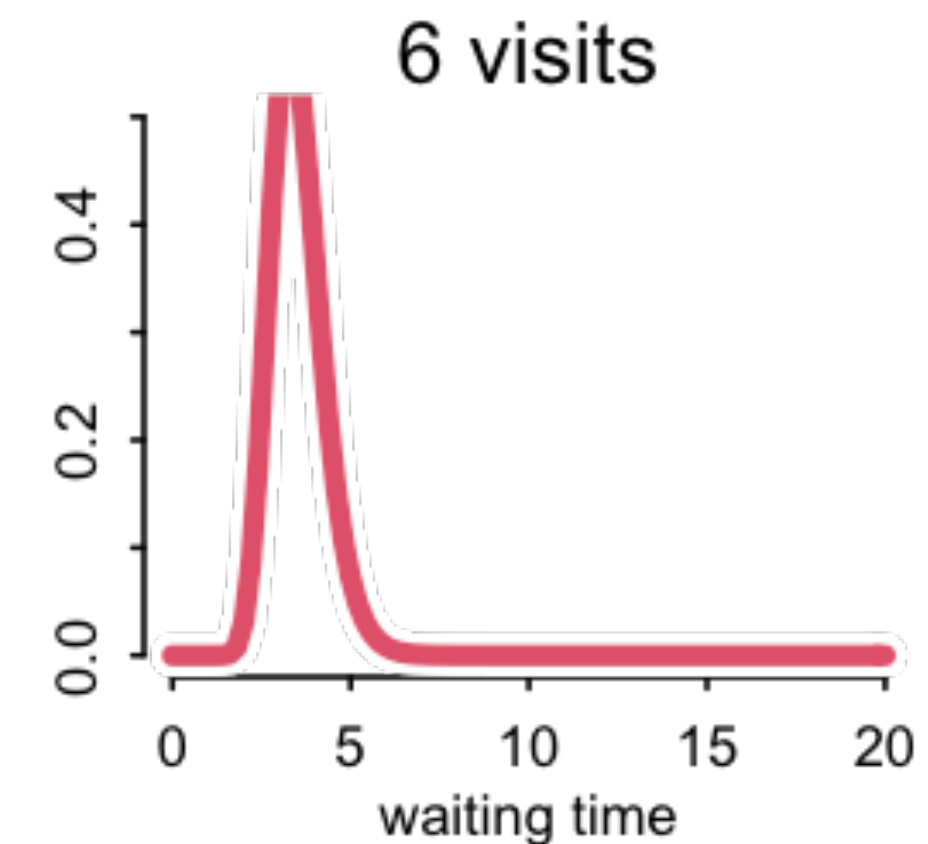
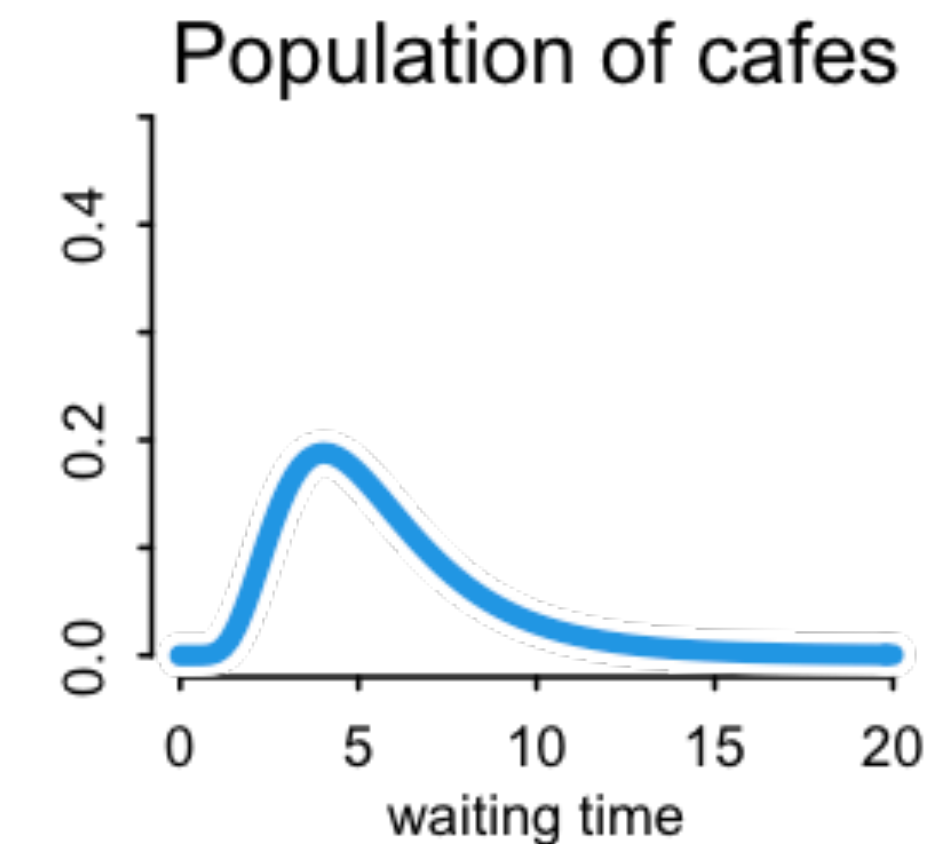
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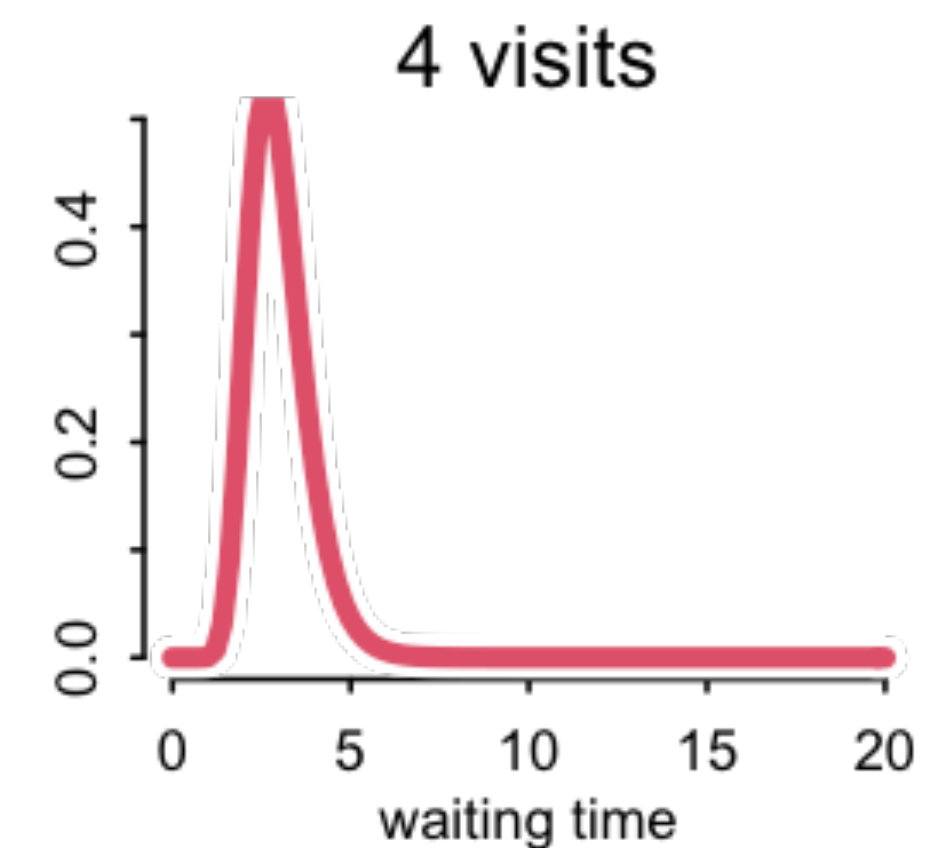
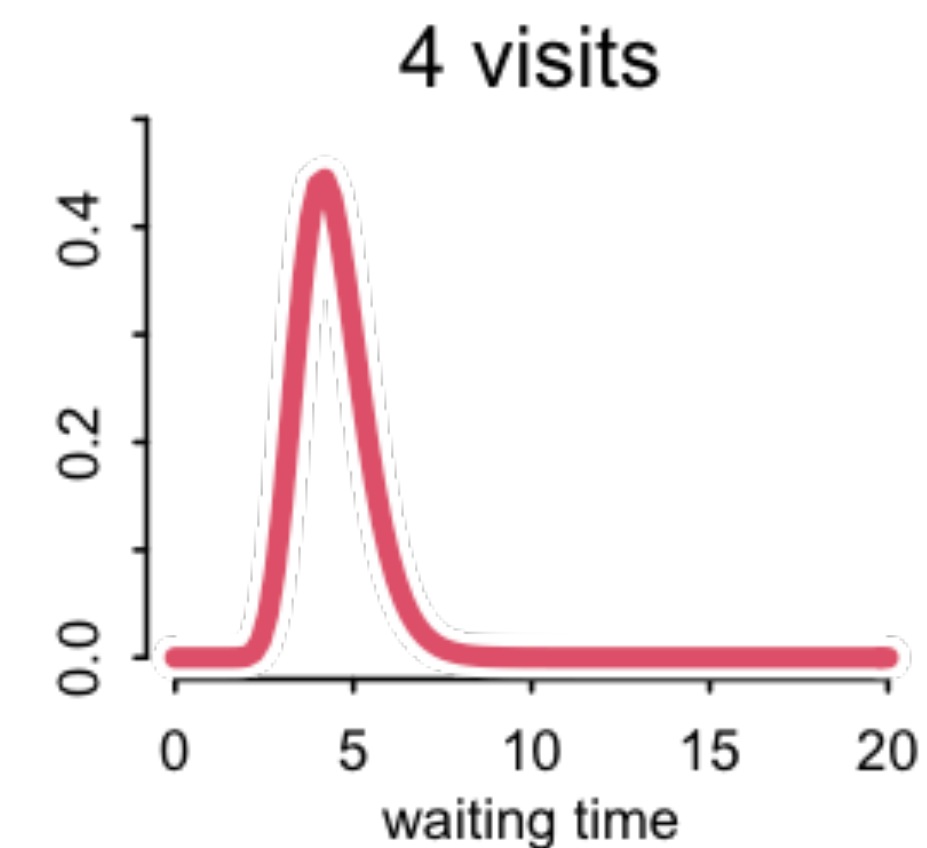
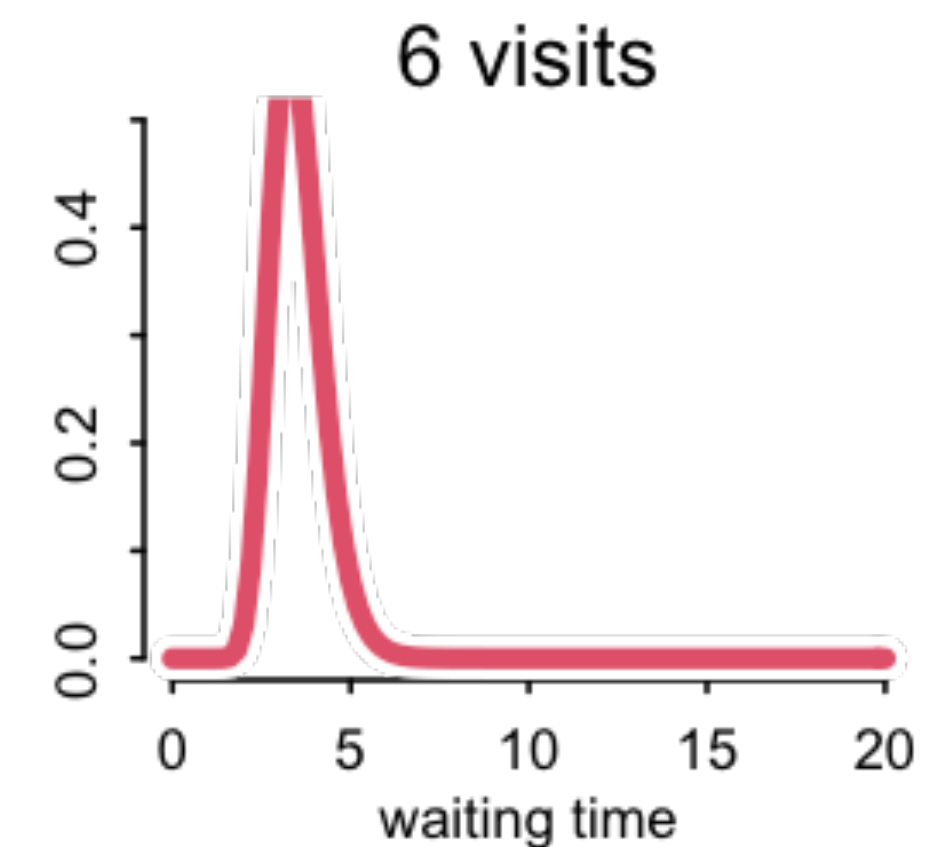
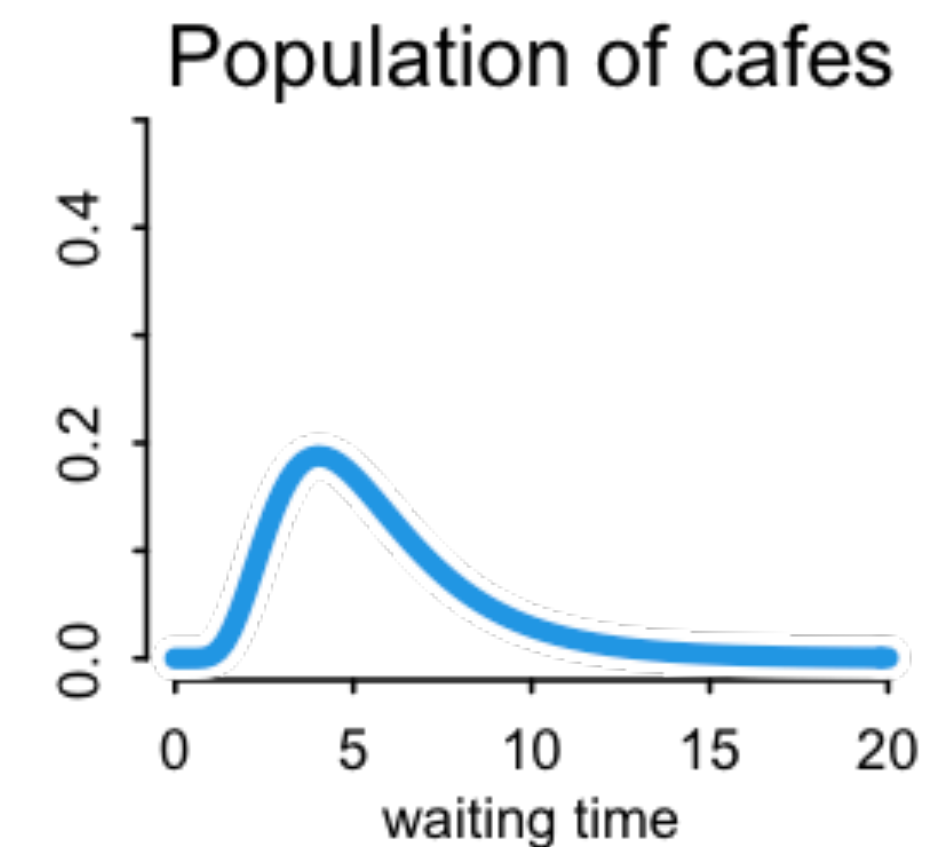
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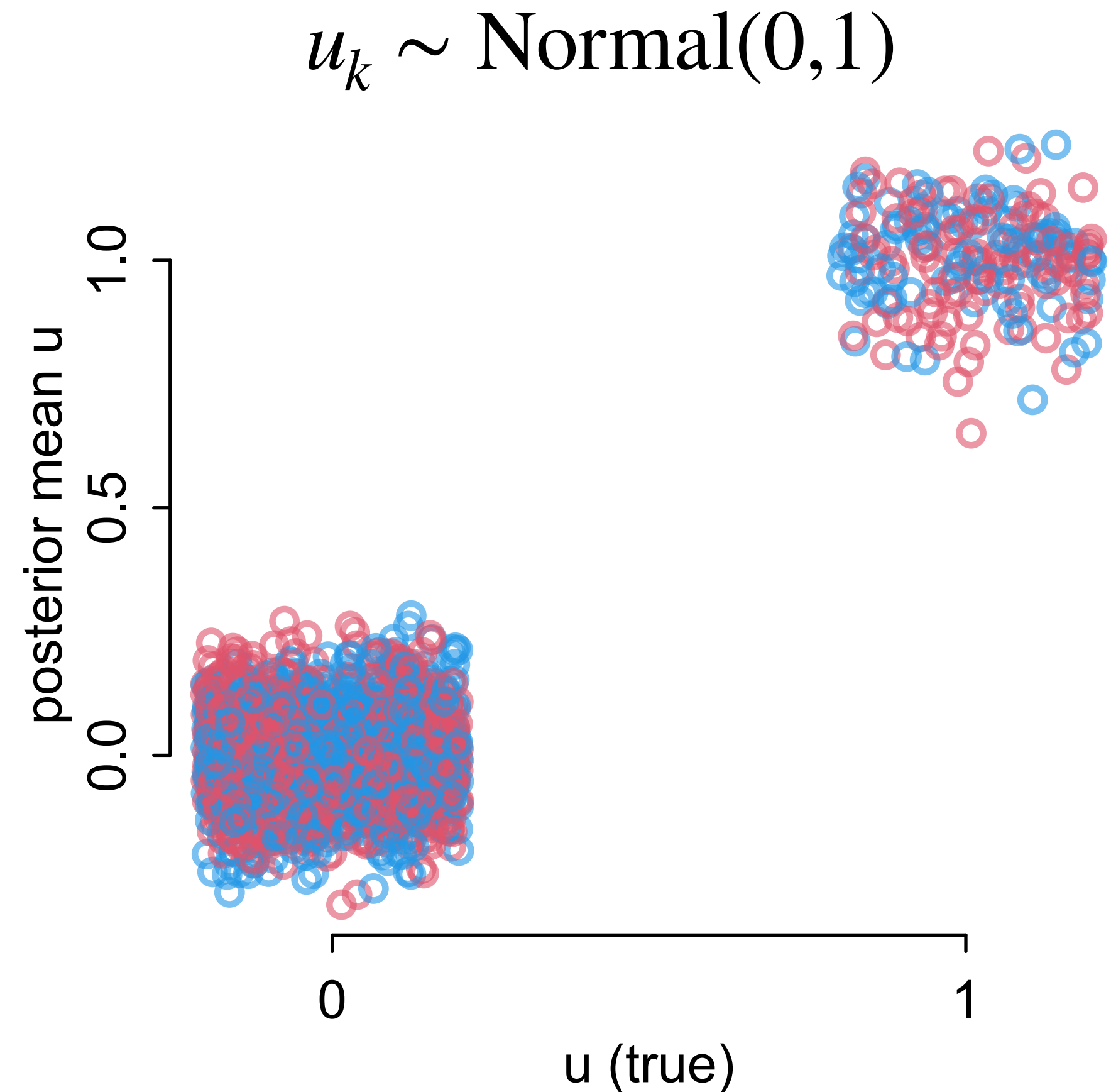
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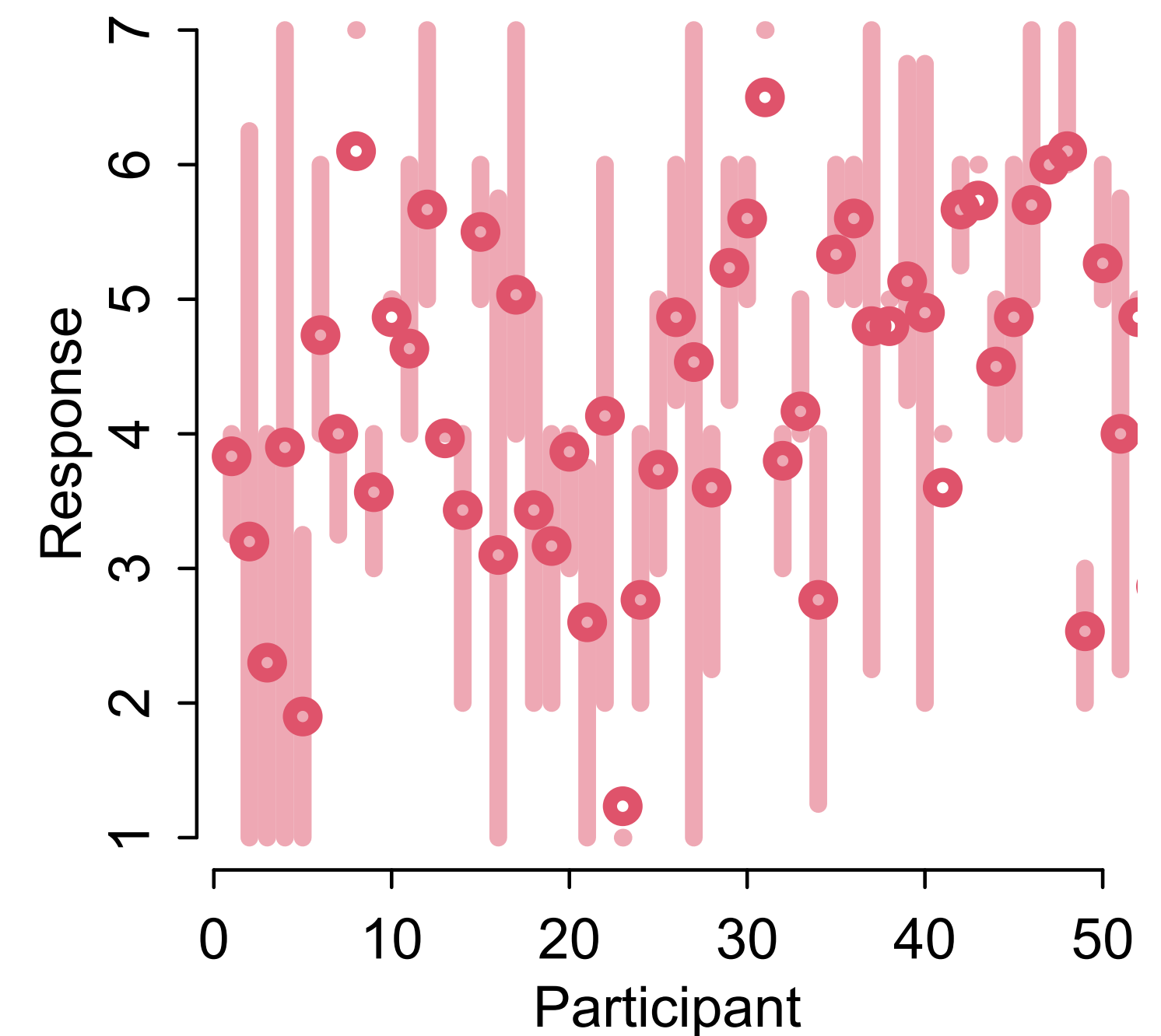
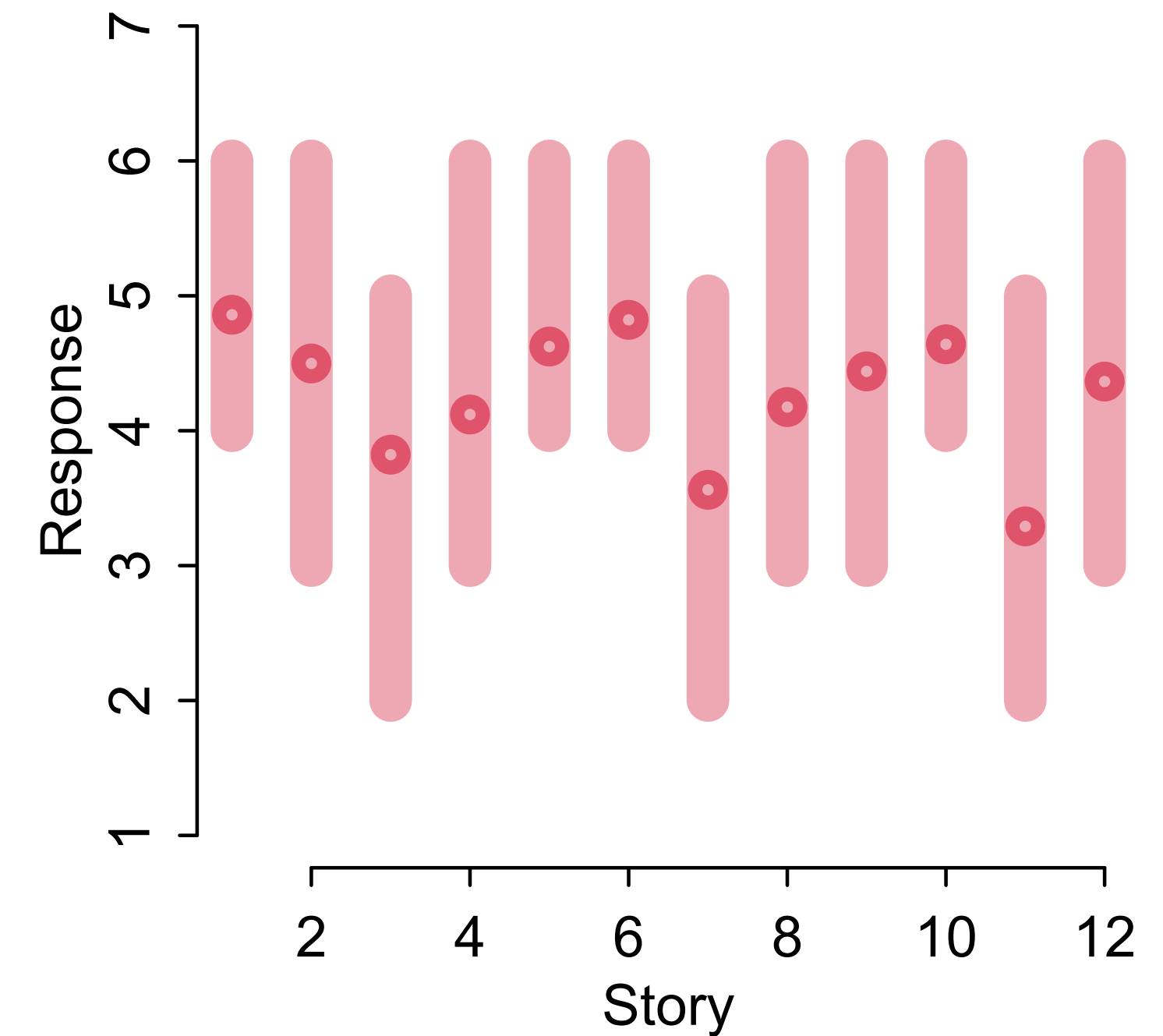
~~(3) Assumes Gaussian variation~~



Practical Difficulties

Varying effects are a good default, but...

- (1) How to use **more than one** cluster type at the same time? For example **stories** and **participants**
- (2) How to sample efficiently
- (3) What about slopes? Confounds?



Fertility & behavior

1989 Bangladesh Fertility Survey

data(bangladesh)

1934 women, 61 districts

Outcome: contraceptive use

age, living children, urban/rural



VISUALISED BY @TERENCE@FOSSTODON.ORG/@RESEARCHREMORA
IN #RSTATS WITH RAYSHADER (@TYLERMORGANWALL)

contraceptive

use

C

district

D

age

A

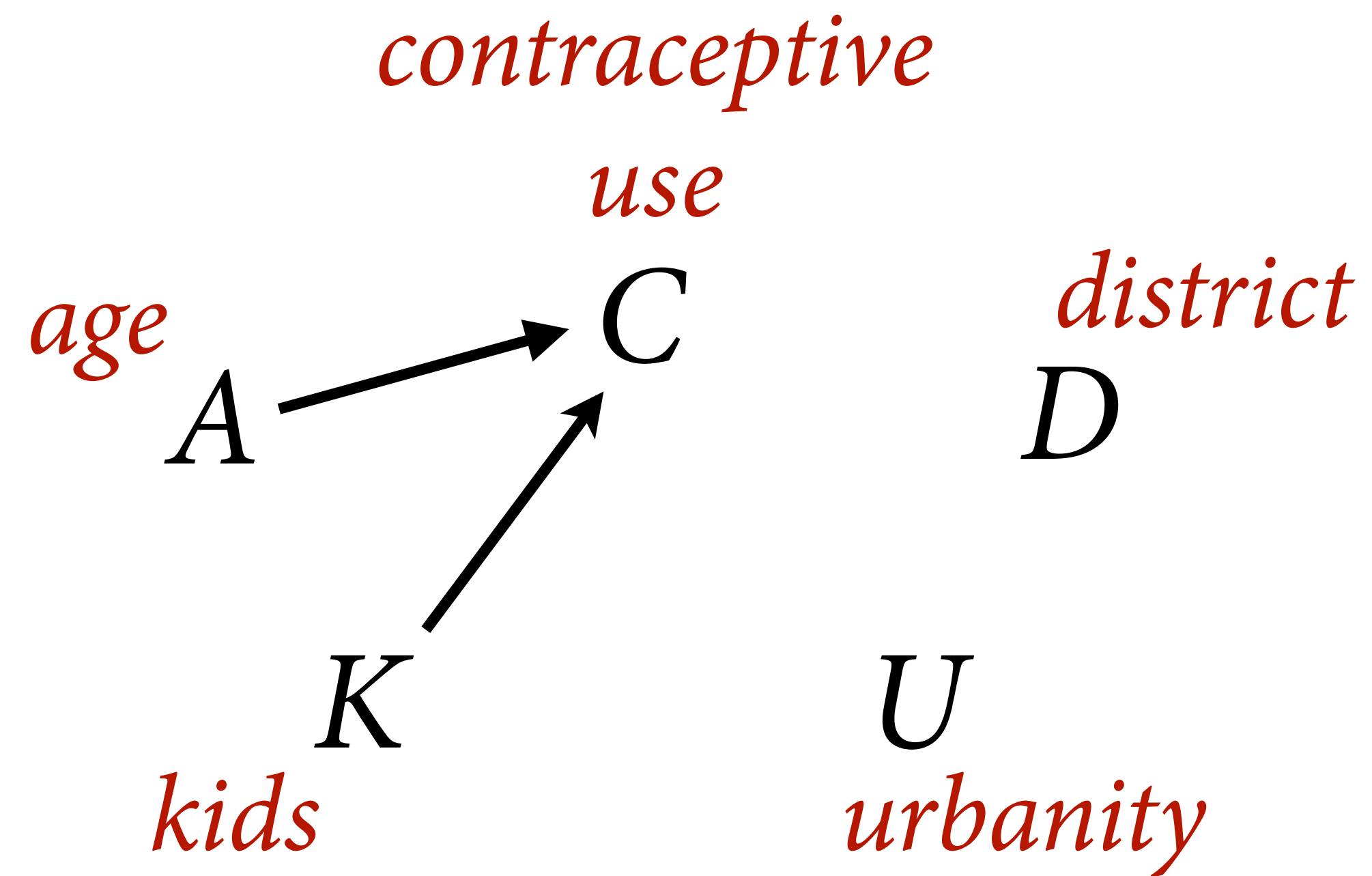
K

kids

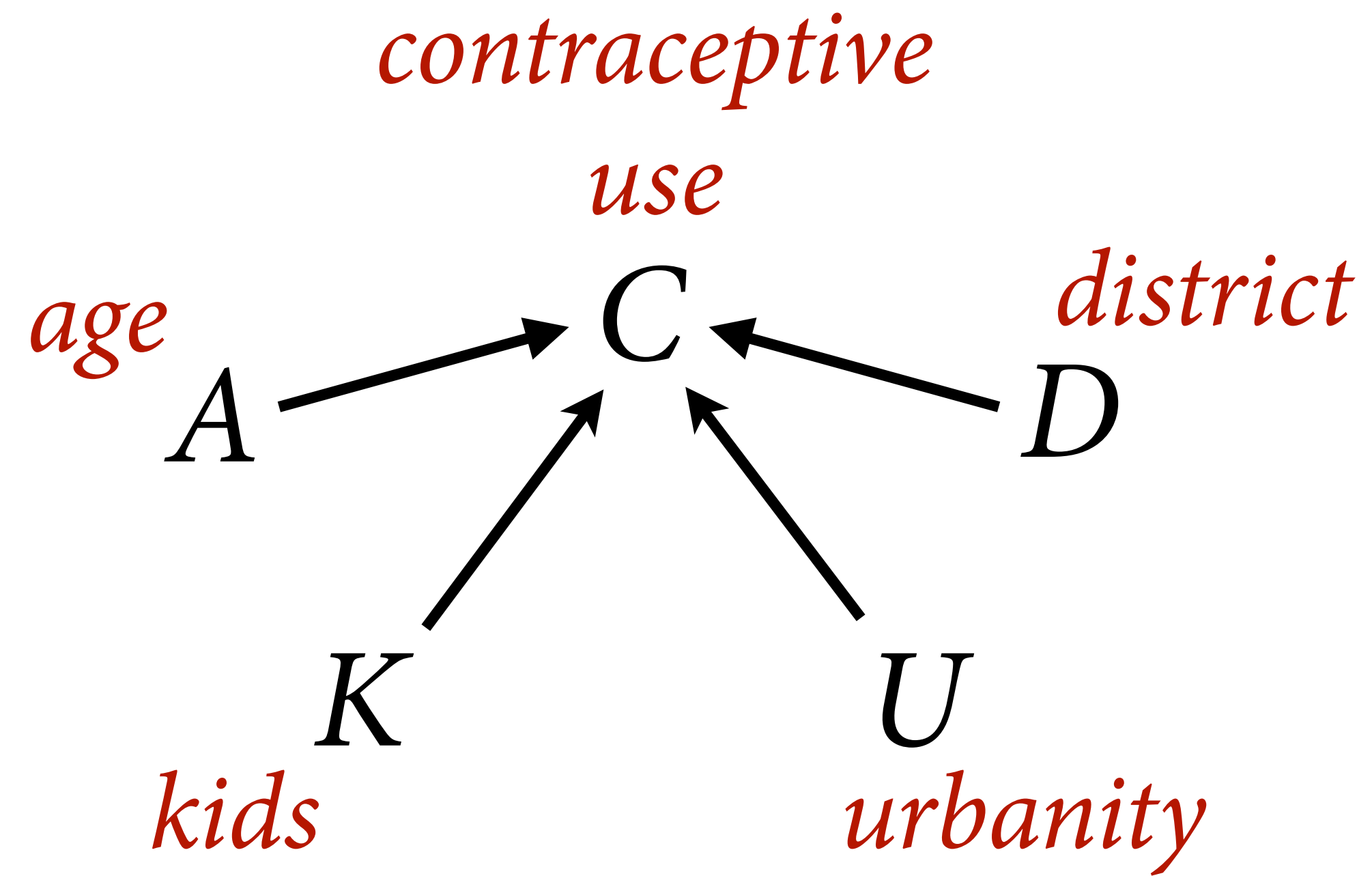
U

urbanity

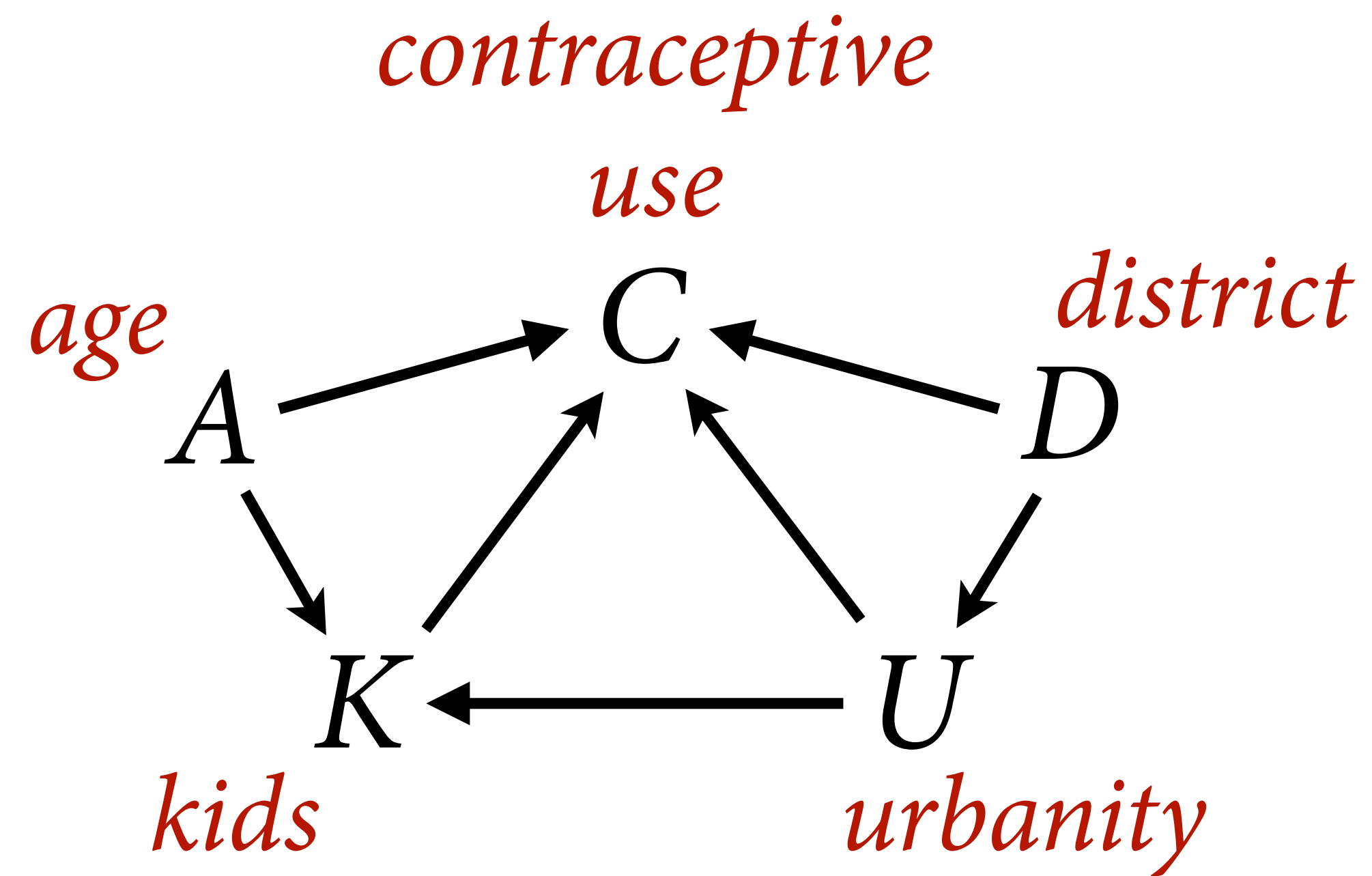
1. Causes of interest



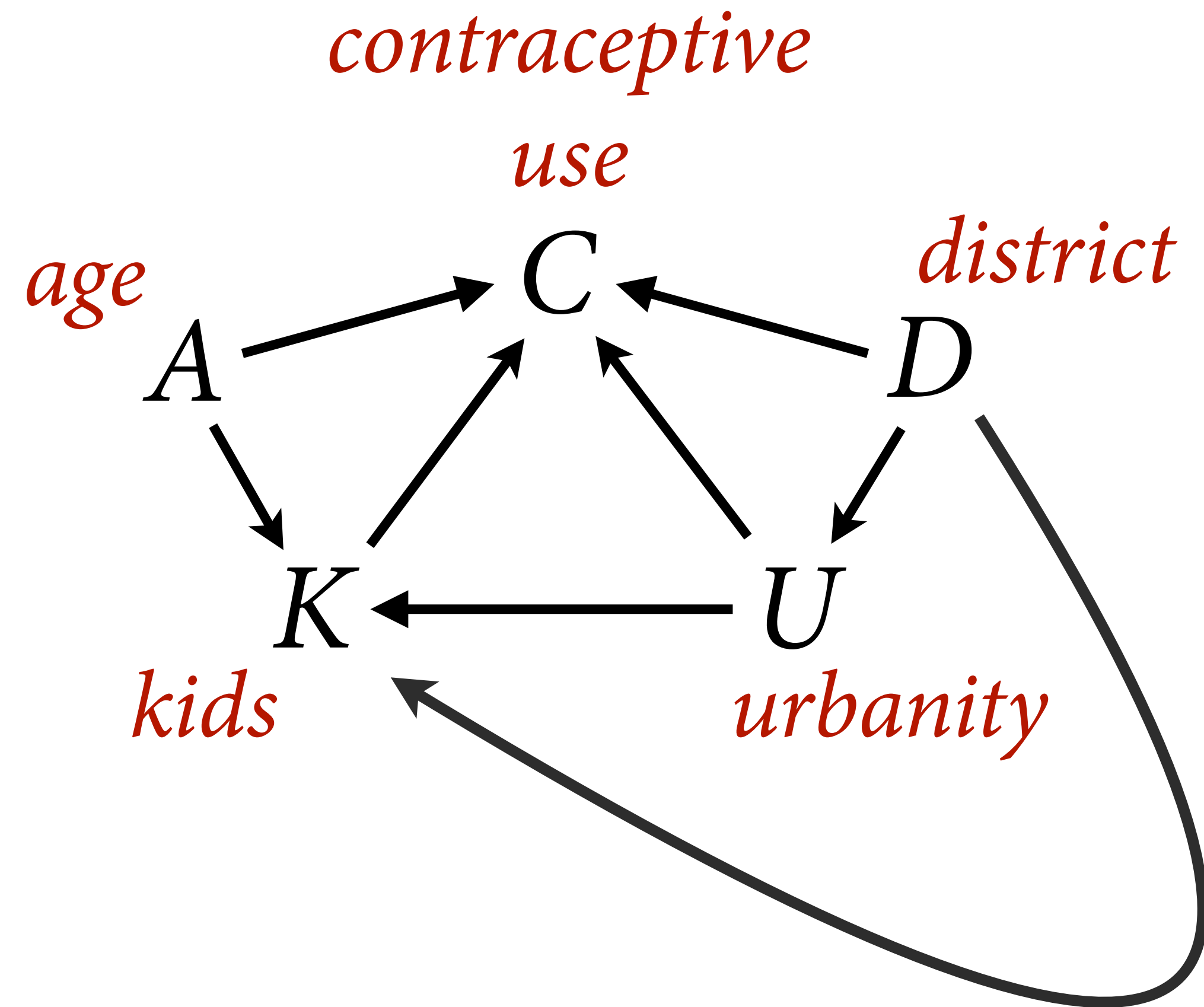
2. Competing causes



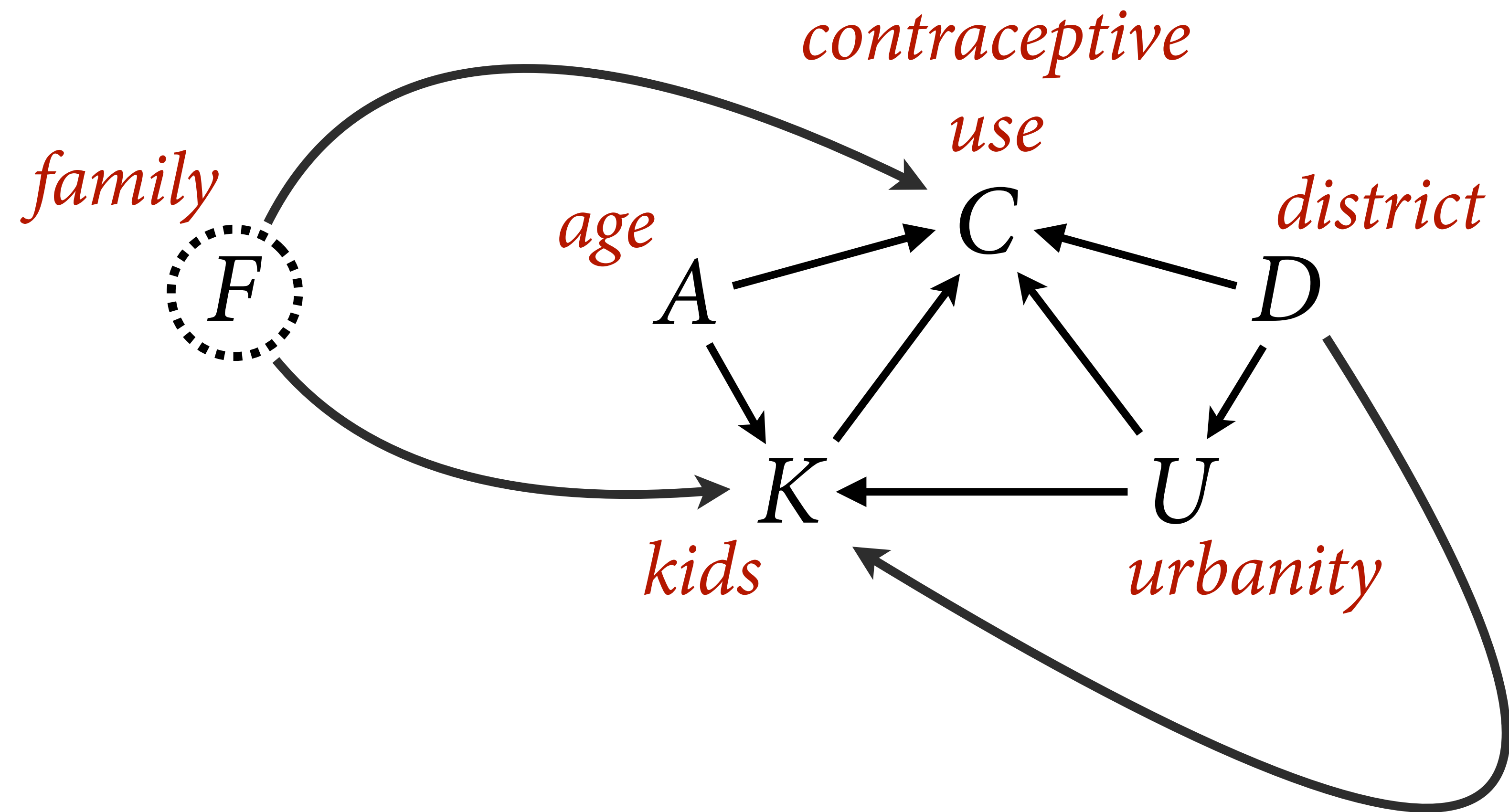
3. Relationships among causes



4. Unfortunate relationships among causes



5. A series of unfortunate relationships among causes

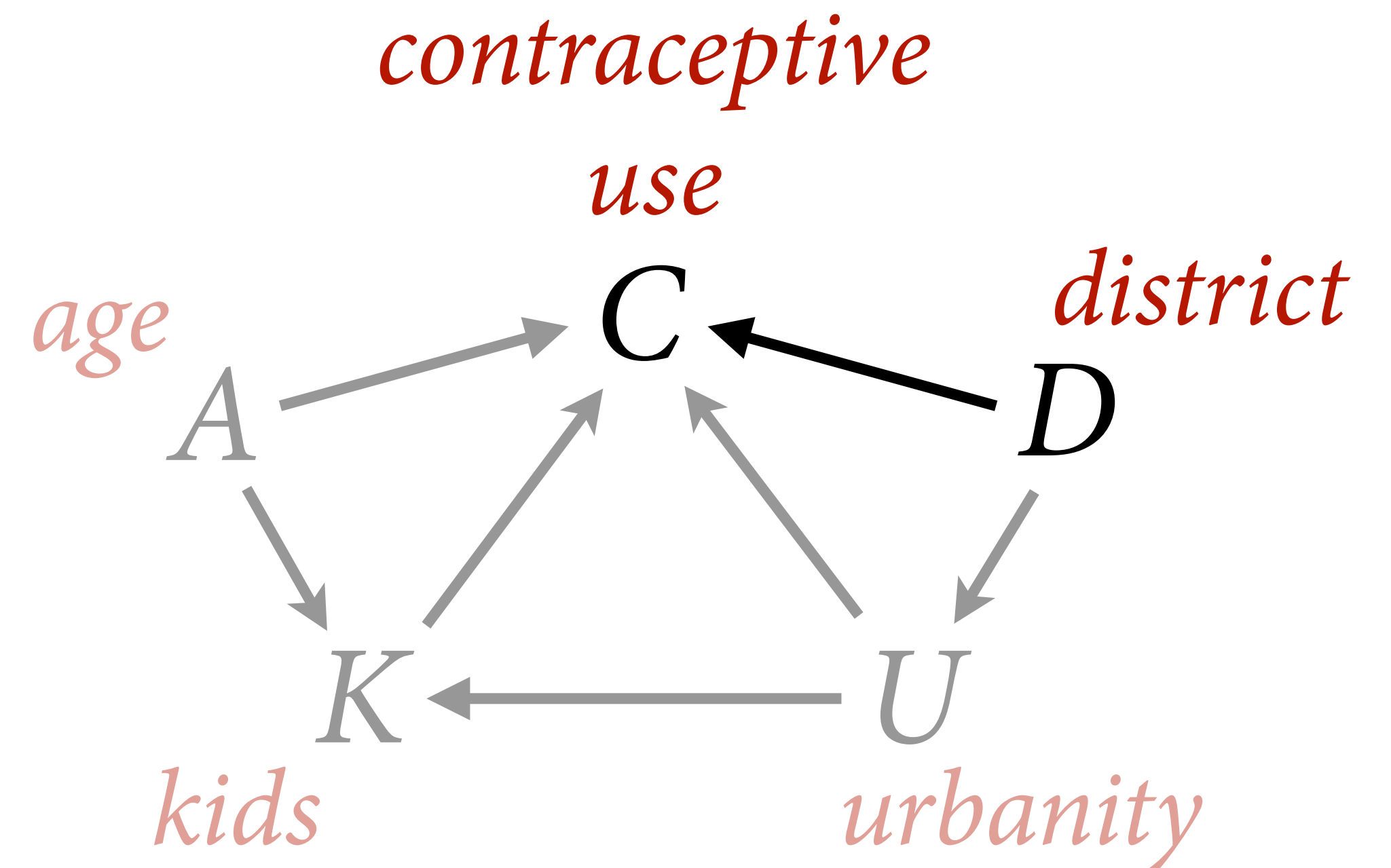


Varying districts

Estimand: C in each district,
partially pooled

Varying intercept on each district

Another chance to understand
partial pooling



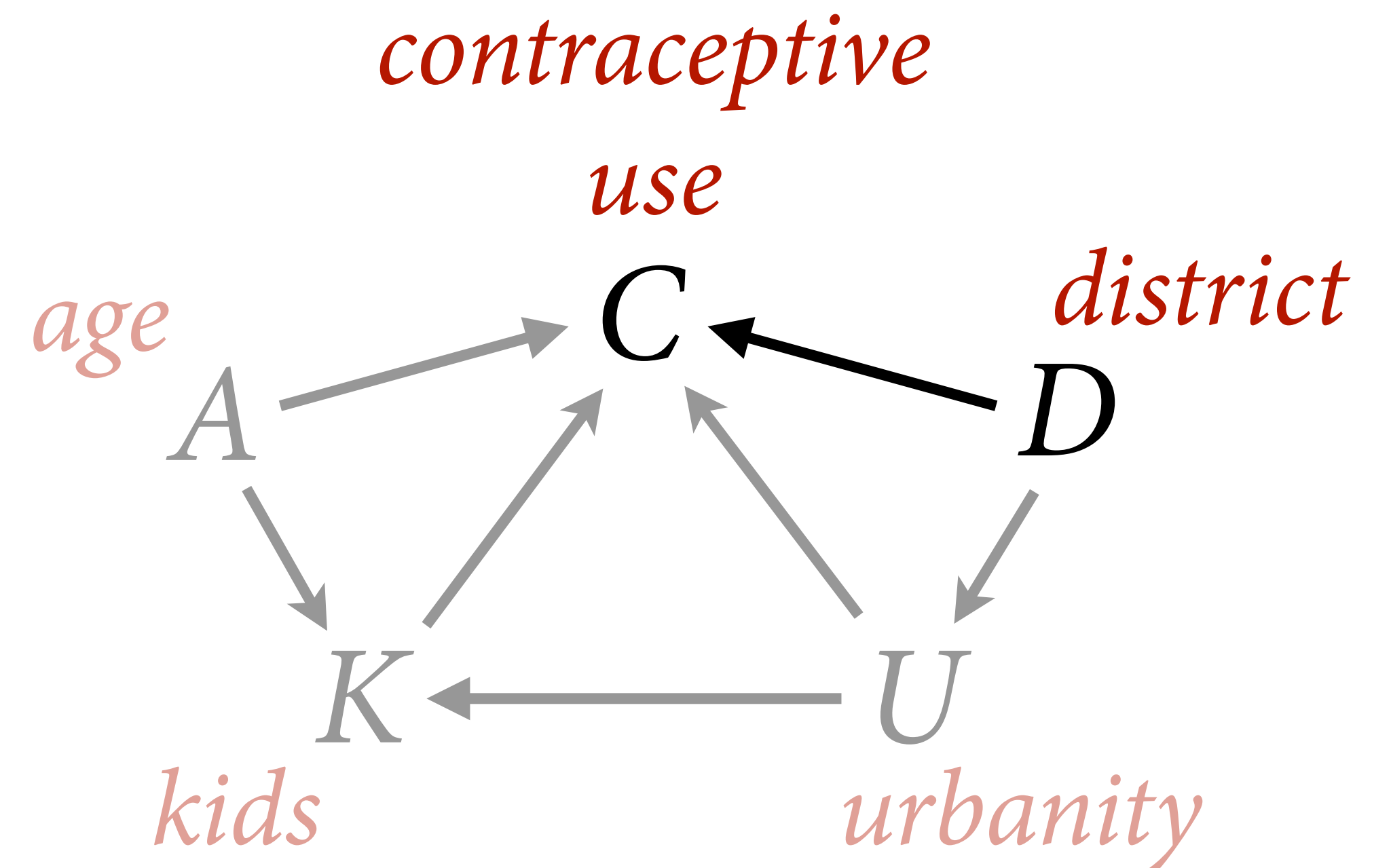
$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\bar{\alpha} \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{Exponential}(1)$$



$$C_i \sim \text{Bernoulli}(p_i)$$

Bernoulli because 0/1 outcome

$$\text{logit}(p_i) = \alpha_{D[i]}$$

log-odds of $C=1$ in each district D

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

Regularizing prior for districts

$$\bar{\alpha} \sim \text{Normal}(0, 1)$$

Average district

$$\sigma \sim \text{Exponential}(1)$$

Standard deviation among districts

```

# simple varying intercepts model
library(rethinking)
data(bangladesh)
d <- bangladesh

dat <- list(
  C = d$use.contraception,
  D = as.integer(d$district) )

mCD <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D],
    vector[61]:a ~ normal(abar,sigma),
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  ) , data=dat , chains=4 , cores=4 )

```

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    sigma ~ exponential(1)
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```

State directly that the vector for a has 61 levels b/c we don't have data for all districts

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

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```

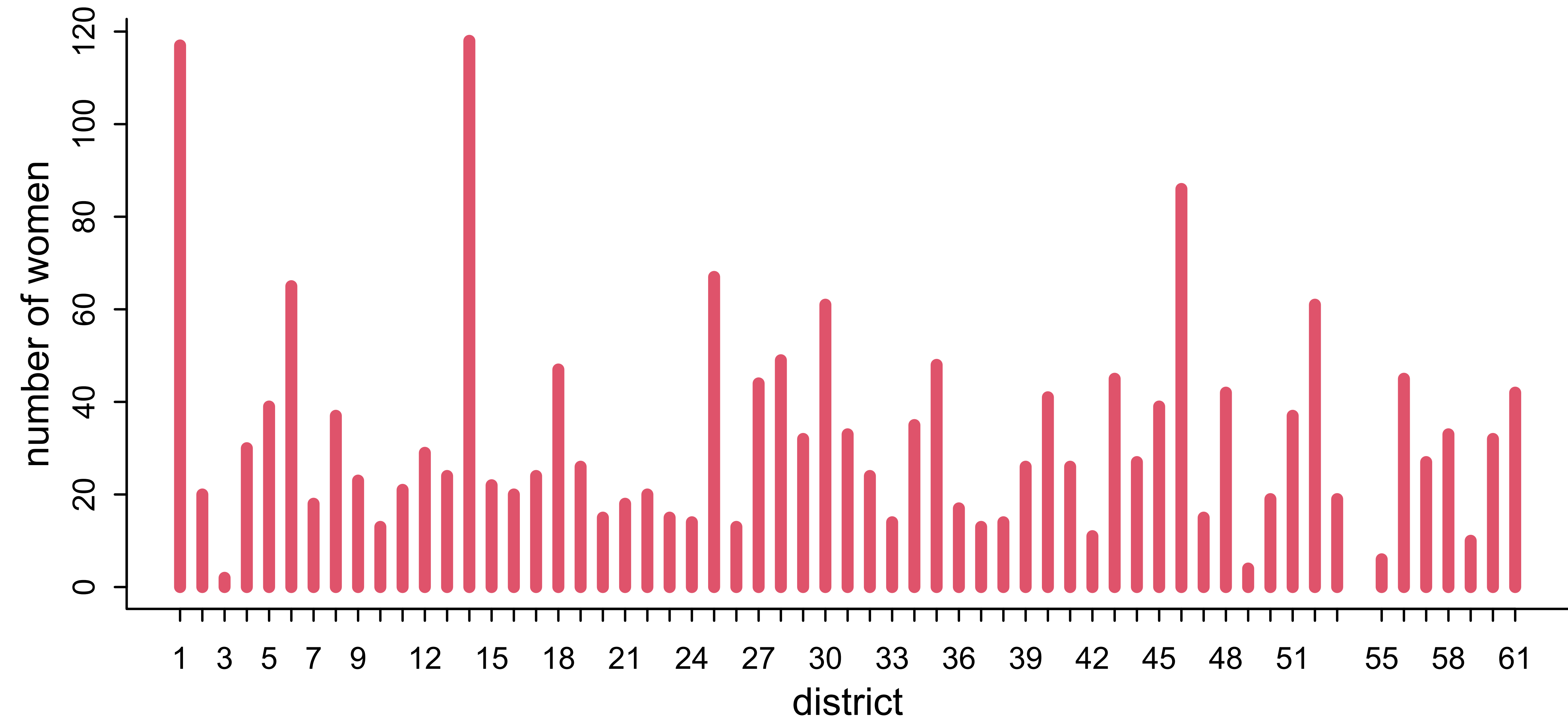
$$C_i \sim \text{Bernoulli}(p_i)$$

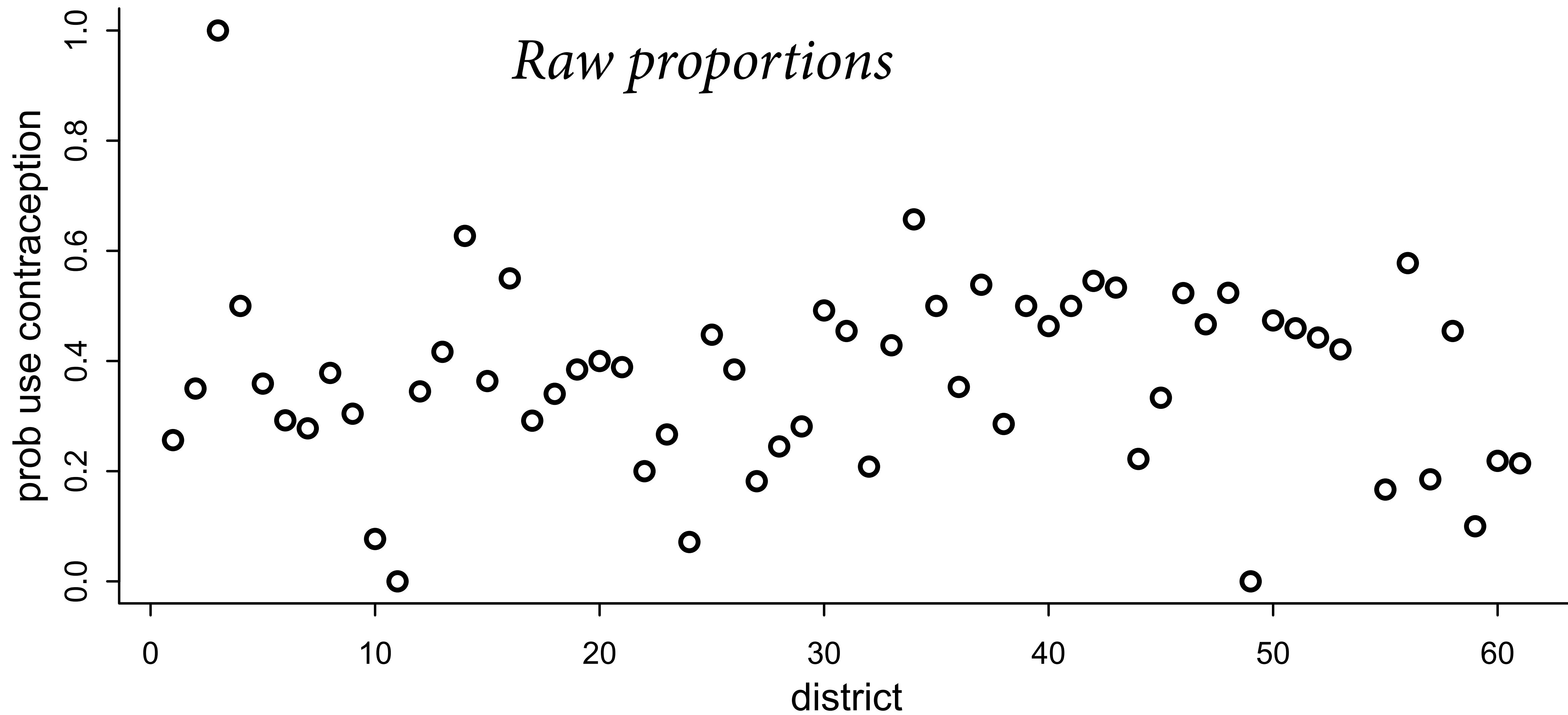
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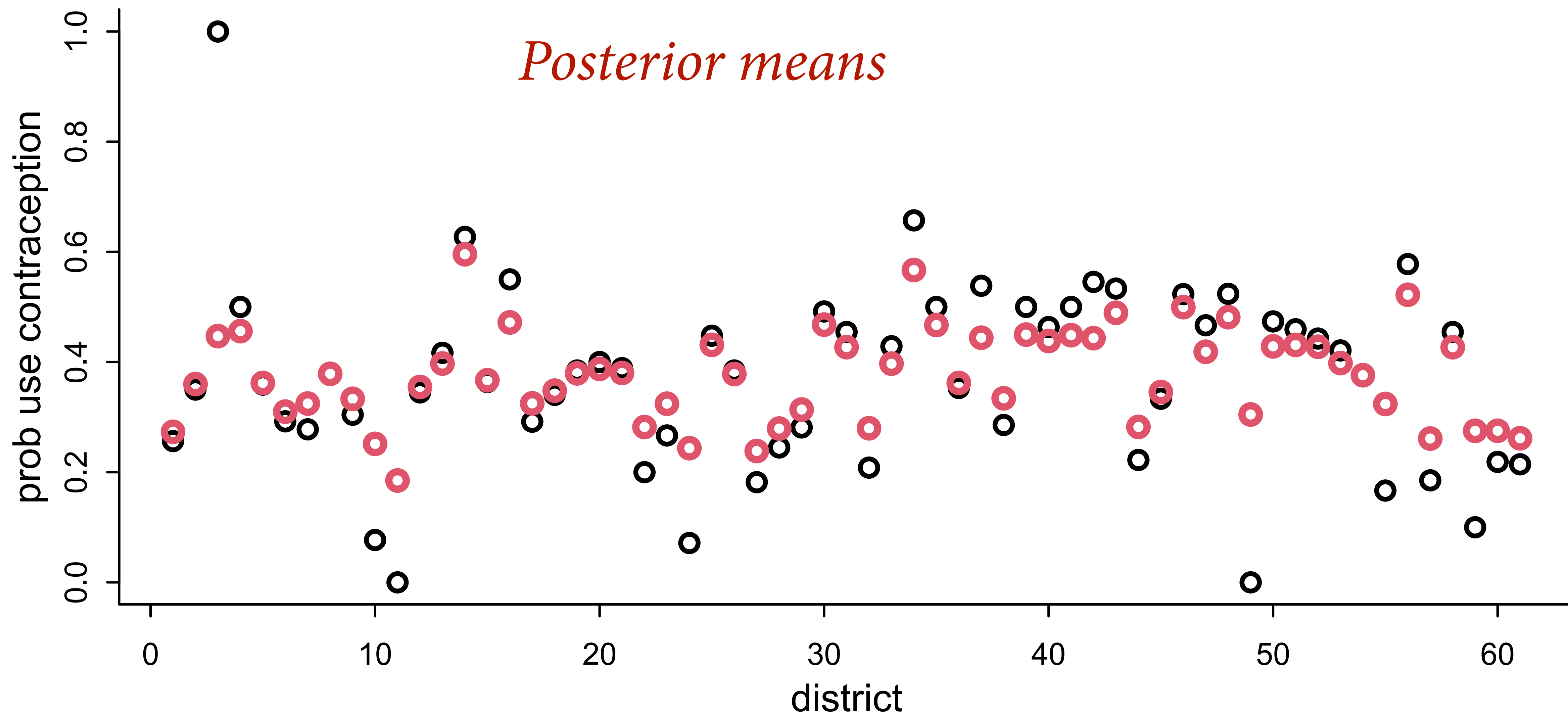
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$$\bar{\alpha} \sim \text{Normal}(0,1)$$

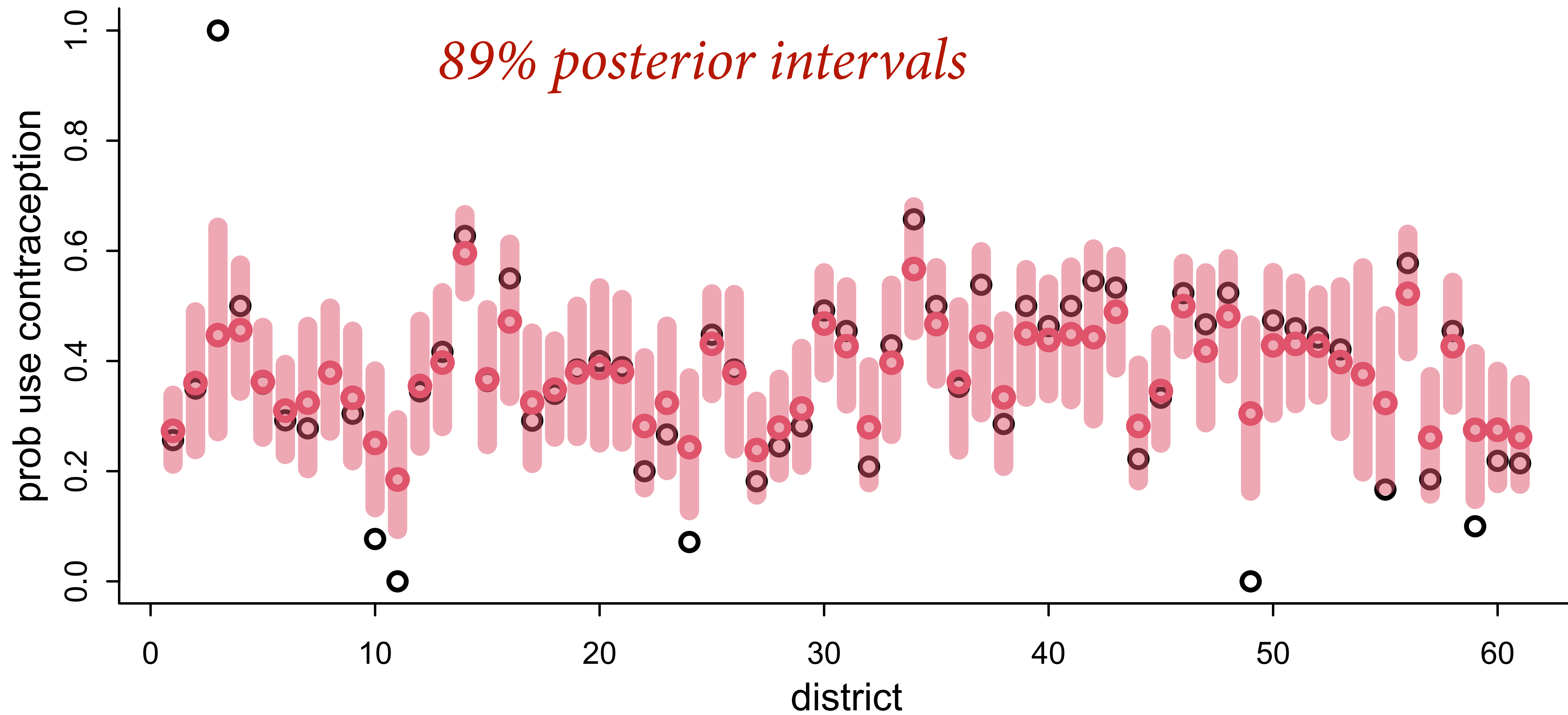
$$\sigma \sim \text{Exponential}(1)$$



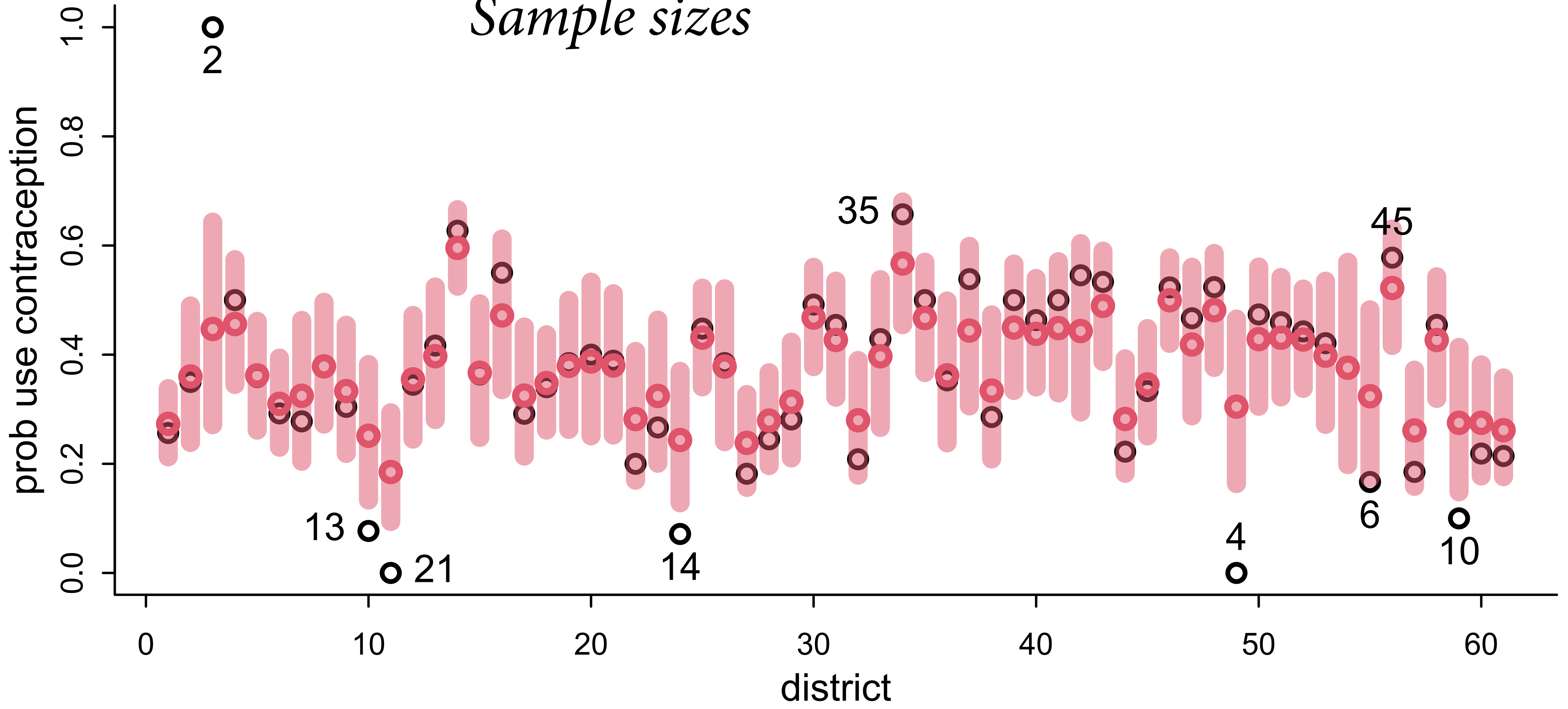


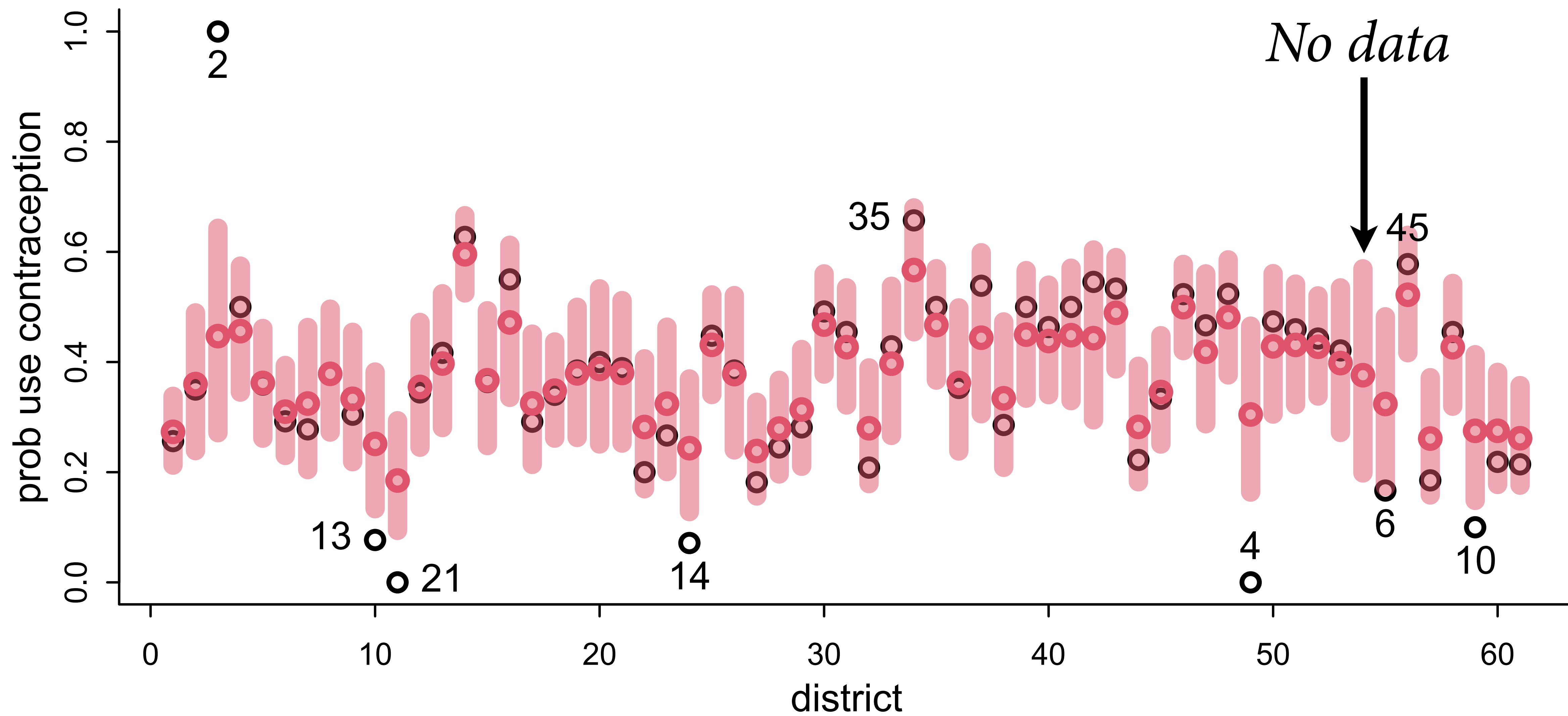


89% posterior intervals



Sample sizes



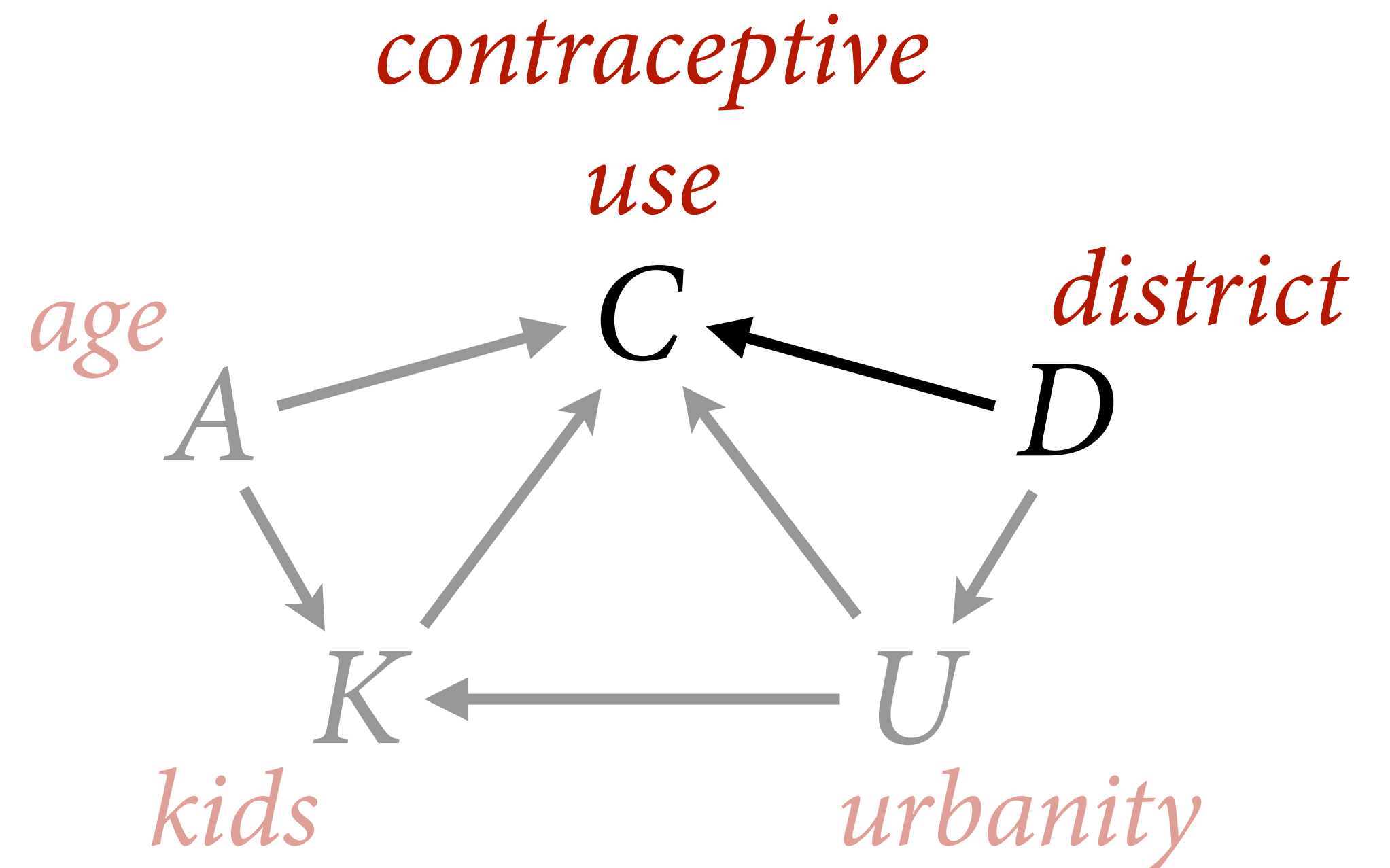
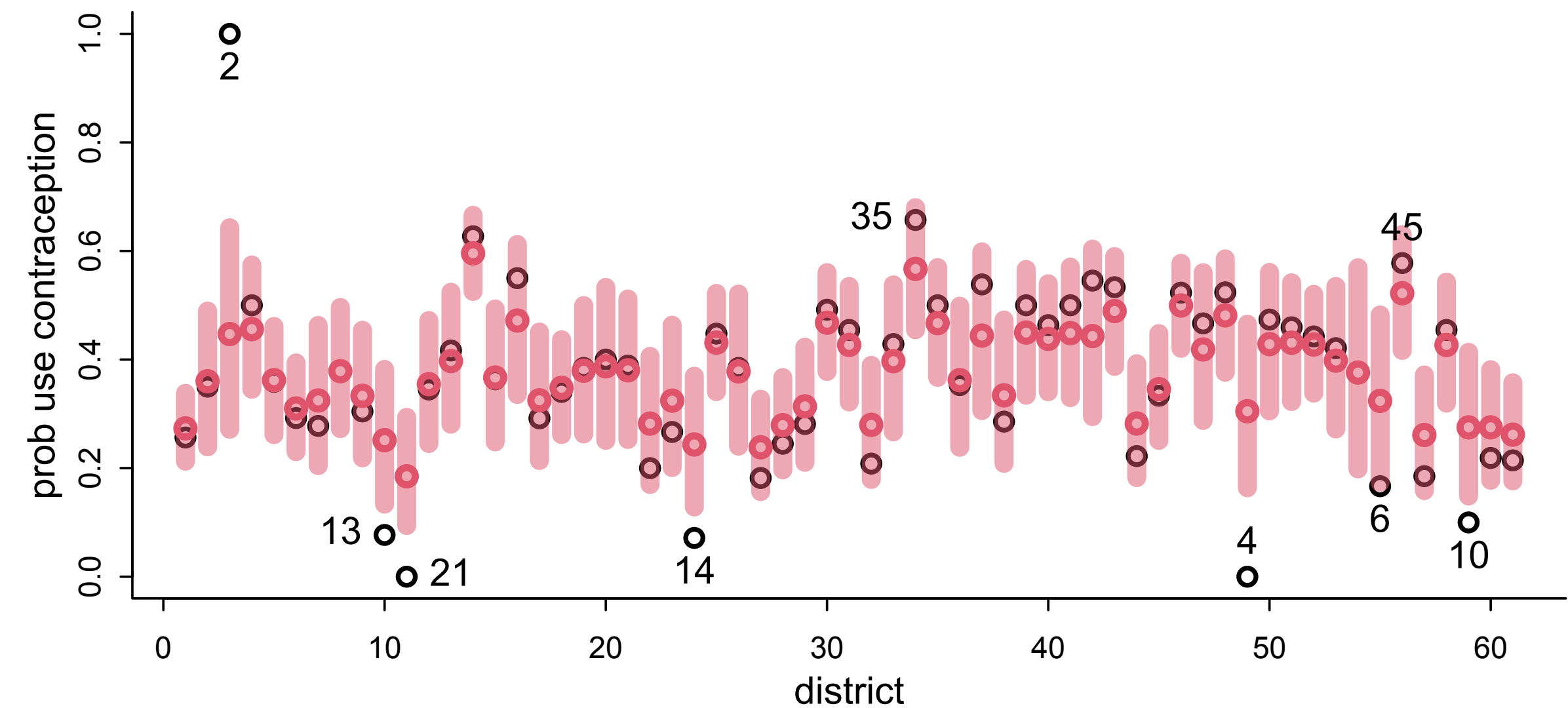


Varying districts

Partial pooling shrinks districts
with low sampling towards mean

Better predictions

No inference yet



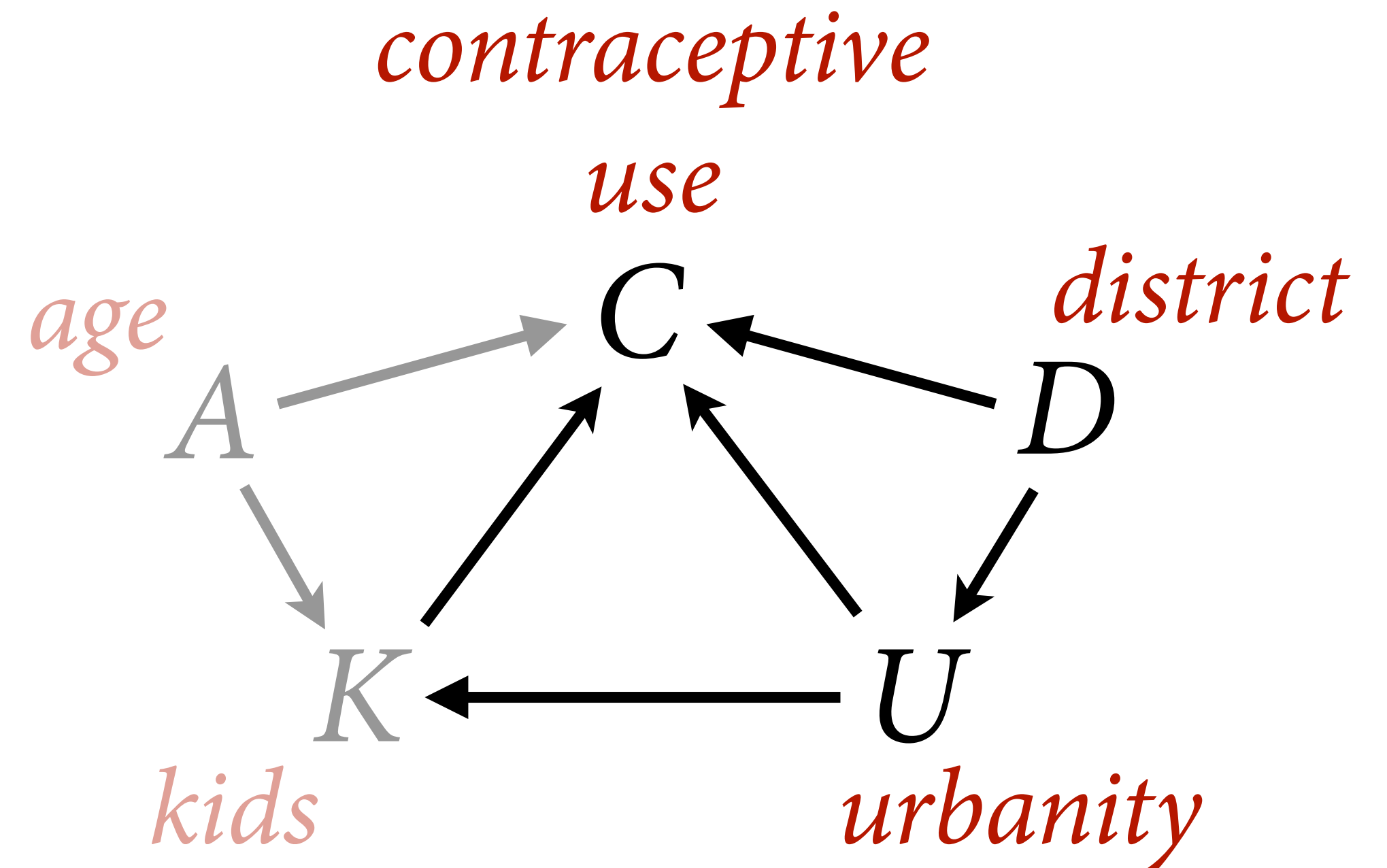
Varying districts + urban

What is the effect of urban living?

District features are potential group-level confounds

Total effect of U passes through K

Do not stratify by K !



$$C_i \sim \text{Bernoulli}(p_i)$$

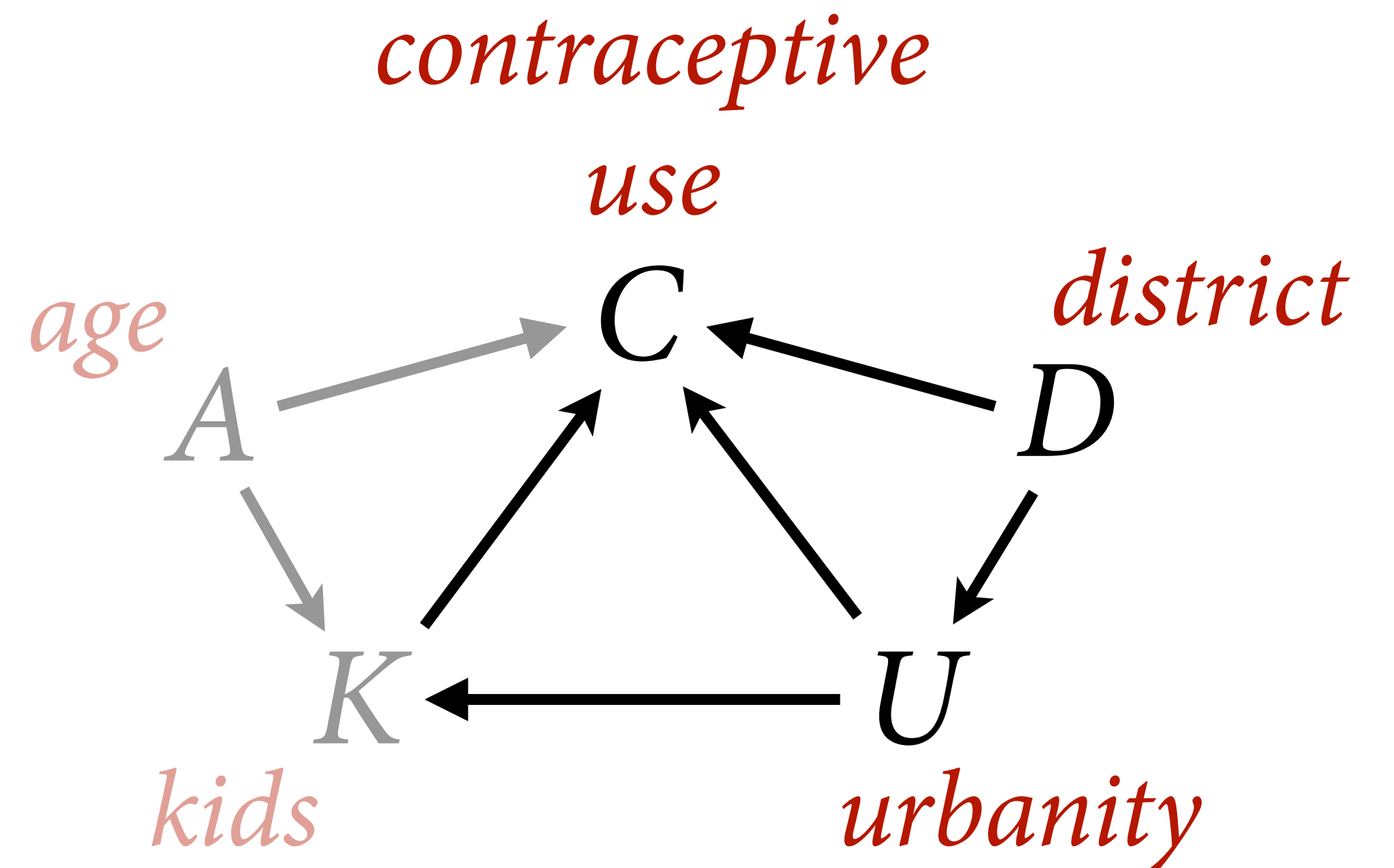
$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$



$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

Regularizing prior for rural

Regularizing prior for urban effect

Averages

Standard deviations

```

dat <- list(
  C = d$use.contraception,
  D = as.integer(d$district),
  U = ifelse(d$urban==1,1,0) )

# total U
mCDU <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D] + b[D]*U,
    vector[61]:a ~ normal(abar,sigma),
    vector[61]:b ~ normal(bbar,tau),
    c(abar,bbar) ~ normal(0,1),
    c(sigma,tau) ~ exponential(1)
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

```

dat <- list(
  C = d$use.contraception,
  D = as.integer(d$district),
  U = ifelse(d$urban==1,1,0) )

# total U
mCDU <- ulam(
  alist(
    C ~ bernoulli(p)
    logit(p) <- a[D]
    vector[61]:a ~
    vector[61]:b ~
    c(abar,bbar) ~
    c(sigma,tau) ~
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

All 4 chains finished successfully.

Mean chain execution time: 4.3 seconds.

Total execution time: 4.7 seconds.

**Warning: 4 of 2000 (0.0%) transitions ended with a divergence.
See <https://mc-stan.org/misc/warnings> for details.**

**Warning: 3 of 4 chains had an E-BFMI less than 0.2.
See <https://mc-stan.org/misc/warnings> for details.**

$$\sigma, \tau \sim \text{Exponential}(1)$$

E-BFMI = Estimated Bayesian Fraction of Missing Information. It quantifies how hard it is to sample level sets at each iteration

```

dat <- list(
  C = d$use.contraception,
  D = as.integer(d$married),
  U = ifelse(d$use.contraception == 1, 1, 0)

# total U
mCDU <- ulam(
  alist(
    C ~ bernoulli(p)
    logit(p) <- a[D]
    vector[61]:a ~
    vector[61]:b ~
    c(abar,bbar) ~
    c(sigma,tau) ~
  ), data=dat , chains=4 , cores=4 )

```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
bbar	0.62	0.16	0.37	0.87	678	1.01
abar	-0.70	0.09	-0.85	-0.56	1123	1.00
tau	0.57	0.24	0.17	0.94	45	1.11
sigma	0.49	0.09	0.34	0.64	330	1.02

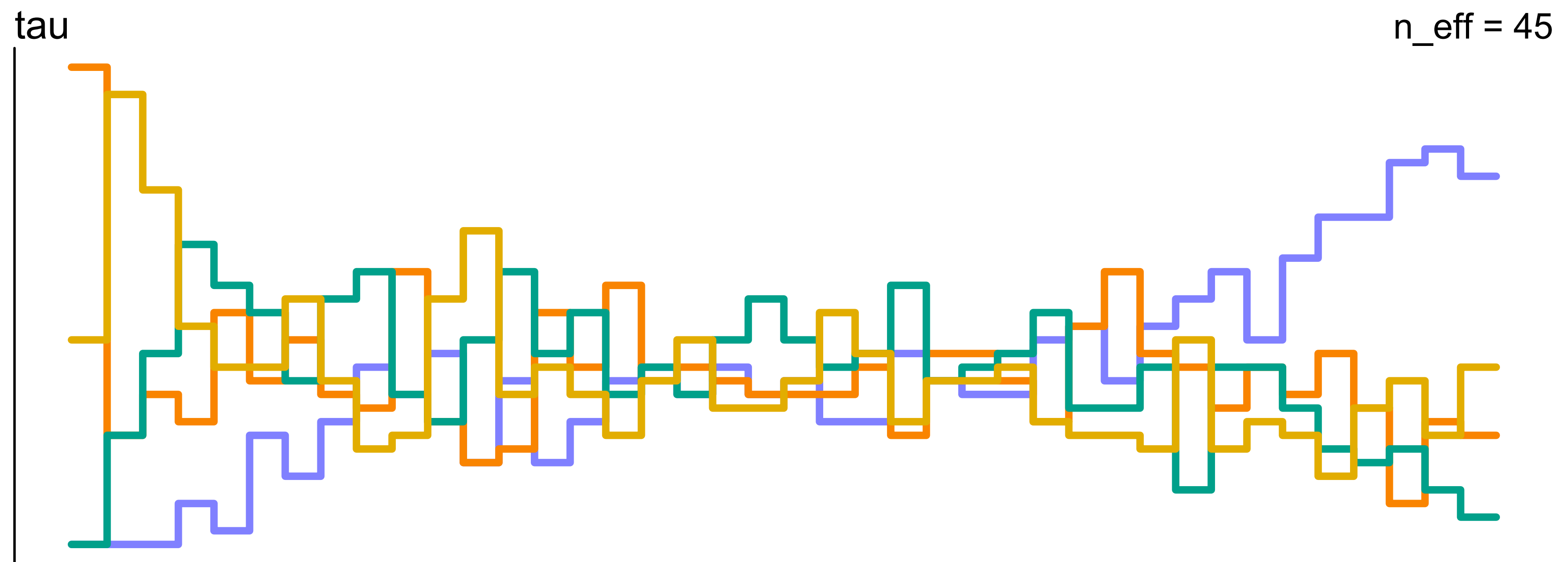
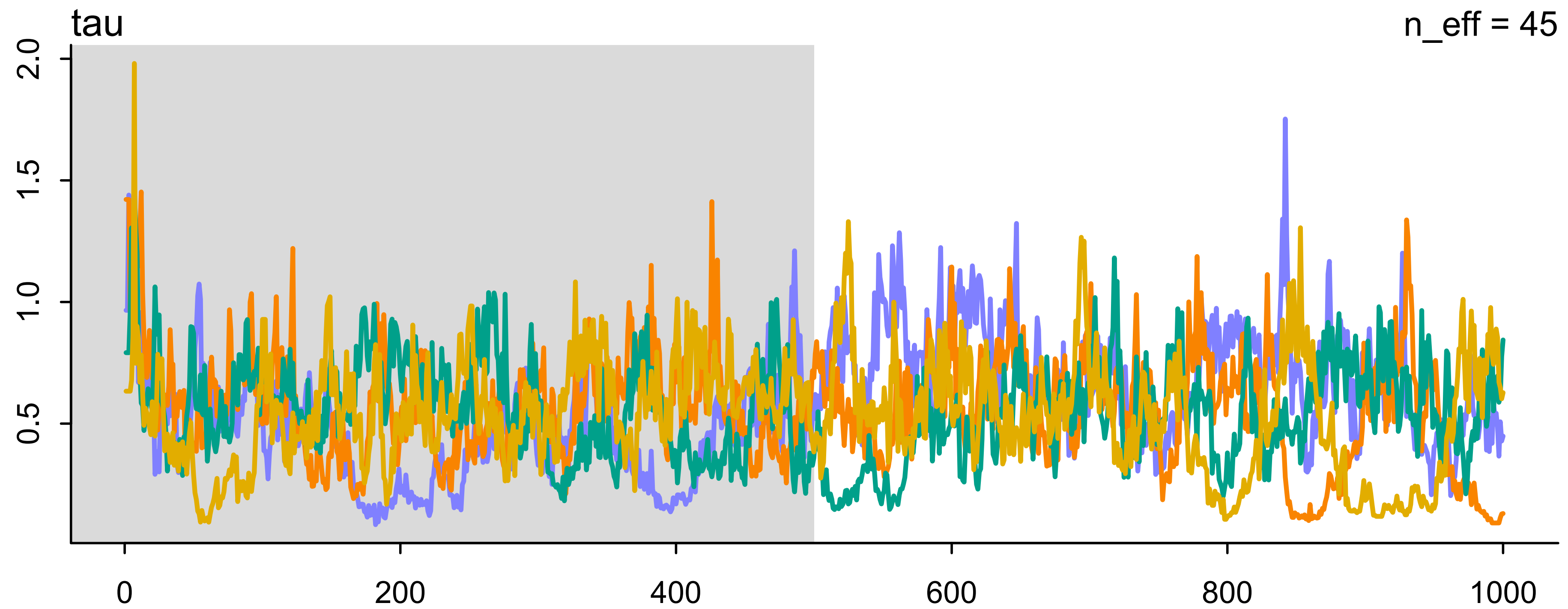
**Warning: 4 of 2000 (0.0%) transitions ended with a divergence.
See <https://mc-stan.org/misc/warnings> for details.**

**Warning: 3 of 4 chains had an E-BFMI less than 0.2.
See <https://mc-stan.org/misc/warnings> for details.**

$$C_i \sim \text{Bernoulli}(p_i)$$

$$p_i = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\sigma, \tau \sim \text{Exponential}(1)$$



More priors, more problems

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

Priors inside priors: “centered”



More priors, more problems

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$z_j = (\alpha_j - \bar{\alpha}) / \sigma$$

More priors, more problems

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$z_j = (\alpha_j - \bar{\alpha}) / \sigma$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

More priors, more problems

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$z_j = (\alpha_j - \bar{\alpha}) / \sigma$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

$$z_{\alpha,j} \sim \text{Normal}(0, 1)$$

Centered varying intercepts

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

Non-centered varying intercepts

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

$$\beta_j = \bar{\beta} + z_{\beta,j} \times \tau$$

$$z_{\alpha,j} \sim \text{Normal}(0, 1)$$

$$z_{\beta,j} \sim \text{Normal}(0, 1)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

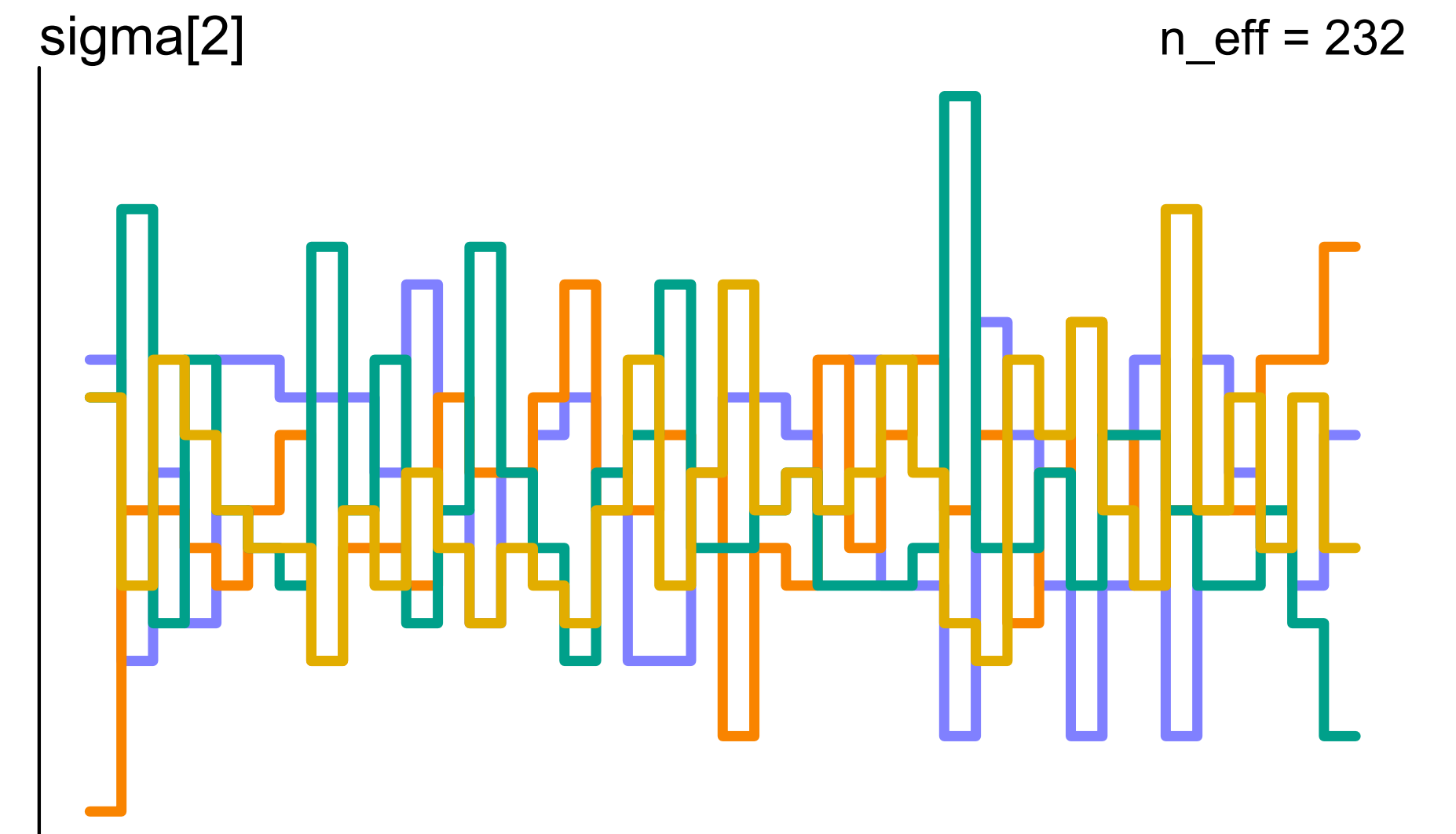
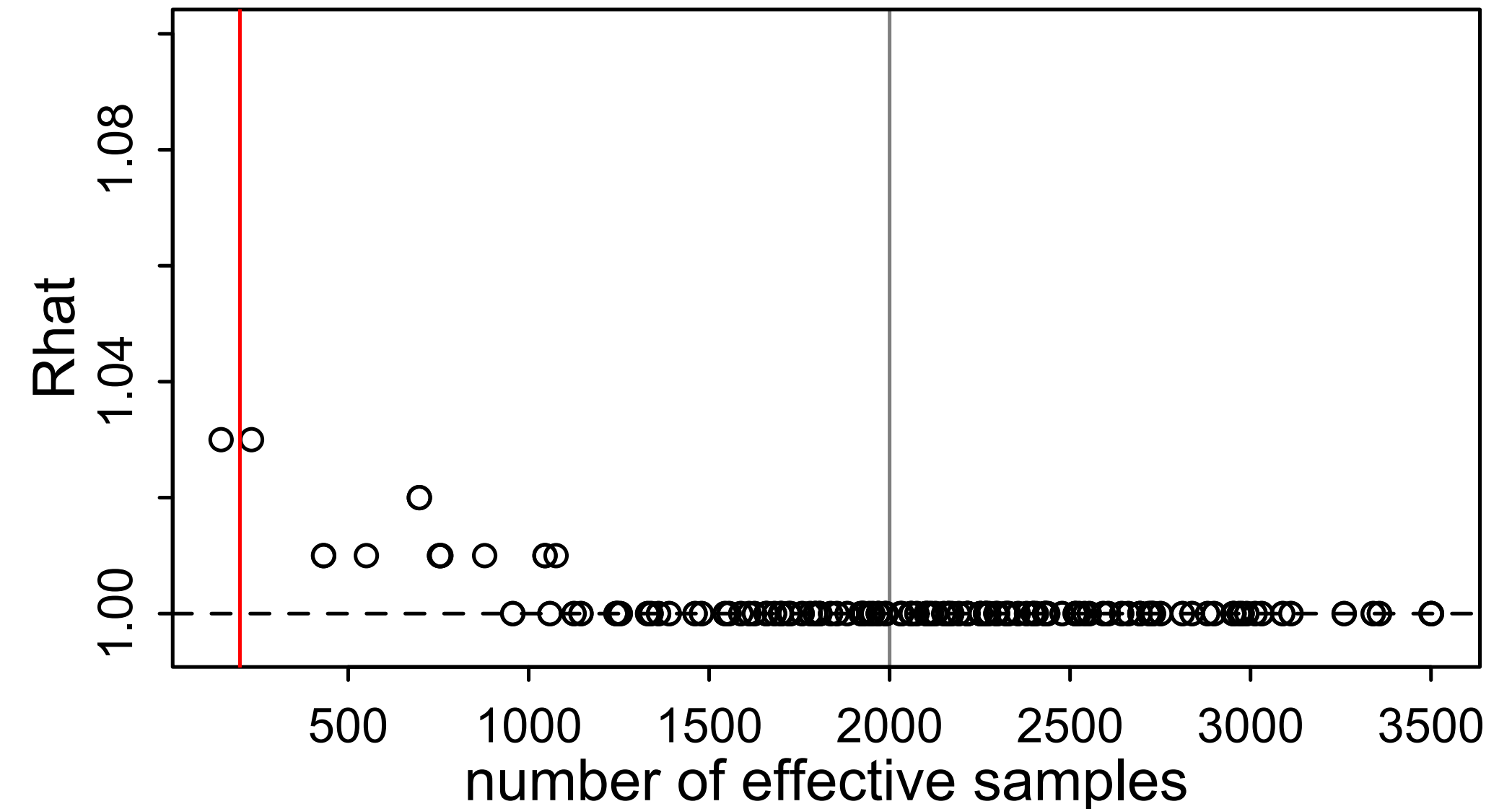
Inconvenient Posteriors

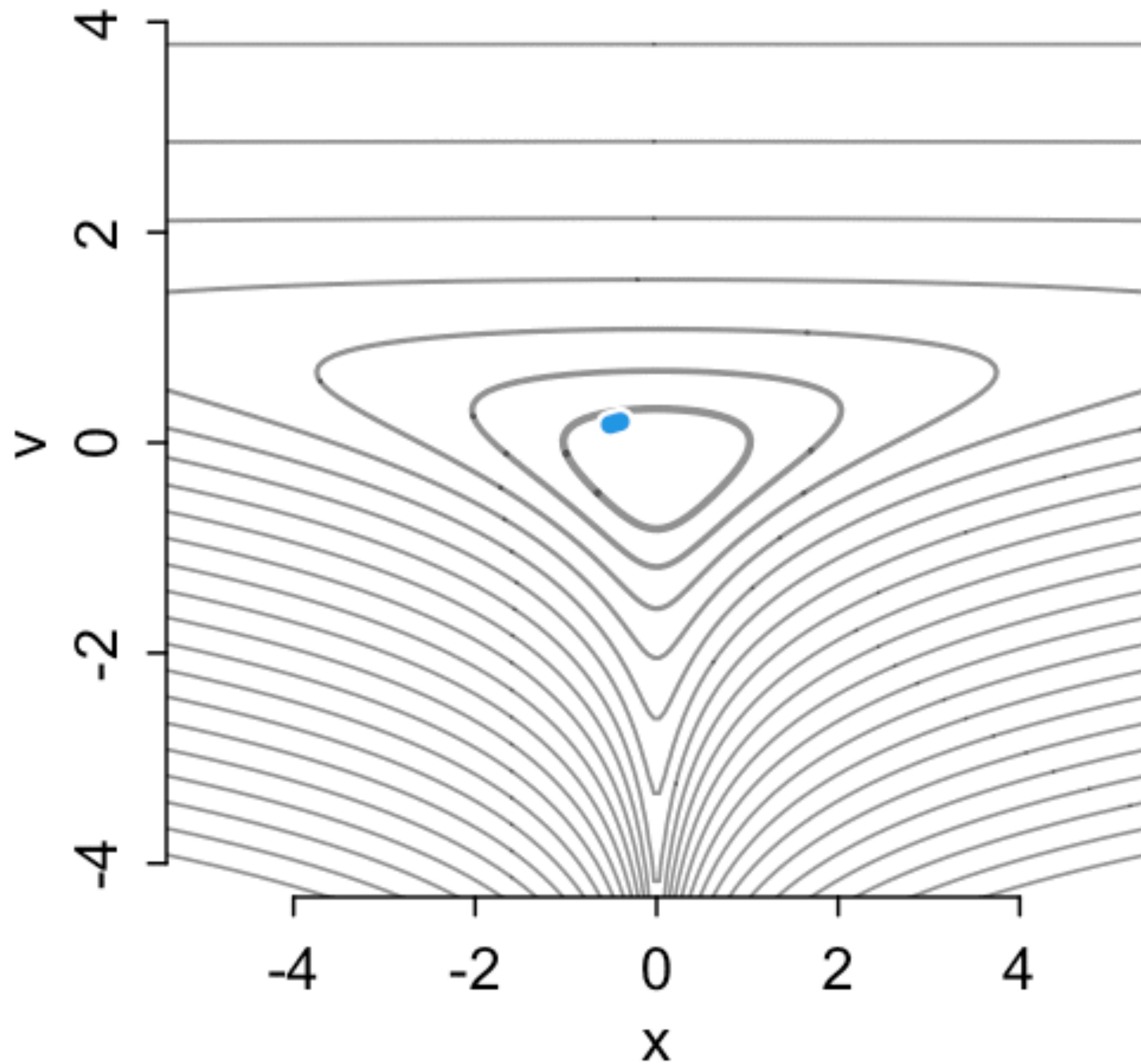
Inefficient MCMC can be caused by steep curvature

Hamiltonian simulation has trouble exploring surface

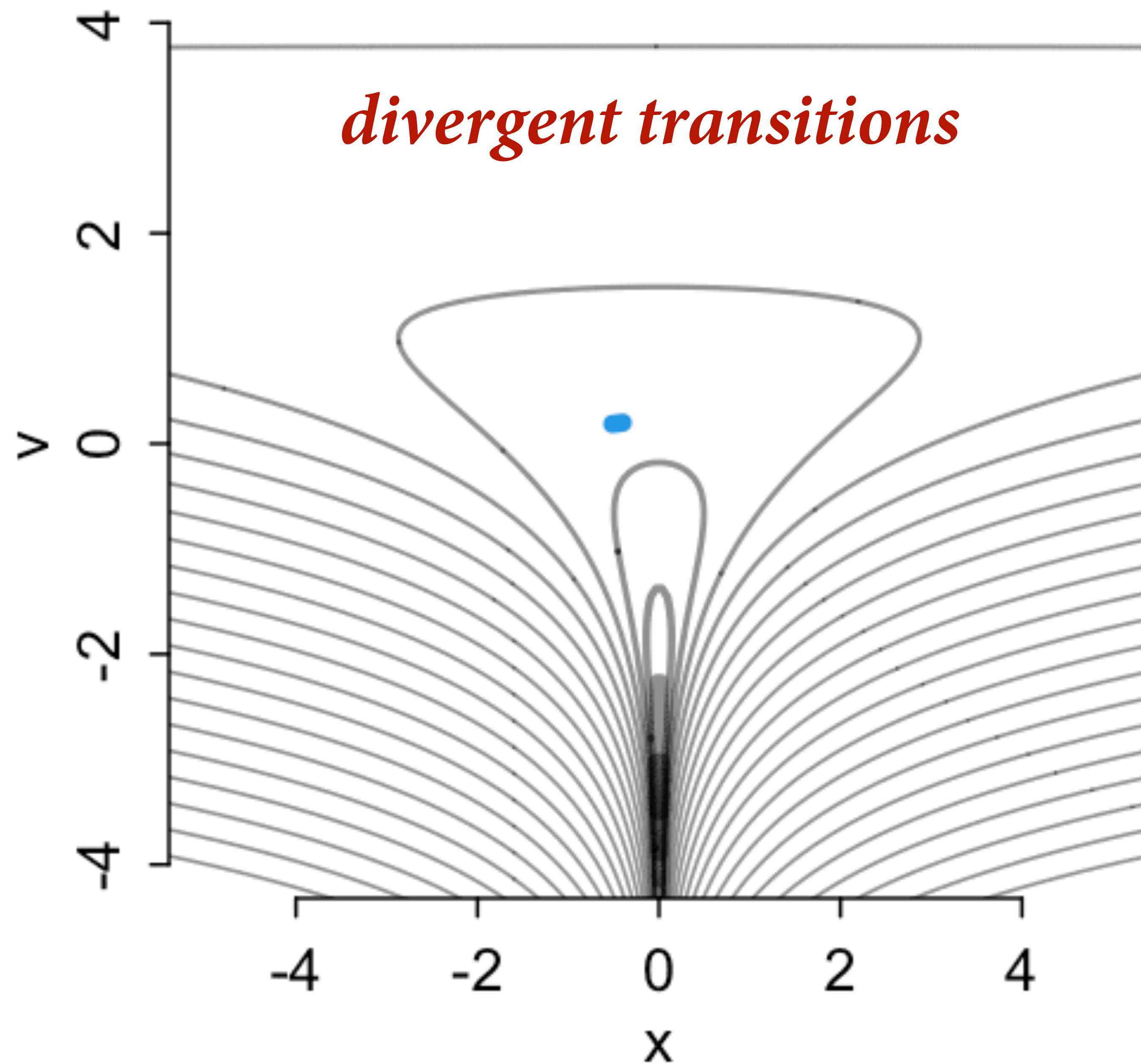
“Divergent transitions”

Transforming priors can help





$$v \sim \text{Normal}(0, 0.5)$$
$$x \sim \text{Normal}(0, \exp(v))$$



$$v \sim \text{Normal}(0,3)$$

$$x \sim \text{Normal}(0, \exp(v))$$



Divergent transitions

Why? Same step size not optimal everywhere

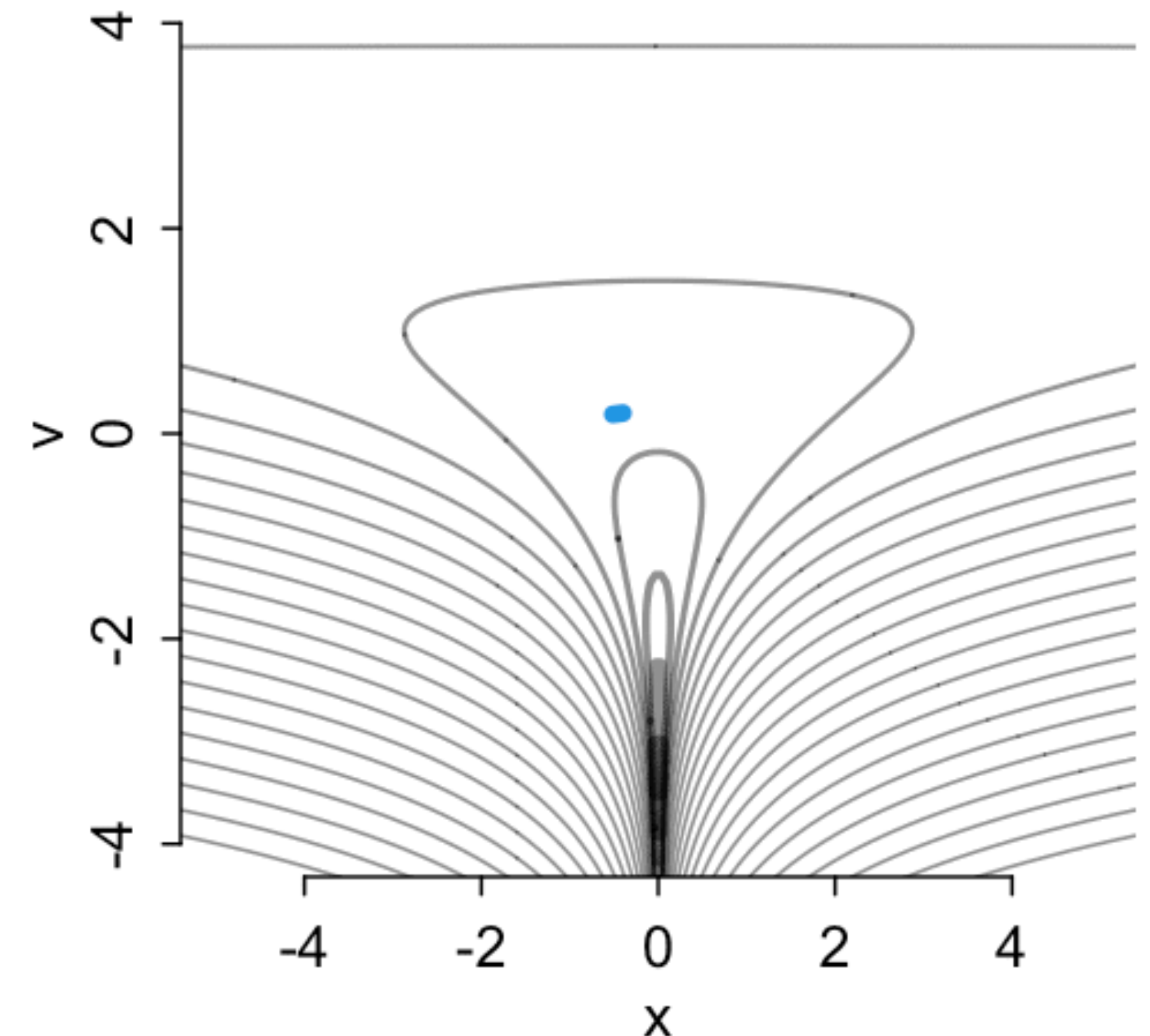
High curvature = simulation cannot follow surface

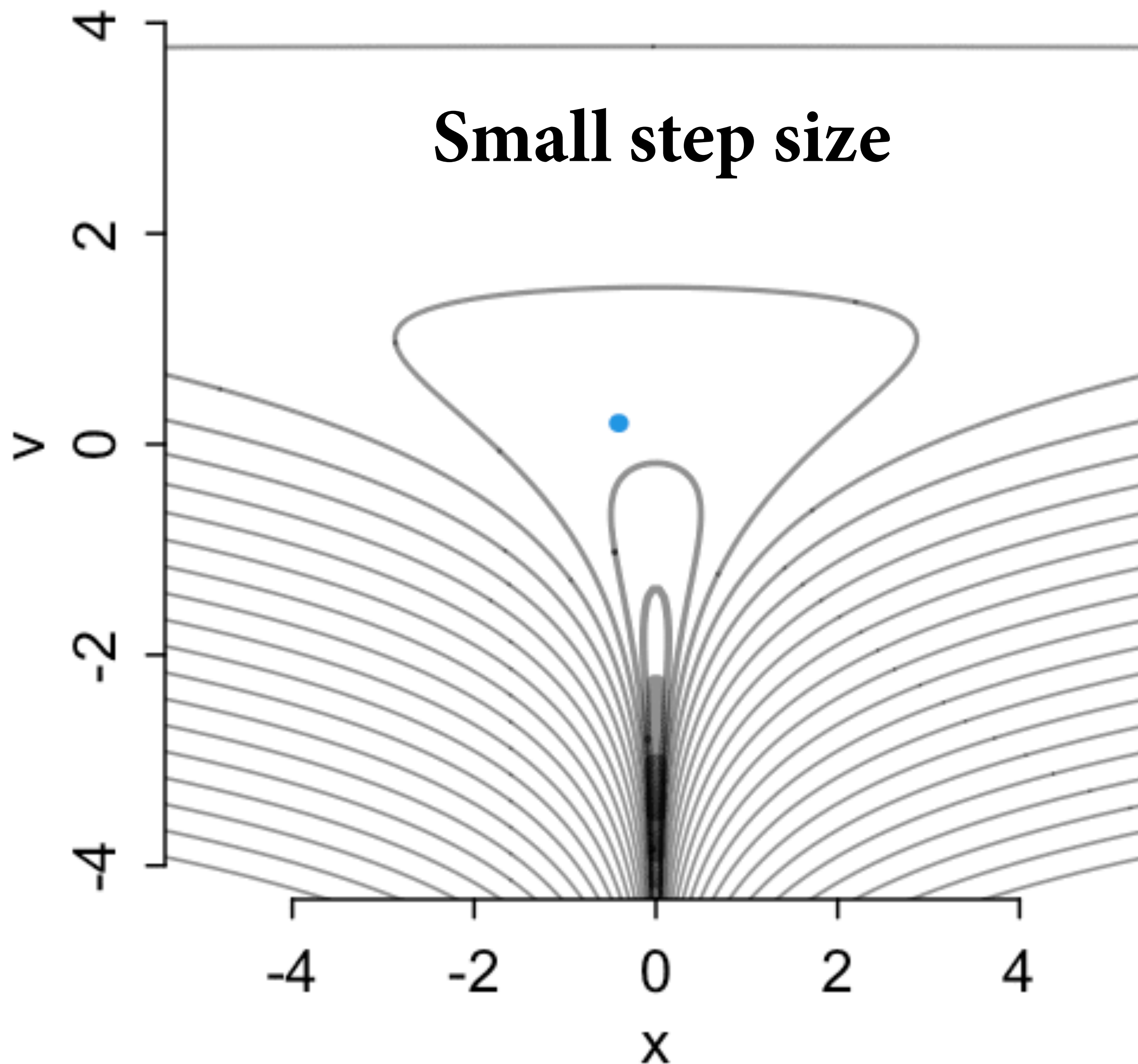
What can we do?

(1) use a smaller step size

(2) reparameterize!

$$v \sim \text{Normal}(0, 3)$$
$$x \sim \text{Normal}(0, \exp(v))$$





$$v \sim \text{Normal}(0, 3)$$

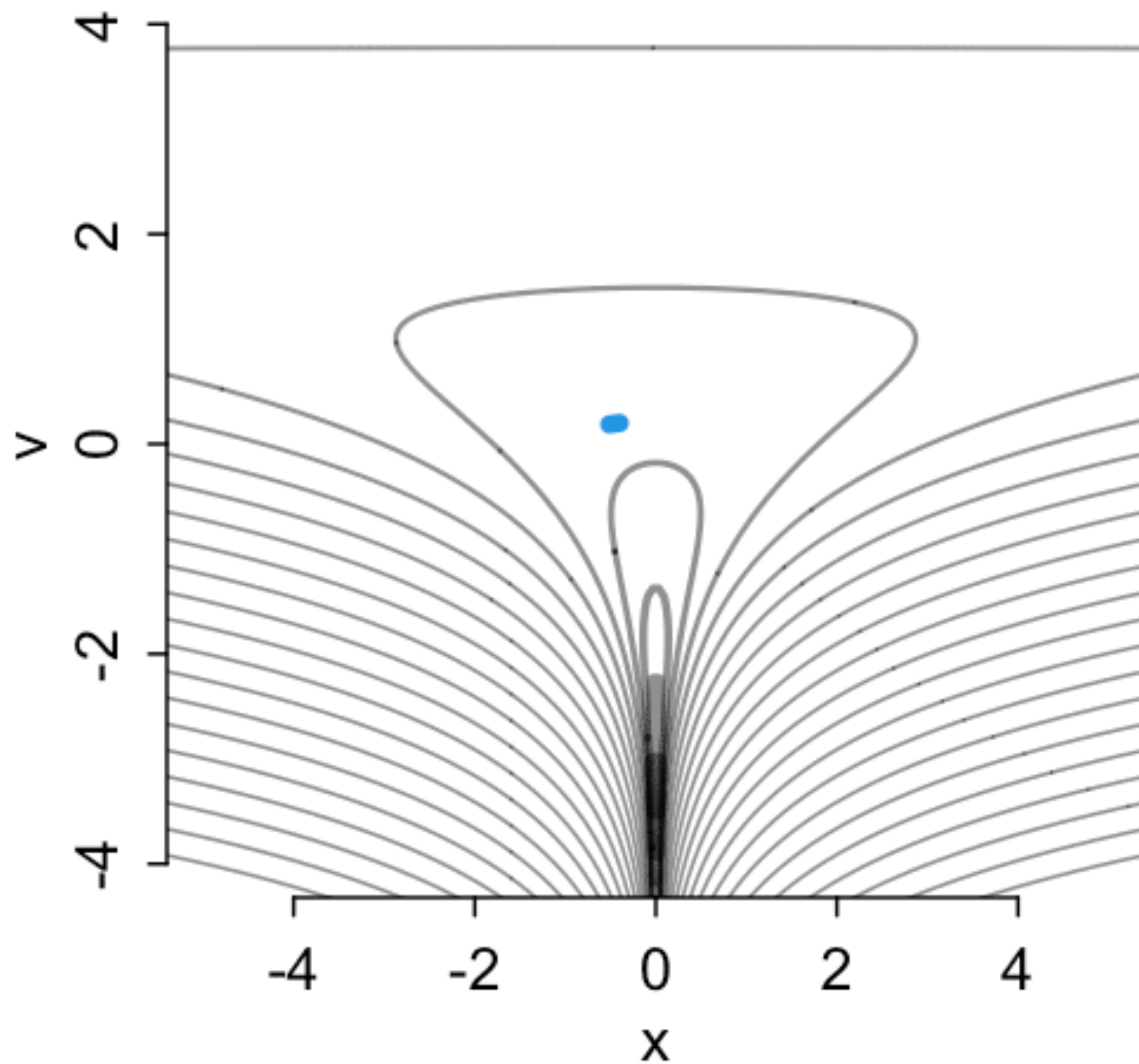
$$x \sim \text{Normal}(0, \exp(v))$$

Small step size helps, but
makes exploration slow

“Centered”

$$v \sim \text{Normal}(0,3)$$

$$x \sim \text{Normal}(0, \exp(v))$$

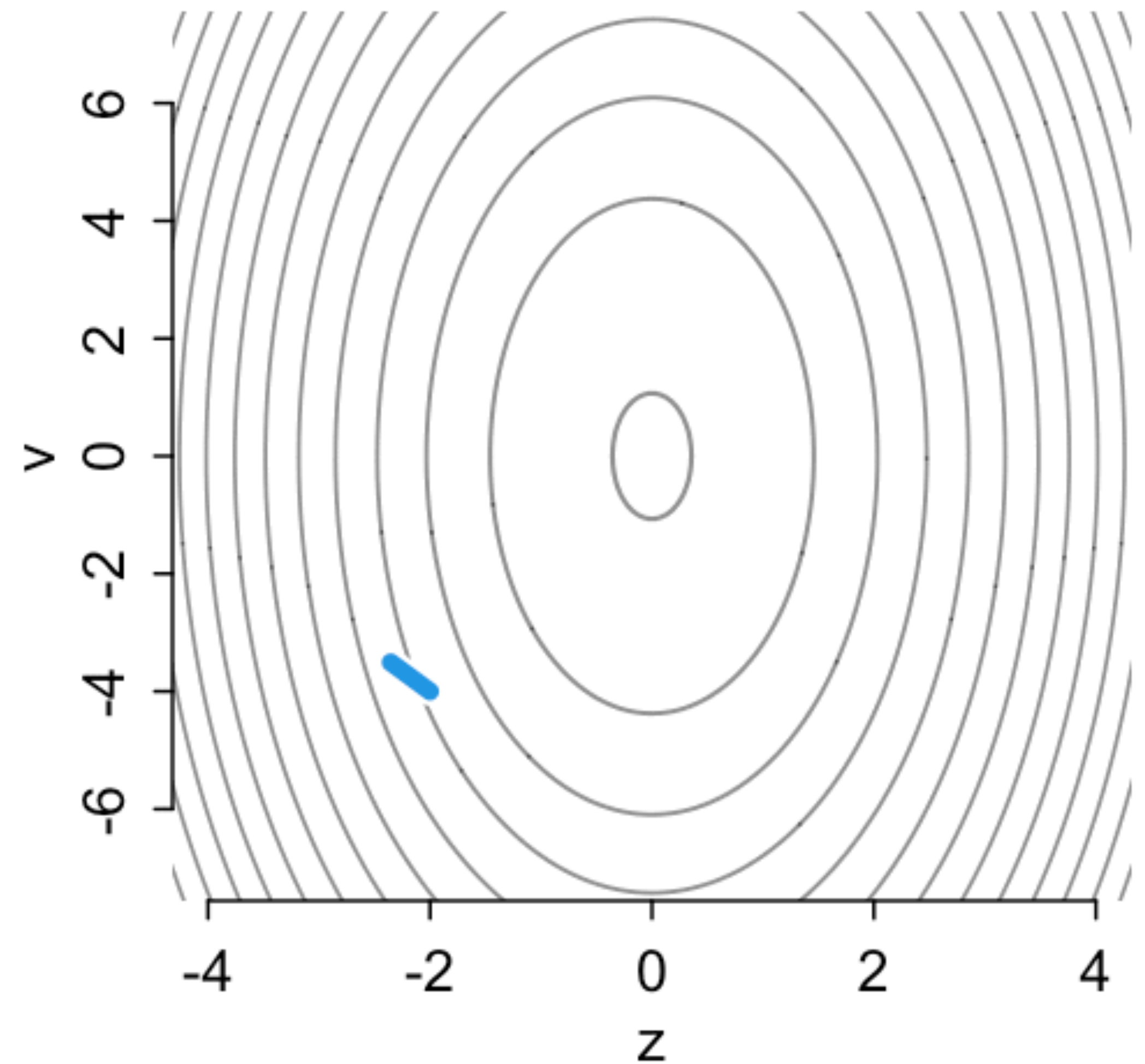


“Non-centered”

$$v \sim \text{Normal}(0,3)$$

$$z \sim \text{Normal}(0,1)$$

$$x = z \exp(v)$$



“Centered”

$$\nu \sim \text{Normal}(0, 3)$$

$$x \sim \text{Normal}(0, \exp(\nu))$$

“Centered”

$$v \sim \text{Normal}(0,3)$$

$$x \sim \text{Normal}(0, \exp(v))$$

“Non-centered”

$$v \sim \text{Normal}(0,3)$$

$$z \sim \text{Normal}(0,1)$$

$$x = z \exp(v)$$

```

mCDUnc <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D] + b[D]*U,
    # define effects using other parameters
    save> vector[61]:a <-< abar + za*sigma,
    save> vector[61]:b <-< bbar + zb*tau,
    # z-scored effects
    vector[61]:za ~ normal(0,1),
    vector[61]:zb ~ normal(0,1),
    # ye olde hyper-priors
    c(abar,bbar) ~ normal(0,1),
    c(sigma,tau) ~ exponential(1)
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

$$\beta_j = \bar{\beta} + z_{\beta,j} \times \tau$$

$$z_{\alpha,j} \sim \text{Normal}(0,1)$$

$$z_{\beta,j} \sim \text{Normal}(0,1)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

	mean	sd	5.5%	94.5%	n_eff	Rhat4
bbar	0.62	0.16	0.37	0.86	1513	1.00
abar	-0.70	0.09	-0.84	-0.56	1457	1.00
tau	0.55	0.23	0.17	0.92	368	1.01
sigma	0.49	0.09	0.36	0.64	753	1.00

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

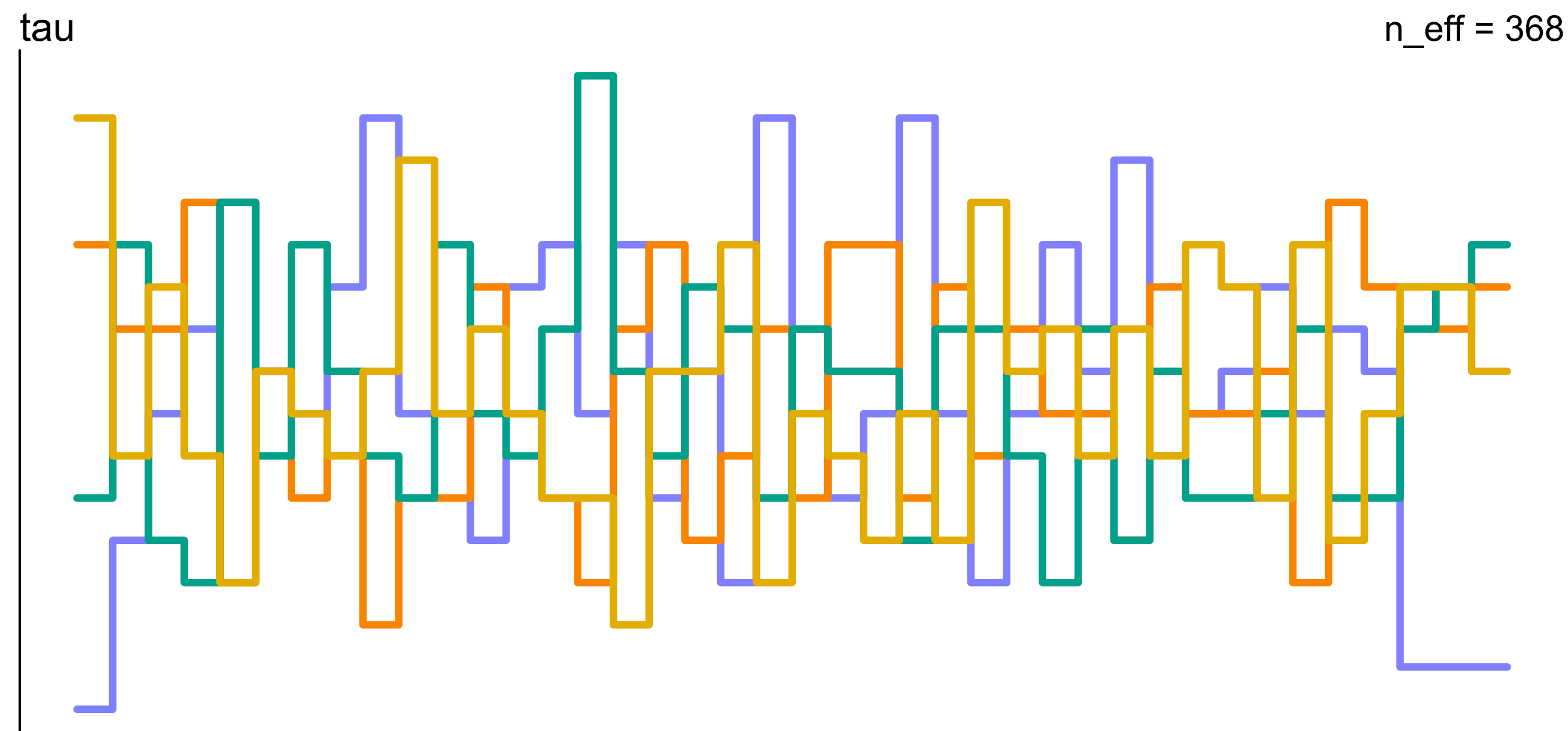
$$\beta_j = \bar{\beta} + z_{\beta,j} \times \tau$$

$$z_{\alpha,j} \sim \text{Normal}(0,1)$$

$$z_{\beta,j} \sim \text{Normal}(0,1)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

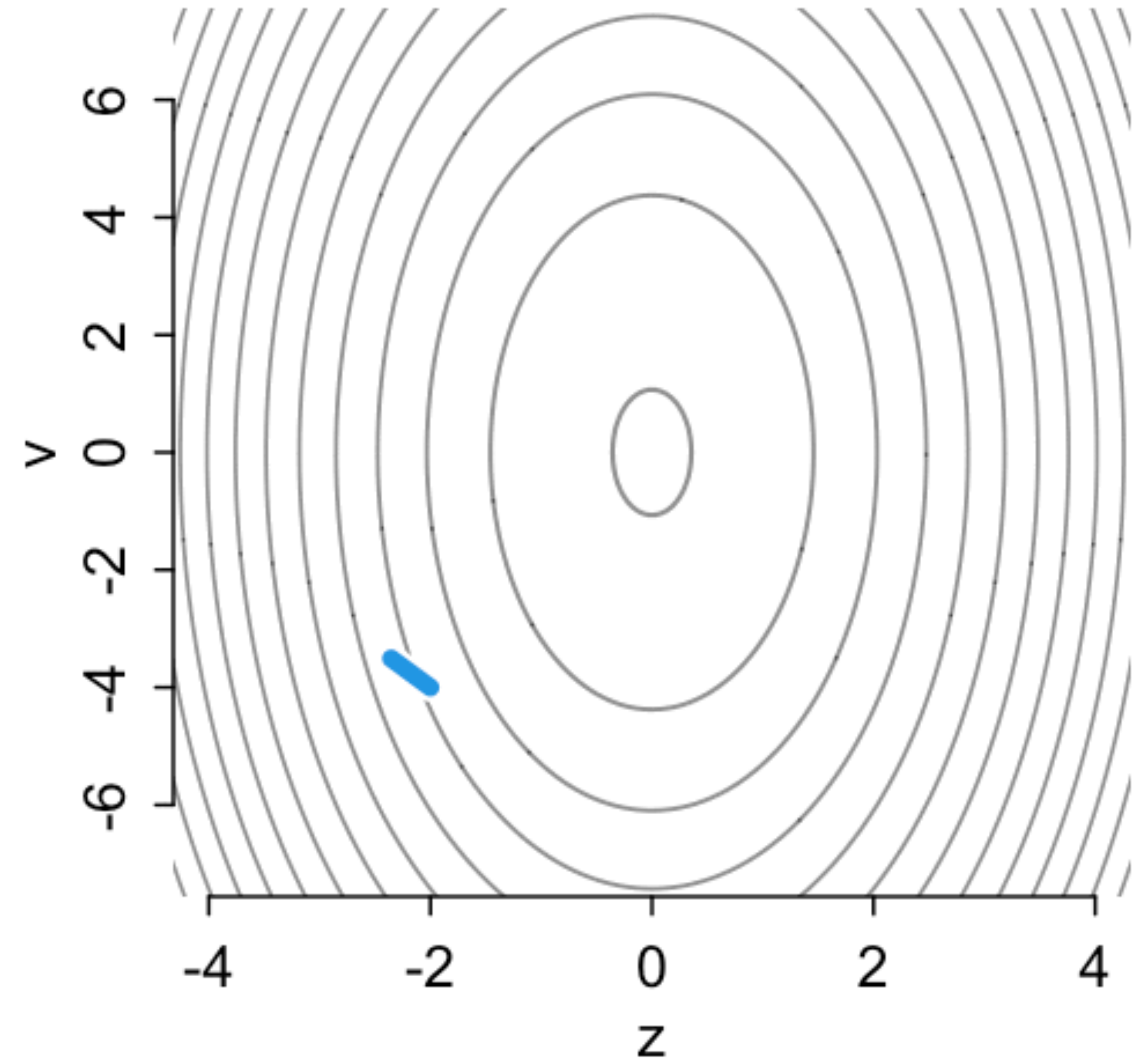


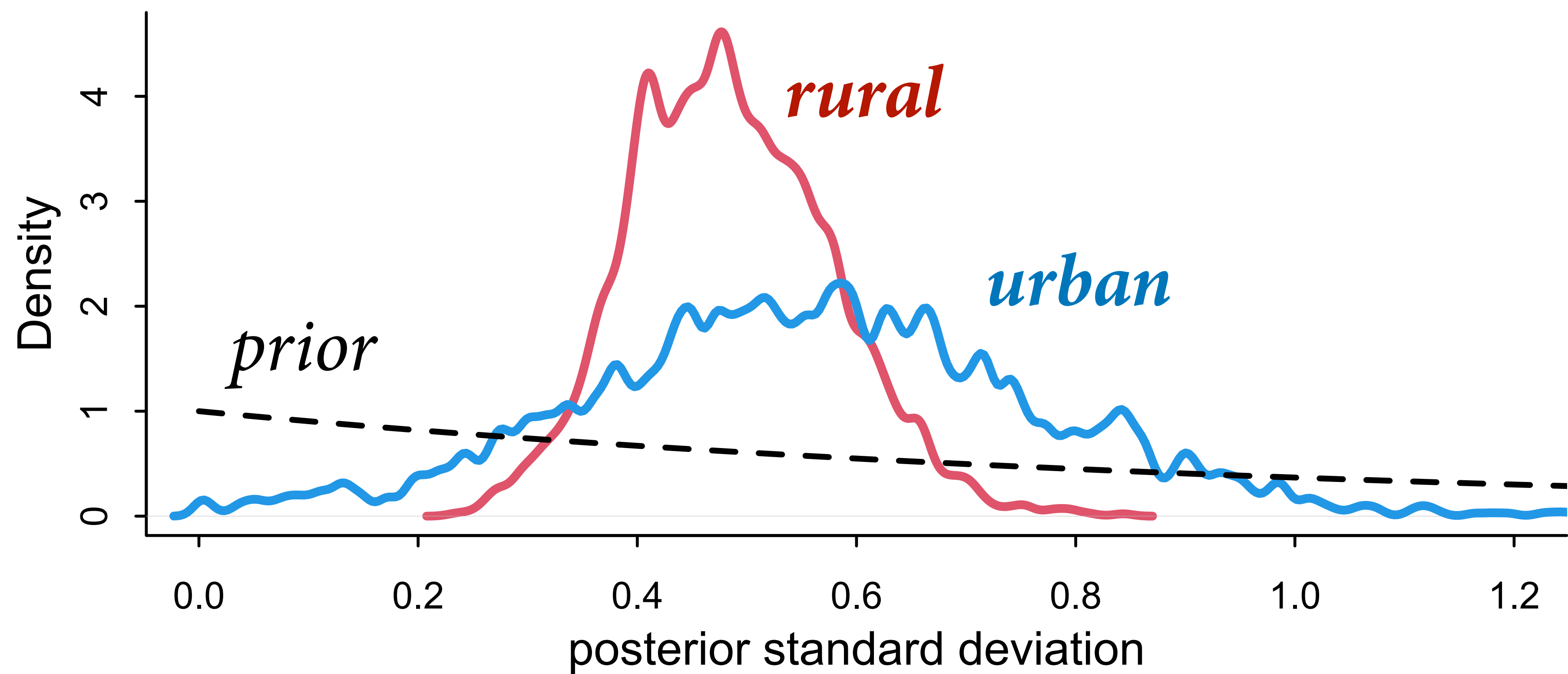
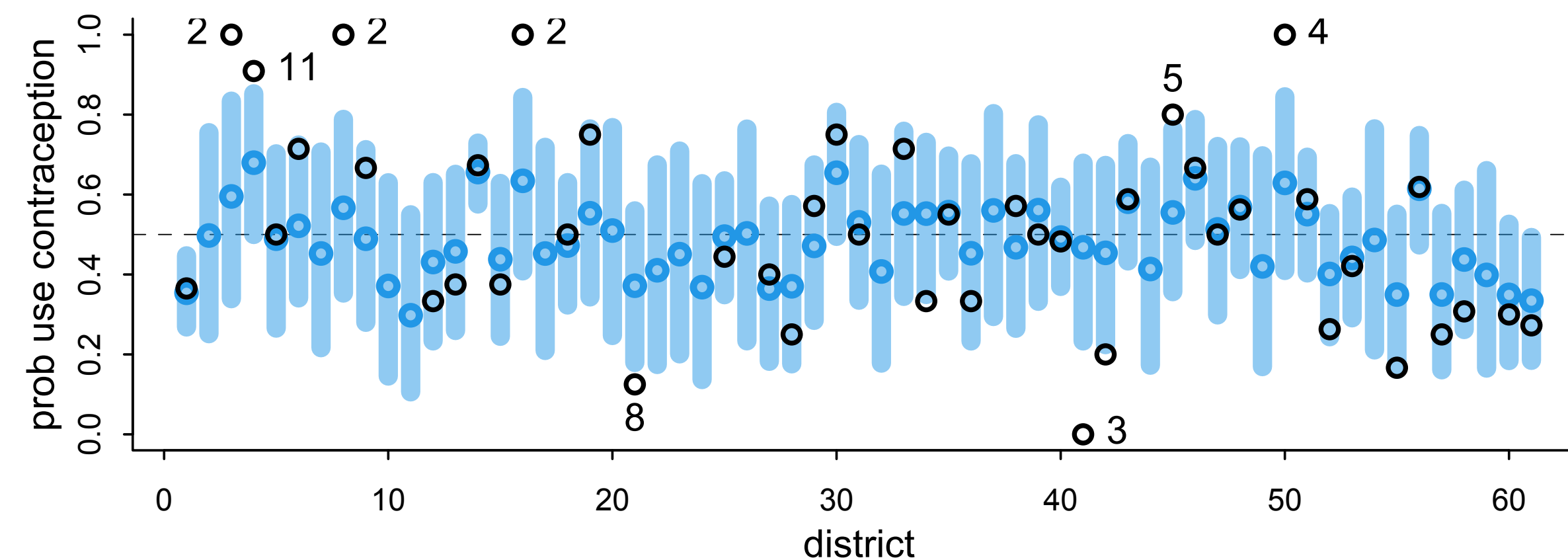
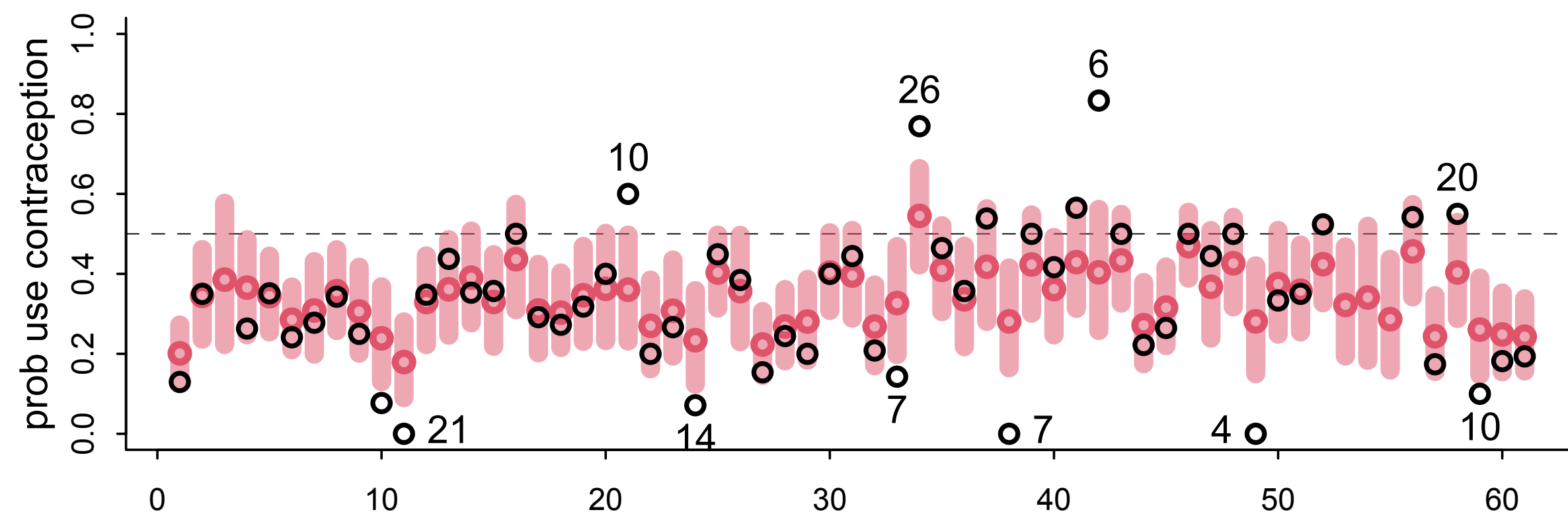
Transformed Priors

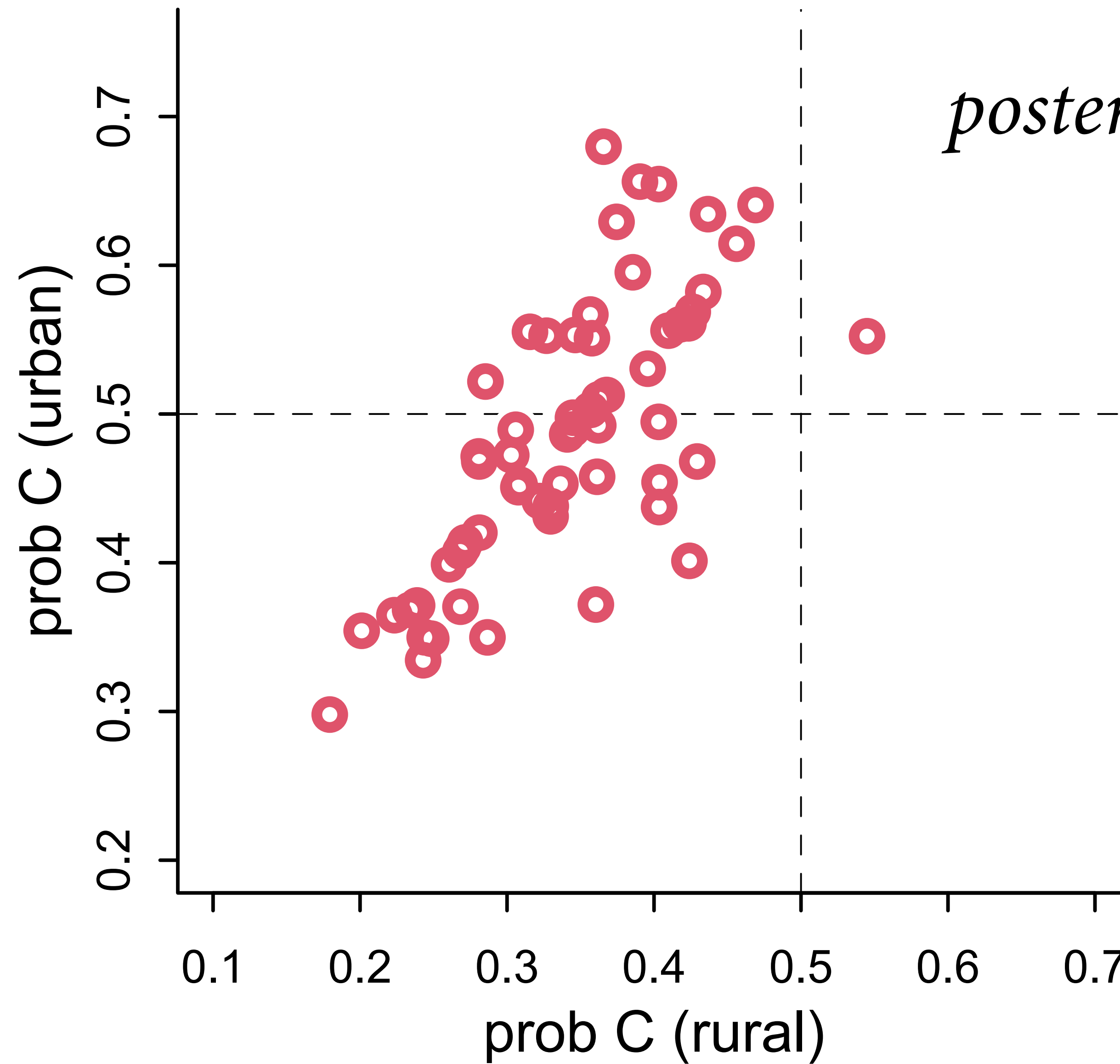
Both centered and non-centered
better in different contexts

Centered: Lots of data in each
cluster (data probability dominant)

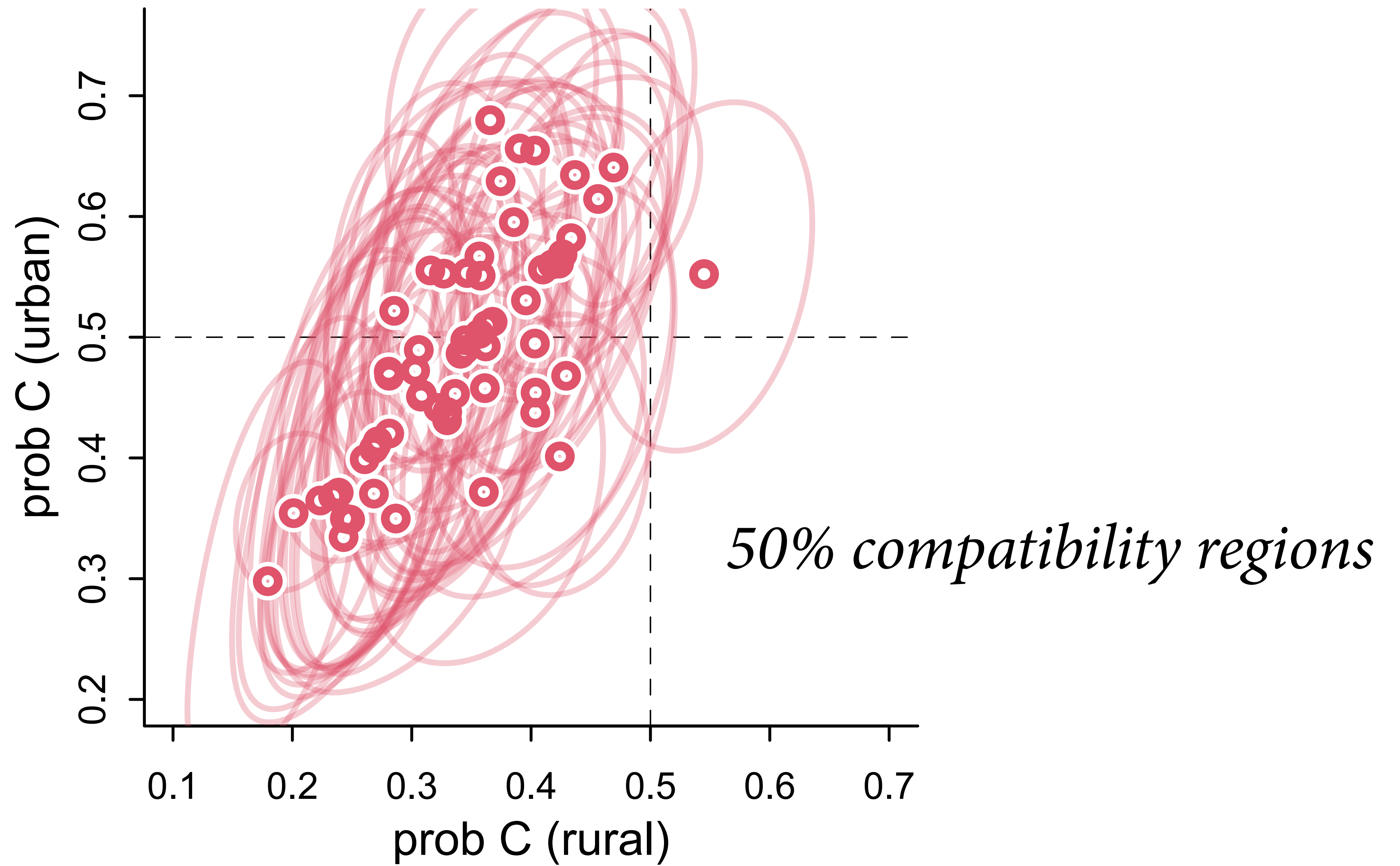
Non-centered: Many clusters, sparse
evidence (prior dominant)

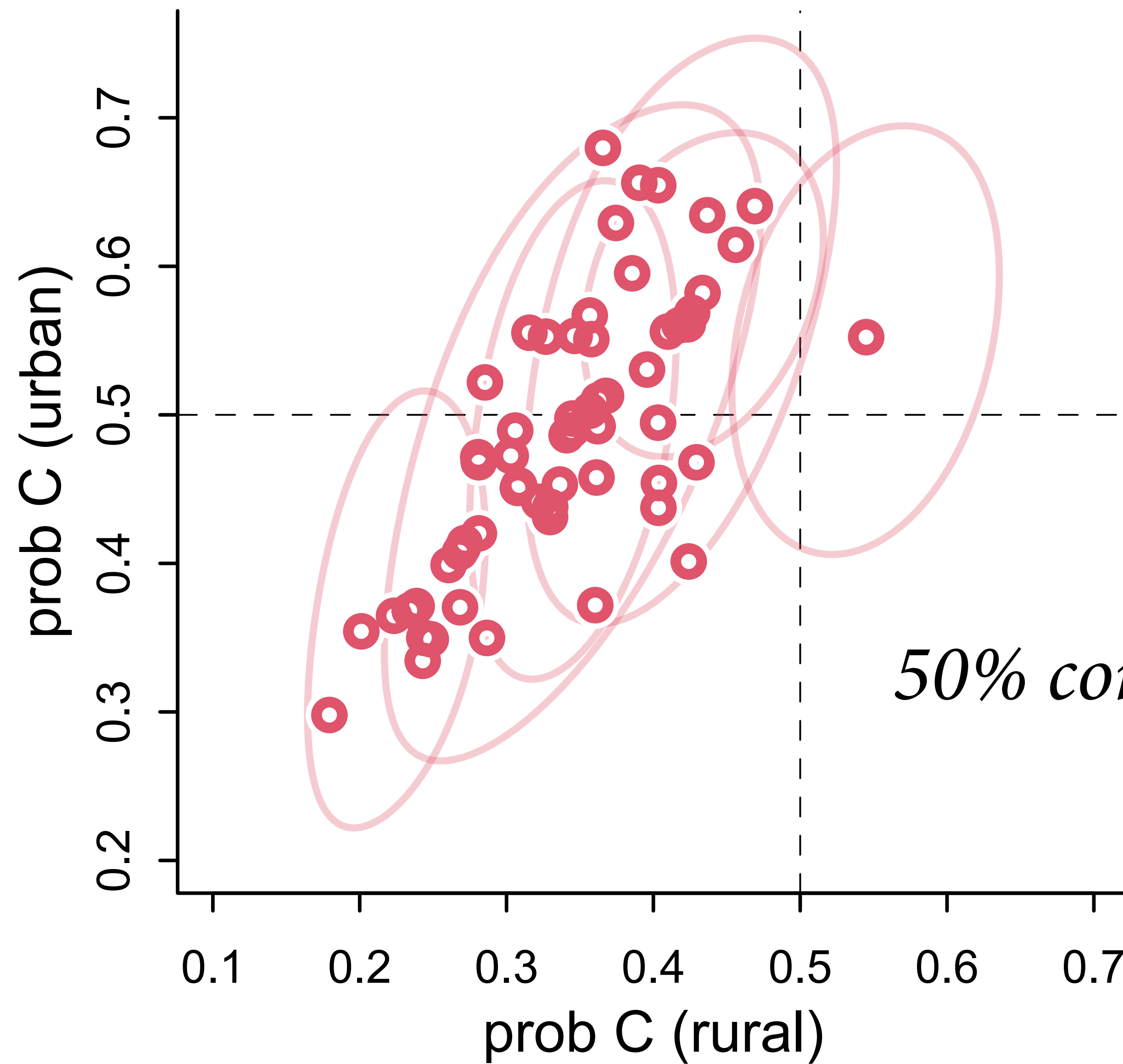






Each dot is a district
-> contraceptive use is
correlated within districts



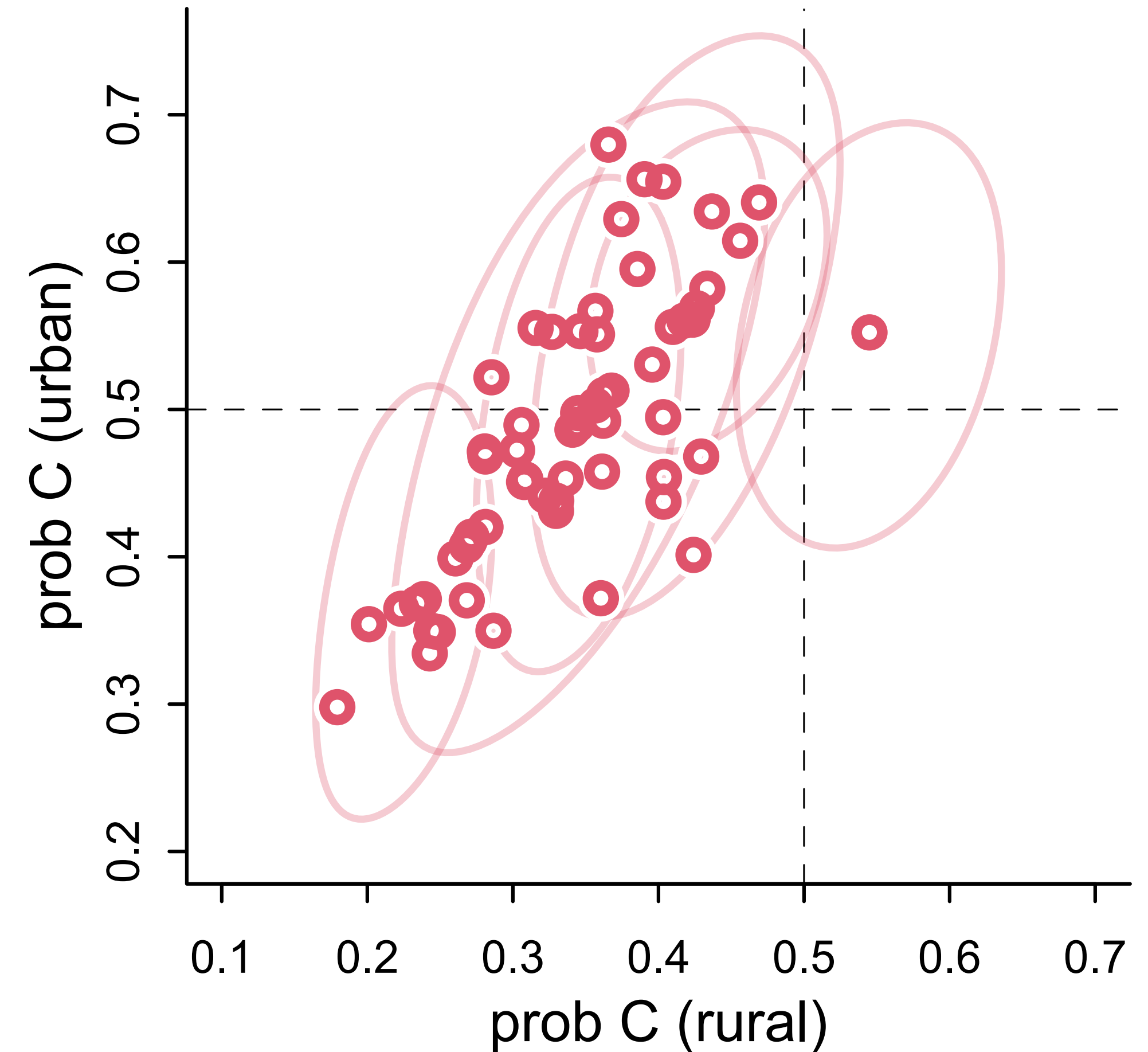


Multilevel adventures

Clusters: Kinds of groups in the data (districts)

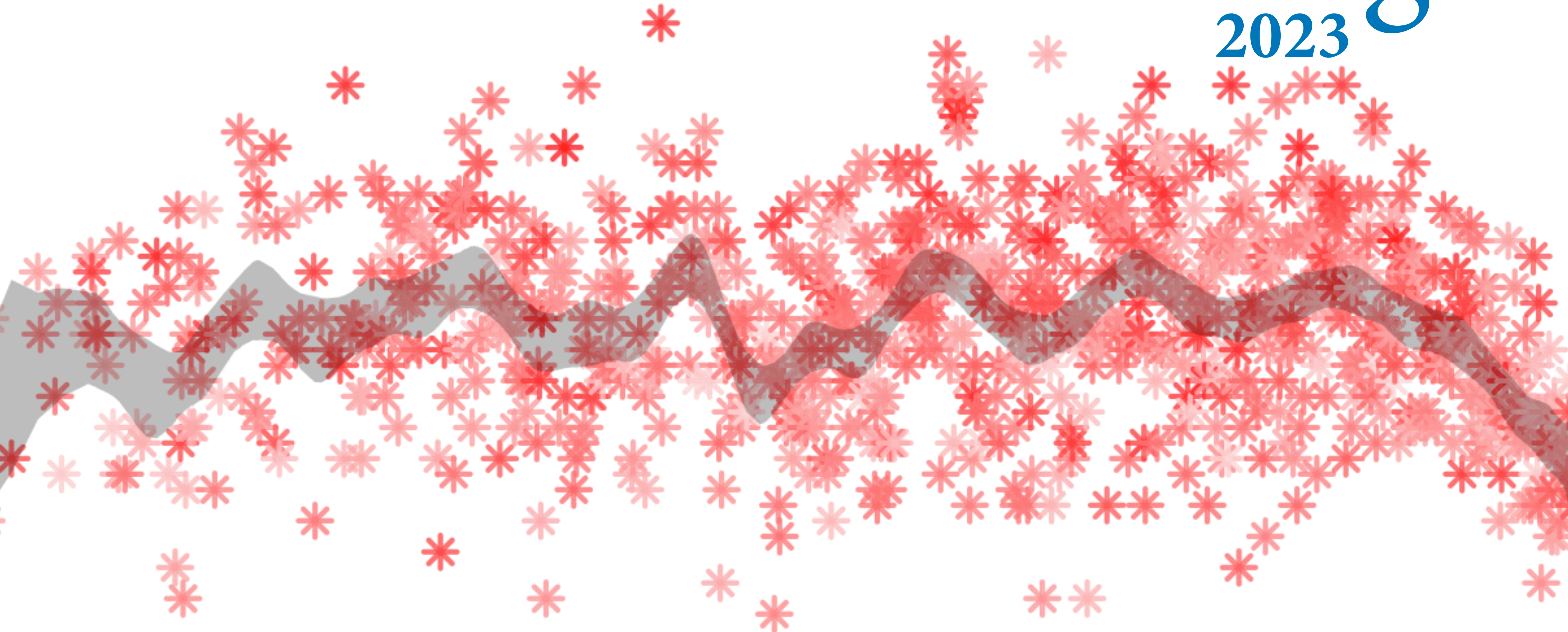
Features: Aspects of the model (parameters) that vary by cluster (rural, urban)

There is useful information to transfer across features



Statistical Rethinking

2023



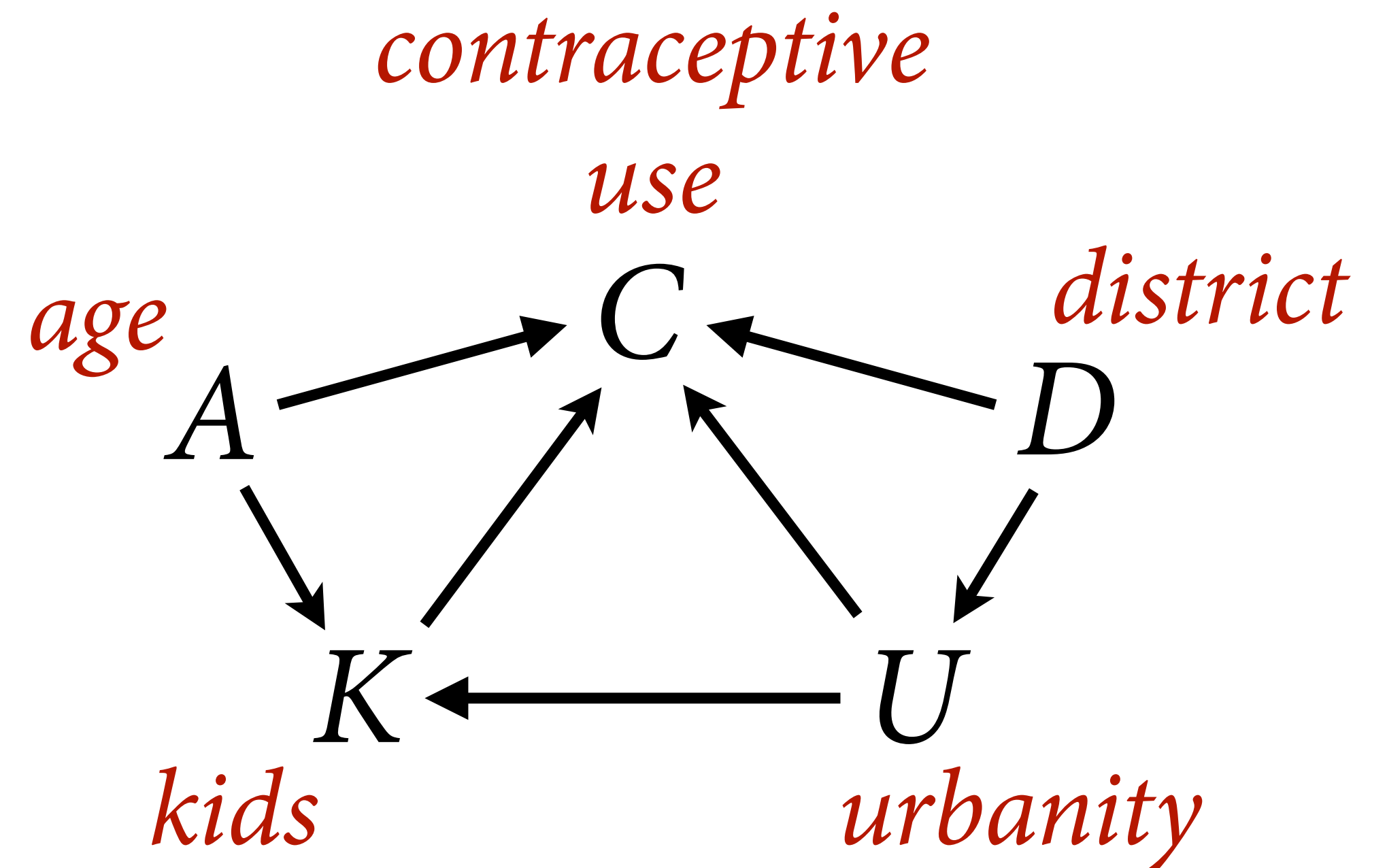
14. Correlated Features

Bangladesh 1989, Party Like It's

Estimand 1: C in each district

Estimand 2: Effect of U

Estimand 3: Effects of K and A

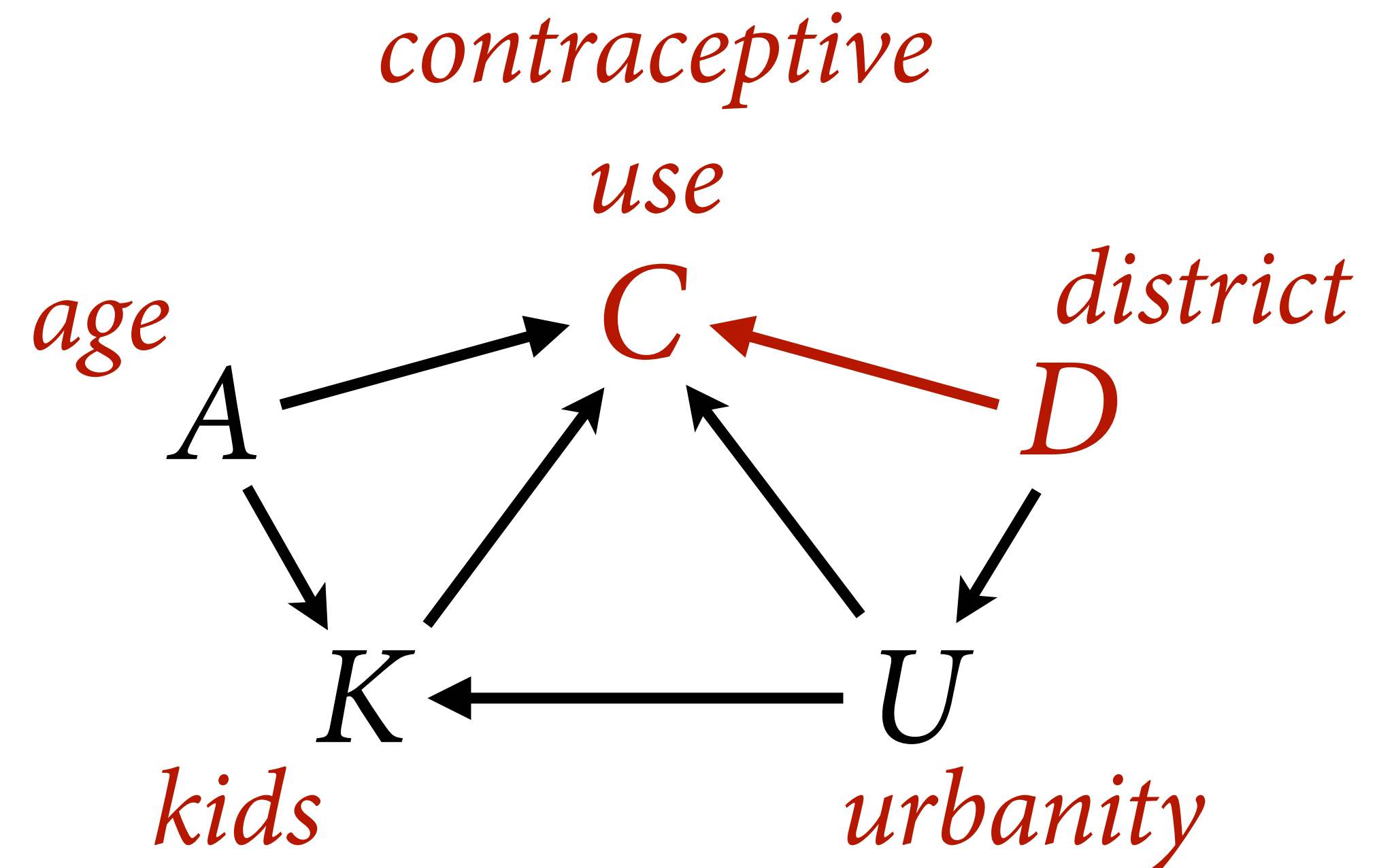


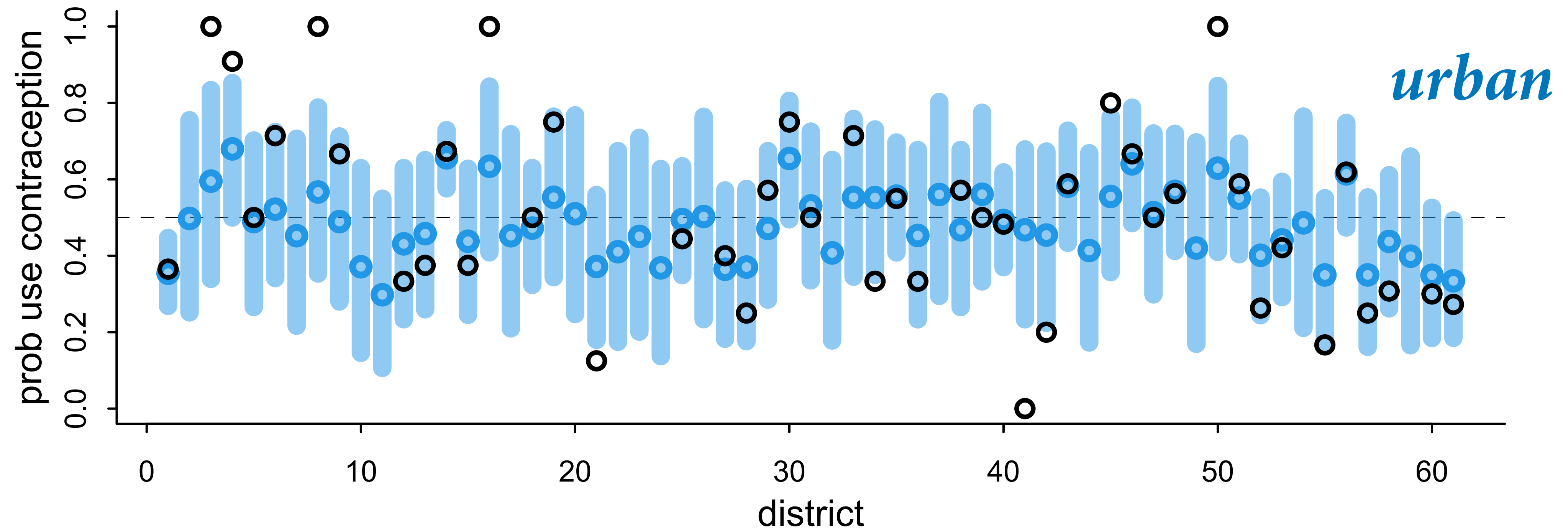
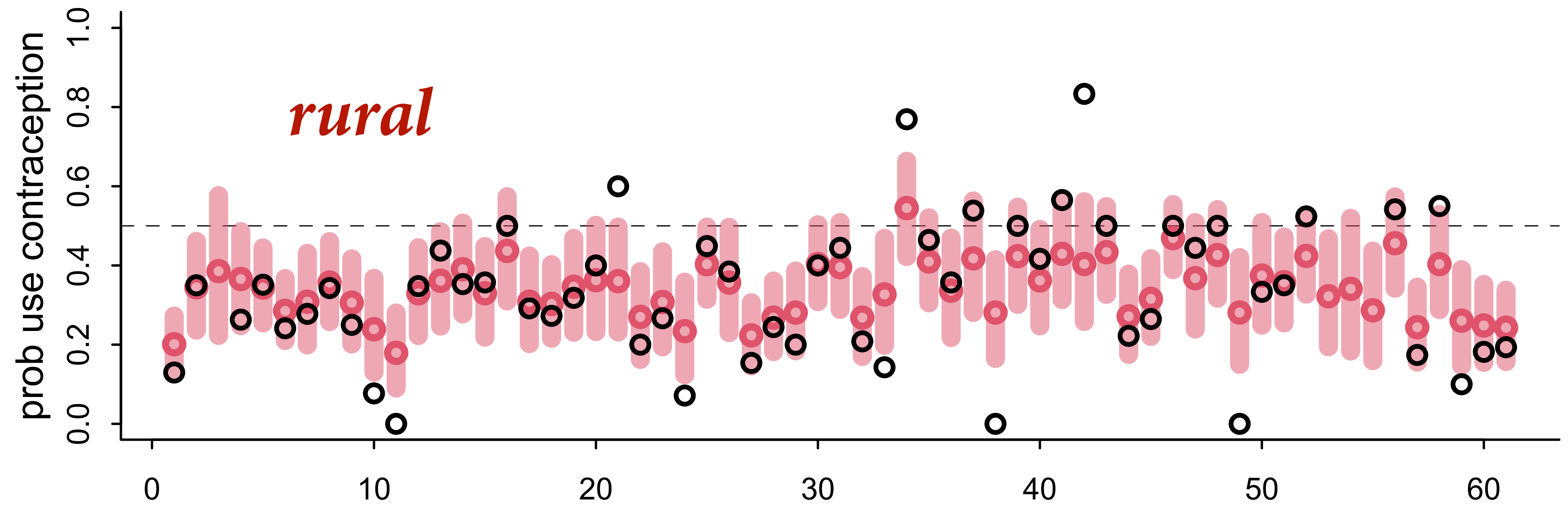
Bangladesh 1989, Party Like It's

Estimand 1: C in each district

Estimand 2: Effect of U

Estimand 3: Effects of K and A



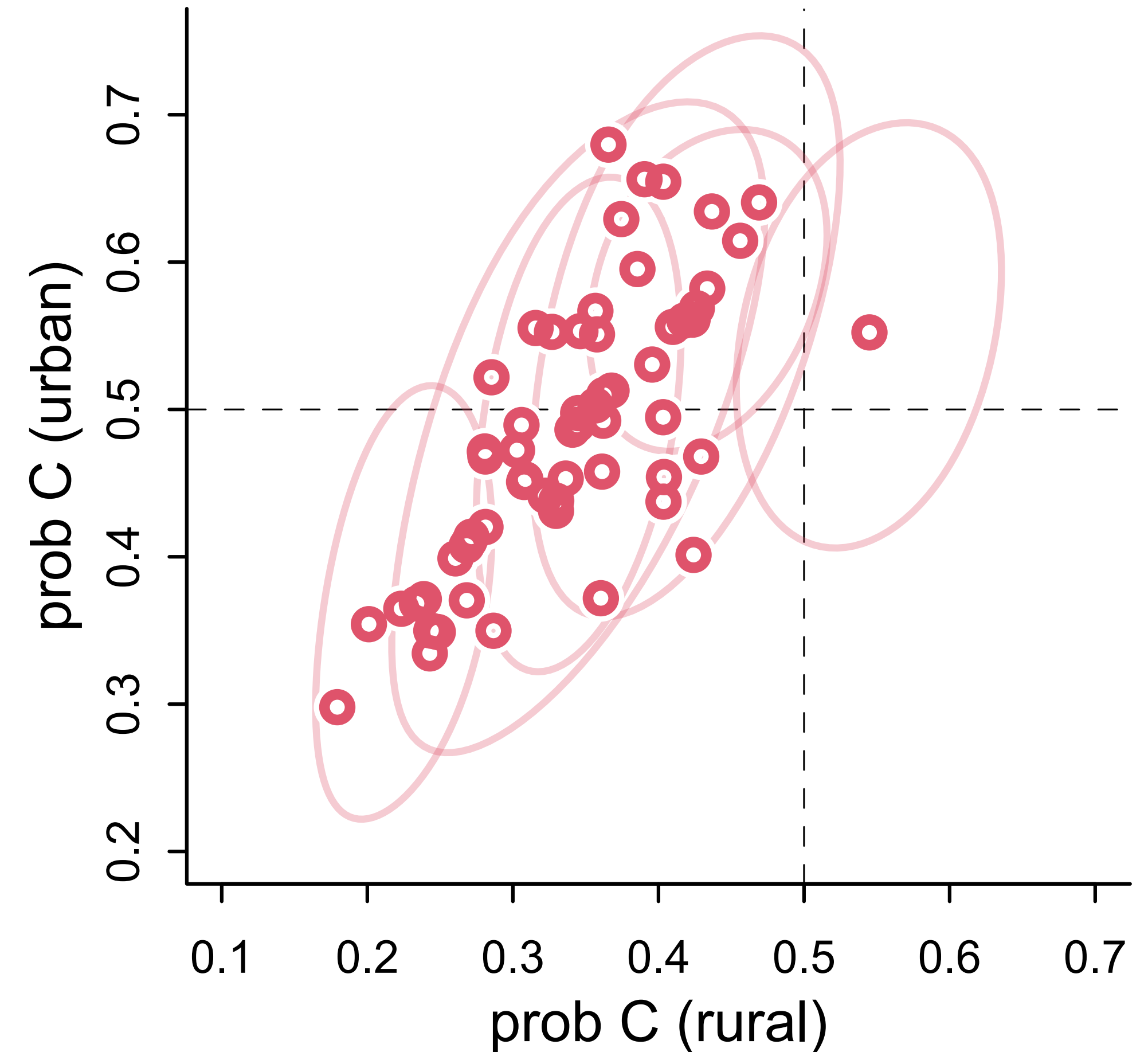


Multilevel Adventures

Clusters: Kinds of groups in the data (districts)

Features: Aspects of the model (parameters) that vary by cluster (rural, urban)

There is useful information to transfer across features



Adding Correlated Features

One prior distribution for each cluster

One feature: One-dimensional distribution

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

Two features: Two-D distribution

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \Sigma)$$

N features: N -dimensional distribution

$$\alpha_{j,1..N} \sim \text{MVNormal}(A, \Sigma)$$

Hard part: Learning associations

```

mCDUnc <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D] + b[D]*U,
    # define effects using other parameters
    save> vector[61]:a <-< abar + za*sigma,
    save> vector[61]:b <-< bbar + zb*tau,
    # z-scored effects
    vector[61]:za ~ normal(0,1),
    vector[61]:zb ~ normal(0,1),
    # ye olde hyper-priors
    c(abar,bbar) ~ normal(0,1),
    c(sigma,tau) ~ exponential(1)
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j = \bar{\alpha} + z_{\alpha,j} \times \sigma$$

$$\beta_j = \bar{\beta} + z_{\beta,j} \times \tau$$

$$z_{\alpha,j} \sim \text{Normal}(0,1)$$

$$z_{\beta,j} \sim \text{Normal}(0,1)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

```

dat <- list(
  C = d$use.contraception,
  D = as.integer(d$district),
  U = ifelse(d$urban==1,1,0) )

# total U
mCDU <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D] + b[D]*U,
    vector[61]:a ~ normal(abar,sigma),
    vector[61]:b ~ normal(bbar,tau),
    c(abar,bbar) ~ normal(0,1),
    c(sigma,tau) ~ exponential(1)
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$C_i \sim \text{Bernoulli}(p_i)$$

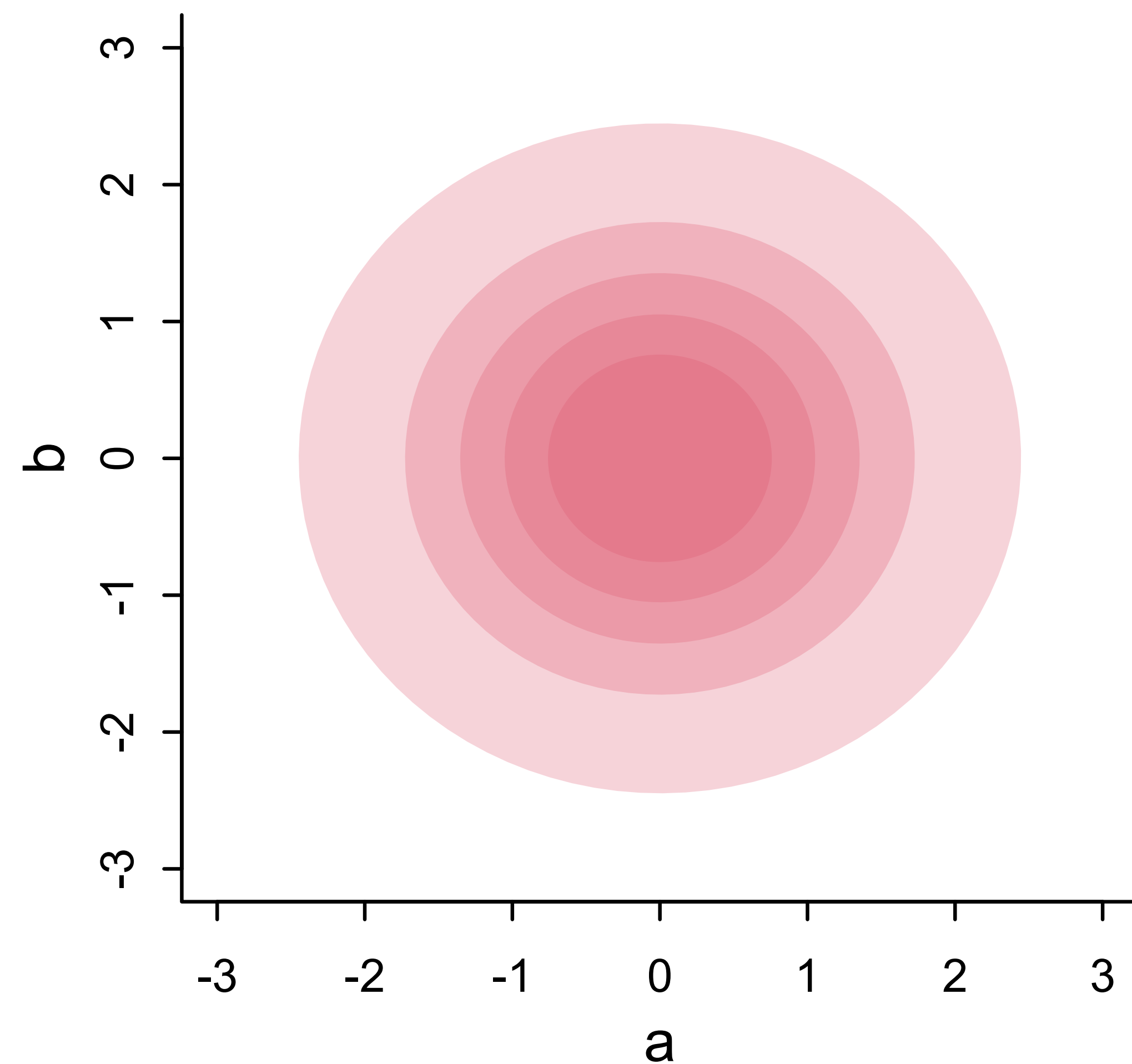
$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$



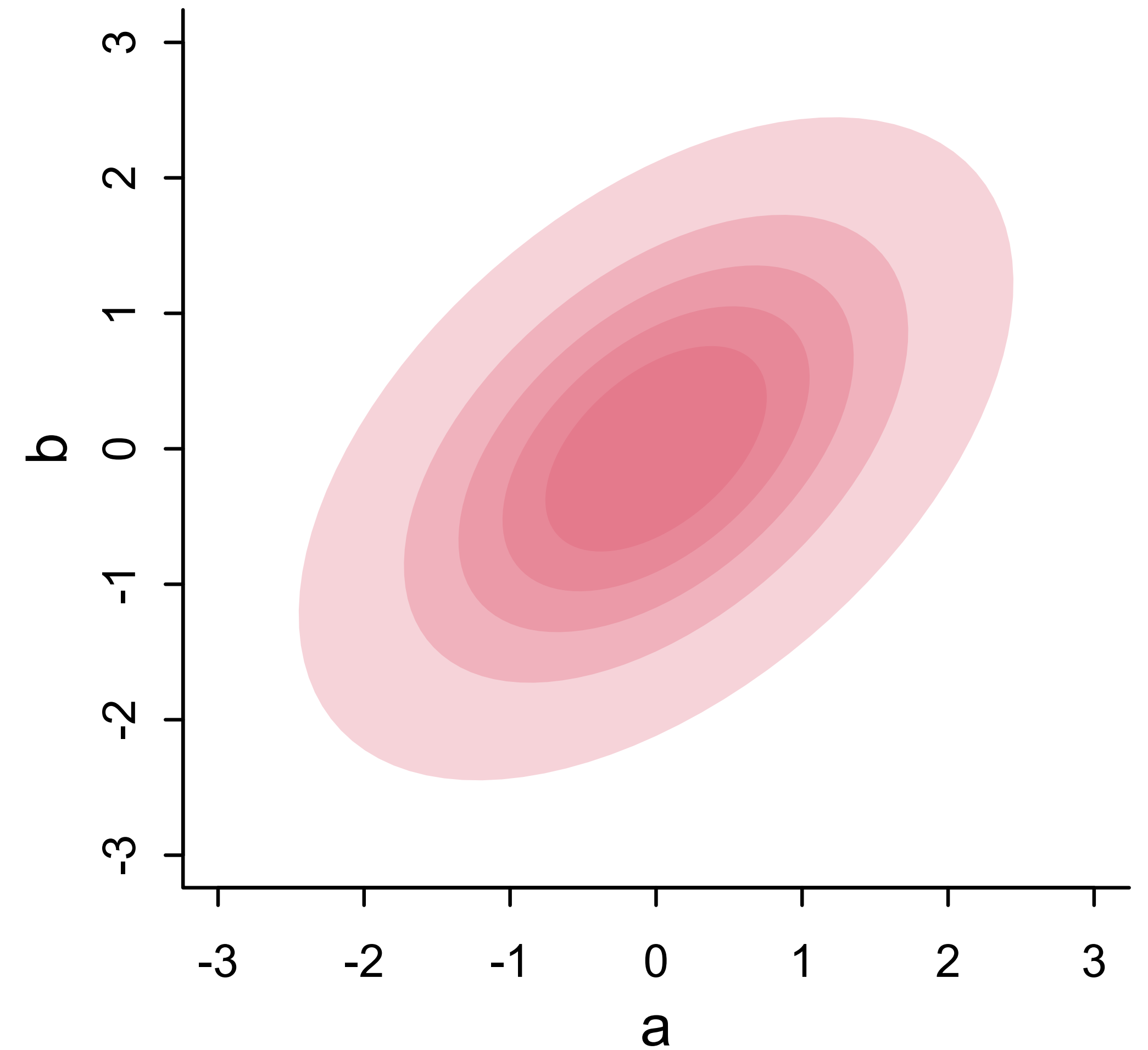
$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \mathbf{R}, [\sigma, \tau])$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$



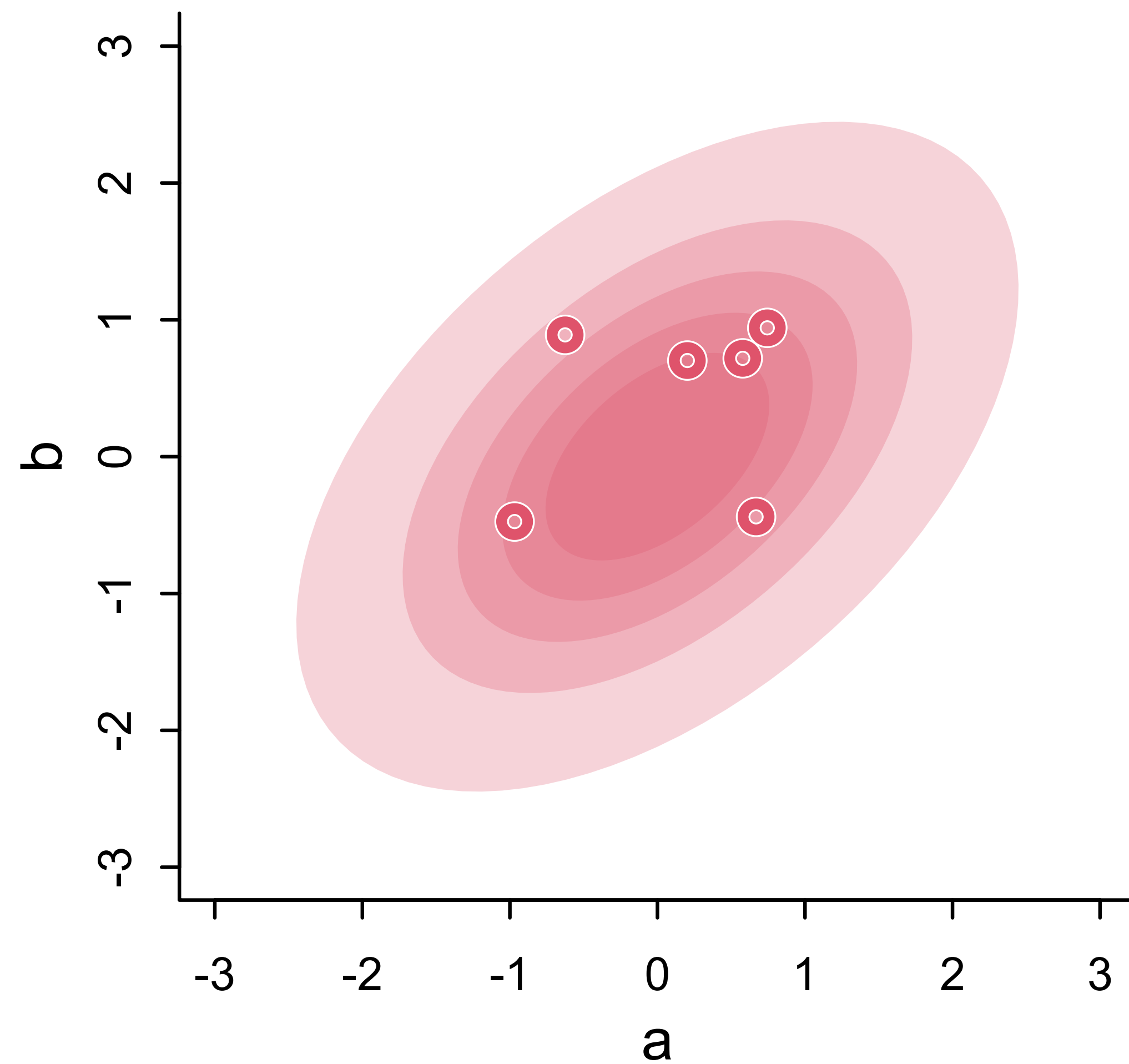
*features for
district j*

*correlation
matrix*

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], R, [\sigma, \tau])$$

*feature
means*

*standard
deviations*



$$C_i \sim \text{Bernoulli}(p_i)$$

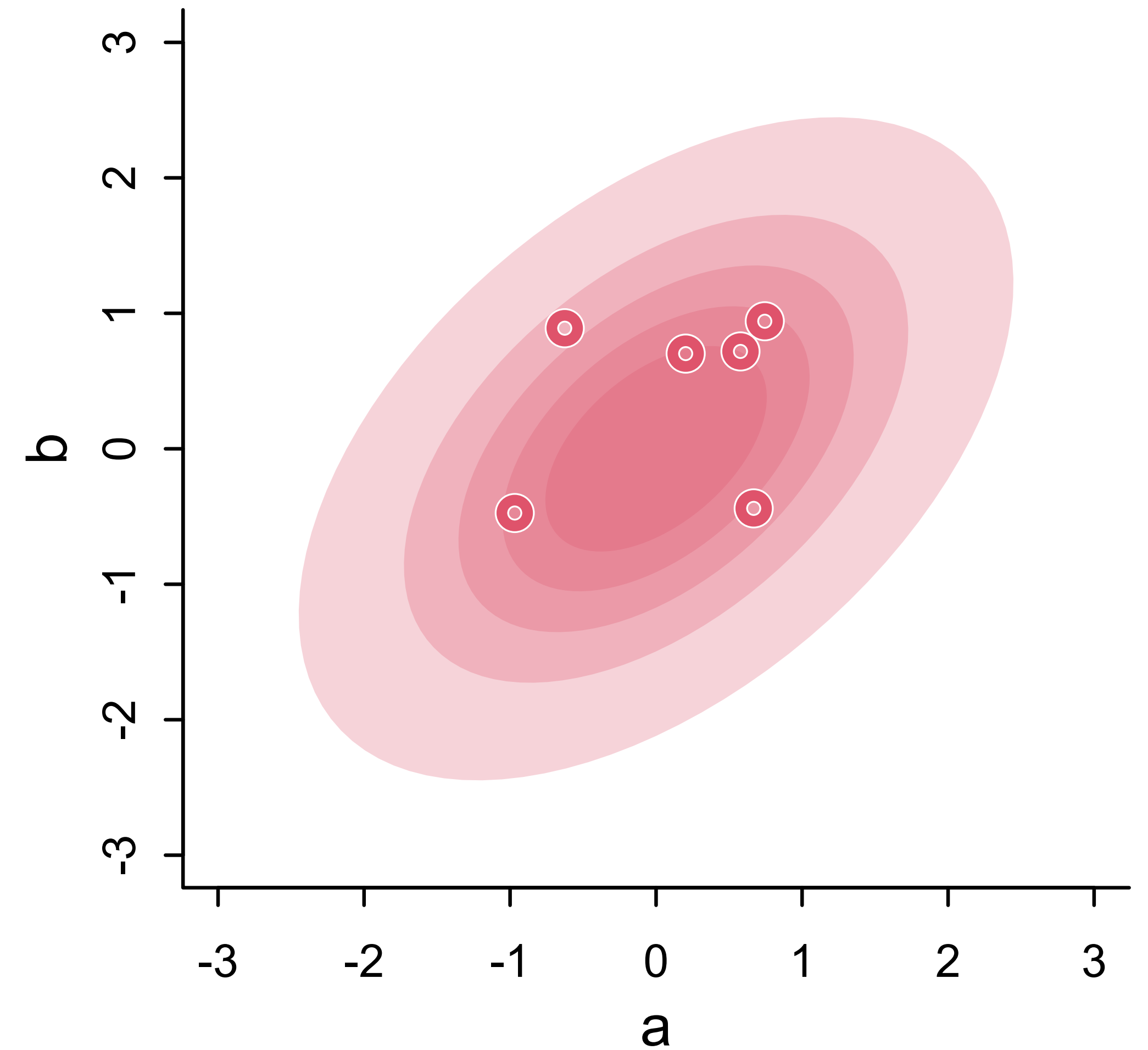
$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \mathbf{R}, [\sigma, \tau])$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$\mathbf{R} \sim ?$$



$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

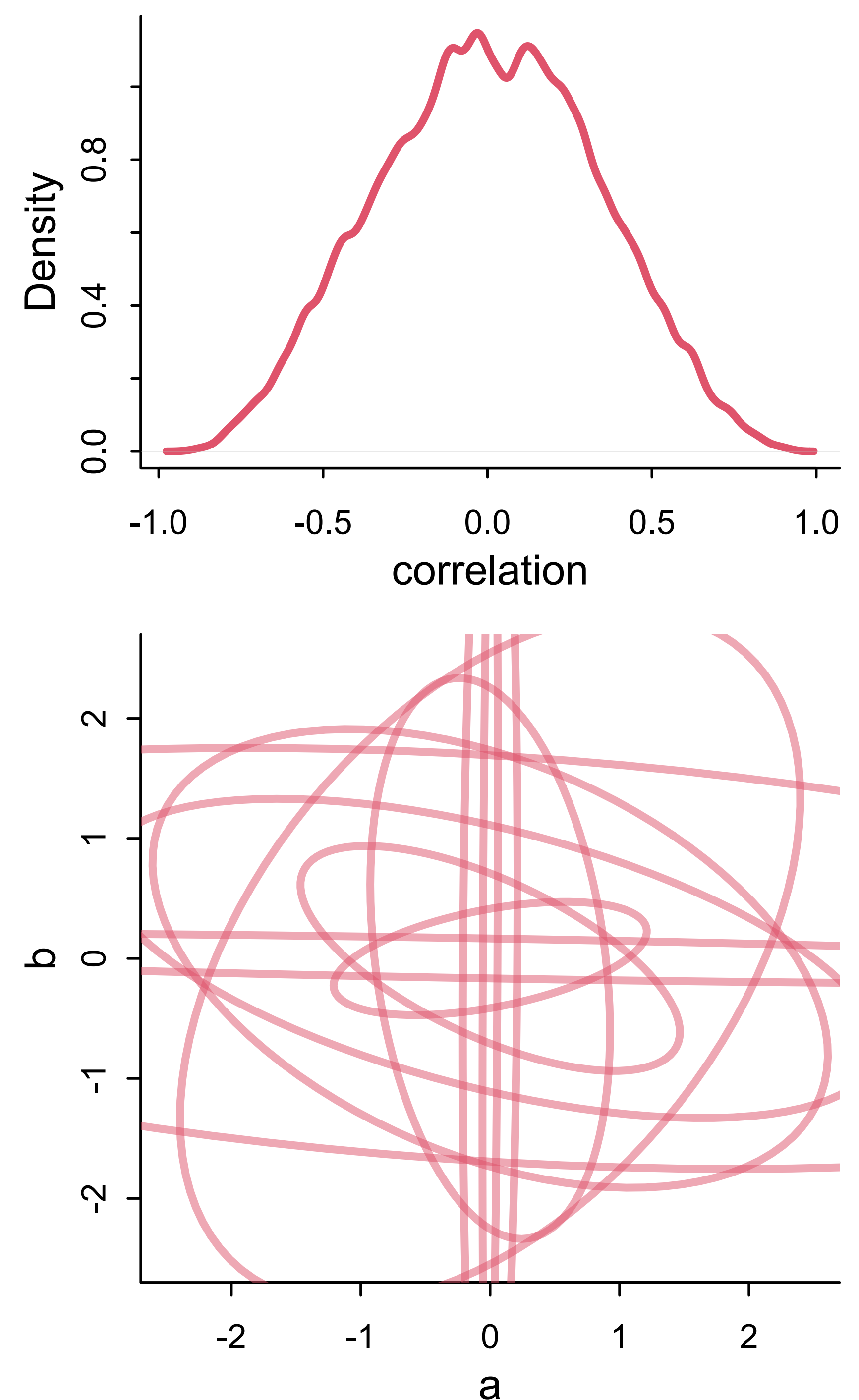
$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \mathbf{R}, [\sigma, \tau])$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0, 1)$$

$$\sigma, \tau \sim \text{Exponential}(1)$$

$$\mathbf{R} \sim \text{LKJCorr}(4)$$

Lewandowski, D., Kurowicka, D. and Joe, H.: 2009, Generating random correlation matrices based on vines and extended onion method, Journal of Multivariate Analysis 100(9), 1989-2001.



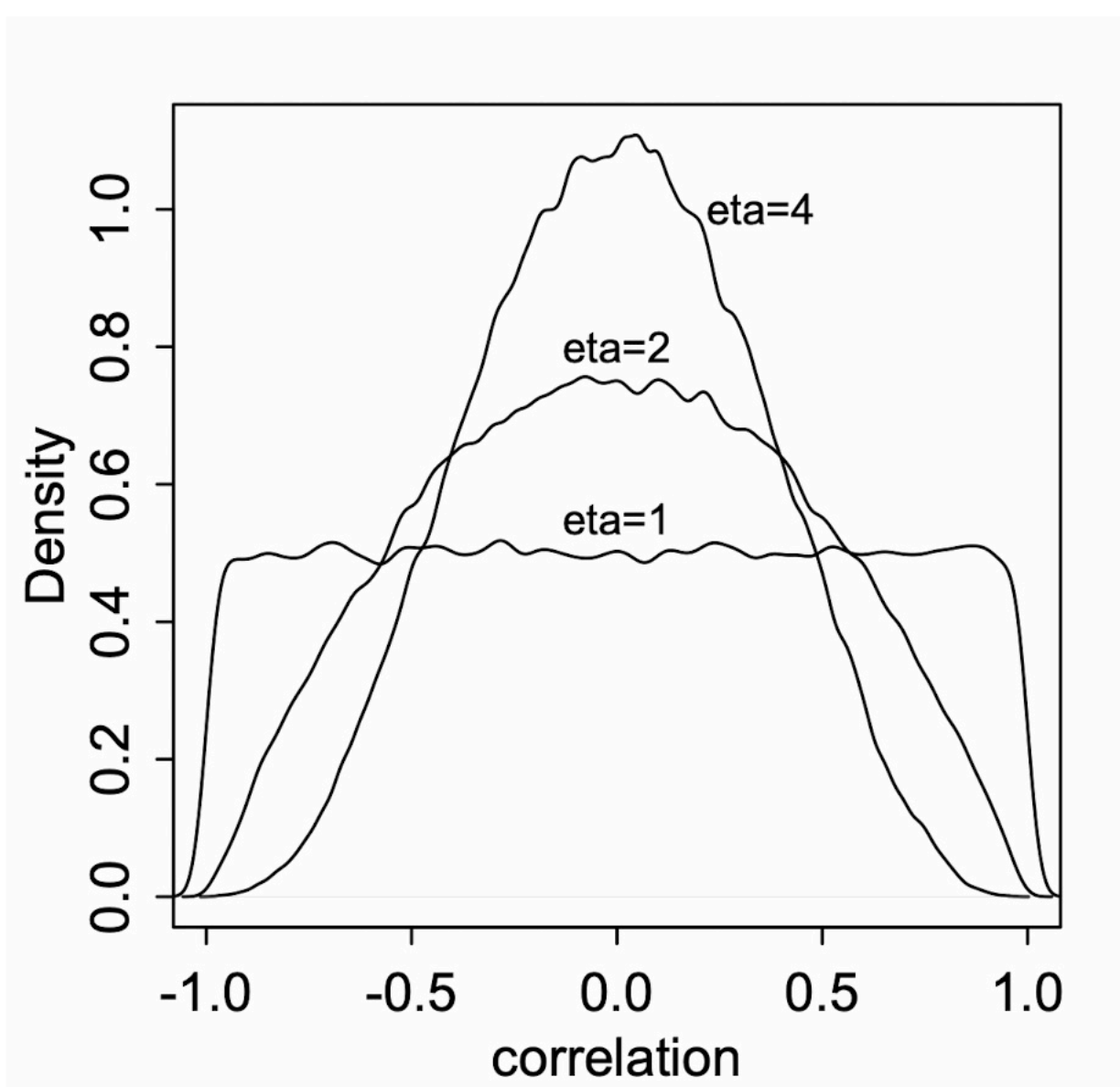


FIGURE 14.3. $\text{LKJcorr}(\eta)$ probability density. The plot shows the distribution of correlation coefficients extracted from random 2-by-2 correlation matrices, for three values of η . When $\eta = 1$, all correlations are equally plausible. As η increases, extreme correlations become less plausible.

```

mCDUcov <- ulam(
  alist(
    C ~ bernoulli(p),
    logit(p) <- a[D] + b[D]*U,

    # define effects using other parameters
    transpars> vector[61]:a <- v[,1],
    transpars> vector[61]:b <- v[,2],

    # priors - centered correlated varying effects
    matrix[61,2]:v ~ multi_normal(abar,Rho,sigma),
    vector[2]:abar ~ normal(0,1),
    corr_matrix[2]:Rho ~ lkj_corr(4),
    vector[2]:sigma ~ exponential(1)
  ) , data=dat , chains=4 , cores=4 )

```

$$C_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{D[i]} + \beta_{D[i]} U_i$$

$$[\alpha_j, \beta_j] \sim \text{MVNormal}([\bar{\alpha}, \bar{\beta}], \mathbf{R}, [\sigma, \tau])$$

$$\bar{\alpha}, \bar{\beta} \sim \text{Normal}(0,1)$$

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61 districts, 2 features each

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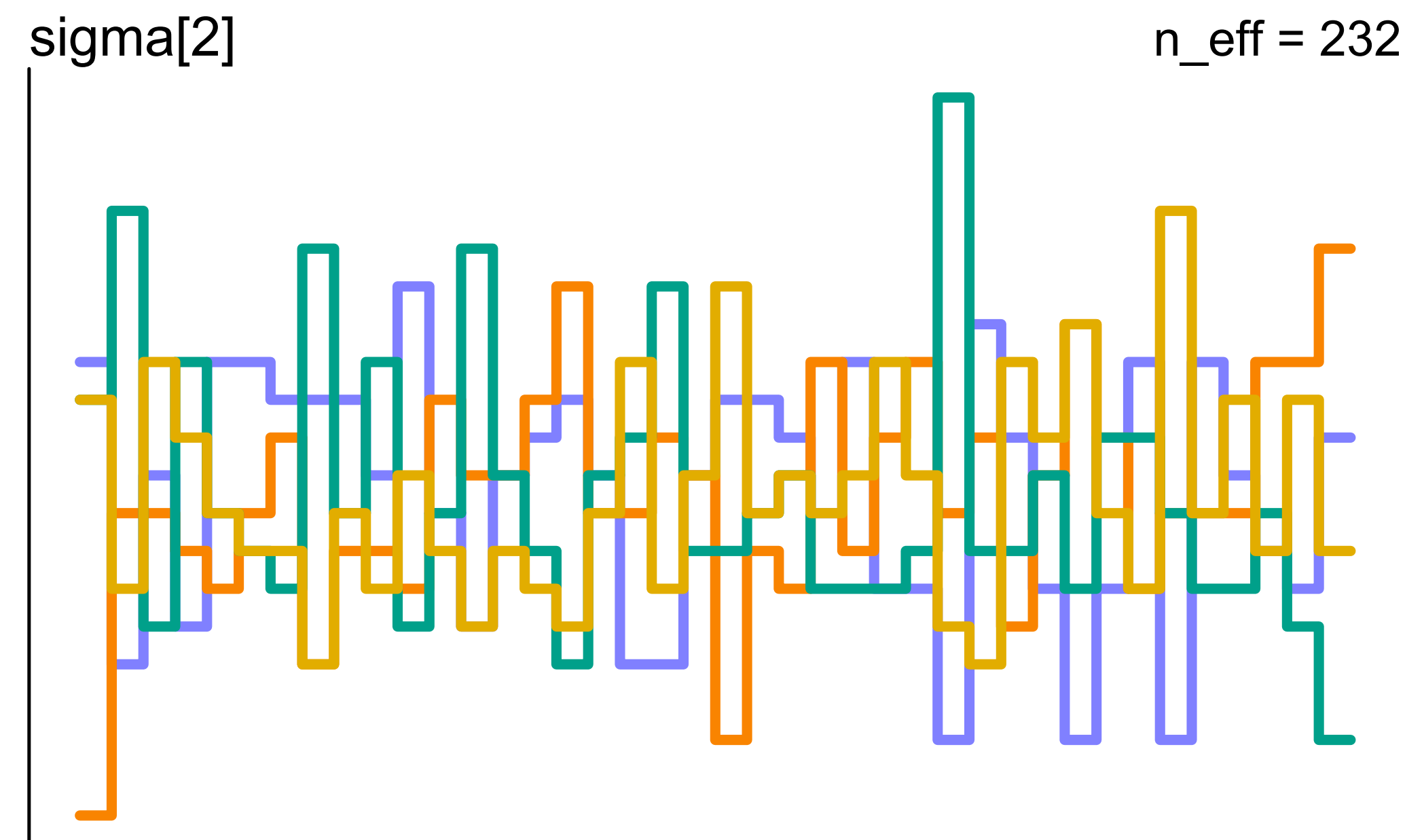
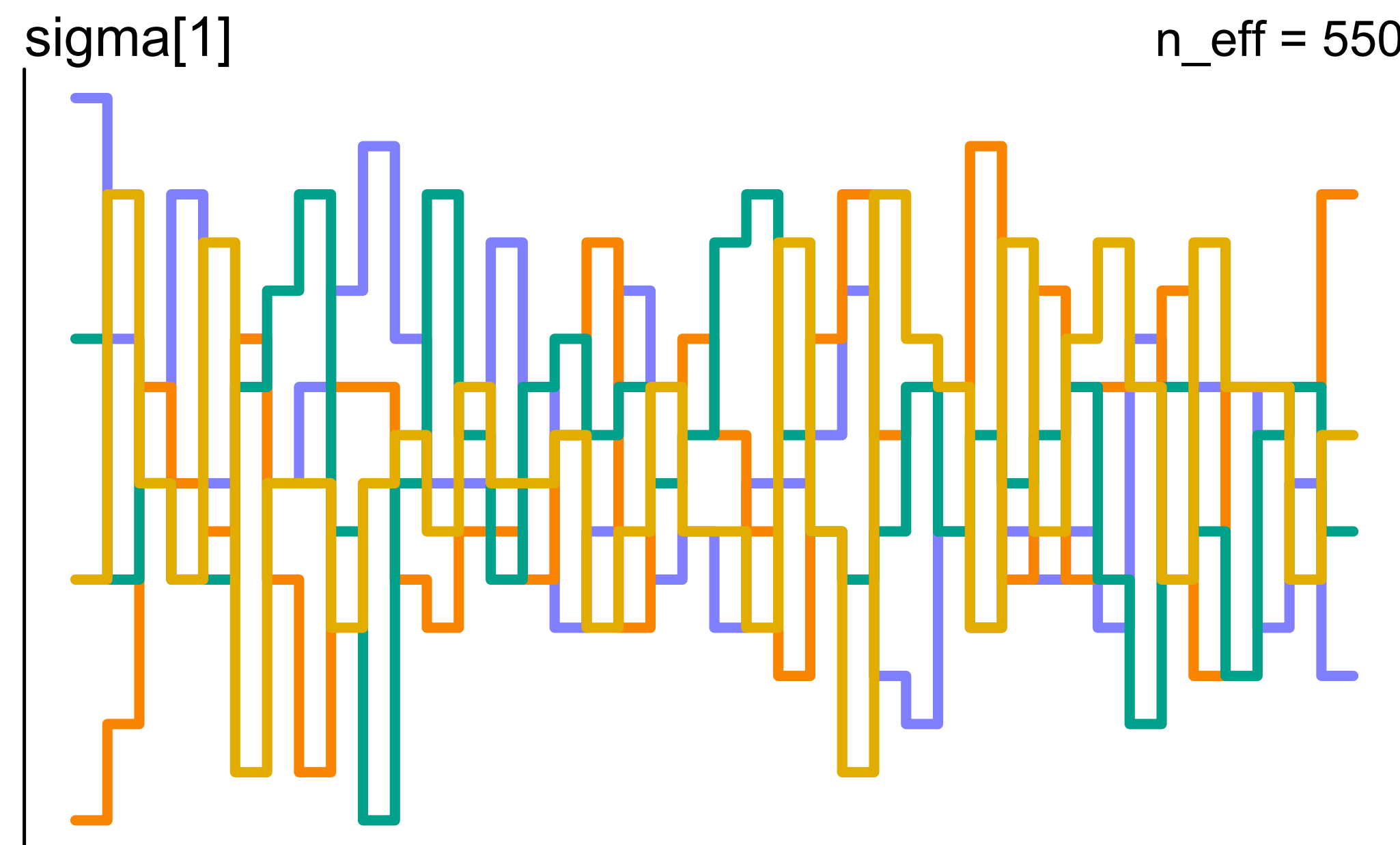
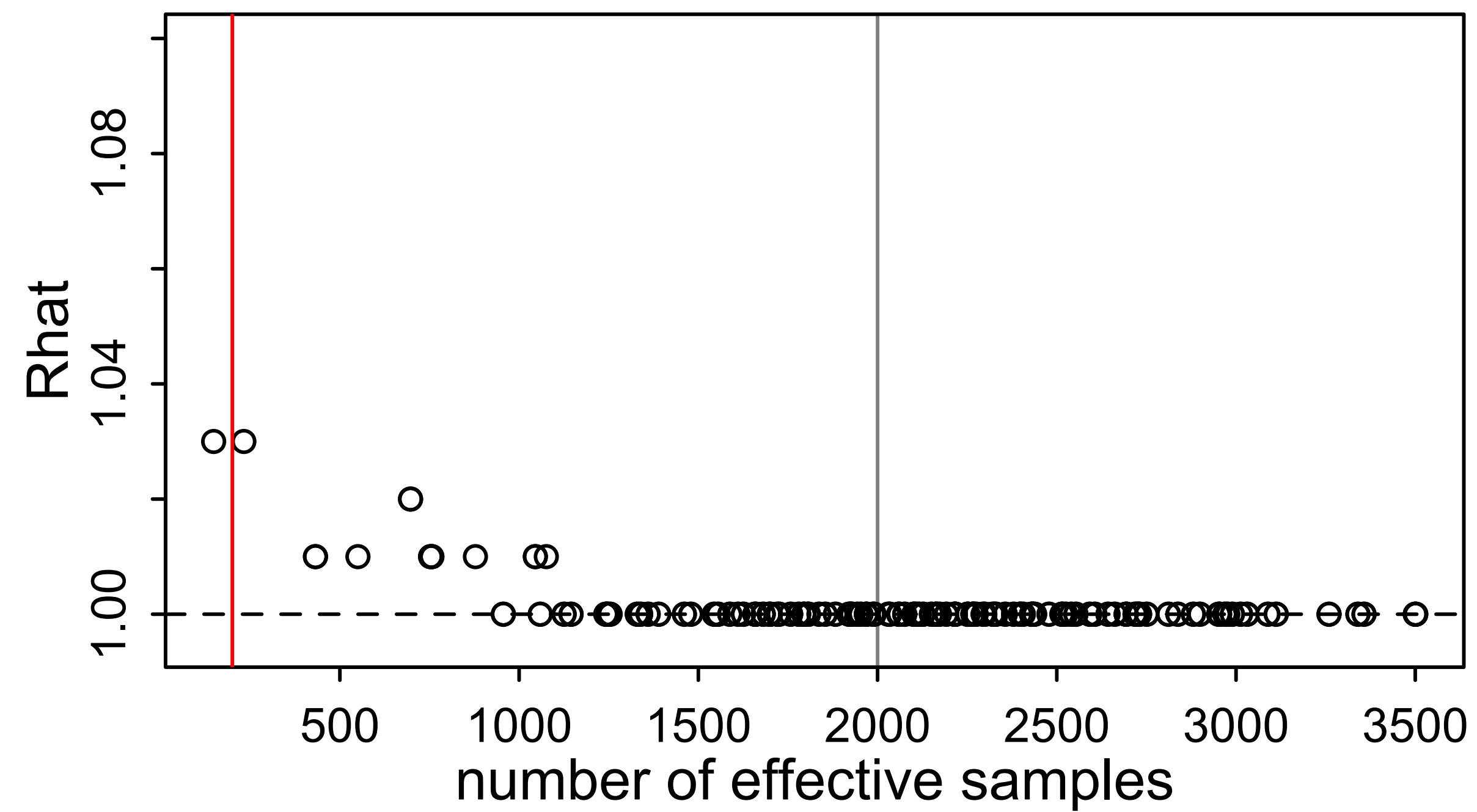
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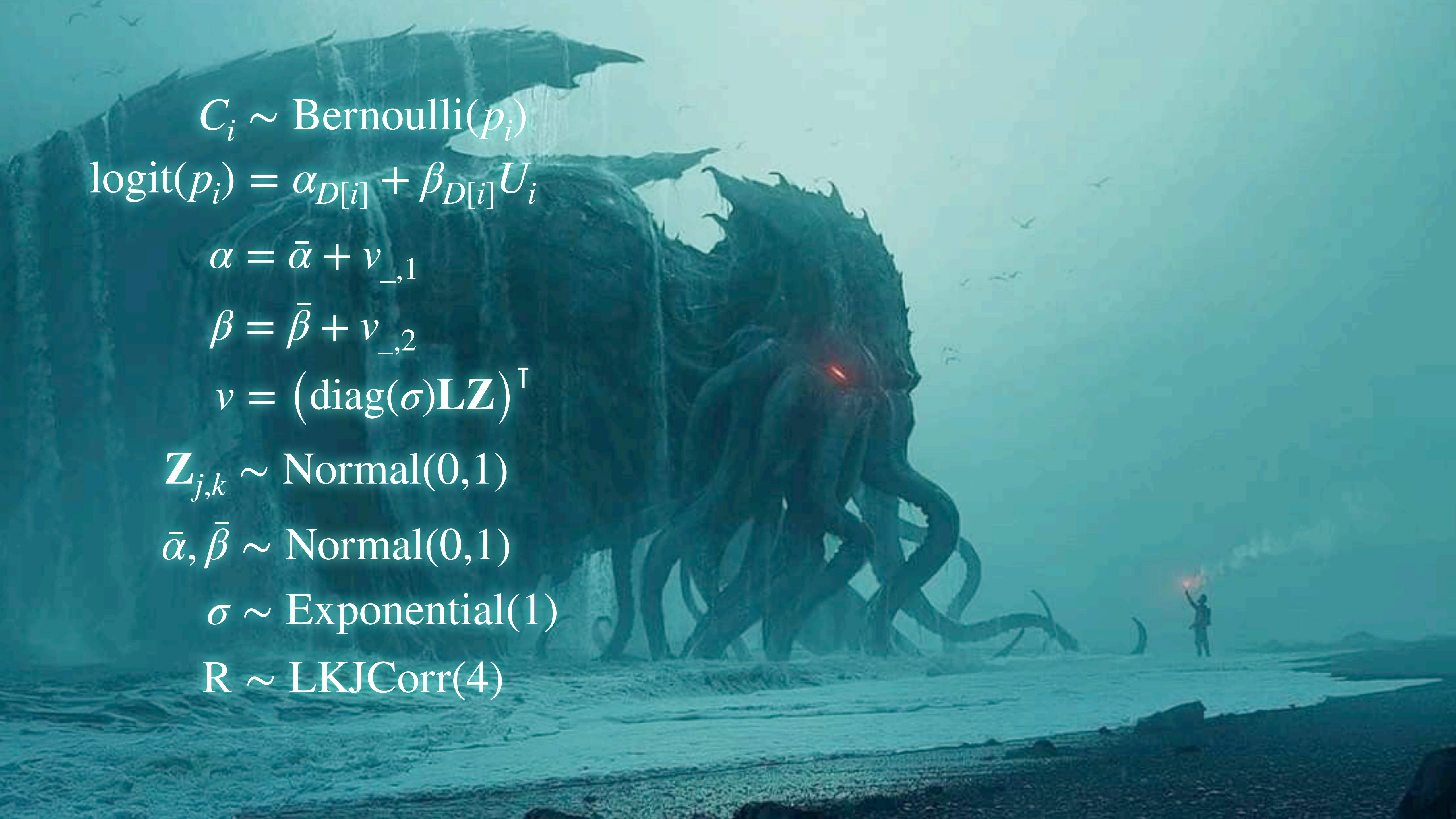
$$\mathbf{v} = (\text{diag}(\sigma) \mathbf{L} \mathbf{Z})^\top$$

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a 61-by-2 matrix $v =$

diagonal matrix of standard deviations

$$\begin{pmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \mathbf{LZ} \end{pmatrix}^T$$

transpose! flips rows and columns

2-by-61 matrix of feature z-scores

Cholesky factor of correlation matrix across features

I. — NOTICES SCIENTIFIQUES

Commandant BENOIT¹.

Note on a method for solving normal equations resulting from the application of the method of least squares to a system of linear equations of smaller number than the unknowns.

(Procédé du Commandant CHOLESKY².)

The artillery commander Cholesky, of the Geographical Service of the army, killed during the Great War, imagined, during research on the compensation of the geodesic networks, a very ingenious process of solving the equations known as normal, obtained by application of the method of the least squares to linear equations in lower number than that of the unknowns.



André-Louis Cholesky (1875–1918)

ques, un procédé très ingénieux de résolution des équations dites *normales*, obtenues par application de la méthode des

```
# define 2D Gaussian with correlation 0.6
N <- 1e4
sigma1 <- 2
sigma2 <- 0.5
rho <- 0.6

# independent z-scores
z1 <- rnorm( N )
z2 <- rnorm( N )

# use Cholesky to blend in correlation
a1 <- z1 * sigma1
a2 <- ( rho*z1 + sqrt( 1-rho^2 ) * z2 ) * sigma2
```

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# use Cholesky to blend in correlation
a1 <- z1 * sigma1
a2 <- ( rho*z1 + sqrt( 1-rho^2 )*z2 )*sigma2
```

```
> cor(z1,z2)
[1] -0.0005542644
> cor(a1,a2)
[1] 0.5999334
> sd(a1)
[1] 1.997036
> sd(a2)
[1] 0.4989456
```

$$\mathbf{v} = \begin{pmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \mathbf{LZ} \end{pmatrix}^T$$

$\text{sqrt}(1 - \text{rho}^2)$ = Cholesky factor for 2 dimensional correlation matrix

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    transpars> vector[61]:b <- abar[2] + v[,2],
    transpars> matrix[61,2]:v <-
      compose_noncentered( sigma , L_Rho , Z ),

    # priors - note that none have parameters inside them
    # that is what makes them non-centered
    matrix[2,61]:Z ~ normal( 0 , 1 ),
    vector[2]:abar ~ normal(0,1),
    cholesky_factor_corr[2]:L_Rho ~ lkj_corr_cholesky( 4 ),
    vector[2]:sigma ~ exponential(1),

    # convert Cholesky to Corr matrix
    gq> matrix[2,2]:Rho <- Chol_to_Corr(L_Rho)
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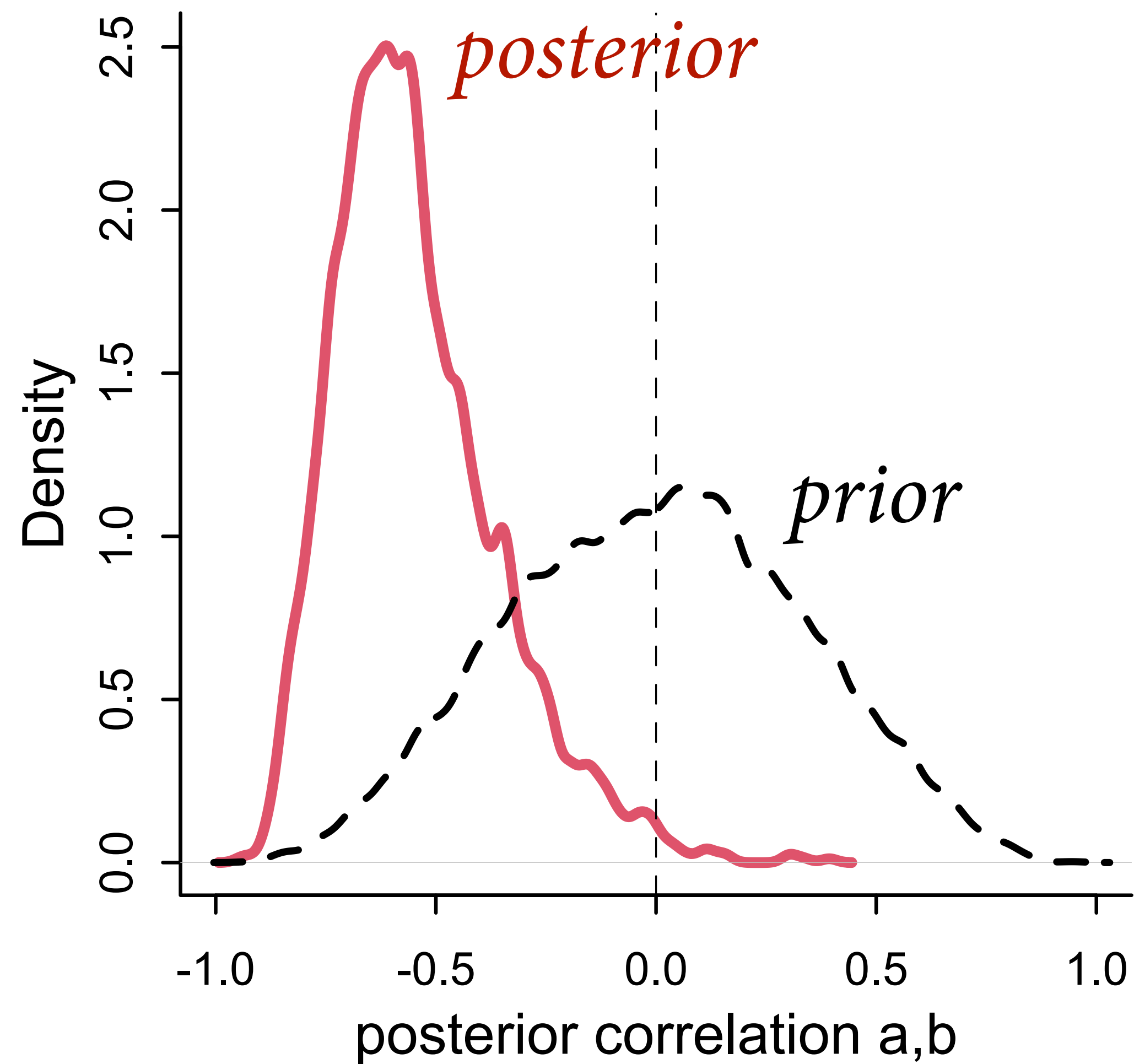
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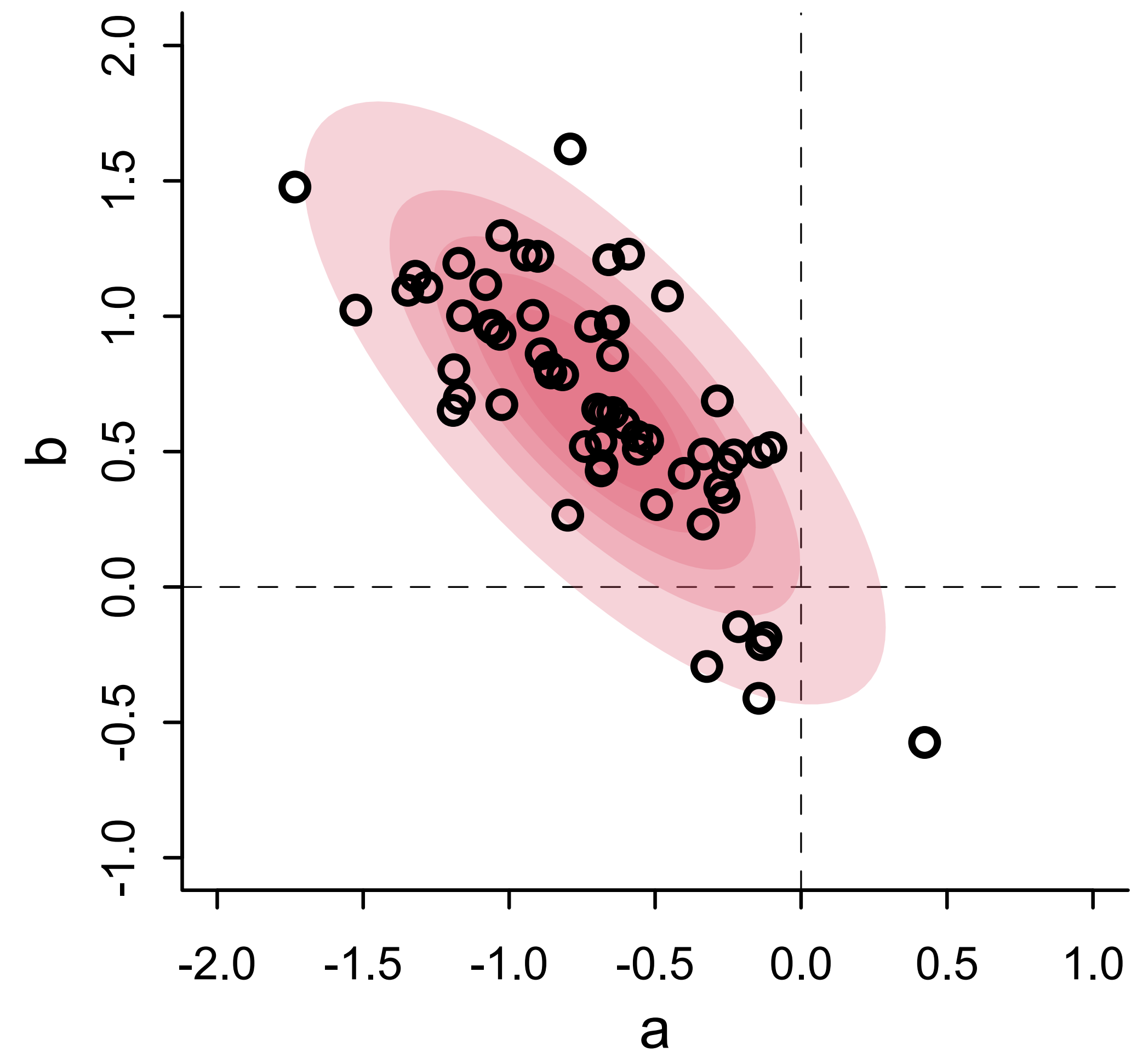
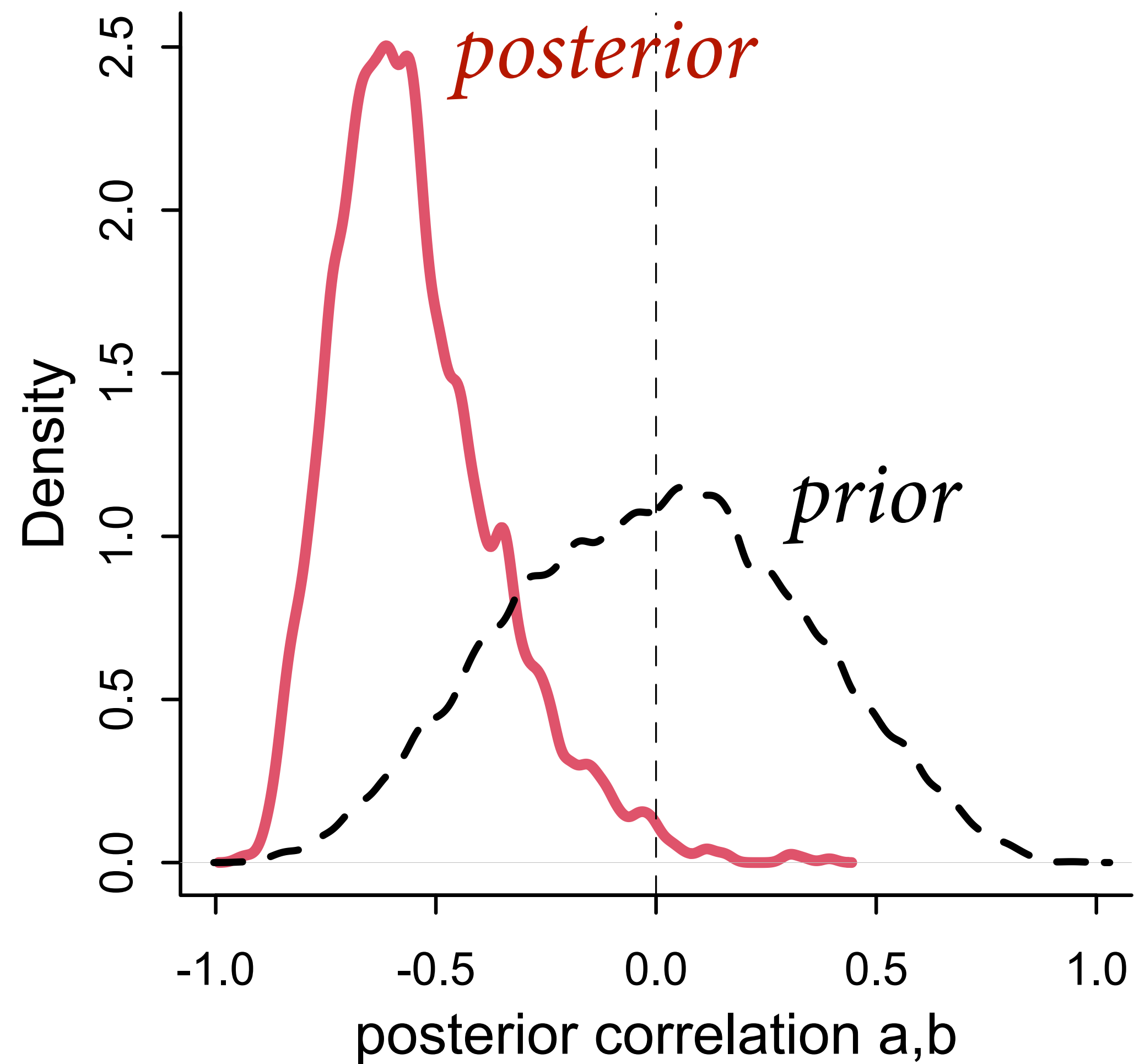
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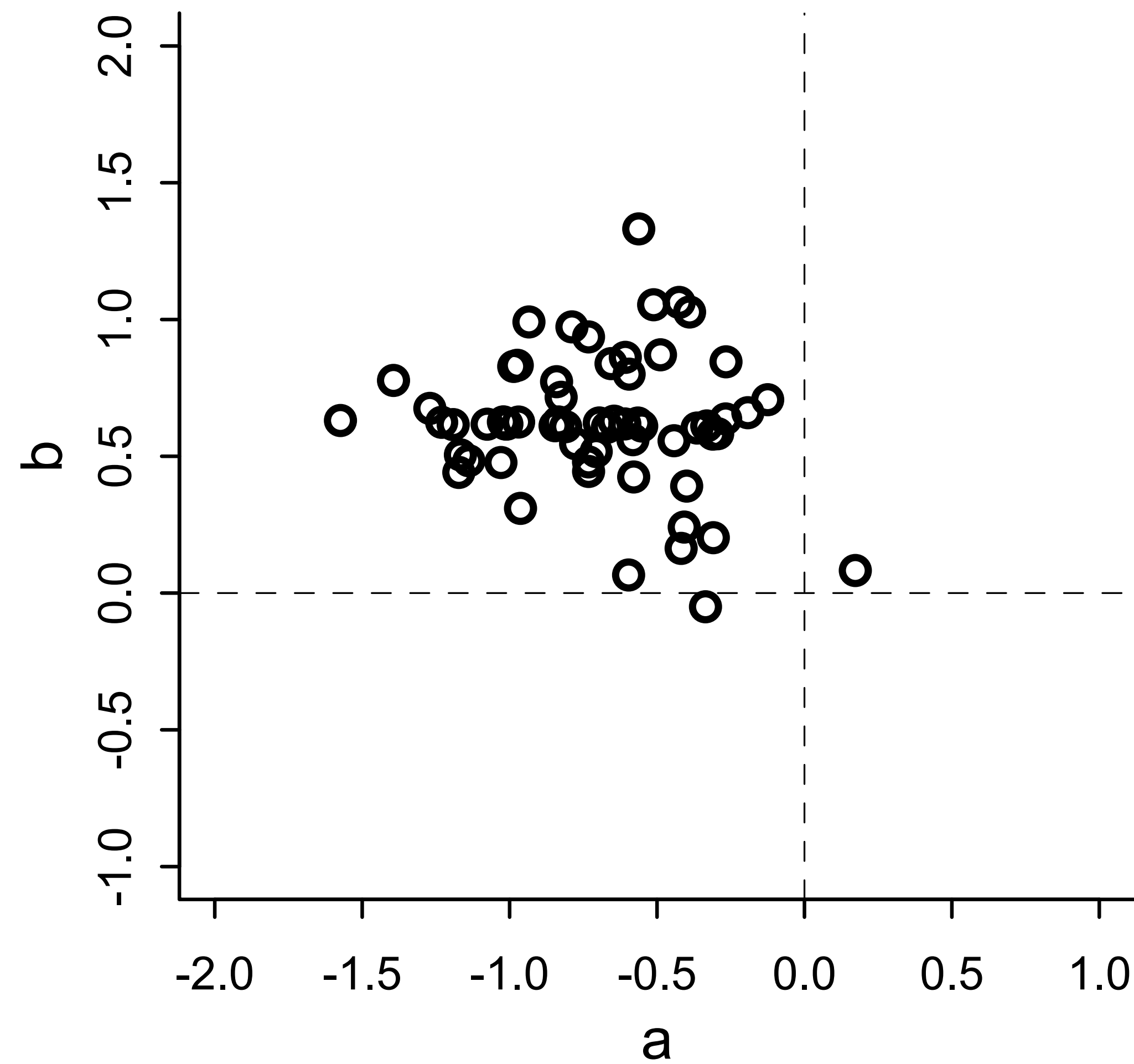
Correlated features



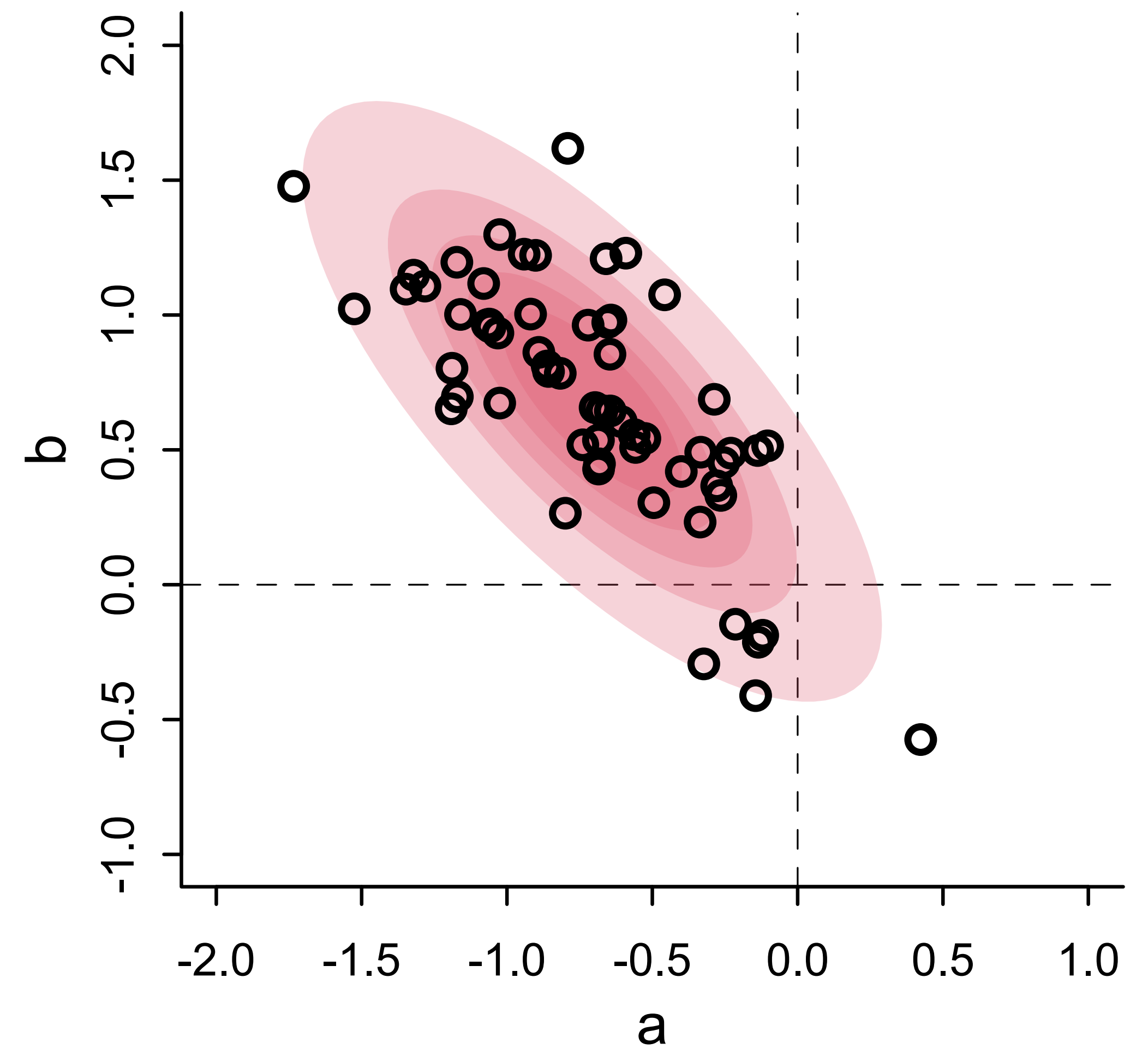
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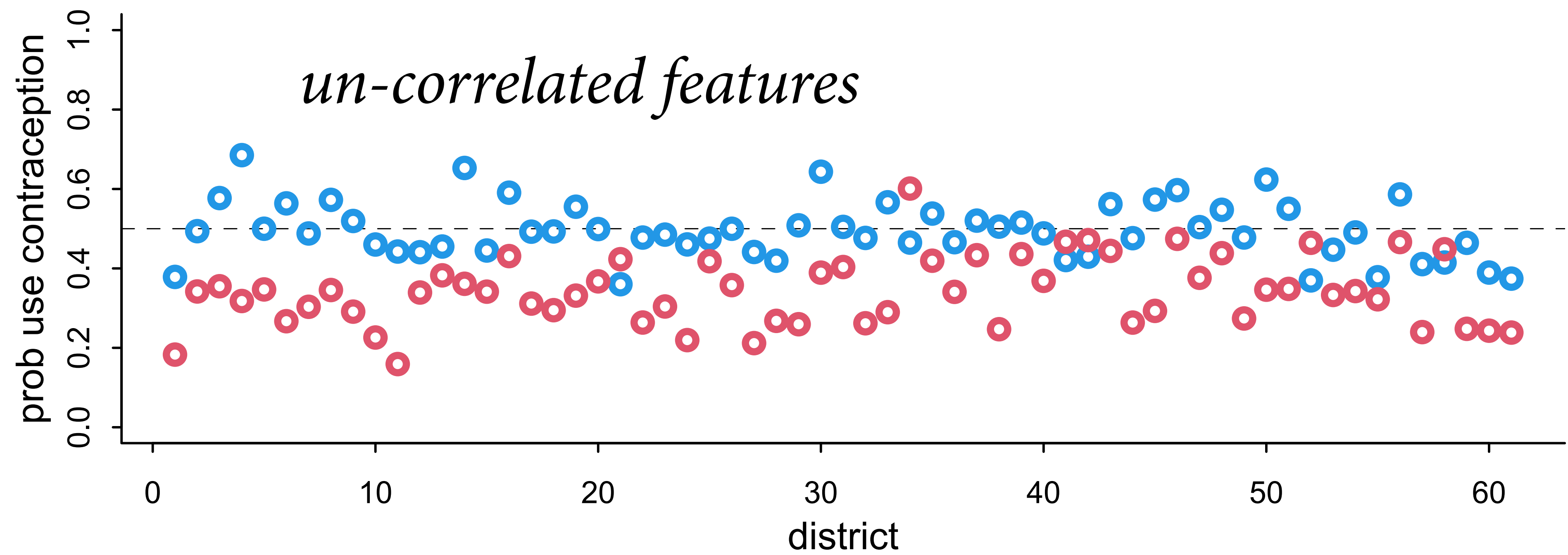
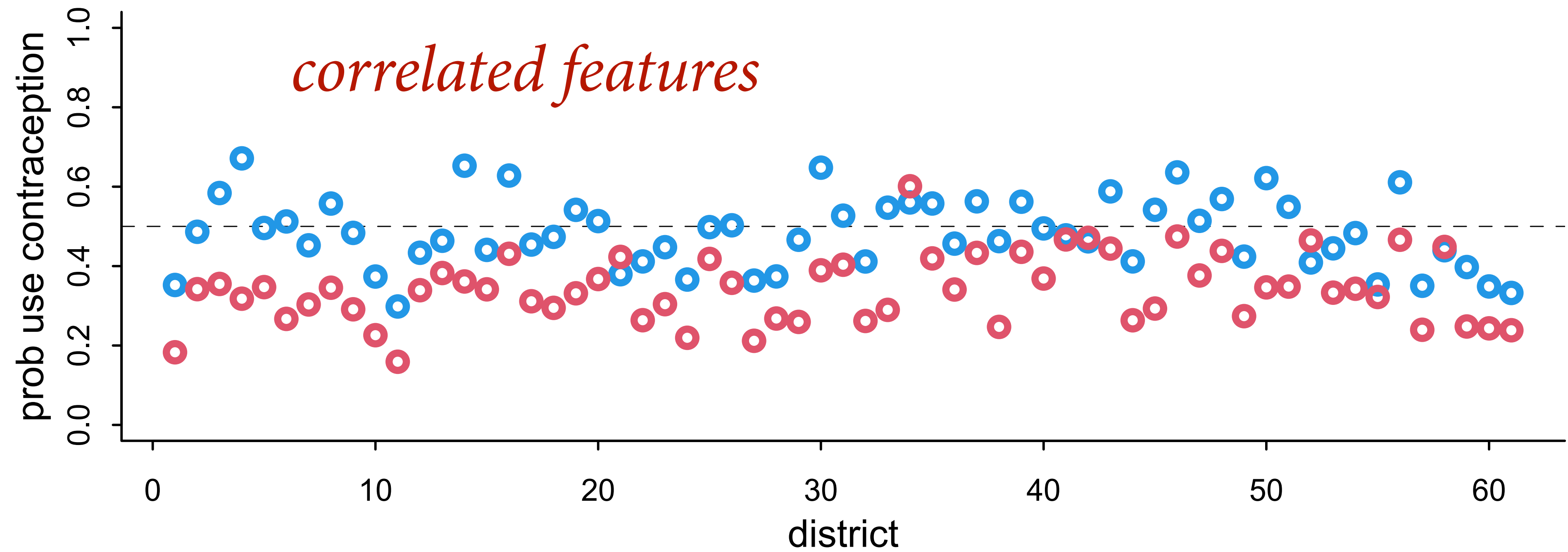
un-correlated features

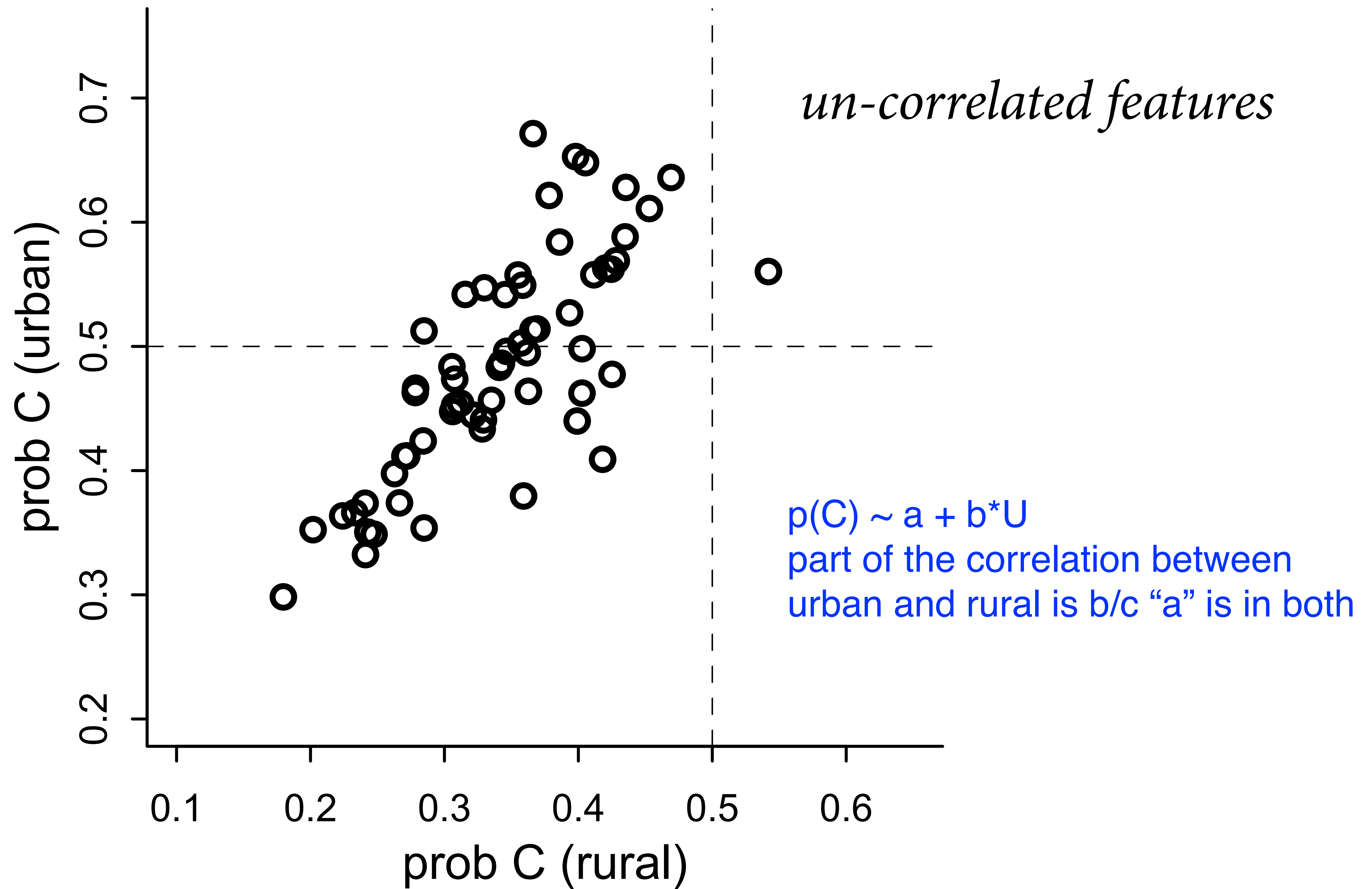


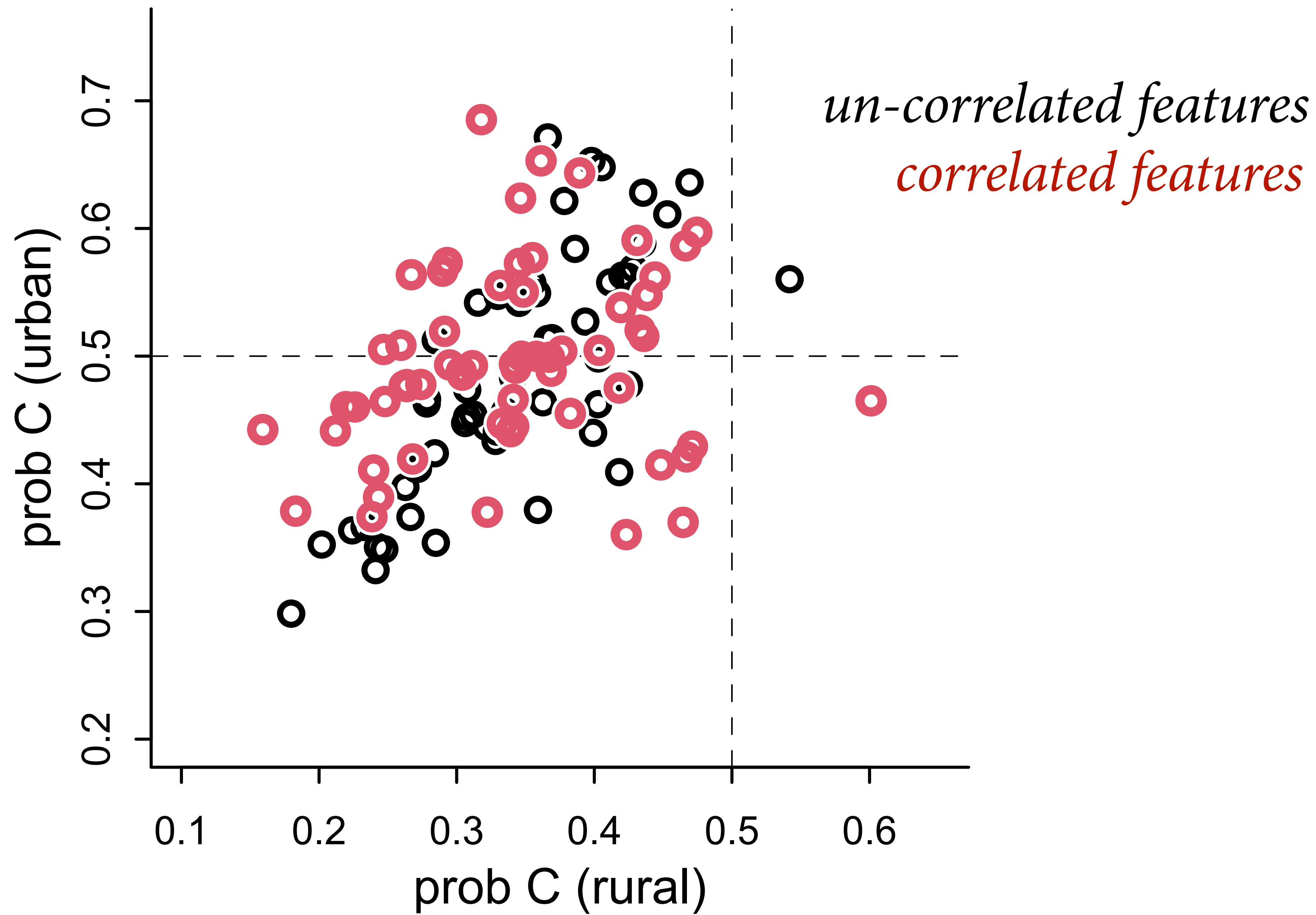
correlated features



$a = \text{rural } p(C); b = \text{difference b/n urban and rural } p(C)$







Correlated Varying Effects

Priors that **learn correlation**

structure:

- (1) partial pooling across features
- (2) learn correlations

$$\alpha_j \sim \text{Normal}(0, 1)$$

$$\beta_j \sim \text{Normal}(0, 1)$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma)$$

$$\beta_j \sim \text{Normal}(\bar{\beta}, \tau)$$

$$\begin{bmatrix} \alpha_j \\ \beta_j \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} \bar{\alpha} \\ \bar{\beta} \end{bmatrix}, \Sigma \right)$$



Correlated Varying Effects

Priors that **learn correlation** structure:

- (1) partial pooling across features
- (2) learn correlations

Varying effects can be correlated even if the prior doesn't learn the correlations

$$\alpha_j \sim \text{Normal}(0, 1)$$

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